GE VERNOVA

ADMS

16.25.1

Distribution Management System (DMS) User Guide

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Change History

Date	Change Description
Jun 1, 2023	ADMS 3.16: • Updated ADMS Toolbar Menus. • Updated Display Symbols. • Updated Finding a Component. • Updated Device Search. • Updated Analyst Search. • Updated Tracing the Network. • Added Trace Upstream. • Added Trace Downstream. • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0 • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Feb 22, 2024	ADMS 16.5.0 • Updated section Topology Processing Results
Apr 05, 2024	ADMS 16.7.0 • Updated document format and styling.
Jun 21, 2024	ADMS 16.9.0 • Added Schematic View Graph Layout Settings
Sep 23, 2024	ADMS 16.12.0 • Updated section adms-tabular-displays. • Updated section Jumpers.

Contents

1. Introduction

ADMS DMS is GE’s solution for real-time management of distribution networks. It is a suite of applications designed specifically to meet the needs of a distribution utility.

DMS presents information in a way that allows better visualization of the status of the distribution network, as well as tools to manage the network effectively.

DMS provides seamless integration of the distribution network view with other systems based on Browser, such as real-time Supervisory Control and Data Acquisition (SCADA) displays.

1.1. An Integrated User Interface for Distribution Operations

The critical feature of an effective real-time network management system is that it must provide the operator with the most comprehensive and accurate view of the actual situation.

In this regard, the combination of the distribution operations model and the distribution user interface plays an important role by providing an integrated environment to collect dynamic data from a variety of sources. It also allows advanced applications to use this data in real time and to present consistent information to the operator.

DMS offers the following features:

- The network view combines a dynamic power system equipment model and geospatial data into an intelligent, real-time electronic mapboard that can be viewed from individual screens or from large-scale displays.
- DMS is fully integrated with GE’s Scada product for monitoring, controlling, and tagging devices.
- Model updates go online without interrupting users.
- DMS leverages the existing asset management model (usually GIS) as its data source for model building and maintenance.
- Study mode allows private "what if" analysis of proposed switching or operational procedures.
- Accurate phase modeling of the distribution network allows all types of network configuration to be supported.
- The Distribution Network Object Model (DNOM) is highly scalable and capable of supporting distribution of all sizes.

DMS uses Browser, GE’s standard user interface environment. Browser is a web-based browser that provides navigation to Distribution Management System (DMS) views, real-time supervisory control and data acquisition (SCADA) displays, archive forms, and any web-enabled applications.

The image below shows an example of how multiple views are presented in DMS integrated user interface.

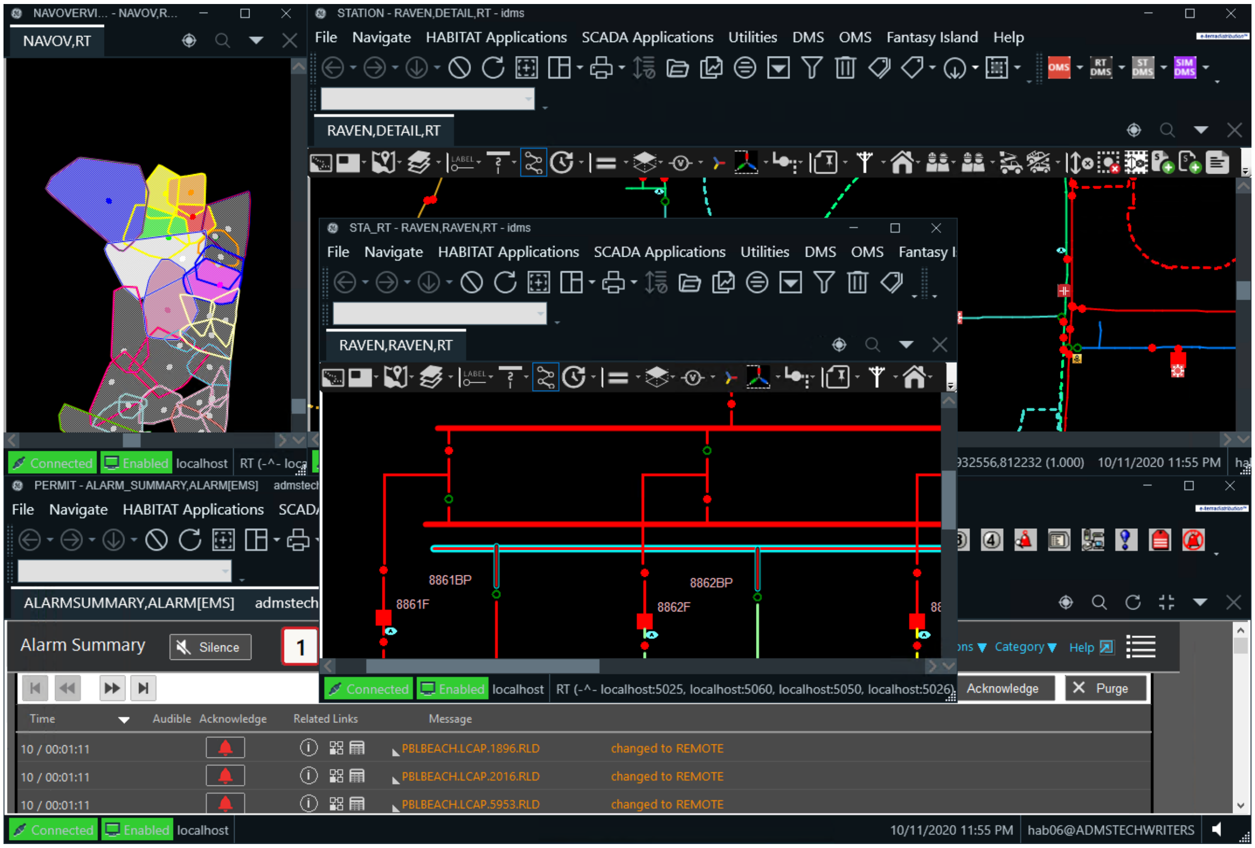


Figure 1. ADMS Integrated User Interface

With DMS, the user interface provides a range of tools that are specifically designed to support a distribution operator, both in functionality and in speed of response. In addition to the static model, the distribution model also integrates dynamic data from SCADA, manual updates, or external systems. Changes to the network topology, such as de-energization or parallel loops, are automatically highlighted. The real-time network model serves all of the user interface clients so that all users see any change to the network immediately. In large systems, users can each be looking at totally separate portions of the network. Overviews of the entire system are also available for easy and quick navigation.

1.2. The Web-Enabled Browser Framework

Browser is the user interface software developed by GE as a solution for control centers. It provides operators with a high-performance, full-graphics, web-based interface to all applications in the real-time control system, along with navigation to nearly all applications featuring a web user interface.

Interfaces to intranet and/or Internet applications using HTML pages or web controls can be natively displayed by Browser and used to enrich the DMS user interface.

The following figure shows the main components of the Browser client/server system and the data/display flow.

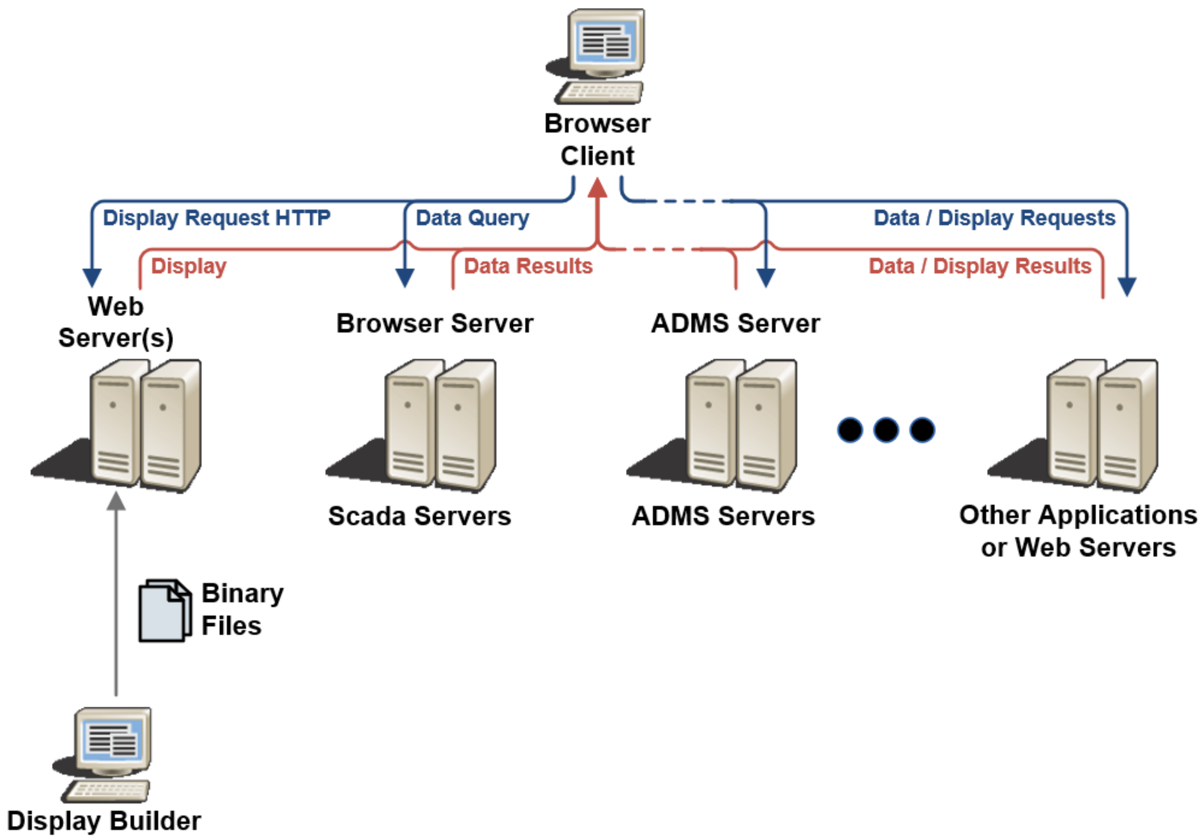


Figure 2. Browser Framework Overview

The image below illustrates the integration of a geographic view of the distribution network showing feeder coloring based on real-time topology, a detailed one-line showing SCADA measurements, and an alarm panel at the bottom that are all in the same set of Browser viewports. This arrangement of viewports can be saved as a room to be recalled later.

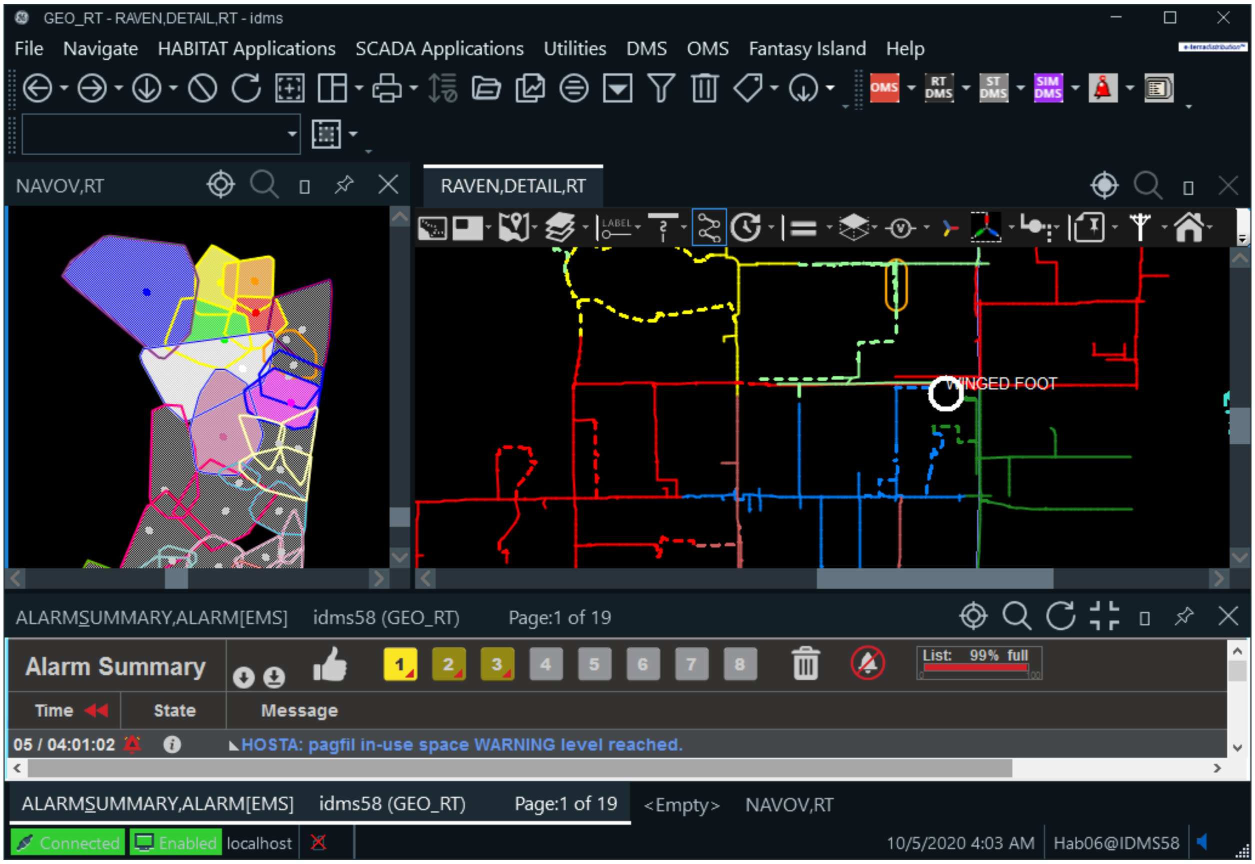


Figure 3. Browser Web Capability

1.3. Dedicated Views for Distribution Operators

DMS extends Browser's functionality by incorporating a number of display types specifically designed for distribution operations. The primary display is the geographic network view, which shows the complete network in a purely geographic view with the addition of real-time status and electrical topology information. This display can be quickly zoomed, panned, or re-centered on another part of the network to provide the operator with a high-performance tool for monitoring network operations.

The DMS geographic network view provides a unique platform for distribution operations. A detailed geographic view of the complete distribution network is combined with real-time data from both the SCADA system and manual operations. The topology processor updates displays to show the effects of switching. The integration of this network view with a full suite of functions provides a complete real-time network management system.

Other display types include automatically generated schematic displays and a full set of tabular displays that are configured for displaying operational exception and power flow information.

2. User Interface Description

This chapter describes the following aspects of ADMS:

- User interface layout
- Toolbars and menus
- Network views
- Tabular displays

2.1. User Interface Layout

2.1.1. Viewports and Panes

All displays in this system are built on the Browser environment. This provides a consistent display structure for all of the control systems, from transmission control Energy Management System (EMS) through SCADA control of substations, to the full DMS and its geographic network views. Each of the different displays brings with it a toolbar that provides a range of special functions.

The overall display environment is built up from a number of viewports and associated panes. For most operational configurations, the displays are spread over a number of monitor screens, as shown in image below.

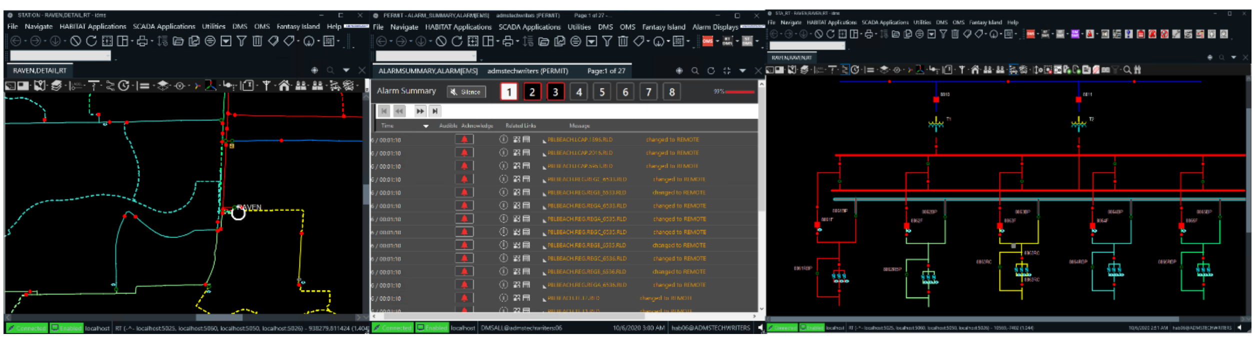


Figure 4. Typical Three-Monitor Display Environment

2.1.2. Browser Main Viewport

The layout of each viewport follows a standard structure, as shown in below image. However, Browser supports a wide range of GE applications; therefore, the actual content of the menus and toolbars varies. The main body of the display is dedicated to the currently running application and may consist of tabular, graphical, or schematic displays with specific toolbars and functionalities.

> **Note**
>
> The status messages field normally shows the status of the connection to the servers, but it

also momentarily shows the status of operations (for example, the number of customers and transformers that have been passed during a trace) after they have been completed.

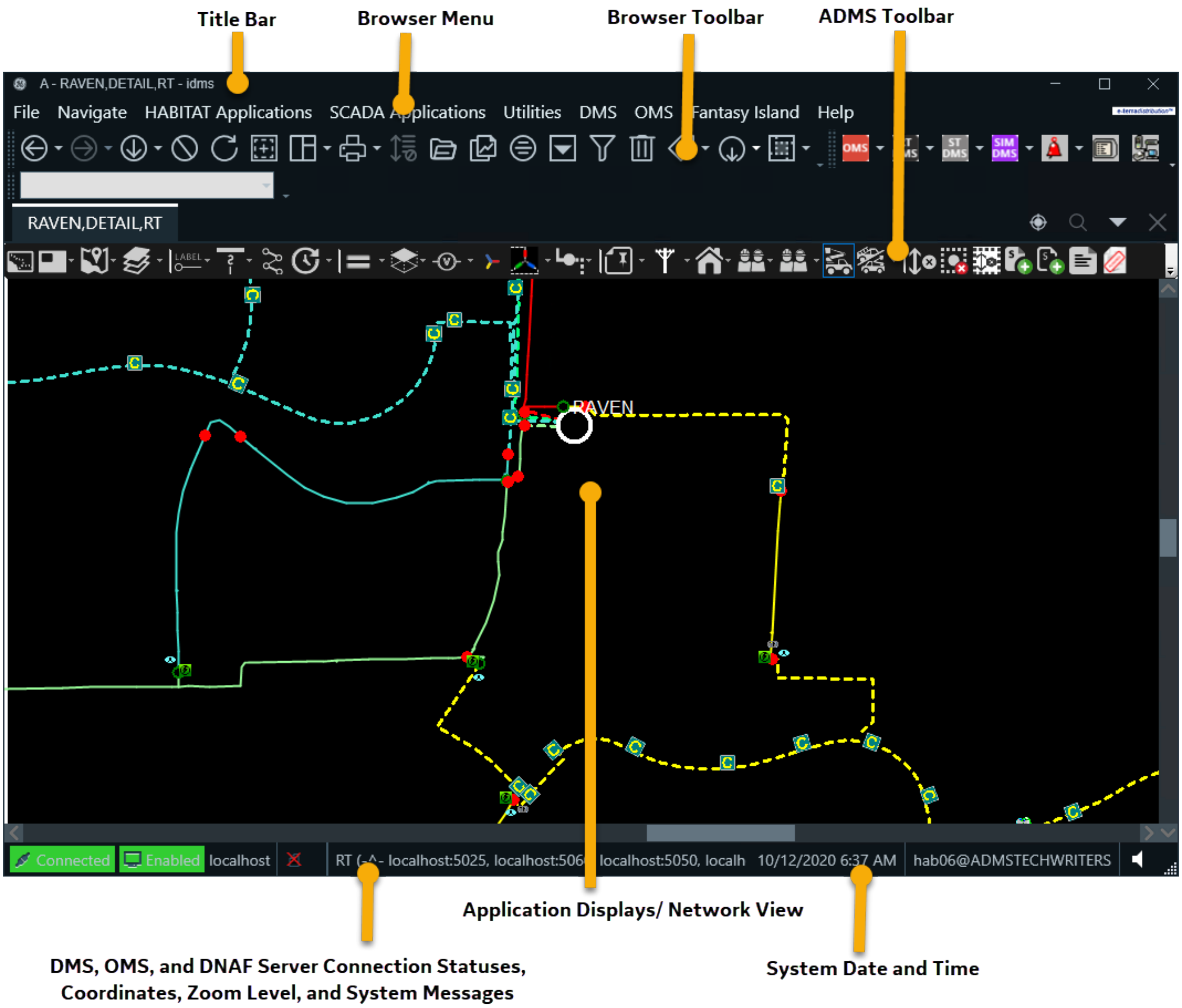


Figure 5. Browser Viewport Layout

2.1.3. ADMS Display Types

There are a number of different types of displays that are used by the ADMS applications. These are both standard tabular and schematic displays used in other GE applications, such as Scada, and a set of graphical network views that have been developed for the ADMS product.

2.1.3.1. Area Overview

The area overview display shows the coverage (footprint) for each station in the given area (service center). Dots indicate station locations. This display provides an easy way to open a new viewport that is directed at the station or area of interest.

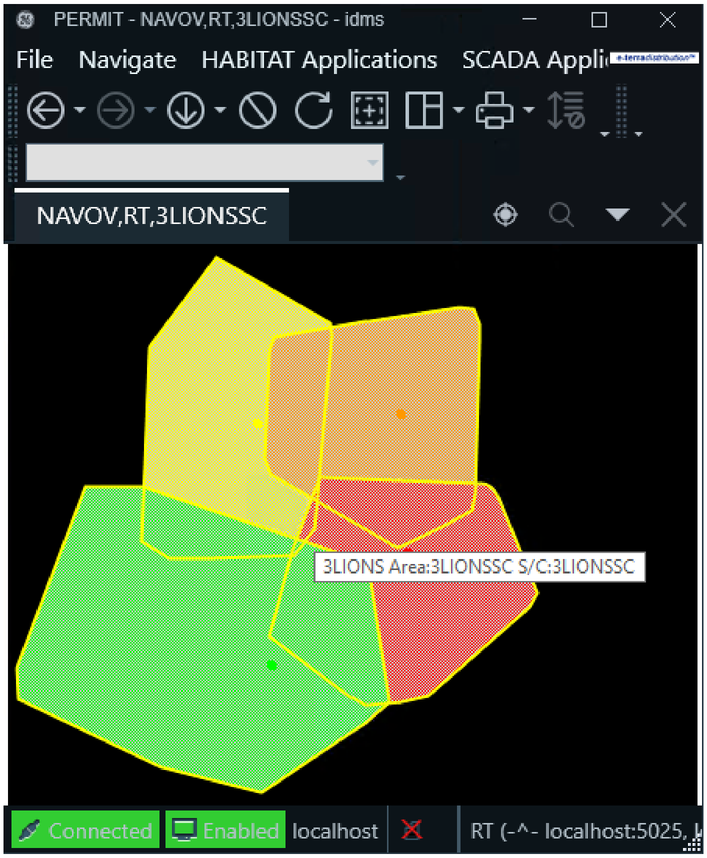


Figure 6. Area Overview

Clicking a point on the area overview display opens a new viewport that shows a geographic view centered on that point.

2.1.3.2. Geographic View

The geographic network view provides a complete and accurate representation of the distribution network outside the station fence. It is an accurate geographic display of the feeders, laterals, and all connected devices.

This display is also capable of showing other physical information such as streets, roads, buildings, waterways, and municipal boundaries.

The geographic view also provides the starting point for most operational tasks. The current states of energization of lines and switches are shown. The operator can navigate to schematic, tabular, and other detailed views of any part of the network.

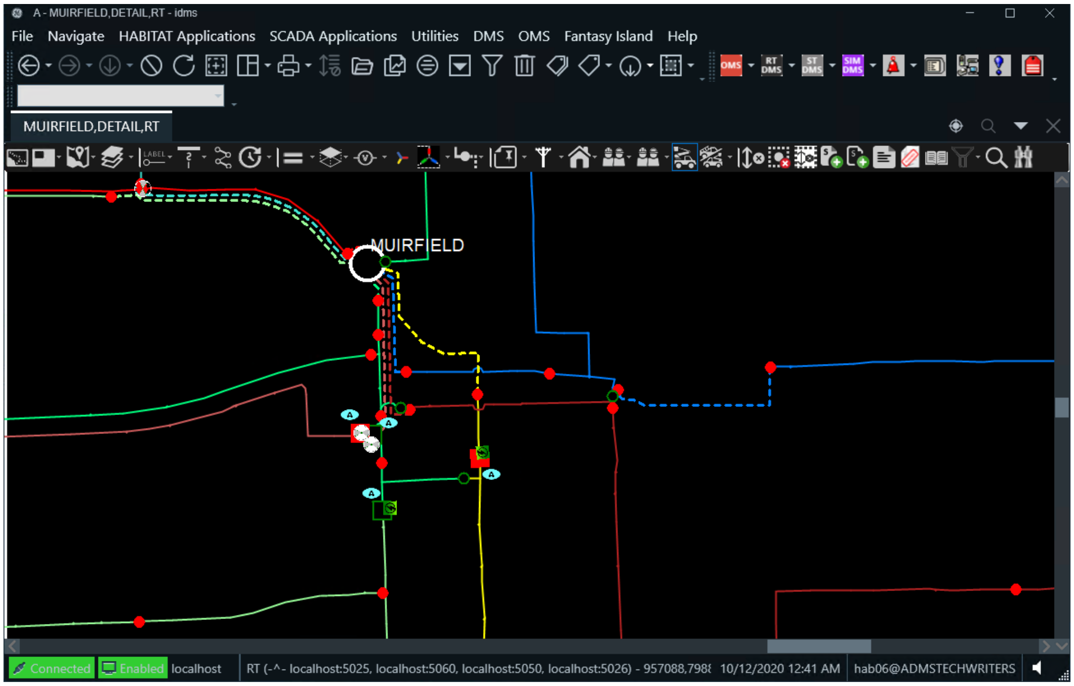


Figure 7. Geographic View of the Distribution Network

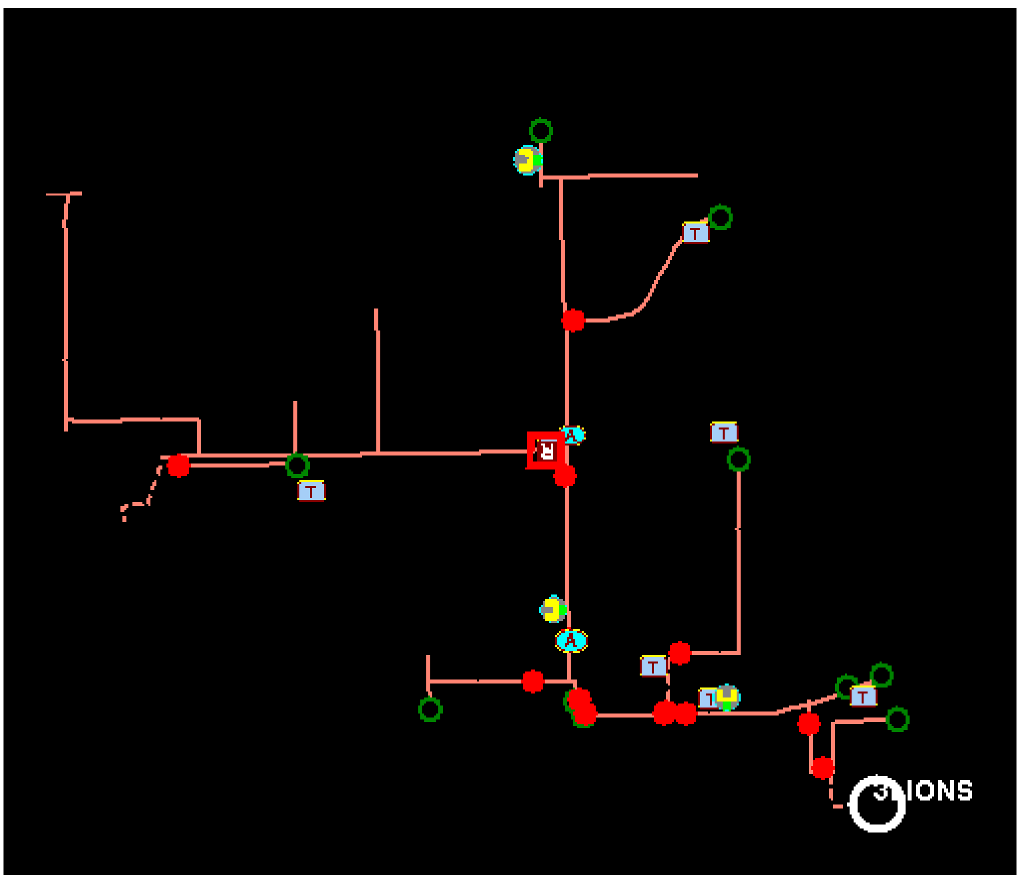


Figure 8. Filtered Feeder View

2.1.3.3. Schematic Views

Schematic views of the network provide an alternate and more easily understood display of devices that are geographically crowded together, such as an Underground Residential Distribution (URD) loop.

Schematic displays are automatically generated "on the fly", and they present the devices in a standardized format.

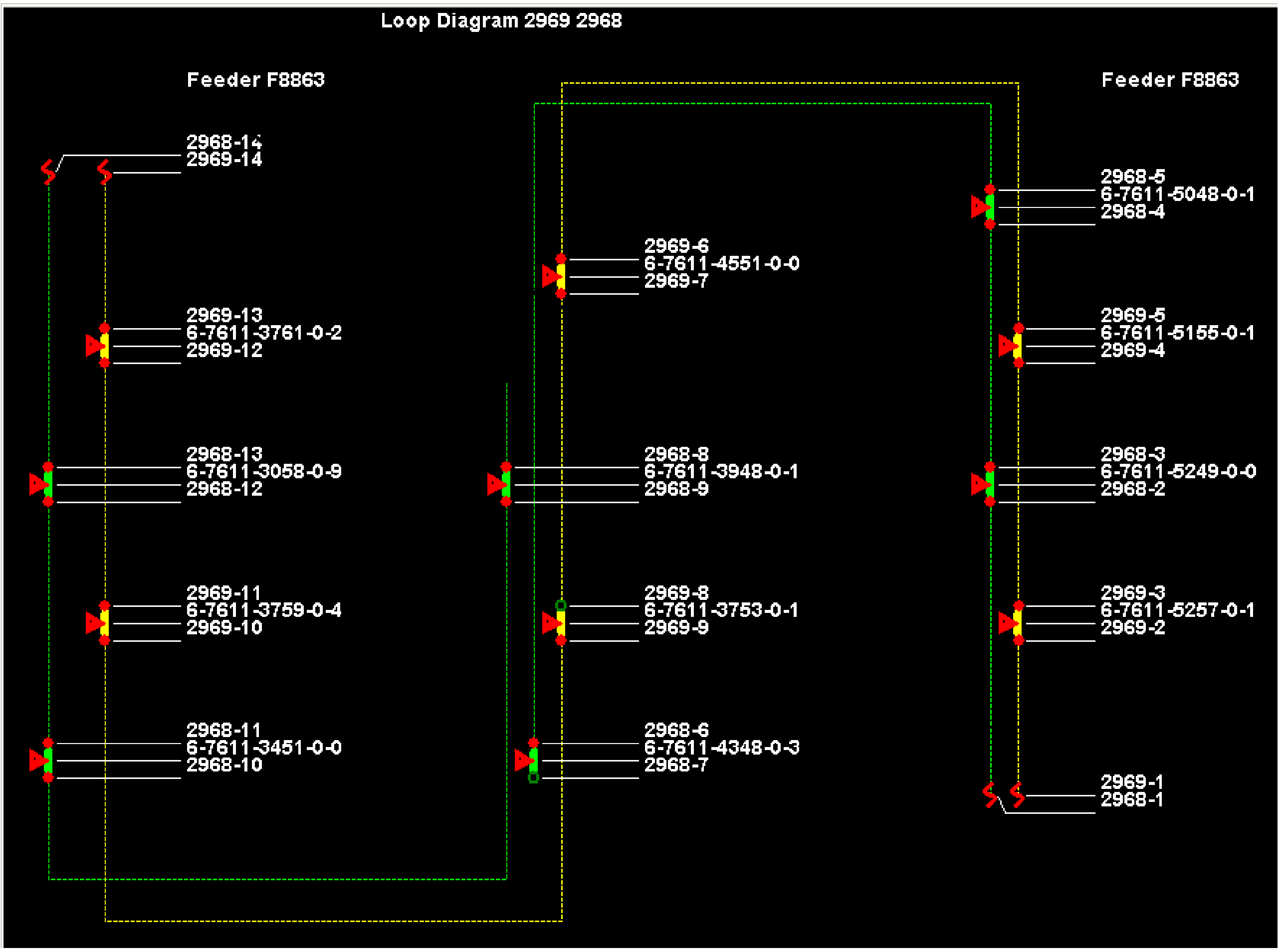
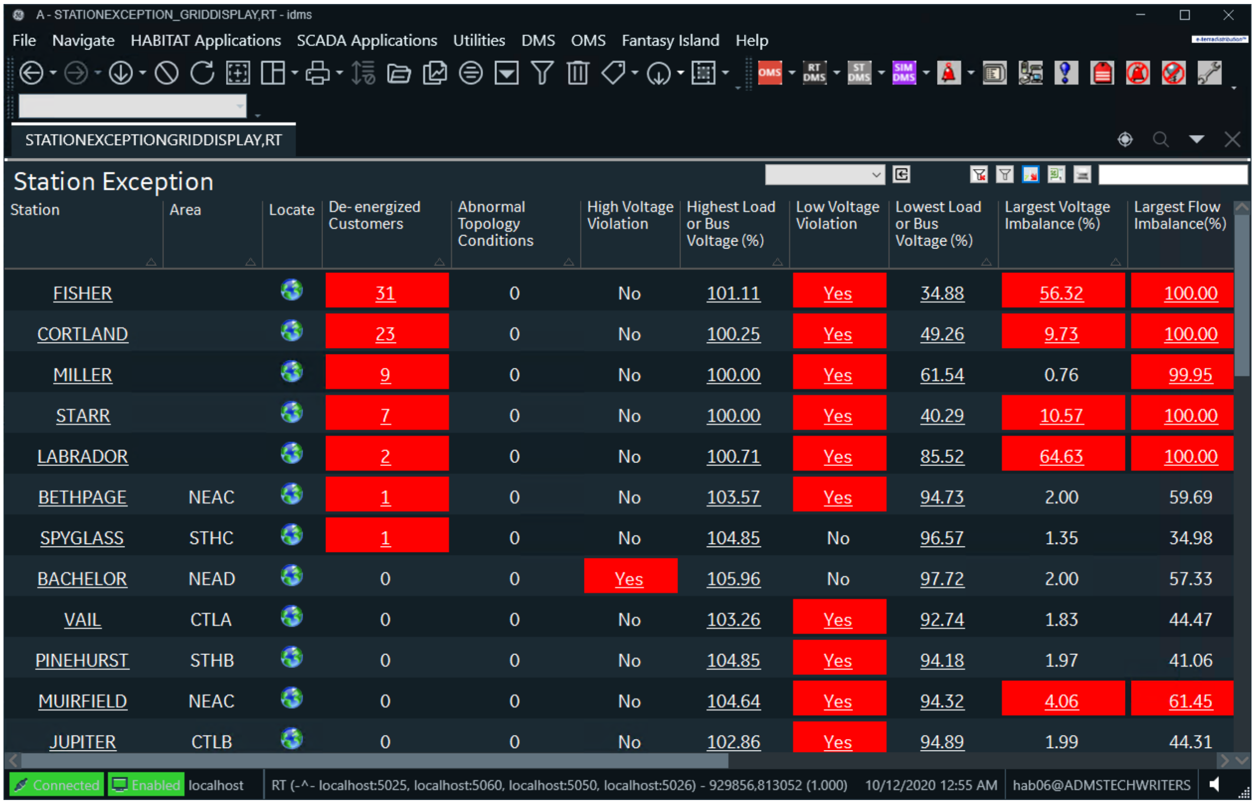


Figure 9. Schematic Display of an Underground Residential Distribution Loop

2.1.3.4. ADMS Tabular Displays



Station	Area	Locate	De-energized Customers	Abnormal Topology Conditions	High Voltage Violation	Highest Load or Bus Voltage (%)	Low Voltage Violation	Lowest Load or Bus Voltage (%)	Largest Voltage Imbalance (%)	Largest Flow Imbalance(%)
FISHER			31	0	No	101.11	Yes	34.88	56.32	100.00
CORTLAND			23	0	No	100.25	Yes	49.26	9.73	100.00
MILLER			9	0	No	100.00	Yes	61.54	0.76	99.95
STARR			7	0	No	100.00	Yes	40.29	10.57	100.00
LABRADOR			2	0	No	100.71	Yes	85.52	64.63	100.00
BETHPAGE	NEAC		1	0	No	103.57	Yes	94.73	2.00	59.69
SPYGLASS	STHC		1	0	No	104.85	No	96.57	1.35	34.98
BACHELOR	NEAD		0	0	Yes	105.96	No	97.72	2.00	57.33
VAIL	CTLA		0	0	No	103.26	Yes	92.74	1.83	44.47
PINEHURST	STHB		0	0	No	104.85	Yes	94.18	1.97	41.06
MUIRFIELD	NEAC		0	0	No	104.64	Yes	94.32	4.06	61.45
JUPITER	CTLB		0	0	No	102.86	Yes	94.89	1.99	44.31

Figure 10. ADMS Tabular Display

ADMS tabular displays are accessed from the ADMS menus. They show information about the distribution network: abnormal switches, voltage violations, and power flow results. ADMS tabulars can display both real-time and study mode results.

> **Note**
>
> ADMS tabulars do not support more than 50,000 rows.

2.1.3.5. Picture-in-Picture View

The picture-in-picture view allows a small window to be opened in one corner of the main screen. The zoom level of this window can be set independently of the main screen. This allows a small cluster of switches to be viewed in detail while still maintaining a broader view of the whole feeder and its neighbors. An example of this is shown in image below.

The picture-in-picture window can be repositioned to any corner of the main screen.

The picture-in-picture window can be set to either center on the currently selected device, or to follow the view at the center of the main screen. The two modes can be selected by toggling the button on the picture-in-picture toolbar.

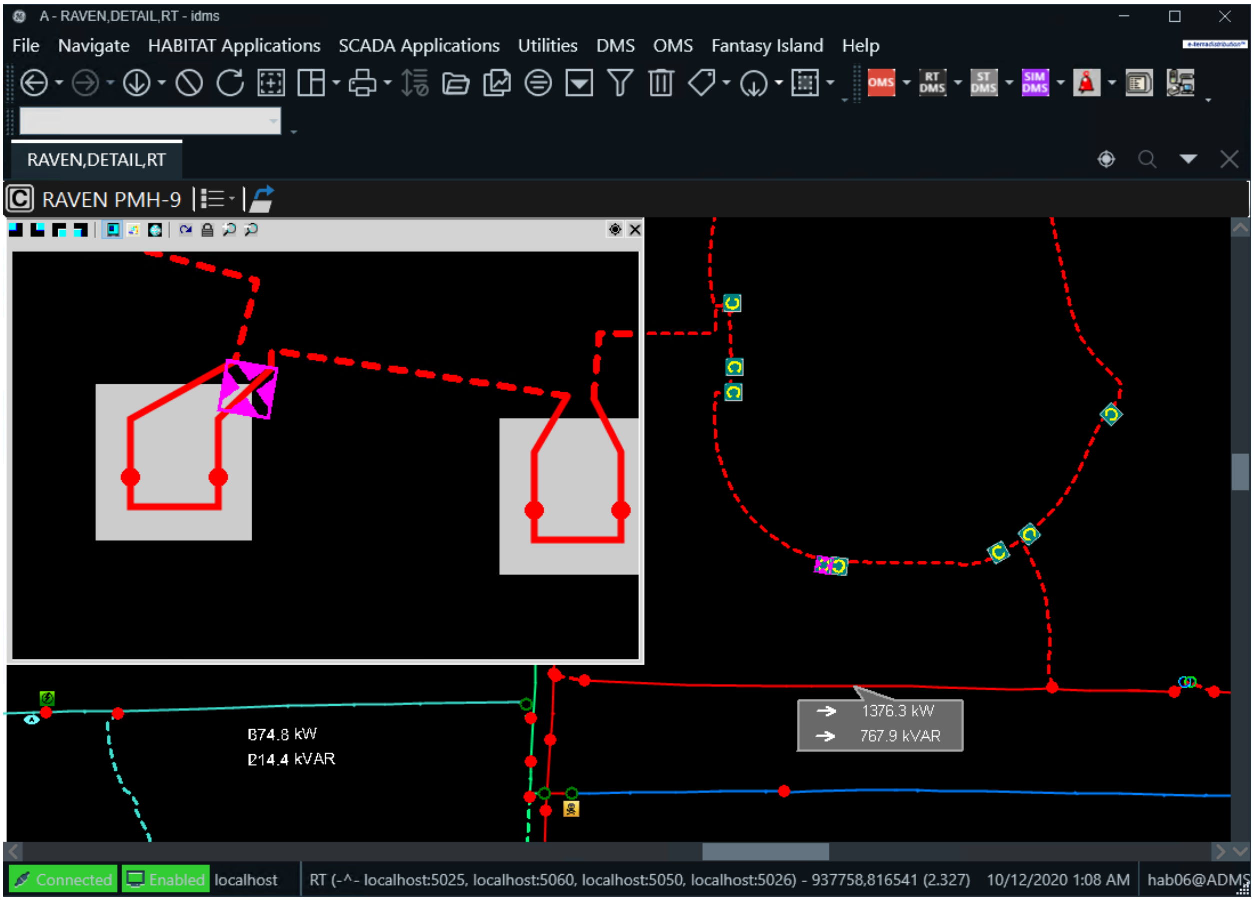


Figure 11. Picture-in-Picture View

2.1.3.6. Browser Tabulars

Browser tabulars are accessed from the Browser menu and the Browser toolbar. They display information about the SCADA system, user permissions, alarms, and tagging. The following example shows the Alarm Summary tabular display.

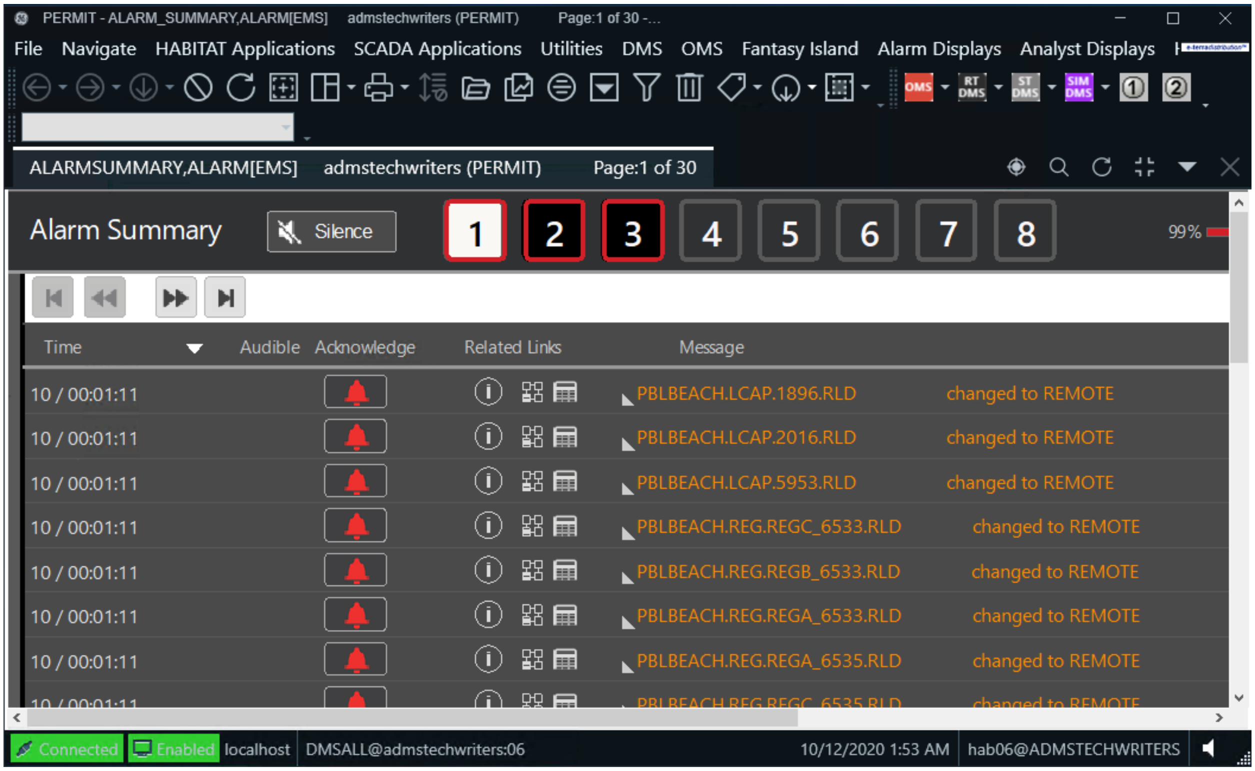


Figure 12. Browser Tabular Display

2.1.3.7. SCADA Station One-Line Display

The Station One-Line display shows the equipment that is contained inside the station fence. Switches and breakers can be directly controlled from this display. In addition, real-time measurements, such as feeder amps, are displayed.

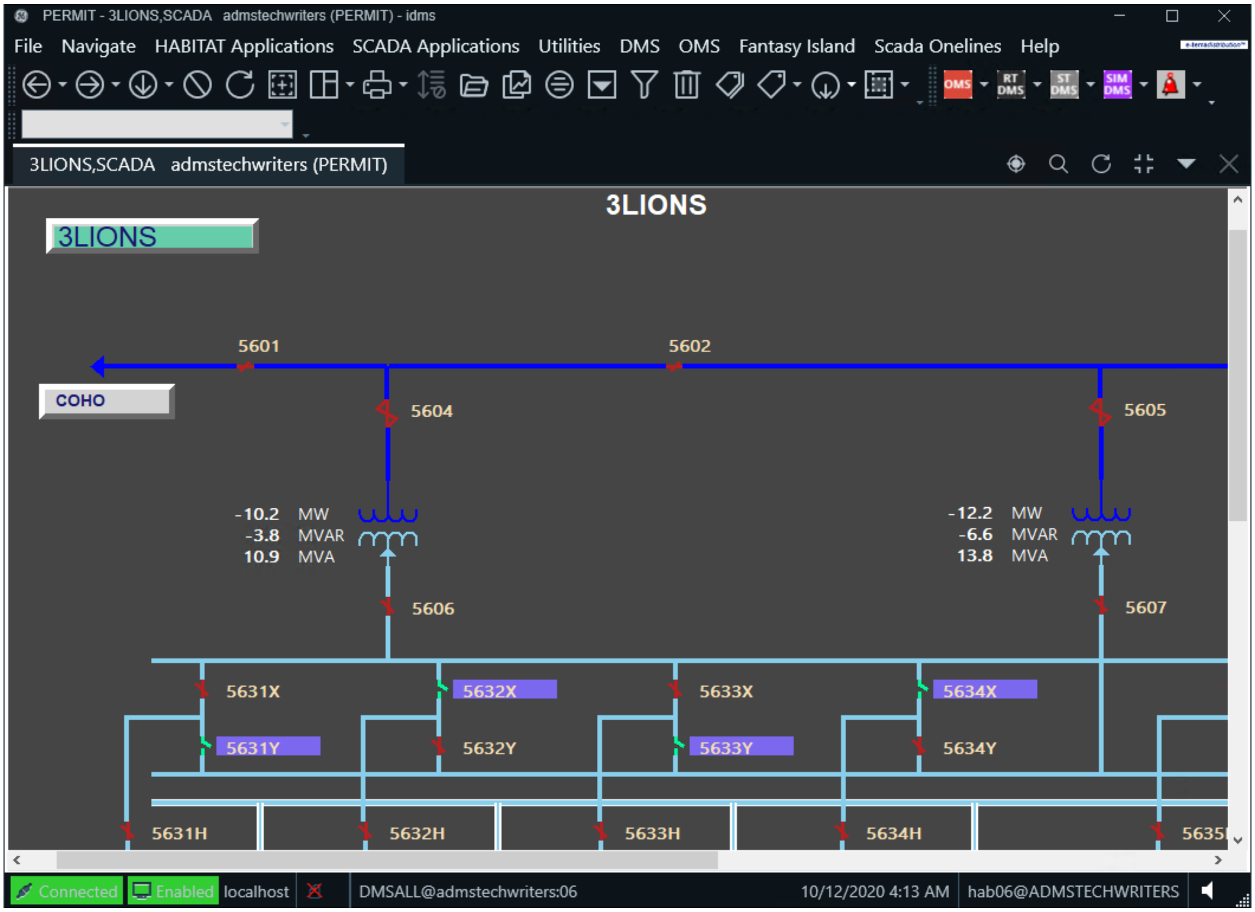


Figure 13. SCADA Station One-Line Display

2.1.3.8. Attribute Panes and Other Pop-Ups

The displays showing device attributes and other information, such as the configuration of temporary modifications, are separate panes that are related to the main displays, but are shown in separate viewports.

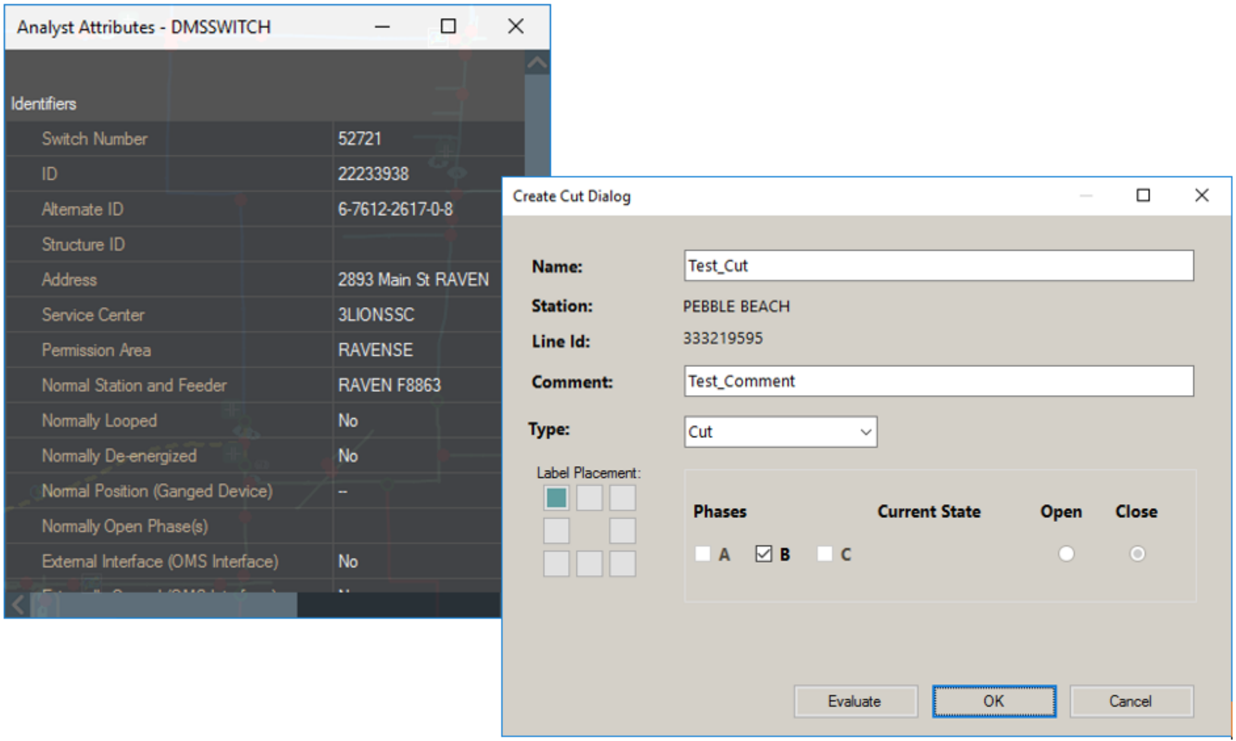


Figure 14. Pop-Up Windows for Switch Attributes and Cut Configuration

2.1.3.9. OMS Summary Display (Optional)

Installations of ADMS include the Outage Management System. This is the main overview of the trouble call and outage management operations. It is a multi-panel tabular display that typically includes an overview of current incidents or tabbed panels showing details about current unplanned and planned outages (see image below). The arrangement of the panels can be changed as required. Outage incidents can be managed from this display or directly from the geographic network view.

For more information about the operation and usage of the Outage Management System, refer to the *Outage Management System User Guide*.

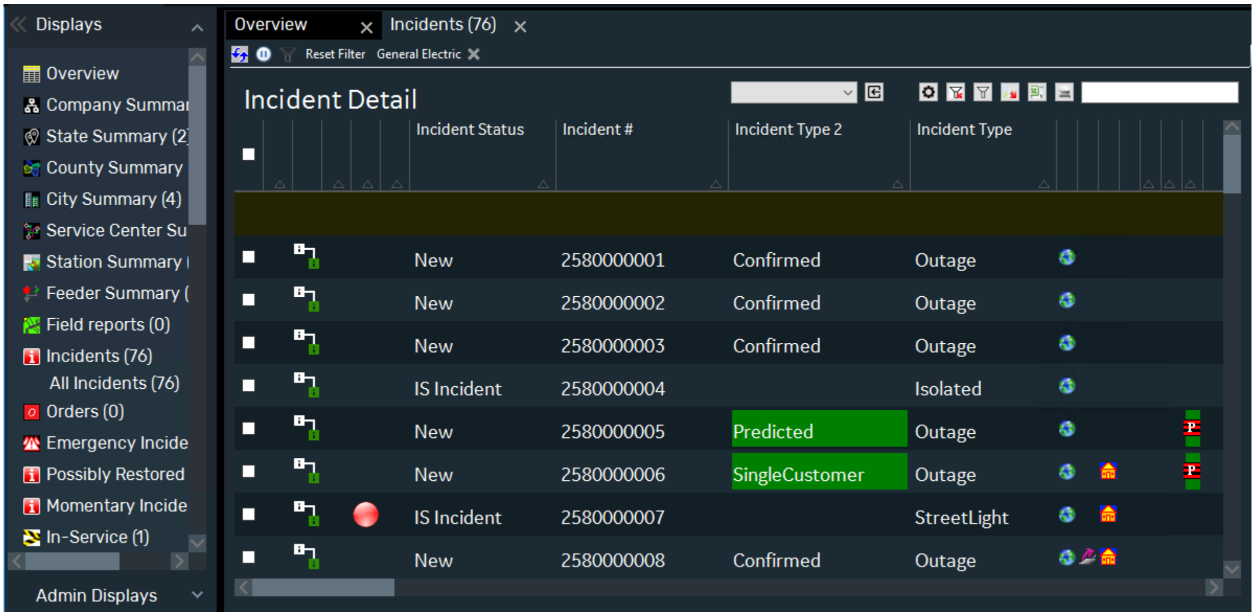


Figure 15. Example Outage Management System Display (Optional)

2.1.4. Toolbars and Menus

The ADMS user interface operates within the Browser display environment; therefore, the standard Browser menu bar and toolbar are available.

- **Browser Menu:** Always visible immediately below the window title bar. Contains Browser, Habitat, Scada, and ADMS menus.
- **Browser Toolbar:** Always visible below the Browser menu. Contains Browser, Habitat, Scada, and ADMS buttons. For more details about Browser menu and toolbar, refer to the *Browser User Guide.*
- **ADMS Toolbar**: ADMS displays the geographic network view (common) or device-specific toolbar to support distribution-specific functions. These are both displayed at the top of the geographic display to maximize the available viewport space for showing the network view.

  The common toolbar displays a standard set of buttons that have been configured to provide access to common functions that apply to the overall ADMS network view. When a device on the network view (geographic or schematic) is selected, the device-specific toolbar is displayed in place of the common toolbar.

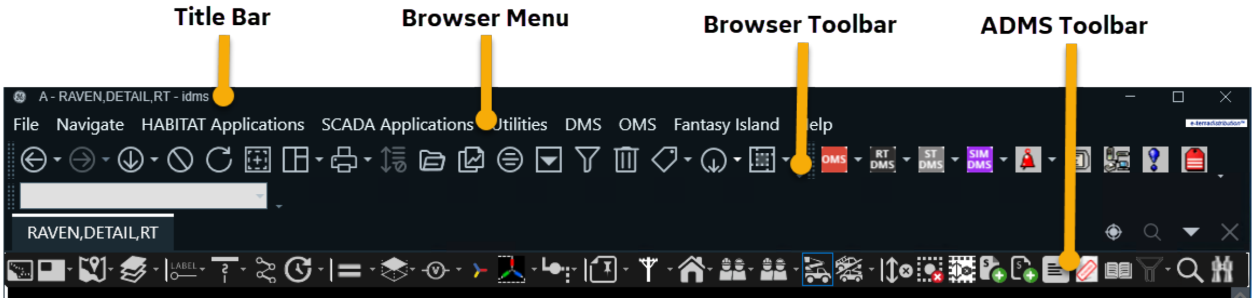


Figure 16. Browser Menu and Toolbar, Main Toolbar, and ADMS Toolbar

2.2. Toolbars and Menus

2.2.1. ADMS Toolbar Menus

The Browser toolbar contains four ADMS-specific drop-down menus: the Outage Management System (OMS) menu, the Real-Time ADMS menu, the study mode ADMS menu, and the simulator mode ADMS menu.

Figure 17. ADMS-Specific Buttons on the Browser Toolbar

OMS Menu

Clicking the OMS button opens the OMS Display Plug In. Clicking the small down arrow next to the button displays a menu of frequently used options.

The Outage Management System is an optional component of ADMS. For more information about the operation and usage of the OMS, refer to the *Outage Management System User Guide*.

RT DMS Menu

Clicking the Real-Time DMS (RT DMS) button shows the real-time geographic network view for the station that is named in the Browser toolbar text entry field in the current viewport.

Clicking the small down arrow next to the button displays a menu of frequently used options. The table below describes the options in the drop-down menu. The list of menu selections is configurable; it can be defined jointly with the end users to meet their real-time operational requirements, and it can be configured by the system administrator.

Table 1. RT DMS Menu Options

Menu Selection	Description
Station Overview	Displays the station overview in the RT mode.
Go to Permission Management Plug-In	Displays the permission mode configured for each substation area.

Menu Selection	Description
Go to Coordinates on Geographic	Centers the current geographic display on the coordinates given in the text entry field.
Go to Station Switch View with Labels	Displays the schematic view of switches with labels for the requested substation. To navigate to this view, type the substation name in the Browser toolbar's text box and select this option from the RTDMS menu.
Go to Station Container View with Labels	Displays the schematic view of all the cabinets for the substation named in the text entry field.
Go to Station Measurement Tabular	Displays the measurement data summary for the substation named in the text entry field.
Go to Station DER Measurement Tabular	Displays the distributed energy resources measurement data summary for the substation named in the text entry field.
Go to Station Meter Measurement Tabular	Displays the meter measurement data summary for the substation named in the text entry field.
Go to System Secondary Measurement Tabular	Displays the secondary voltage measurement data summary for the entire system.
Go to Station Static Schedule Tabular	Displays the static schedule data summary for the substation named in the text entry field.
Go to Station Dynamic Schedule Tabular	Displays the dynamic schedule data summary for the substation named in the text entry field.
DMS Site Overview Display	Displays the dynamics synchronization and secondary server information to be used for the planned failover.
Advanced Application Plan Status View	Displays the execution summary for advanced application plans.
Station Exception	Displays the station exception list tabular.
Station Exception (Refresh)	Displays the station exception list tabular that periodically refreshes the presented data.
Stations List	Displays the tabular menu of substation displays.
Areas List	Displays a tabular list of all the currently defined areas.
SCADA/DER Devices Tree View Display	Displays the SCADA/DER Devices Tree View display.
Station Output Summary	Displays the tabular summary of power flow results.
Station Protection Settings Group Summary	Displays the tabular menu for the substation protection settings groups.
Station Settings Group Templates Summary	Displays the tabular menu for the substation protection settings group templates.
Mobile Stations List	Displays a tabular list of mobile substations.
Tag Summary	Displays a tabular list of currently applied tags.
Orphan Tag Summary	Displays a tabular list of orphan tags.
DER Manual Dispatch	Shows measured and aggregated values for Distributed Energy Resources (DERs) and allows managing DERs.
Annotation Summary	Displays a list of annotation types.
Abnormal Summary	Displays a list of temporarily added elements: cuts, jumpers, and grounds.
Orphan Summary	Displays the list of orphan temporary elements: cuts, jumpers, and grounds.

Menu Selection	Description
Stations Containing Third Party Equipment	Displays a tabular summary of stations that contain equipment that is owned by a third-party company.
DNAF Display Plug-In	Opens the DNAF Display plugin, where you can use the LVM System Monitor functionality and view DNAF application results, measurement information, station exceptions, and the demand aggregation summary.
Base Application Status	Displays the execution status summary for base DNAF applications (DPF, PRV, and FL).
Advanced Application Status	Displays the execution status summary for advanced DNAF applications (LVM, FISR, AFR, PLOS, and FISRCA).
Application Input Summary	Displays the summary of device input tabulars. These tabulars allow an operator to enter data to be used by base DNAF applications.
BLA Station Summary	Displays the bus load allocation data for each substation.
LVM Station Summary	Opens the Load and Volt-VAR Management Station Summary display with the summary of LVM solutions for each station.
LVM Device Exclusions List	Opens the LVM Device Exclusions List display with a summary of all devices in the system that are excluded for LVM.
Manual Input Summary	Opens the Manual Entry Input Summary display, which shows the number of manually entered parameters for each substation and equipment type. All fields are clickable and navigate to relevant displays.
Manual Input Parameter Entry	Opens the Manual Input Parameter Entry pop-up display, where you can change voltage, load, and temperature settings for a single network element or the entire system.
DNAF System Configuration	Opens the DNAF System Configuration pop-up display, where you can configure, enable, or disable Distribution Network Analysis Functions on a station, service center, or optimization group basis.
Load Shed Overview	Opens the Load Shed Overview display with the summary of load shed devices for each station.
Automation Schemes List	Displays the tabular summary for automated local peer-to-peer switching schemes used to automate the fault isolation and service restoration functions.
Pending Switches	Displays a tabular list of switches that are pending a change in their normal status.
Stations Commissioning and Decommissioning	Displays a tabular summary of whether each station currently contains equipment that is modeled in the distribution electrical network but not active.
DPF Analyst Results	Displays the tabular DPF analyst summary.

ST DMS Menu

Clicking the Study DMS (ST DMS) button displays the study mode geographic network view for the station that is currently named in the Browser toolbar text entry field in the current viewport.

Clicking the small down arrow next to the button displays a pull-down menu with a range of options. Typically, these are the most frequently used functions in the study mode. The menu options listed in the table below serve as examples only. The actual list of menu selections is configured by the system administrator.

Table 2. ST DMS Menu Options

Menu Selection	Description
Station Overview	Displays the station overview in the ST mode.
Go to Coordinates on Geographic	Centers the current geographic display on the coordinates given in the text entry field.
Go to Station Switch View with Labels	Displays the schematic view of switches with labels for the substation named in the text entry field.
Go to Station Container View with Labels	Displays the schematic view of all the cabinets for the substation named in the text entry field.
Go to Station Measurement Tabular	Displays the measurement data summary for the substation named in the text entry field.
Go to Station DER Measurement Tabular	Displays the distributed energy resources measurement data summary for the substation named in the text entry field.
Station Exception	Displays the substation exception list tabular.
Station Exception (Refresh)	Displays the station exception list tabular that periodically refreshes the presented data.
Stations List	Displays the tabular menu of substation displays.
SCADA/DER Devices Tree View Display	Displays the SCADA/DER Devices Tree View display.
Areas List	Displays a tabular list of all the currently defined areas.
Station Output Summary	Displays the tabular summary of power flow results.
Station Protection Settings Group Summary	Displays the tabular summary for the substation protection settings group.
Station Settings Group Templates Summary	Displays the tabular menu for the substation protection settings group templates.
Mobile Stations List	Displays a tabular list of mobile substations.
Tag Summary	Displays a tabular list of currently applied tags.
Orphan Tag Summary	Displays a tabular list of orphan tags.
Annotation Summary	Displays a list of annotation types.
Abnormal Summary	Displays a list of temporarily added elements: cuts, jumpers, and grounds.
Stations Containing Third Party Equipment	Displays a tabular summary of stations that contain equipment that is owned by a third-party company.
Base Application Status	Displays the execution status summary for base DNAF applications (DPF, PRV, and FL).
Advanced Application Status	Displays the execution status summary for advanced DNAF applications (LVM, FISR, AFR, PLOS, and FISRCA).
Application Input Summary	Displays the summary of device input tabulars. These tabulars allow an operator to enter data to be used by base DNAF applications.
BLA Station Summary	Displays the bus load allocation data for each substation.
LVM Station Summary	Opens the Load and Volt-Var Management Station Summary display with the summary of LVM solutions for each station.

Menu Selection	Description
Manual Input Summary	Opens the Manual Entry Input Summary display, which shows the number of manually entered parameters for each substation and equipment type. All fields are clickable and navigate to relevant displays.
Manual Input Parameter Entry	Opens the Manual Input Parameter Entry pop-up display, where you can change voltage, load, and temperature settings for a single network element or the entire system.
DNAF System Configuration	Opens the DNAF System Configuration pop-up display, where you can configure, enable, or disable Distribution Network Analysis Functions on a station, service center, or optimization group basis.
Load Shed Overview	Opens the Load Shed Overview display with the summary of load shed devices for each station.
Automation Schemes List	Displays the tabular summary for automated local peer-to-peer switching schemes used to automate the fault isolation and service restoration functions.
Pending Switches	Displays a tabular list of switches that are pending a change in their normal status.
Stations Commissioning and Decommissioning	Displays a tabular summary of whether each station currently contains equipment that is modeled in the distribution electrical network but not active.
DNAF Analyst Results	Displays the tabular DNAF analyst summary.

SIM DMS Menu

Clicking the Simulator DMS (SIM DMS) button shows the simulator mode geographic network view for the station that is currently named in the Browser toolbar text entry field in the current viewport.

For more information about the simulator mode, see the *Distribution Operator Training Simulator (DOTS) User Guide.*

Clicking the small down arrow next to the button shows a pull-down menu with a range of options. Typically, these are the most frequently used functions in the simulator mode. The menu options listed in the table below serve as examples only. The actual list of menu selections is configured by the system administrator.

Table 3. SIM DMS Menu Options

Menu Selection	Description
Station Overview	Displays the station overview in the SIM mode.
Go to Coordinates on Geographic	Centers the current geographic display on the coordinates given in the text entry field.
Go to Station Switch View with Labels	Displays the schematic view of switches with labels for the substation named in the text entry field.
Go to Station Container View with Labels	Displays the schematic view of all the cabinets for the substation named in the text entry field.

Menu Selection	Description
Go to Station Measurement Tabular	Displays the measurement data summary for the substation named in the text entry field.
Go to Station DER Measurement Tabular	Displays the distributed energy resources measurement data summary for the substation named in the text entry field.
Station Exception	Displays the station exception list tabular.
Station Exception (Refresh)	Displays the station exception list tabular that periodically refreshes the presented data.
SCADA/DER Devices Tree View Display	Displays the SCADA/DER Devices Tree View display.
Areas List	Displays a tabular list of all the currently defined areas.
Station Output Summary	Displays the tabular summary of power flow results.
Station Protection Settings Group Summary	Displays the tabular summary for the substation protection settings group.
Station Settings Group Templates Summary	Displays the tabular menu for the substation protection settings group templates.
Mobile Stations List	Displays a tabular list of mobile substations.
Tag Summary	Displays a tabular list of currently applied tags.
Annotation Summary	Displays the list of annotation types.
Abnormal Summary	Displays the list of temporarily added elements: cuts, jumpers, and grounds.
Stations Containing Third Party Equipment	Displays a tabular summary of stations that contain equipment that is owned by a third-party company.
Base Application Status	Displays the execution status summary for base DNAF applications (DPF, PRV, and FL).
Advanced Application Status	Displays the execution status summary for advanced DNAF applications (LVM, FISR, AFR, PLOS, and FISRCA).
Application Input Summary	Displays the summary of device input tabulars. These tabulars allow an operator to enter data to be used by base DNAF applications.
Manual Input Summary	Opens the Manual Entry Input Summary display, which shows the number of manually entered parameters for each substation and equipment type. All fields are clickable and navigate to relevant displays.
Manual Input Parameter Entry	Opens the Manual Input Parameter Entry pop-up display, where you can change voltage, load, and temperature settings for a single network element or the entire system.
Automation Schemes List	Displays the tabular summary for automated local peer-to-peer switching schemes used to automate the fault isolation and service restoration functions.
Pending Switches	Displays a tabular list of switches that are pending a change in their normal status.
Stations Commissioning and Decommissioning	Displays a tabular summary of whether each station currently contains equipment that is modeled in the distribution electrical network but not active.
DPF Analyst Results	Displays the tabular DPF analyst summary.
DOTS Evaluation Time	Opens the DOTS Evaluation Time display to specify the system time for running DPF in SIM mode.

2.2.2. Common Toolbar Functions

The ADMS Geographic Openspace (common) toolbar contains various tools that can be used to change the geographic view and show or hide different data layers.

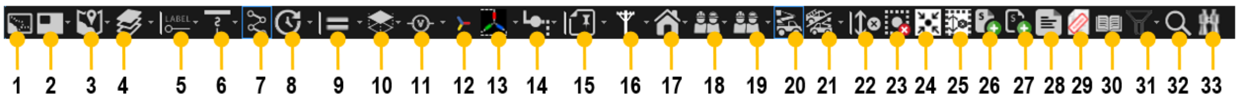


Figure 18. ADMS Geographic Common Toolbar

Common Toolbar Key for the above image					
1.	Load Surrounding Substations (Update Partial Model View)	**2.**	Picture in Picture	**3.**	Web Maps
4.	Background Layers	**5.**	Show/Hide Labels	**6.**	Show/Hide Laterals
7.	Show/Hide Non-Reconfigurable Sections	**8.**	Show/Hide Future	**9.**	Show Halos
10.	Show/Hide Fault Indicator Halos	**11.**	Show/Hide Voltage Levels	**12.**	Show/Hide Phase Colors
13.	Show/Hide Phases (Nominal)	**14.**	Show/Hide Overhead/Underground	**15.**	Show/Hide Annotations
16.	Show/Hide Structures	**17.**	Show/Hide Customers	**18.**	Show/Hide Crews
19.	Filter Crews by Type	**20.**	Show/Hide Vehicles	**21.**	Filter Vehicles by Type
22.	Clear Trace	**23.**	Clear All Marks	**24.**	Show Marked Devices
25.	Clear All Marks and Traces	**26.**	Add Marked Devices to SWO	**27.**	Add Marked Devices to Daily Log
28.	Switch Order	**29.**	New Planned Outage Request	**30.**	Study Case Setup
31.	Feeder Filter	**32.**	Find Tool	**33.**	Switching Order Search
34.	Start SWO Recording (appears in study mode only)	**35.**	Stop SWO Recording (appears in study mode only)		

The icons on the common toolbar allow operators to view the display in different formats, layers, and sizes.

Many of the buttons are toggleable. A button becomes highlighted when it is enabled.

The following table describes the typical functions that are available on the geographic (common) toolbar. Each function and how to use it is discussed in more detail throughout this user guide. Note, however, that each implementation of ADMS can be configured to include or exclude particular functions as required.

Table 4. Geographic Toolbar Icons Functions and Descriptions

Icon	Function	Description
	**Load Surrounding Substations (Update Partial Model View)**	Refer to Partial Model View.

Icon	Function	Description
	Picture in Picture	Contains options to show the picture-in-picture window with geographic display, station overview, and navigation overview. Refer to Navigating with the Picture-in-Picture View.
	Web Maps	Displays a menu to show/hide web maps. (Connection to a web map service is requires.) Refer to show/Hide Web Maps.
	Background Layers	Displays a menu to show/hide land-based information layers. Refer to Showing Land-Based Information.
	Show/Hide Labels	Toggles device labeling on and off. Refer to Show/Hide Labels.
	Show/Hide Laterals	Toggles the display of laterals on and off. Refer to Show/Hide Laterals.
	Show/Hide Non-Reconfigurable Sections	Refer to Reconfigurable Line Section View.
	Show/Hide Future	Displays a menu to show/hide devices under commissioning. Refer to Showing Equipment Under Commissioning.
	Show Halos	Toggles the display of the violation, capacity margin, and third-party owned equipment halos on affected lines and/or devices. Refer to Show/Hide Voltage Violations, Show/Hide Overloads, Show/Hide Potential Fault Halos, and Show/Hide Third-Party Owned Equipment Halos for details.
	Show/Hide Fault Indicator Halos	Displays a menu to show/hide fault location halos based on fault indicator types. Refer to Show/Hide Fault Indicator Halos.
	Show/Hide Voltage Levels	Displays a menu to show/hide devices based on voltage level. Refer to Show/Hide Voltage Levels.
	Line Coloring	Toggles between showing lines with feeder and phase colors. Refer to Show/Hide Phase Colors.
	Show/Hide Phases (Nominal)	Displays a menu to show/hide network elements by phase. Show/Hide Phases.
	Show/Hide Overhead/Underground	Displays a menu to show/hide overhead or underground devices. Refer to Show/Hide Overhead/Underground.
	Show/Hide Annotations	Displays a menu to show/hide annotations by type. Refer to Show/Hide Annotations.
	Show/Hide Structures	Displays a menu to show/hide structure objects (for example, poles or lamps) by type. Refer to Show/Hide Structures.
	Show/Hide Customers	Displays a menu to show/hide customers on the geographic view.

Icon	Function	Description
	Show/Hide Crews	Displays a menu to show/hide crews on the geographic view.
	Filter Crews by Type	Displays a menu to show/hide specific crew types on the geographic view.
	Show/Hide Vehicles	Toggles the display of vehicles on the geographic view.
	Filter Vehicles by Type	Displays a menu to show/hide specific vehicle types on the geographic view.
	Clear Trace	Clears all traces in all viewports. Refer to Tracing the Network.
	Clear All Marks	Clears the "mark" halo from all switches in all viewports. Refer to Marking Switches.
	Show Marked Devices	Shows all marked devices on the geographic view. Refer to section 4.1.2 Marking Switches.
	Clear All Marks and Traces	Clears the "mark" halo from all switches and also clears all traces in all viewports.
	Add Marked Devices to SWO	Includes the marked devices in a new switch order.
	Add Marked Devices to Daily Log	Includes the marked devices in the daily log.
	Switch Order	Opens the Switching Order application.
	New Planned Outage Request	Opens a new planned request in the Switching Order application.
	Study Case Setup	Displays a pop-up window for the creation and retrieval of savecases. Refer to Study Case Setup.
	Feeder Filter	Displays a menu to show/hide feeders on a geographic network view. Refer to Feeder Filtered Displays.
	Find Tool	Displays the Find pop-up, which is used to locate network devices, customers, or land-based features on the current display or anywhere in the system. Refer to Finding a Component.
	Switching Order Search	Opens the Switching Order Search pop-up, which is used to filter and open switching orders for edit or review.
	Start SWO Recording	Starts the Switching Order recording session to record the operator's actions as switching steps. This button is only available in study mode.
	Stop SWO Recording	Stops the Switching Order recording session. This button is only available in study mode.

2.2.3. Device-Specific Toolbars

A device-specific toolbar appears whenever a device is selected on a distribution display. These toolbars provide a set of buttons to access functions that are specific to the type of device. As an example, image below shows the device-specific toolbar for a standard overhead switch. Each device-specific toolbar is identified by a device-type icon at the left side of the toolbar.

Figure 19. Distribution Switch Toolbar

Device-specific toolbars combine basic common functions that appear on most toolbars (such as attribute information and tracing) with a set of functions that are specific to the type of device. For detailed descriptions, refer to the *ADMS Display Reference Manual*.

Bitmaps for all network elements are shown in the table below.

Table 5. Device Type Icon and Description

Icon	Device Type
	Auto-Transformer, Regulator, Power Transformer
	Breaker
	Bus
	Capacitor Bank
	Cut
	Distributed Energy Resource (DER)
	Distribution Transformer (Overhead)
	Distribution Transformer (Underground)
	Fault Indicator / Fault
	Fuse
	Generator
	Incident
	Jumper (Temporary Switch)
	Line (Overhead/Underground)
	Power Transformer
	Primary Meter (Load)
	Reactor
	Recloser
	Reference Bus Interface
	Reserved Capacity
	Switch (Overhead/Underground/Disconnect/Switch-Fuse/Recloser)
	Switch, Multi-Position, Composite Switch
	Substation
	Switch Cabinet
	Temporary Ground
	UI Container
	Vault

2.2.4. Browser Controls

The standard Browser navigation toolbar can be used with ADMS displays, which allows users to move back and forth between previously selected displays.

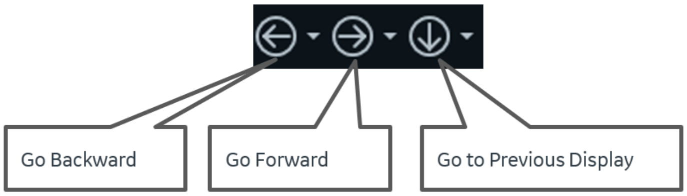


Figure 20. Browser Display Navigation Buttons

2.2.5. Right-Click Menu

Clicking the right mouse button displays a pop-up menu of relevant functions. The right-click menu is context-sensitive based on the location of the cursor when the mouse button is clicked. If the cursor is positioned over the blank space of the geographic network view, a generic menu is displayed. If the cursor is positioned over a device such as a switch, a menu of switch-related functions is displayed.

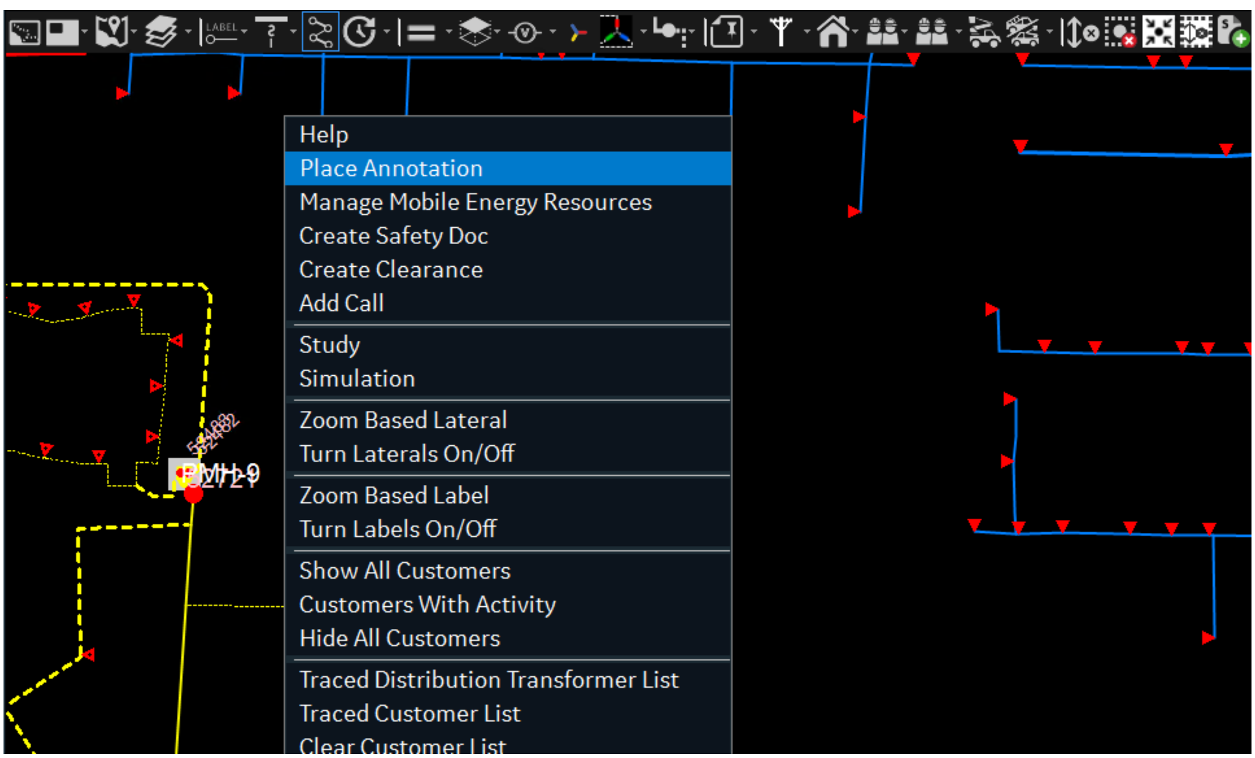


Figure 21. Right-Click Menu

The right-click menus of the network elements are described in the *ADMS Display Reference Manual*.

2.3. Network Views

ADMS is capable of displaying views of the network in geographic and schematic forms. There is a high level of consistency between these displays in terms of function, behavior, and symbology.

2.3.1. Geographic Network View

The geographic network view shows the complete details of the electrical distribution network outside of the substation. The display is made up of a number of layers to control the information that is shown and to minimize clutter. This display provides the user with a real-time visualization of the status of the network. Network management functions such as switch operations can be performed directly on the network view, and the results of the changes are automatically presented.

Any status changes made to the real-time network view are immediately updated for all users on the ADMS system; however, navigation guides such as network tracing are only shown on the individual user's workstation.

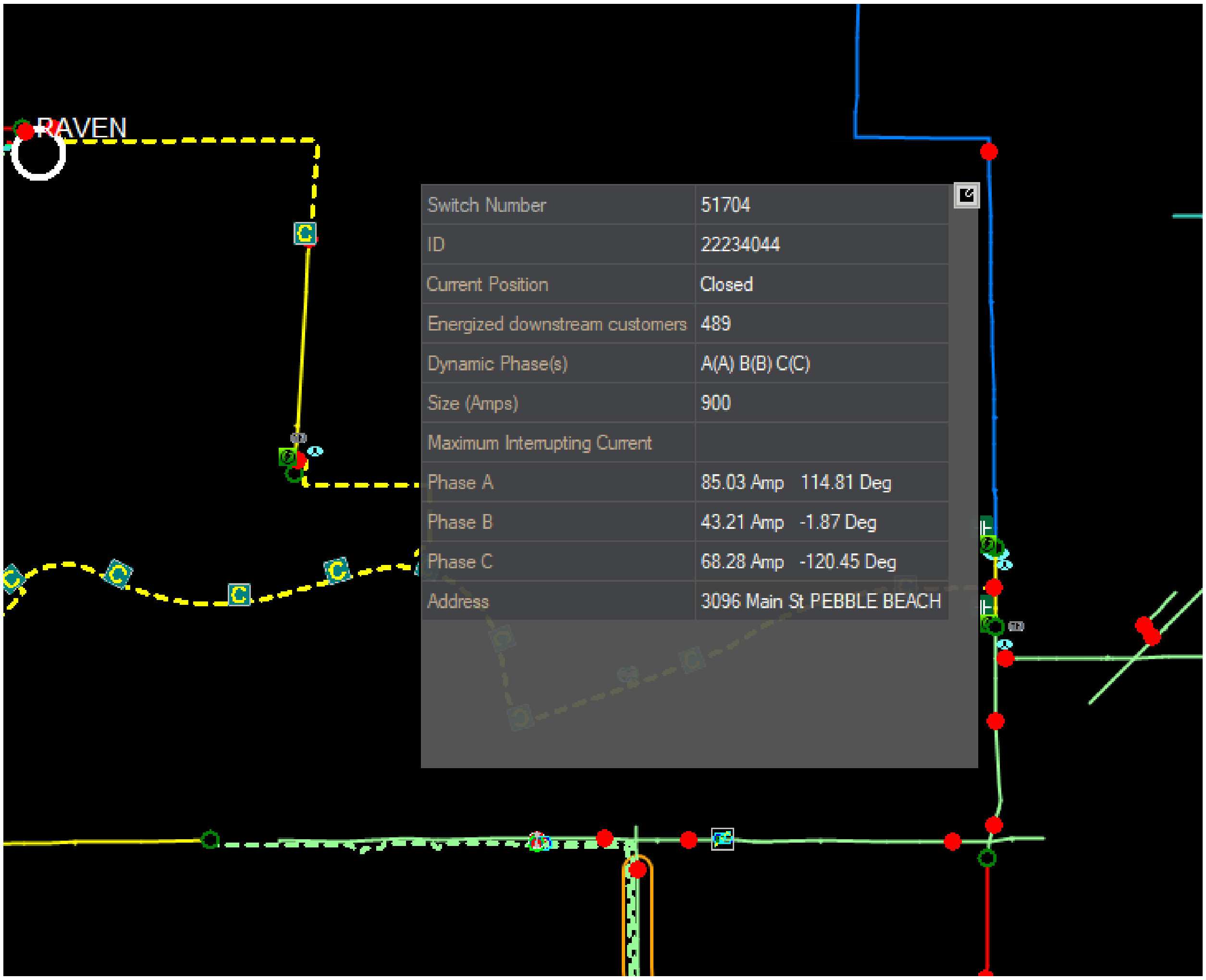


Figure 22. Geographic Network View with a Pop-Up Display

In addition to the electrical distribution network, a range of landbase information can also be displayed on the geographic network view. This information includes roads, streets, street names, railways, buildings, waterways, municipal boundaries, and so on.

The landbase information is normally not displayed, but it can be turned on as a set of individual layers. For each layer, there can also be additional levels of details displayed depending on the zoom level.

Figure 23. Display of Land-Based Layers on a Geographic View

2.3.2. Substation Internals

In addition to the standard SCADA one-line diagrams to show the devices within the substation, you can produce alternate schematic views of the substation internals (see image below) using the Substation Editor. These displays have the same functionality as the geographic network views. In this way, all standard ADMS functions can be applied to the devices within the substation in the same manner as the feeders.

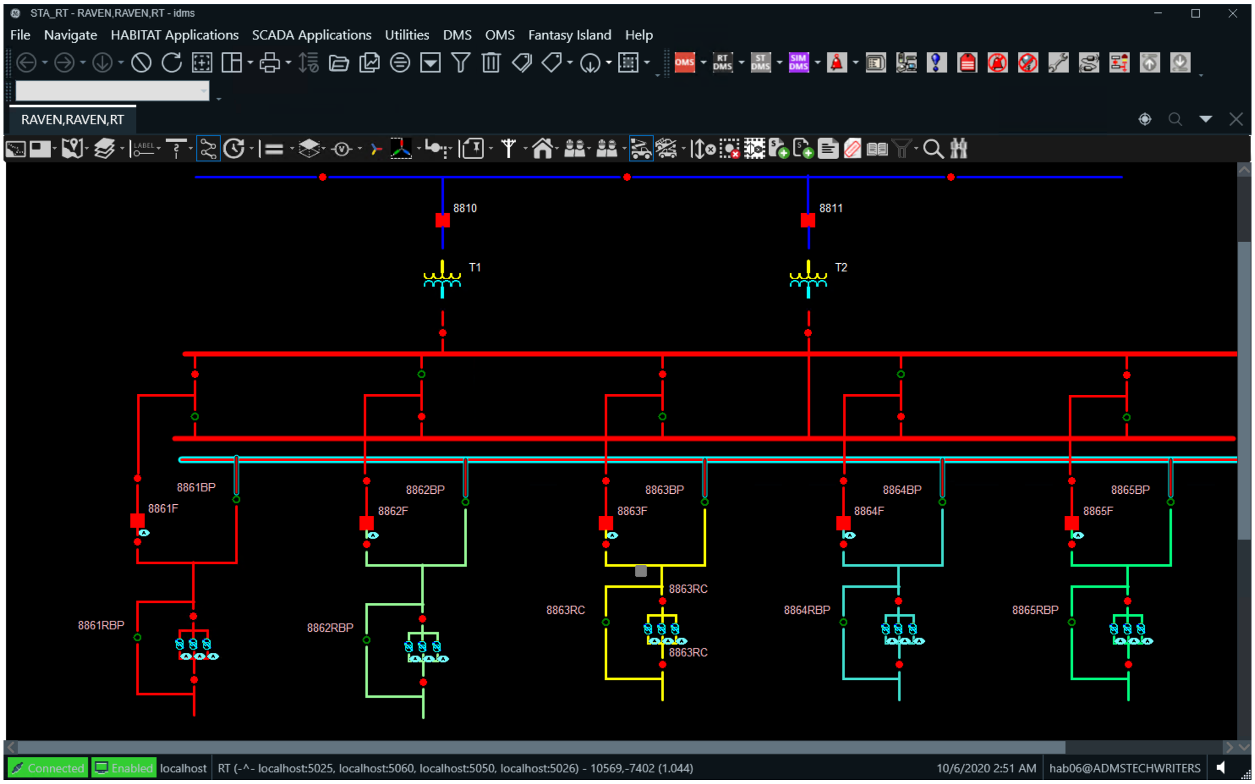


Figure 24. Internal View of a Substation

2.3.3. Schematics

There are many instances within the distribution network where a geographic representation of the devices is not the best view for effective operation. The devices can be geographically coincident or connected in a way that is difficult to interpret. In these cases, a schematic view is needed to ensure safe operation of the network.

The Underground Residential Distribution (URD) loops are a case where the connectivity of the network is difficult to interpret from the geographical view, and the elbow switches associated with each transformer make the display very cluttered.

URD loop schematics (see image below) are automatically generated to provide an easily interpreted display of the configuration of each phase and to show the associated elbow switches. Additional information such as cable lengths is also displayed. This display is fully operational.

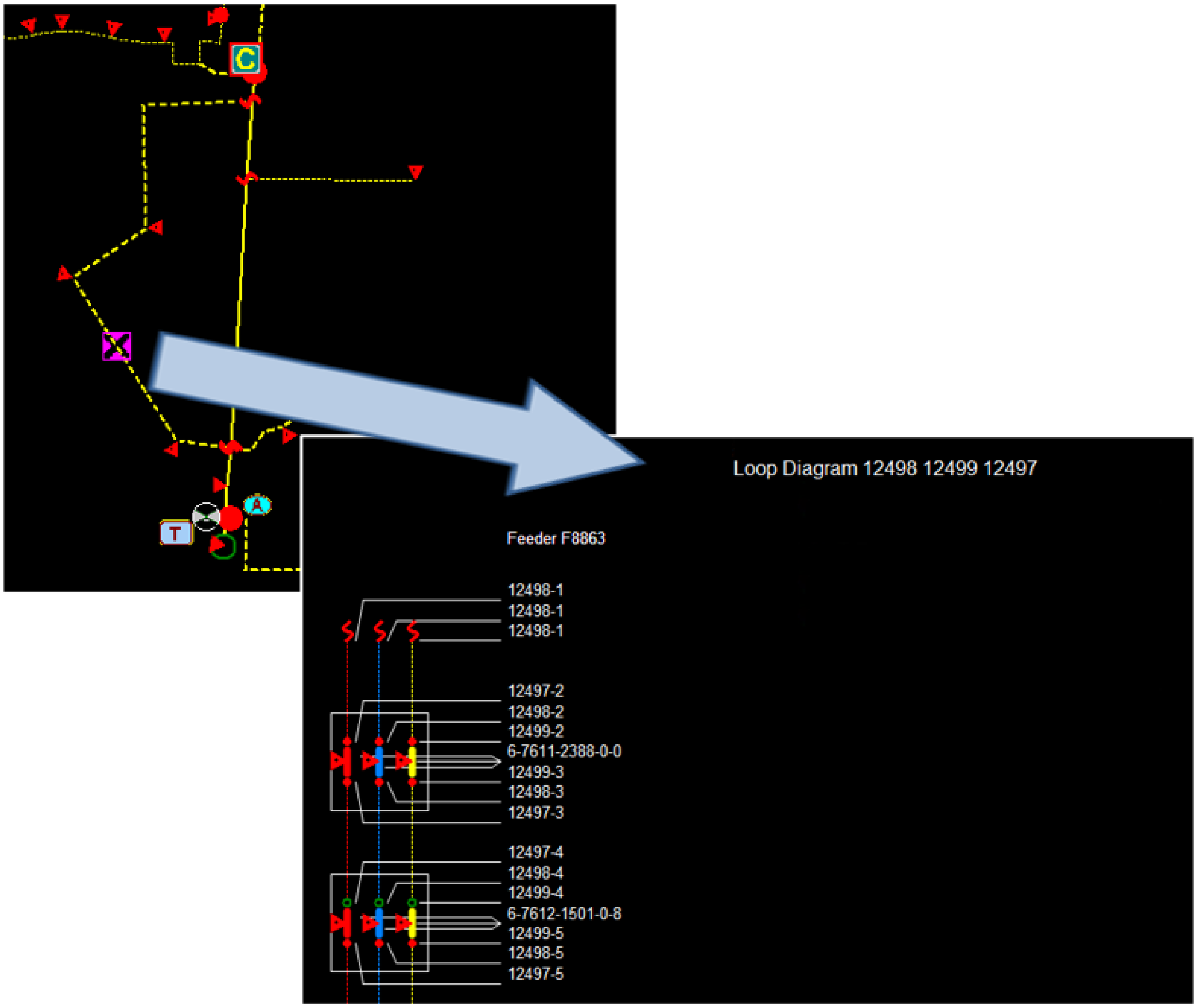


Figure 25. Geographic and Schematic Views of a URD Loop

2.3.4. Internal World Views

The visualization of the current state of the network is the critical function of the network view. In many cases, some of these switches are contained within UI containers, cabinets, or vaults. The internal switching of containers can be presented either as part of the overall network view or as separate internal world view displays.

The image below shows how the internal view of a vault can be maximized by zooming the geographic network view. At a defined zoom level, the display of the vault changes from a single symbol to an automatically generated schematic view of the switches and connections within the vault.

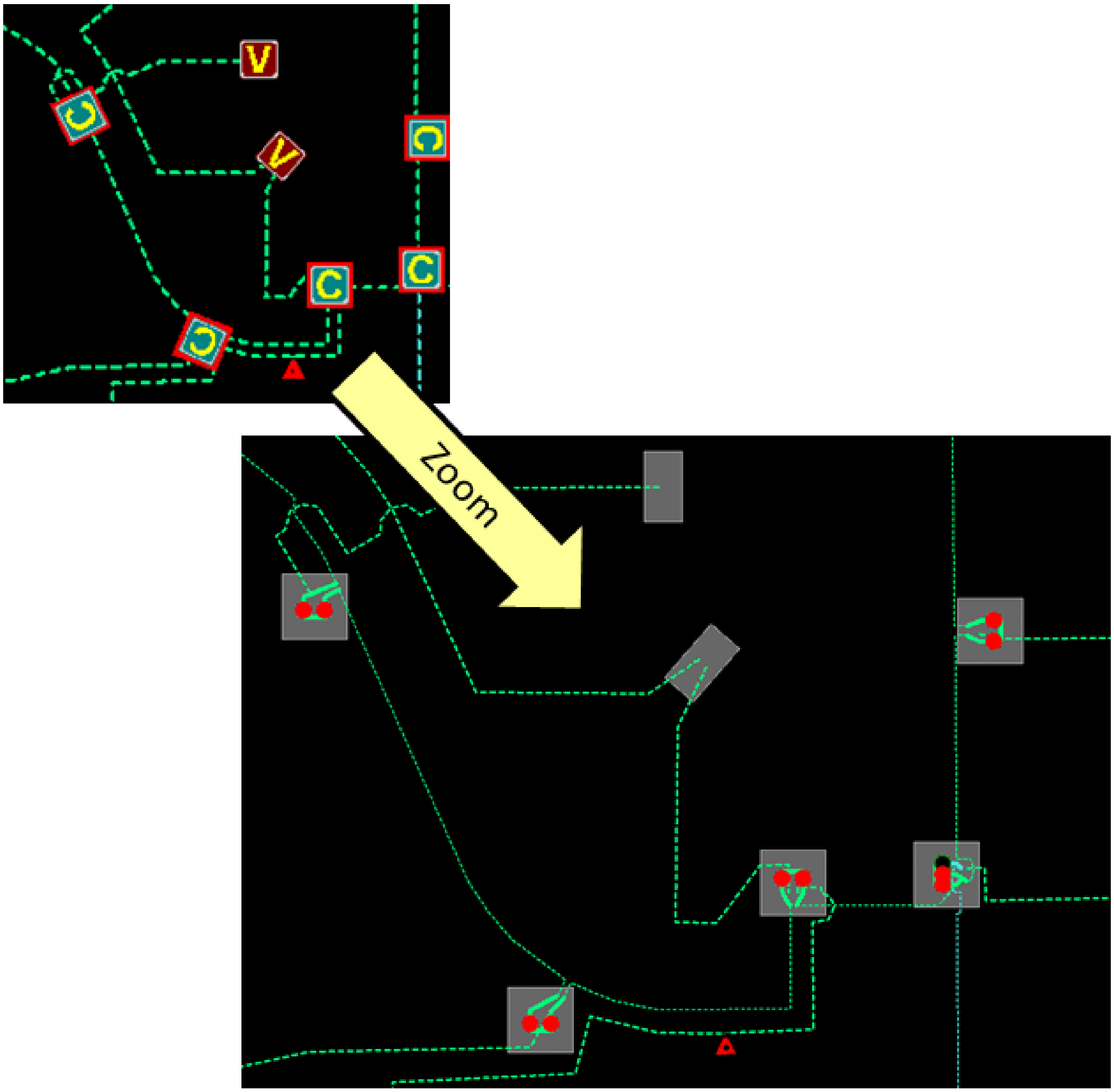


Figure 26. Internal View of Cabinets and Vaults Maximized by Zooming In

Alternatively, container internals can be presented as a separate schematic view as shown in image below. This internal view is presented in a separate viewport, which is linked to the symbol on the geographic view.

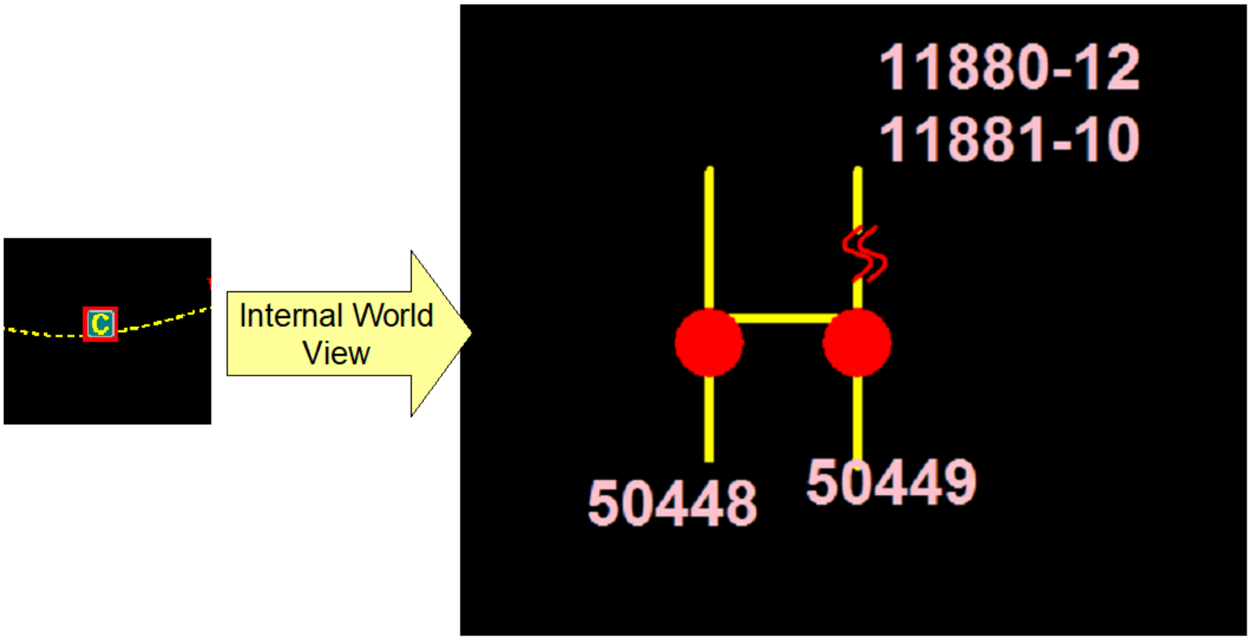


Figure 27. Schematic Internal World View of a Switch Cabinet

2.3.5. Display Symbols

This section describes the default set of symbols used by ADMS.

Display symbols for each type of device in the distribution model can be redefined to suit specific customer needs using the Viewer Configuration Editor, which is further discussed in the *Viewer Configuration Editor User Guide*.

For complete descriptions of the network view symbology, refer to the *Geographic Network View Symbology Reference Guide*.

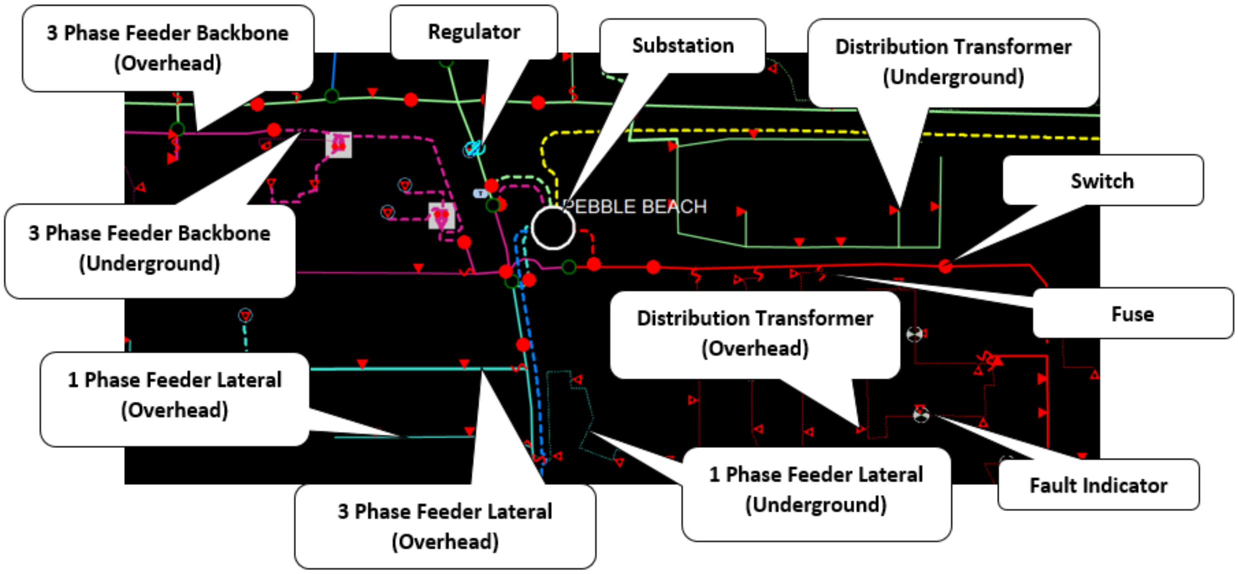


Figure 28. Basic Network Symbols

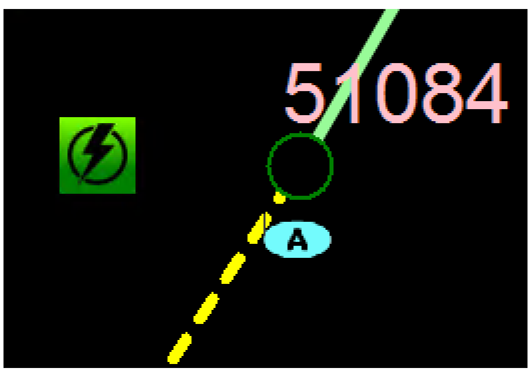


Figure 29. SCADA-Controlled Switch Symbols

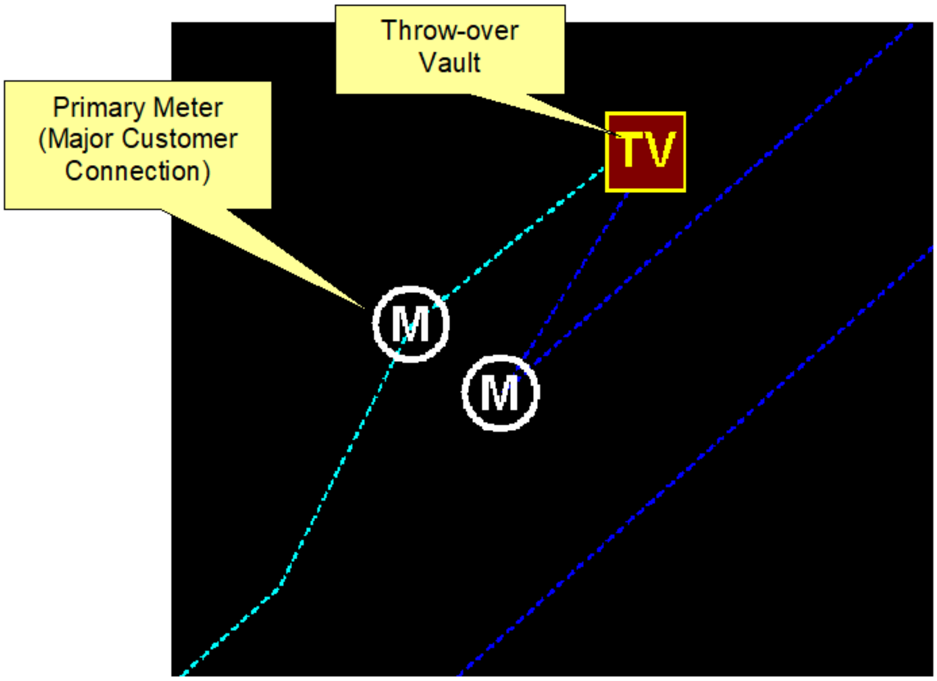


Figure 30. Primary Meter and Vault Symbols

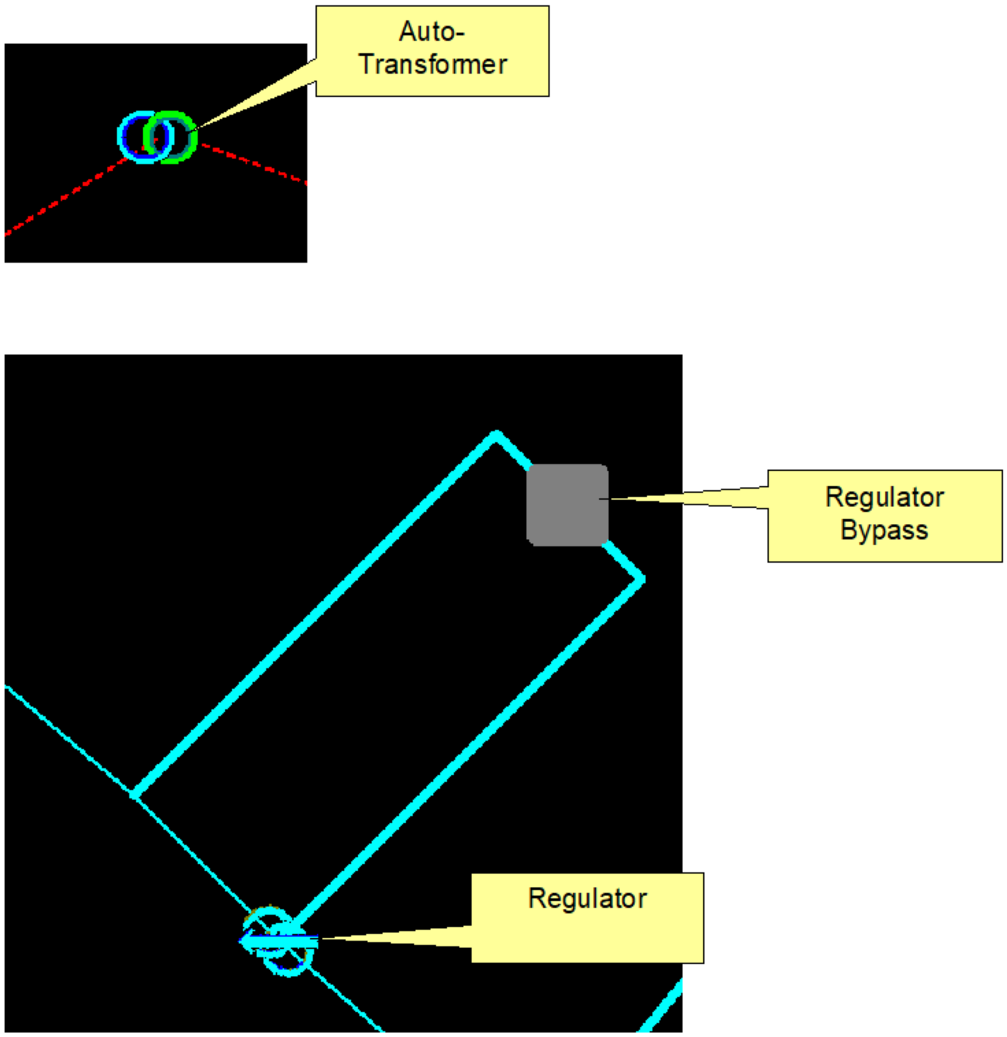


Figure 31. Auto-Transformer and Regulator Symbols

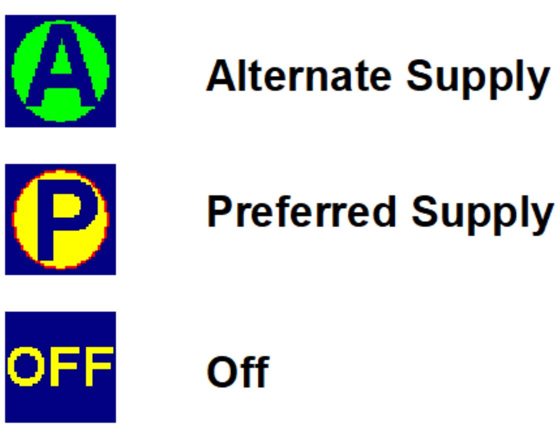


Figure 32. Throw-Over Switch Symbols

Figure 33. Multi-Position Switch Symbols

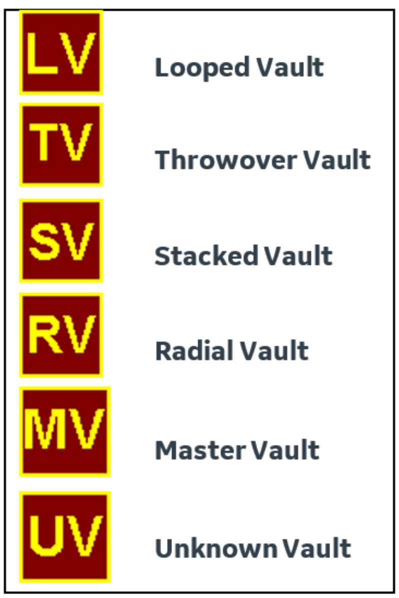


Figure 34. Vault Symbols

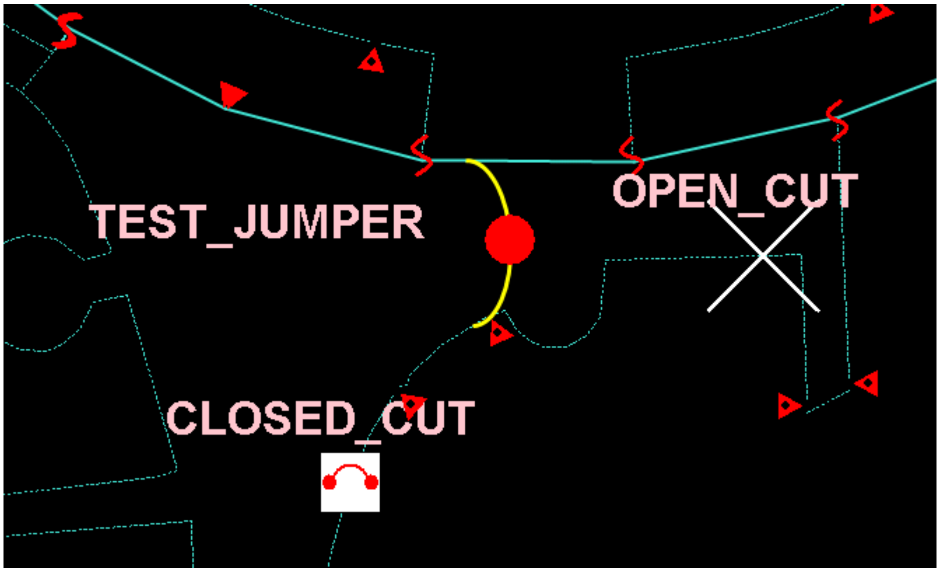


Figure 35. Temporary Modification Symbols

2.3.6. Display Tooltips

The basic attributes for switches, transformers, and other key devices on network displays are shown as tooltips. Tooltips are small pop-up windows that are displayed when the cursor is positioned over a symbol.

Tooltips appear on geographic displays and schematic displays, as well as tabular displays.

Tooltips on geographic displays appear after the mouse pointer hovers over a point placement object for two seconds. The tooltip disappears after 5 seconds, or when the mouse pointer moves. To fix a tooltip, click the button at the upper-right corner of the tooltip.

Figure 36. Display Tooltip

2.3.7. Display Annotations

Display annotations are available to the operator for annotating geographic views. These are special icons that can be located anywhere on the screen or associated with a particular device (see image below). They can be used to mark the position of faults, line crews, key customers, police stations, or other operational items that are not connected to the electrical network. A text comment can be associated with the annotation, which is displayed as both a label and a tooltip.

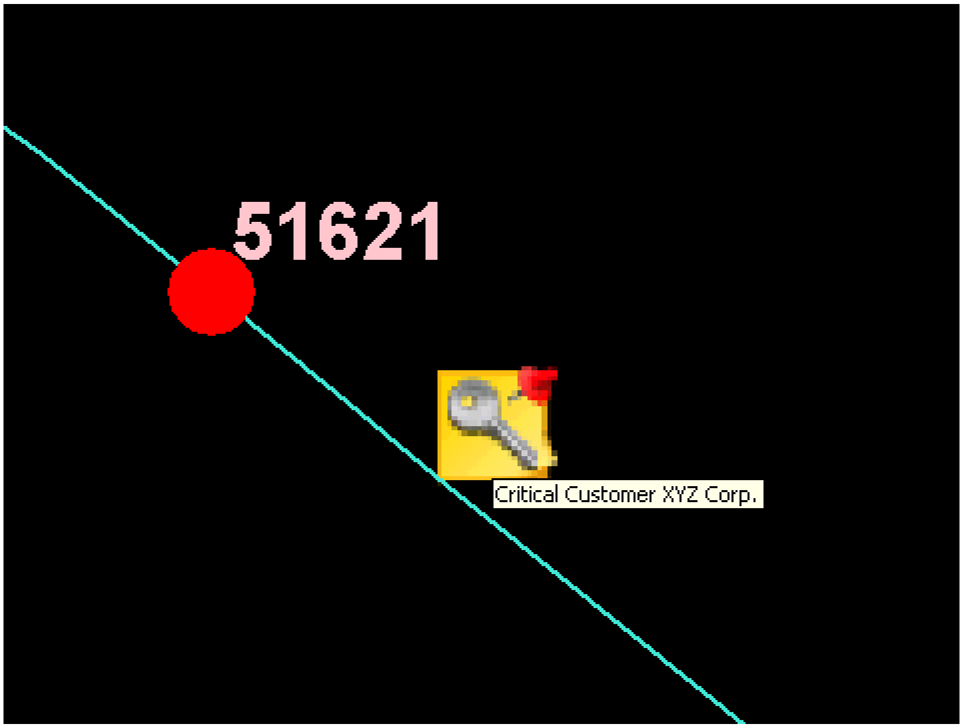


Figure 37. Example of a Display Annotation for a "Key" Customer

2.4. Tabular Displays

Tabular displays are used to show summary and detailed information about the state of the network that is not practical or efficient to display on the geographic diagrams.

The main tabular displays are called up by selecting them from the RTDMS, STDMS, or SIMDMS menus. The values in the displays are linked to other tabular displays to provide greater detail as appropriate. Many displays also include links to geographic views in order to locate the relevant device.

Tabular displays provide powerful sorting and filtering functions to allow easy access to the required information. For more information about working with tabular displays, refer to Working with Tabular Displays.

By default, the maximum number of rows in a grid tabular display is limited to 10,000. Therefore, note that tabular displays with a large number of entries (for example, the Measurement Data display) might not show all existing data.

Tabular displays can be customized by adding or removing details or navigation points for operators or dispatchers. For more details, refer to the *Grid Tabular Builder User Guide*.

The following are brief descriptions of the key tabular displays. For detailed descriptions of all ADMS tabular displays, refer to the *ADMS Display Reference Manual*. Note that each implementation of ADMS is likely to include additional custom-built tabular displays.

2.4.1. Station Exception

The Station Exception display summarizes abnormalities and violations by station and is used as a basis for navigating to more-detailed information about the exceptions. Separate station exception summaries are provided for real-time mode and study mode.

The display can be accessed from the RTDMS, STDMS, and SIMDMS menus, or by pressing Ctrl+S on the keyboard.

The station exception summary provides a high-level perspective of the activities within a dispatcher's Areas of Responsibility. The summary only displays the stations where the dispatcher has been granted Write access.

By default, stations are sorted and presented to the user to show the worst conditions in the following priority order:

- De-energization (number of de-energized customers)
- Abnormal topology
- High-voltage violations
- Low-voltage violations
- Voltage imbalance violations
- Overload violations
- Flow imbalance violations
- Ground return violation
- DPF solution status

For voltage and flow conditions, the worst result for each station is shown. A violation is shown as a flag with a red or blue background. Underlined text indicates that linked displays provide more details. Clicking the Locate button navigates to the corresponding device on the geographic display.

Selecting the substation name navigates to an exception summary for each feeder of the substation.

2.4.2. De-Energization Display

The De-Energization display shows de-energized loads and transformers along with the associated customer count and the date and time of de-energization. This display can be accessed via the station exception or the feeder exception summary. Clicking the Locate button navigates to the specified device on the geographic display.

2.4.3. Voltage Violation Displays

The Voltage Violation displays show violations calculated by the Limit Monitor function that is run as part of the Distribution Power Flow solution. These displays can be accessed via the Station or Feeder Exception summary.

Separate voltage violation displays are provided for high voltage, low voltage, and voltage imbalance violations.

Voltage violations are evaluated for all loads. The violations are ordered such that the most severe violations are presented first. Clicking the Locate button navigates to the offending equipment on the geographic display.

Dispatchers are also alerted to feeder voltage violations through alarms that are shown on the Alarm Summary display.

2.4.4. Overload Violation

The Overload Violation display shows overloads calculated by the Limit Monitor function that is run as part of the Distribution Power Flow solution. This display can be accessed via the Station or Feeder Exception summary.

Overload conditions are examined for lines, switching devices, capacitors, transformers, and reactors. The violations are ordered such that the most severe violations are presented first. Clicking the Locate button navigates to the geographic display with the selected component centered and highlighted on the display.

Dispatchers are also alerted to feeder overloads through alarms that are shown on the Alarm Summary display.

2.4.5. Ground Return Violation

The Ground Return Violation display shows limit current, imbalance current, and ground return current violation levels for a particular switch, calculated by the Limit Monitor function running as part of the Distribution Power Flow solution. This display can be accessed via the Station Exception display.

Clicking the Locate button navigates to the specified device on the geographic display.

2.4.6. Abnormal Topology

The Abnormal Topology display shows switches and permanent jumpers in abnormal positions, abnormally energized loads, temporary grounds, cuts, and jumpers. This display can be accessed via the Station or Feeder Exception summary.

The date and time when the abnormal condition was detected are provided for switches, permanent jumpers in abnormal positions, temporary grounds, cuts, and jumpers. Each abnormal condition also provides a reference to the area, substation, and type of change that occurred.

Clicking the Locate button navigates to the specified device on the geographic display.

The Tag column indicates whether any tags have been applied to the listed components. A gray symbol indicates that no tag has been applied.

2.4.7. Measurement Data

The Measurement Data display provides a list of SCADA measurements known by ADMS for the selected station. In the real-time environment, these measurements are used as inputs to the DPF solution.

2.4.8. Station Output Summary

The Station Output Summary display provides navigation to the distribution network advanced function results for a particular station. The display can be accessed from the RTDMS menu, or by pressing Ctrl+O on the keyboard.

2.4.9. DPF Station Results

The Distribution Power Flow (DPF) Station Results display provides high-level DPF output for the substation and can be accessed from the Station Output Summary display. The DPF Station Results display contains the following information:

- Information about DPF convergence for the station
- Value of supply bus voltage in kV
- Power transformer LTC tap positions and loadings
- Substation capacitor steps
- Voltage regulator tap positions and loadings
- Feeder capacitor status (ON/OFF)

2.4.10. Station Losses Summary

The Station Losses Summary display can be accessed from the DPF Station Results display by clicking Loss/PQI Summary. The information on this display is recalculated each time the DPF solves for a station. Instantaneous losses and Power Quality Indices (PQI) are presented. Loss and PQI results for individual components are also shown on Operator Output displays that can be opened from the geographic network view.

2.4.11. Short Circuit Calculation Summary

The Short Circuit Calculation Summary display, accessible from the Station Output Summary display, provides the results of the short circuit calculations for each feeder breaker of the selected station. This is a study-mode display.

2.4.12. Mobile Substations

The Mobile Substation Summary display lists all mobile substations that are currently connected to the distribution network.

2.4.13. Tag Summary

The Tag Summary display lists all the tags currently applied to the distribution network.

2.4.14. Annotation Summary

The Annotation Summary display shows a list of annotation types that can be applied to the distribution network. Clicking any of the annotation types navigates to a summary of the currently applied annotations of that type.

2.4.15. Abnormal Summary

The Abnormal Summary display shows a list of temporary modification types that can be applied to the distribution network. It also displays abnormal situations that might exist on feeders with customer counts, fault indicator settings, and so on. Clicking any of the modification types navigates to a summary of the currently applied modifications of that type.

2.4.16. Advanced Application Status

The Advanced Application Status (previously - DSO Control) display shows the status of the network optimization functions, such as FISR, AFR, and PLOS for each substation.

For more information, refer to the *Distribution Network Analysis Functions (DNAF) User Guide*.

2.4.17. Application Input Summary

The Application Input Summary (previously - DOAN Control) display provides the summary for inputs and measurements used for DNAF applications.

Detailed information is available in the *Distribution Network Analysis Functions (DNAF) User Guide*.

2.4.18. BLA Station Summary

The Bus Load Allocation (BLA) Station Summary display provides a summary of the bus load allocation calculations for each substation.

For more information, refer to the *Distribution Network Analysis Functions (DNAF) User Guide*.

2.4.19. Manual Input Summary

The Manual Entry Input Summary display shows a summary of all network elements with parameters set via a device's manual input display or the Application Input Summary display.

2.4.20. Manual Input Parameter Entry

The Manual Input Parameter Entry pop-up display allows an operator to manually set parameters for the whole system or system sub-elements (service centers, stations, power transformers, and feeders) using a tree-like structure.

2.4.21. Station Commissioning and Decommissioning

The Station Commissioning and Decommissioning display provides an overview for stations with the equipment that is modeled in the distribution electrical network but not active (for example, equipment planned to be put in service, planned to be removed, or was recently removed).

2.4.22. Pending Switches

This summary lists all switches that currently have "pending permanent" status. For more details, see Specific Device State Decorations. Clicking the Locate button navigates to the specified device on the geographic display.

3. Working with the User Interface

3.1. Getting Started

ADMS works within the overall Browser environment. This means that operators have a single startup and login to access both the ADMS functions and the SCADA functions.

To start the ADMS system:

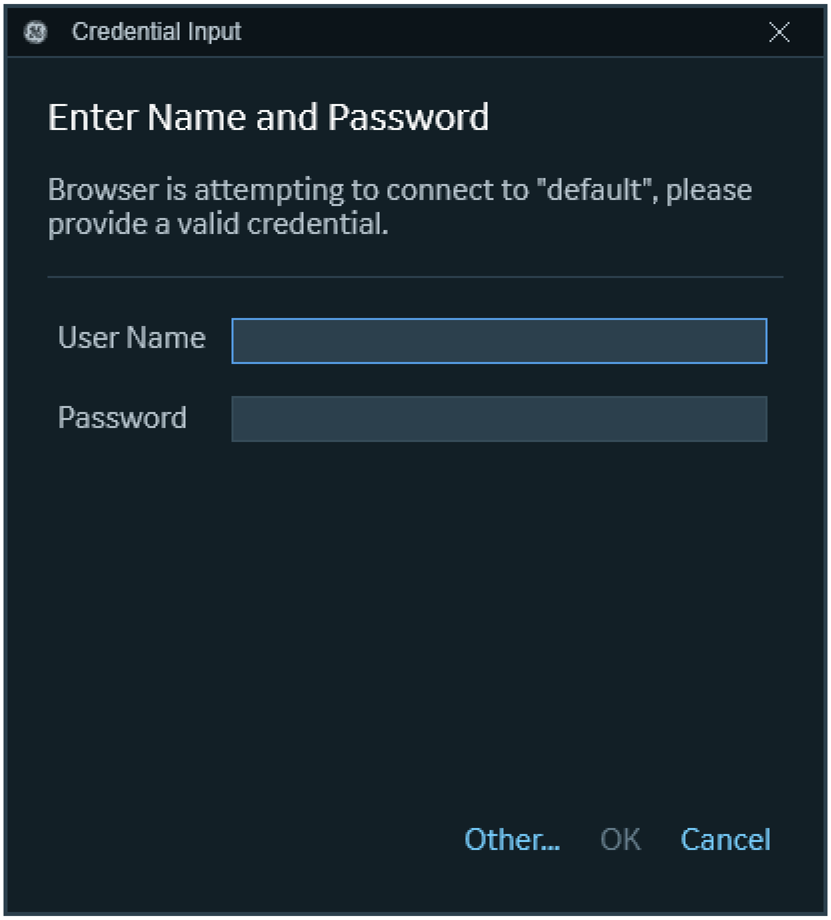
- Click the shortcut available in the Start menu or on the desktop.

This shortcut opens the standard viewports, establishes connections to the servers, and presents the login screen.

3.1.1. Logging In and Out

To log in to ADMS:

1. Launch the ADMS client shortcut.
2. In the Credential Input dialog box, enter your user name and password.



3. Click Login.

To log off the ADMS system:

> **Caution**
>
> The following action completely shuts down the application.

- On the Browser toolbar, click the Logout button.

3.1.2. Client Operation Modes

Several operation modes define the client's functionality. Different display background colors are used to differentiate client modes. Network display, pop-up, and tabular background colors are configurable for client modes. For more information about configuring background colors, refer to the *Net Dynamics Configuration Editor User Guide*.

Real-Time Mode

In this mode, users can operate the SCADA system, distribution management system, and outage management system functions within their areas of responsibility. User actions are reflected on the entire network and can be seen from all clients.

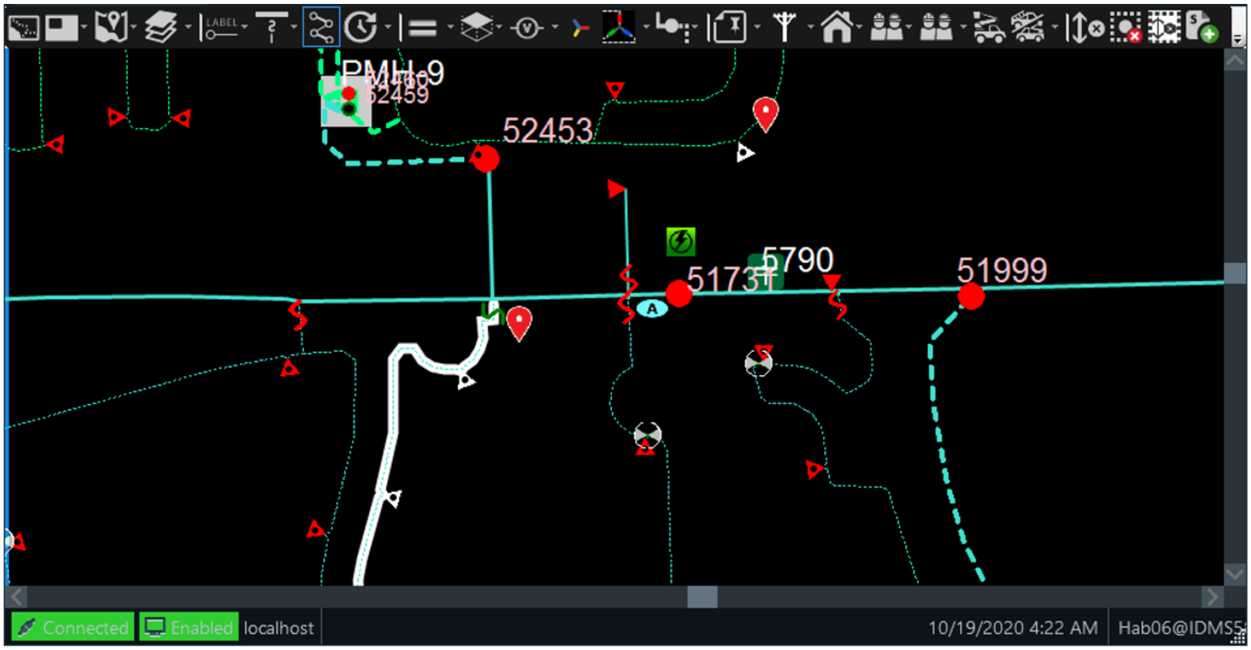


Figure 38. Real-Time Mode View

Study Mode

This mode allows users to implement various study scenarios on a separate client workstation without affecting the real-time distribution network or other users.

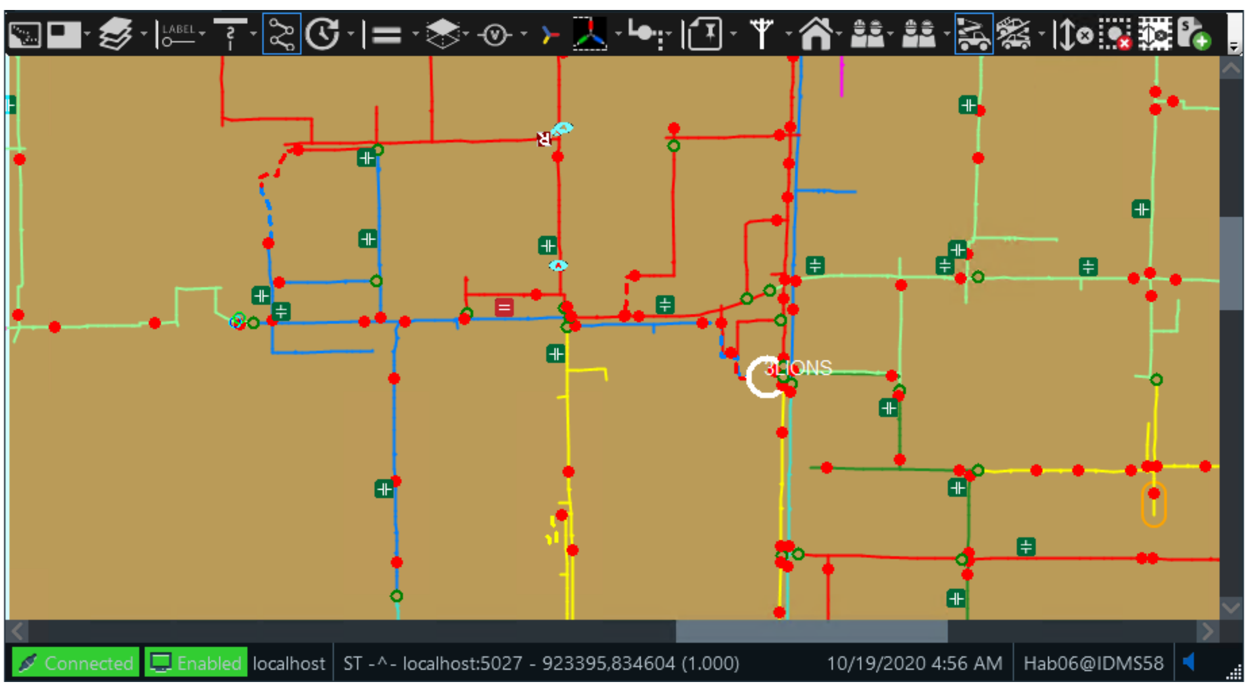


Figure 39. Study Mode View

Simulator Mode

This mode operates in a separate training or testing environment. Changes made in the simulator mode are immediately reflected in the real-time mode. For example, placing a fault on a line produces an actual fault in the distribution system, which results in protective device tripping, fault indicator halos, line de-energization, and customer calls.

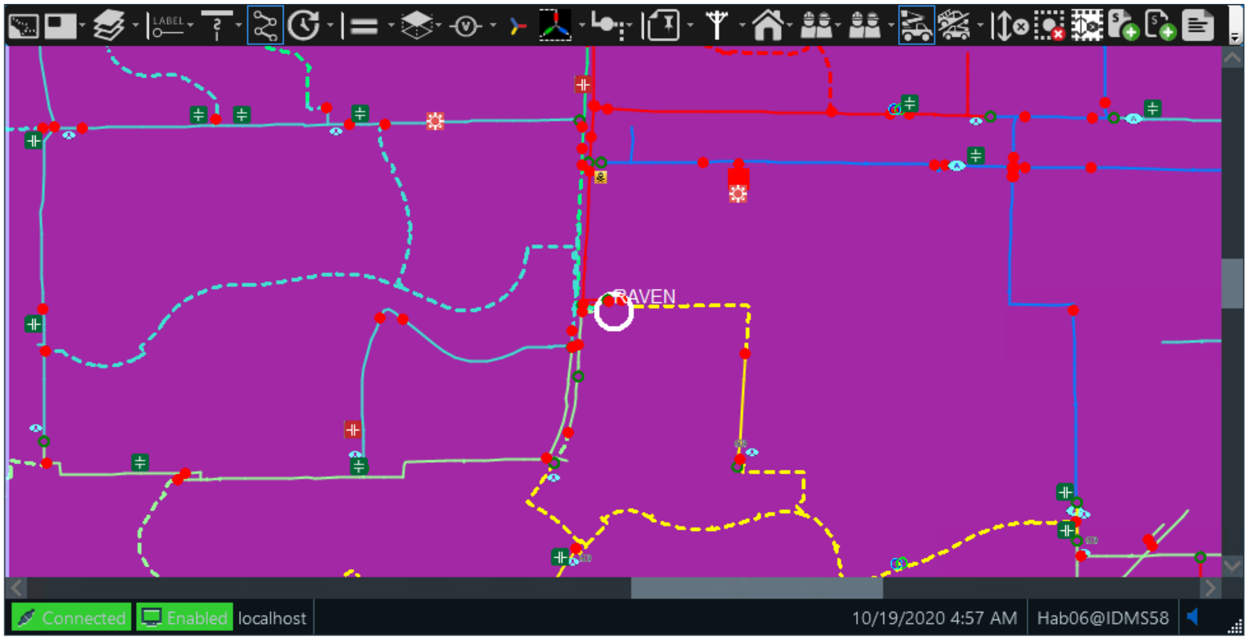


Figure 40. Simulator Mode View

For more information, refer to the *Distribution Operator Training Simulator (DOTS) User Guide*.

Read-Only Mode

In read-only mode, users have read-only access to the distribution management system data.

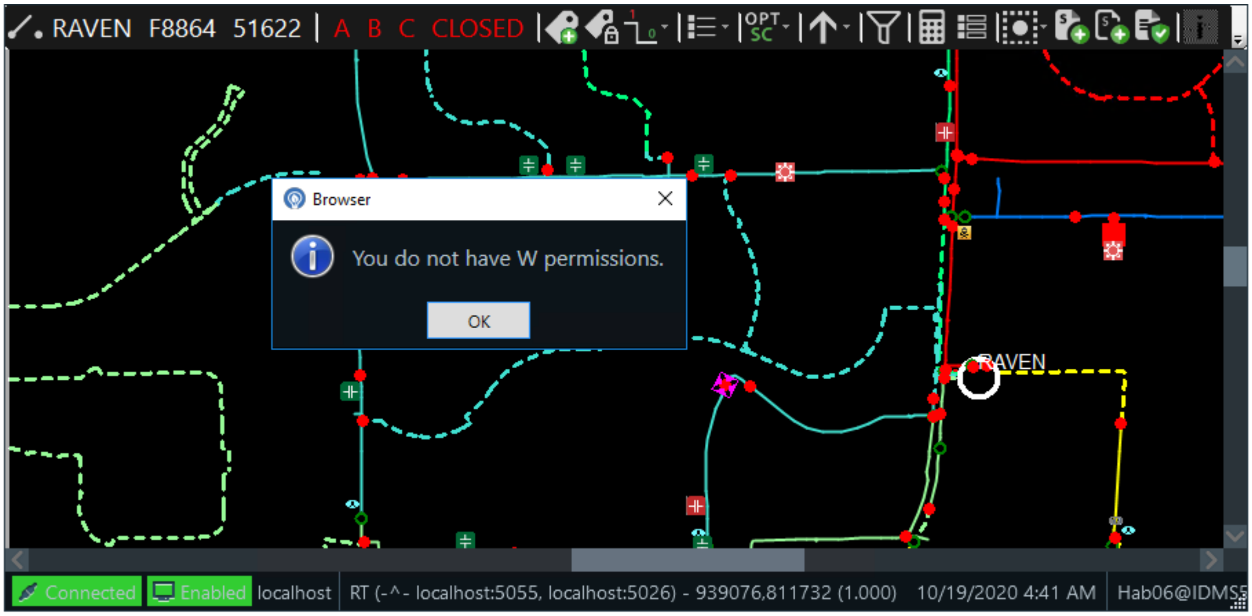


Figure 41. Read-Only Mode View

OMS-Specific Mode

In this mode, users can perform OMS functions. DMS data and operations are read-only and DNAF data and operations are not available.

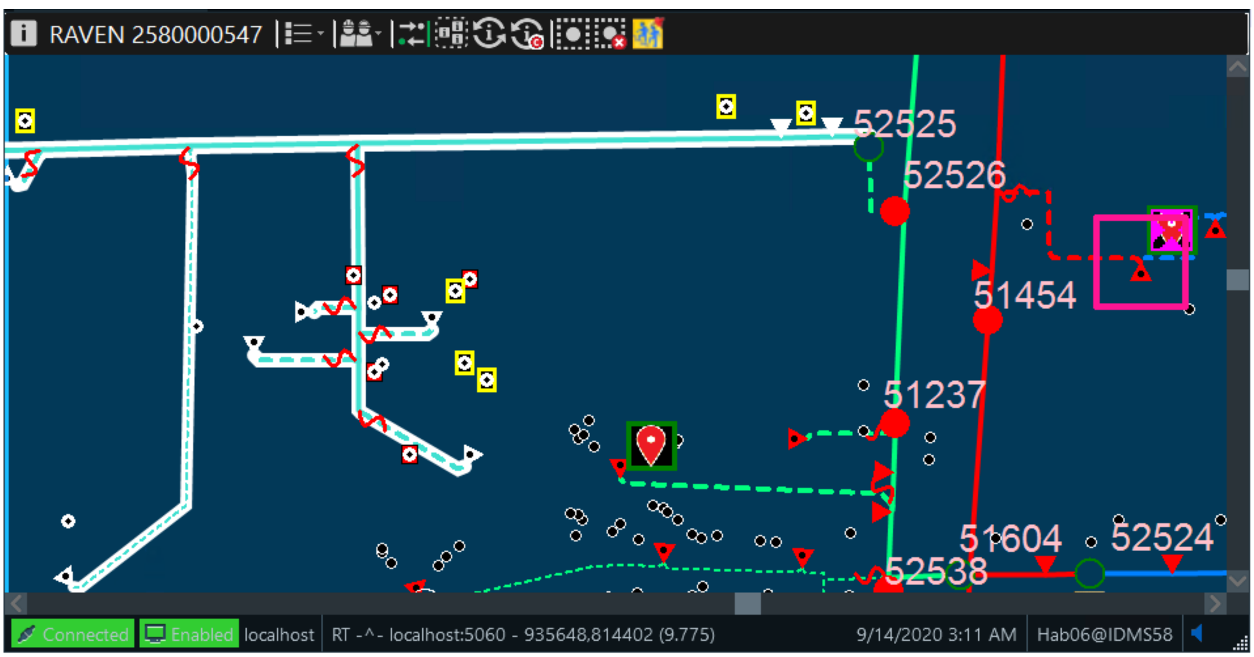


Figure 42. OMS-Specific Mode View

DNAF-Specific Mode

In this mode, users can perform DNAF functions. DMS data and operations are read-only and OMS data and operations are not available.

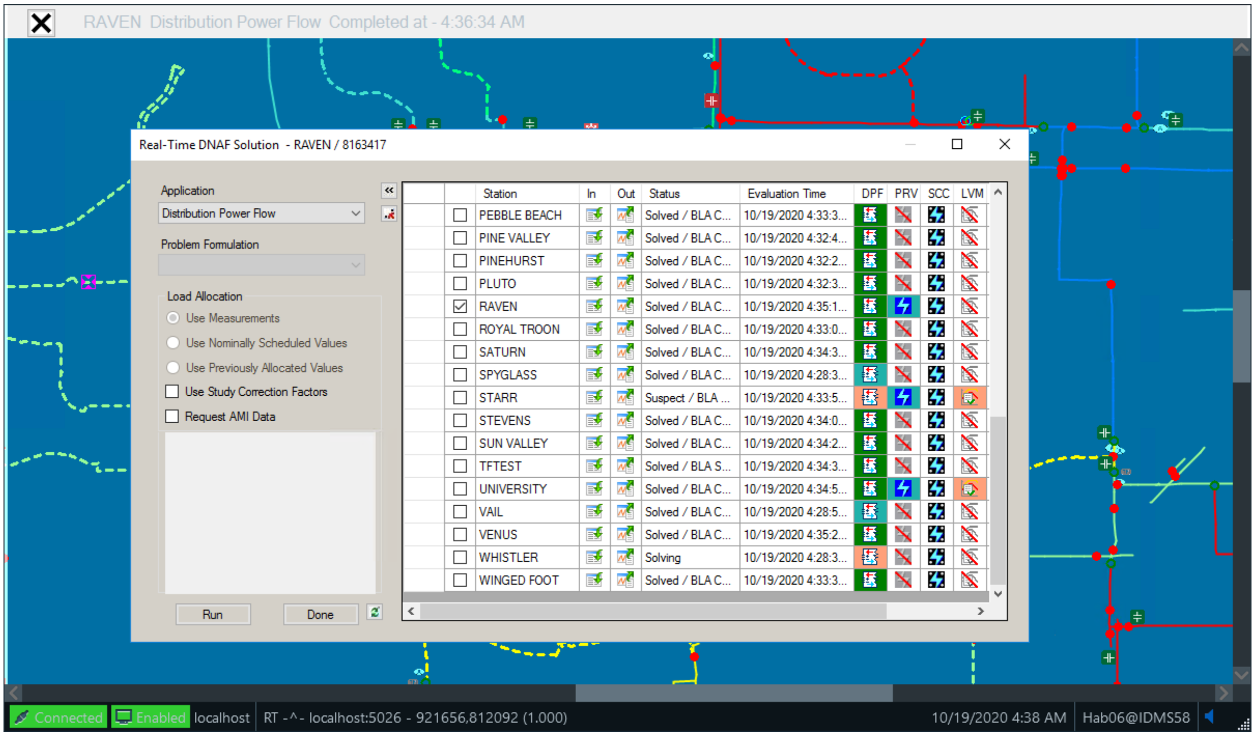


Figure 43. DNAF-Specific Mode View

3.1.3. Confirming That the System Is Up and Running

Once the system starts up, it is important to ensure that the system is running correctly and that the client is connected to the server.

The primary indication that the DMS (Main) server is not connected is a gray mesh pattern over the geographic network view (see image below). In addition, the status bar in the lower-left corner the viewport shows the names of the servers to which the client is connected or if there is no connection.

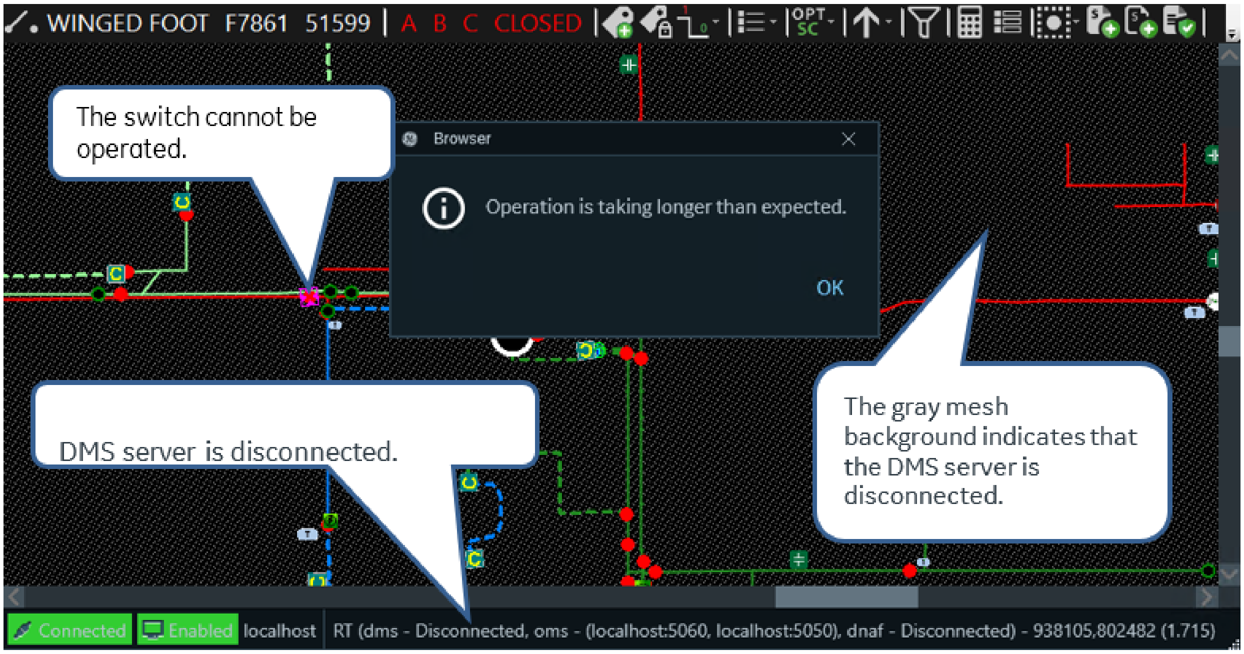


Figure 44. DMS (Main) Server Disconnected Status Indication

In the case where the system loses its connection to the DMS server after a period of operating normally, the gray mesh pattern appears and the server Disconnected message appears. It is not possible to operate any devices that cause a status change in this condition. An error message appears, as shown in image above. The gray mesh pattern automatically disappears as soon as the connection to the server has been restored and any dynamic data changes have been re-synchronized.

The system also checks that the versions of the station models on the client are exactly the same as those on the server. Since the models are updated regularly (usually every night), there are brief periods when the server has updated but the client is still waiting to receive the new versions of the station models. When this situation occurs, a yellow border is displayed on the geographic network view and the background of the display shows a yellow mesh pattern, as shown in image below. The yellow mesh pattern only appears for the station that is not in step with the server, and so it is only when the client is focused on the particular station that the effect of the model update process is visible. Normal operations are permitted while the yellow mesh pattern is displayed, but users are warned to be aware that some part of the model may change in the next few minutes. The yellow border and mesh pattern disappear automatically when the models are up to date.

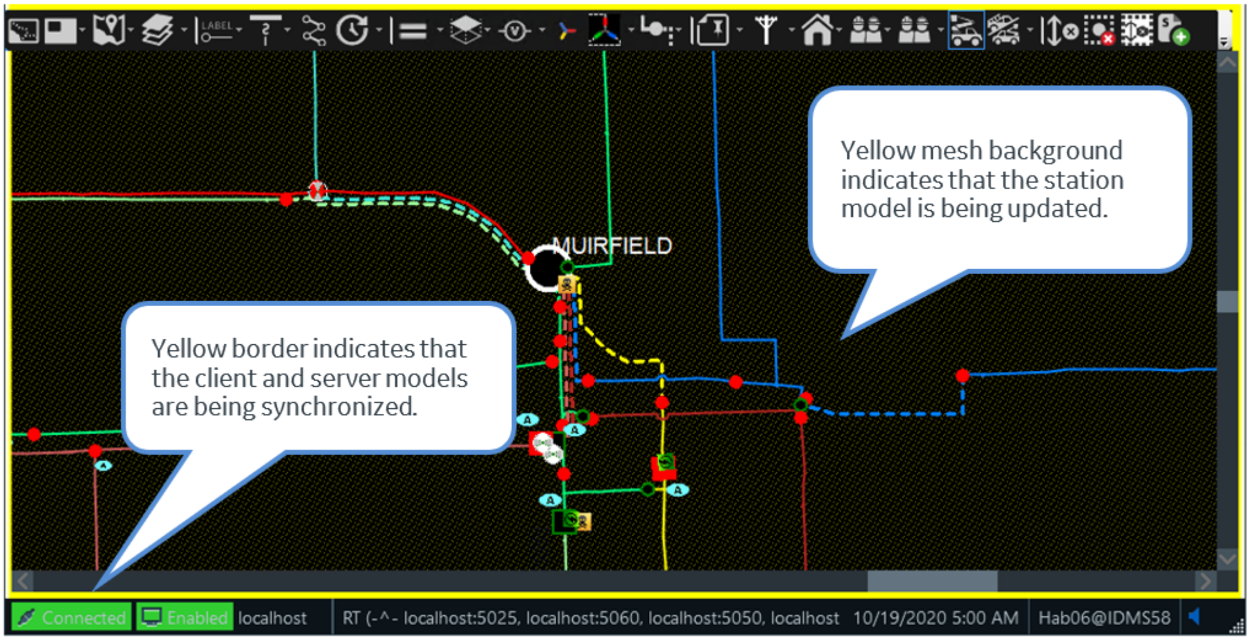


Figure 45. Pending Model Update Indications

3.1.4. Setting or Changing Your Area of Responsibility

Areas of Responsibility (AOR) are used by the system to manage the portion of the network that each user is expected to control at any one time. Setting the AOR affects both the SCADA system and the distribution management functions. This means that the Alarm displays, SCADA controls, and manual switch operations are all restricted to the same area of the network.

The default settings for AOR are normally set by a combination of both the permissions assigned to the individual user name and the specific workstation that is being used.

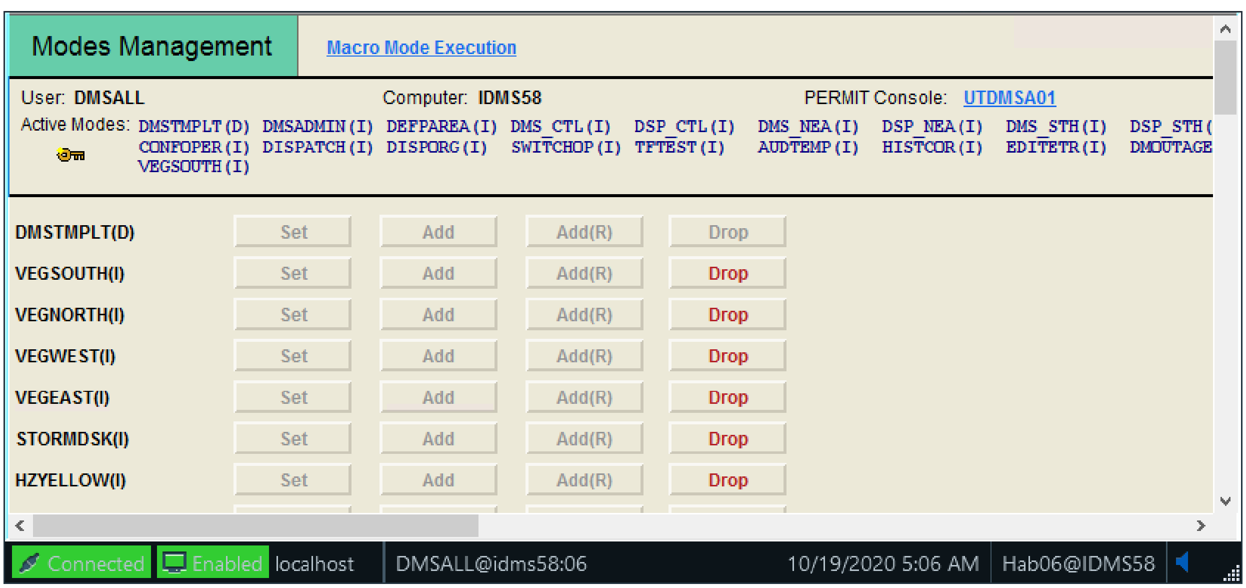


Figure 46. AOR Modes Management Display

To review or modify AOR settings, on the Browser menu bar, select HABITAT Applications > Permit > Mode Management. The Modes Management display opens, as shown in image above. This display shows the list of control areas that are available and active for the current login. Click the Add button to add another area to the current permissions, or click the Drop button to remove an area. The Add® button is used to add an area in view-only (no controls) mode.

> **Tip**
>
> A quick way to see the list of stations you currently have in your AOR is to call up the Station Exception display. The stations that are shown on this display are those in your current AOR.

3.1.5. Managing the Multi-Screen Desktop

The standard operational client is a multi-screen- computer with up to four (or more) monitors connected. This means that the monitors form a single virtual screen that is controlled by the mouse and the keyboard. Viewports or windows can be displayed anywhere within this virtual screen. There is just one mouse pointer, which can be moved to any screen in a single movement. Viewports can be opened and moved to any monitor screen.

Viewports can be stretched across multiple monitor screens. However, if a viewport is maximized, it is re-sized to fully occupy the current monitor screen.

When working with maximized viewports, it is very easy to lose track of viewports as they become hidden behind each other. To avoid this problem, check the Microsoft Windows task bar for a complete list of all currently open viewports.

It is important to recognize, when working on a multi-screen environment, only one viewport can have focus at a time (that is, respond to mouse clicks or key presses). Operators must be sure that they are looking at the correct screen before entering data or clicking the mouse buttons.

3.1.5.1. Rooms

Browser allows operators to save a particular configuration of displays so it can be recalled as required. This can greatly speed up operations that require a number of displays to be opened and arranged on the screen. Each saved configuration of displays is called a "room".

To create a room:

1. Arrange the displays on all screens as required.
2. Open the Browser Console panel by clicking on the entry on the taskbar.
3. Select the Rooms tab.
4. Enter a name for the room in the text field, and click the Save Private Room button.

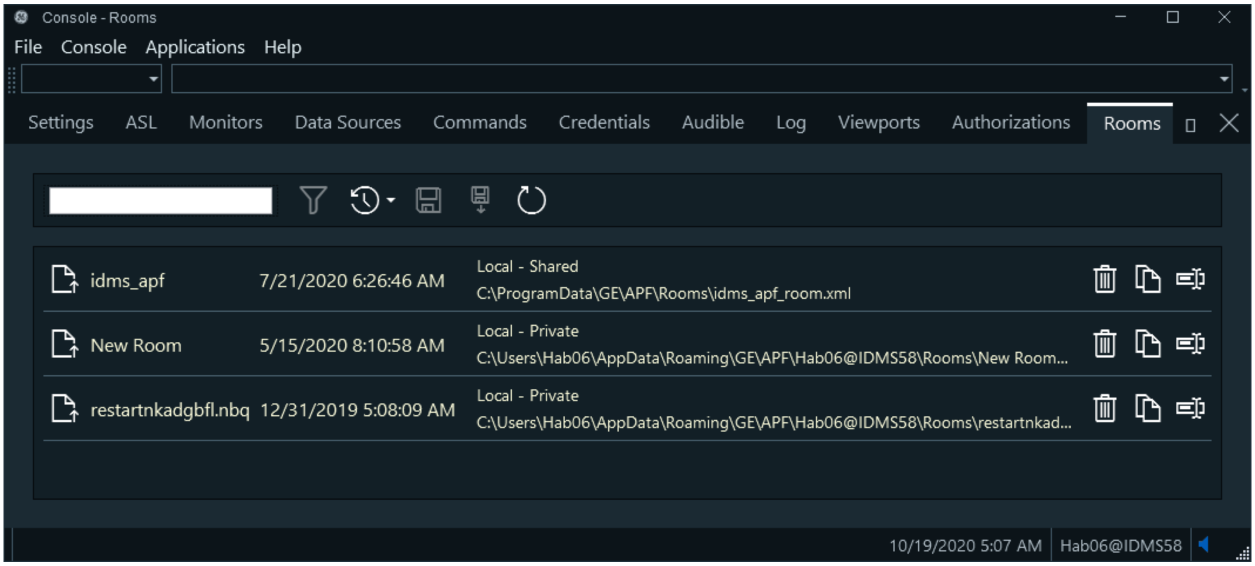


Figure 47. Rooms Tab

To recall a saved room:

1. Go the Rooms tab on the Browser Console panel.
2. Select the required room from the list, and click the Callup button.

3.1.5.2. Startup Room

A startup room is defined to give the initial group of viewports that is displayed upon login. This room is configured by the system administrator.

3.1.5.3. Frames

Frames are a function of Browser that allow multiple views to be displayed within a single viewport. This can be useful as a method of displaying all the information relating to a current operation in one viewport. For example, close-up geographic views of two points on the network that are involved in a switching operation can be displayed as frames in a viewport that can be easily minimized and restored as required during the switching process.

Another example of frames is shown in image below, where a geographic network view, alarm list, and station overview are combined in a single viewport.

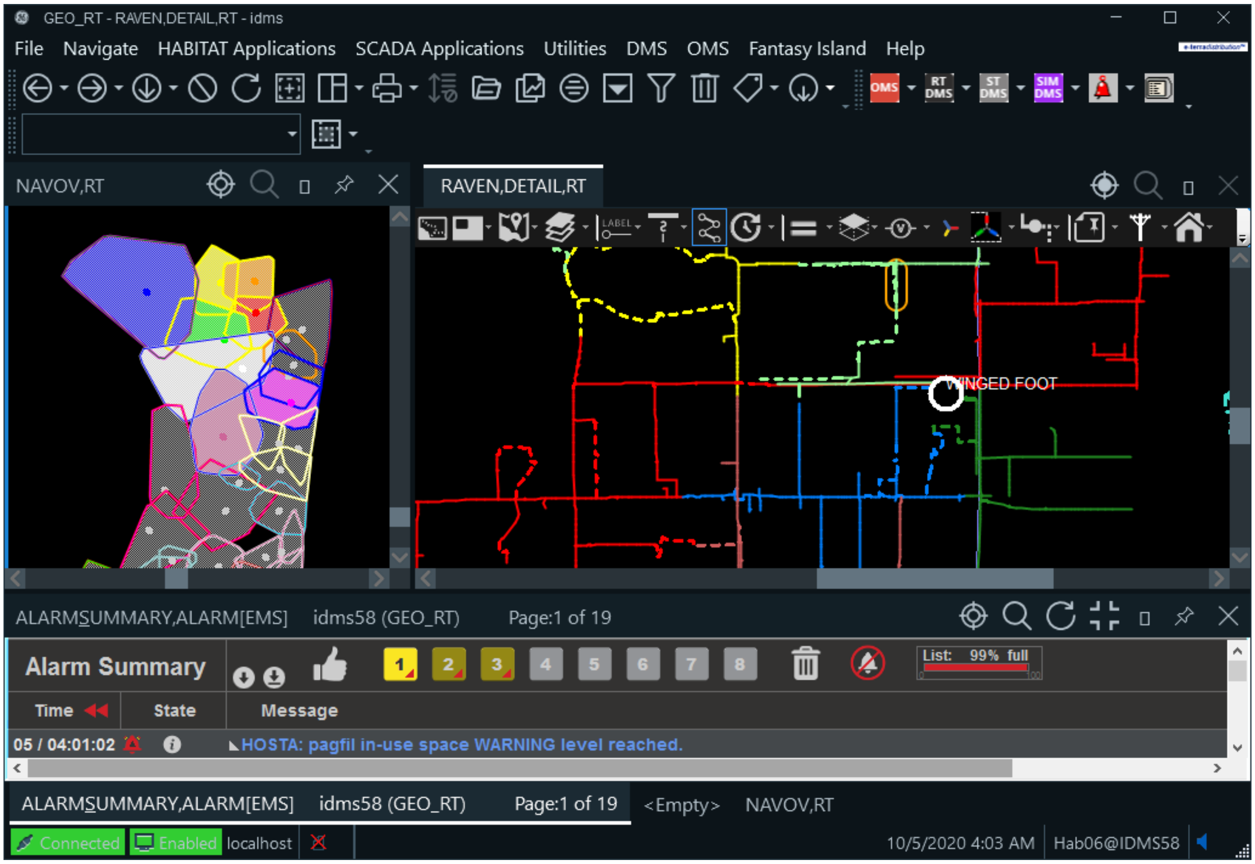


Figure 48. Example of Frames within a Viewport

To insert a frame into a viewport, click the Frames menu to the right of the Browser toolbar (see image below). Click the appropriate menu item to insert the frame in the desired position. Once the frame has been inserted, it can be re-sized by dragging the separator bar. In a similar fashion, additional frames can be inserted in the viewport.

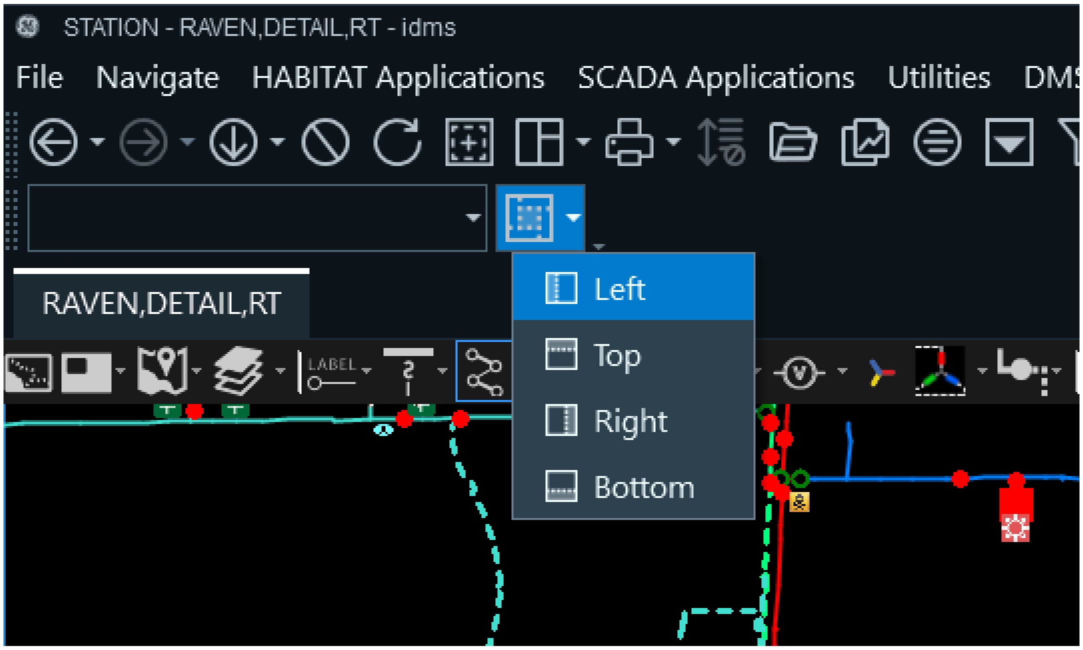


Figure 49. Frames Menu

To load a display into a frame, it must be selected by clicking on the header bar at the top of the frame. The header bar is colored dark gray to indicate that the frame has focus. Any selection from the menus or toolbars is then loaded into that frame.

A frame can be deleted by clicking the "X" in the right corner of the header bar. Frame configurations can be saved in rooms, which allows for a specialized display to be easily retrieved when required.

3.1.6. Display Navigation - Back, Forward, and Previous

Navigating between previously shown displays is performed by clicking the browser control buttons on the Browser toolbar. These buttons are grayed out if there is no display history available. Clicking the Back button moves to the previous display shown in the viewport. Clicking the Forward button moves through the history to the most recently shown display. In addition to the Back and Forward buttons, there is a Previous Display button that allows the user to step back to the previous different type of display.

The lists maintain the last 40 display selections.

> **Tip**
>
> When a viewport is first opened, the Back and Forward buttons are grayed out. When there are other displays/views available to be selected, the buttons are shown in green. Changes of zoom level are not considered to be a different display.

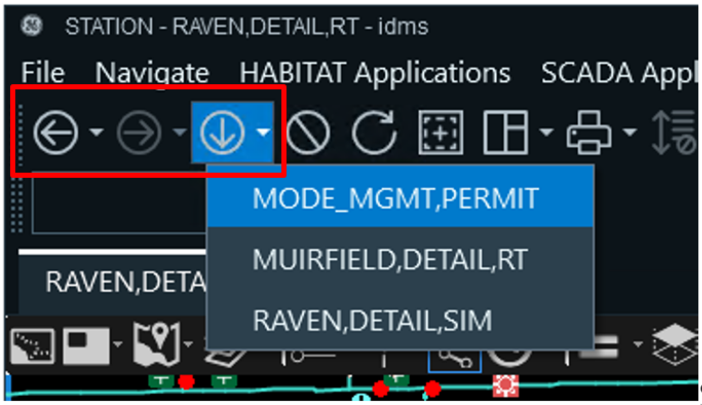


Figure 50. Browser Navigation Buttons

3.1.7. Opening a New Viewport

To open a new viewport, click the Create New Viewport button on the Browser toolbar. This creates a new viewport with the ADMS toolbars and menus.

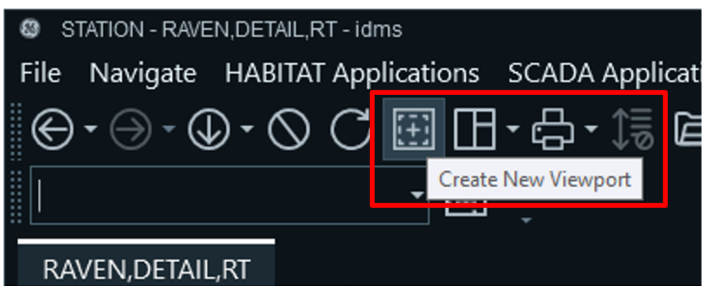


Figure 51. Creating a New Viewport

3.2. Basic Operations with Network Views

3.2.1. Partial Model View

The Partial Model View (PMV) refers to the function that presents a view of only the requested station and its electrically adjacent stations (that is, those stations that can be tied via switching to the requested station). Other stations are not loaded and are not shown on the network view, which reduces the client memory footprint.

When a station network view is called, the resulting display is centered on that station along with a number of stations that are also shown surrounding it.

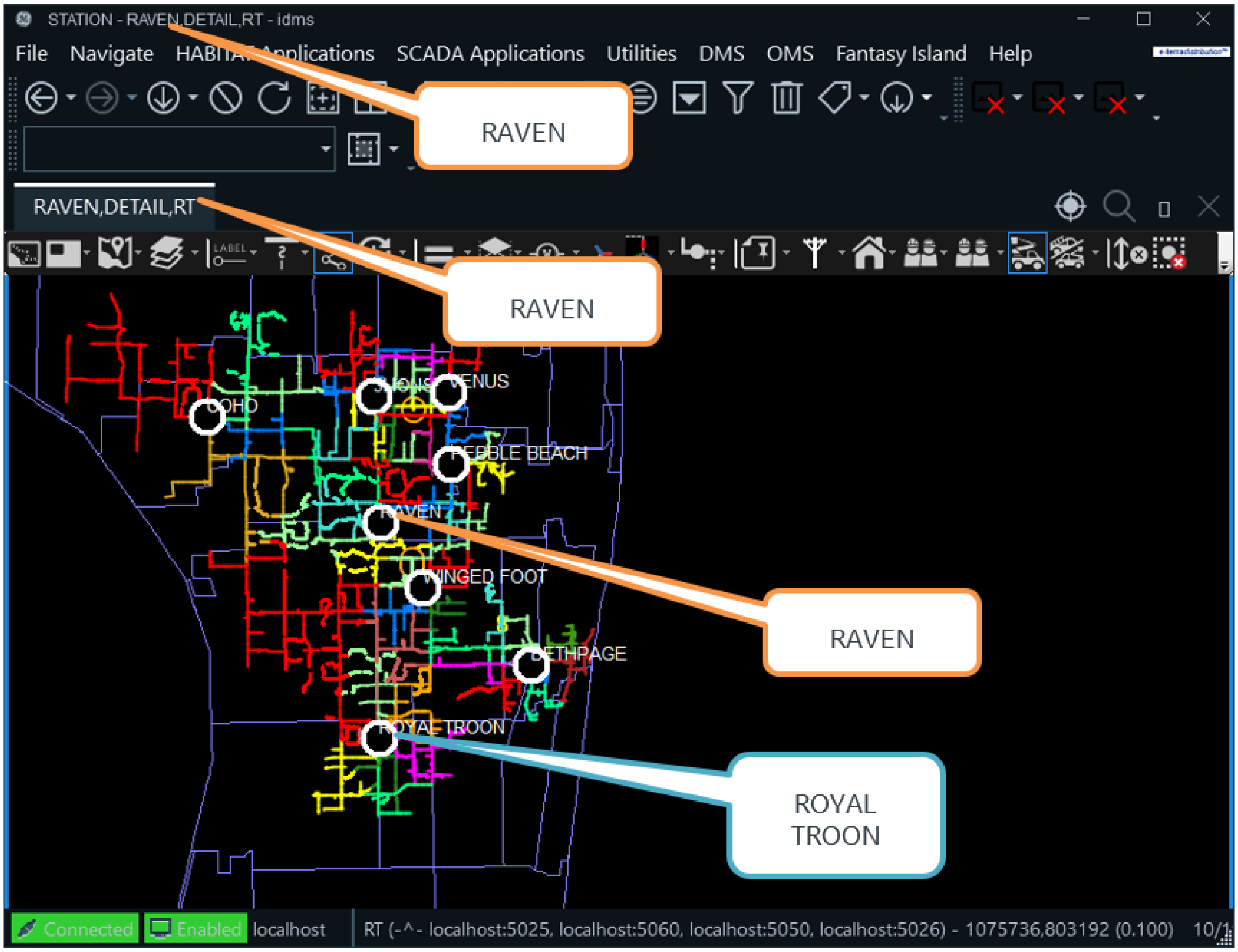


Figure 52. Partial Model View of the RAVEN Substation

In image above, RAVEN is selected, but the resulting display also includes COHO, 3LIONS, VENUS, PEBBLE BEACH, WINGED FOOT, BETHPAGE, and ROYAL TROON. These are all stations that can potentially be tied to RAVEN.

The PMV can be easily re-centered by clicking the Load Surrounding Substations (Update Partial Model View) button on the common toolbar. This uses the center of the current display to determine the selected station and the neighbors to be displayed.

If you find that you have scrolled to the edge of a station and there is no neighbor being displayed, re-center the display. Note that both the main geographic display and the navigation view, if showing, are updated. Some of the stations that were displayed as part of the previous partial model view can be unloaded as part of this process.

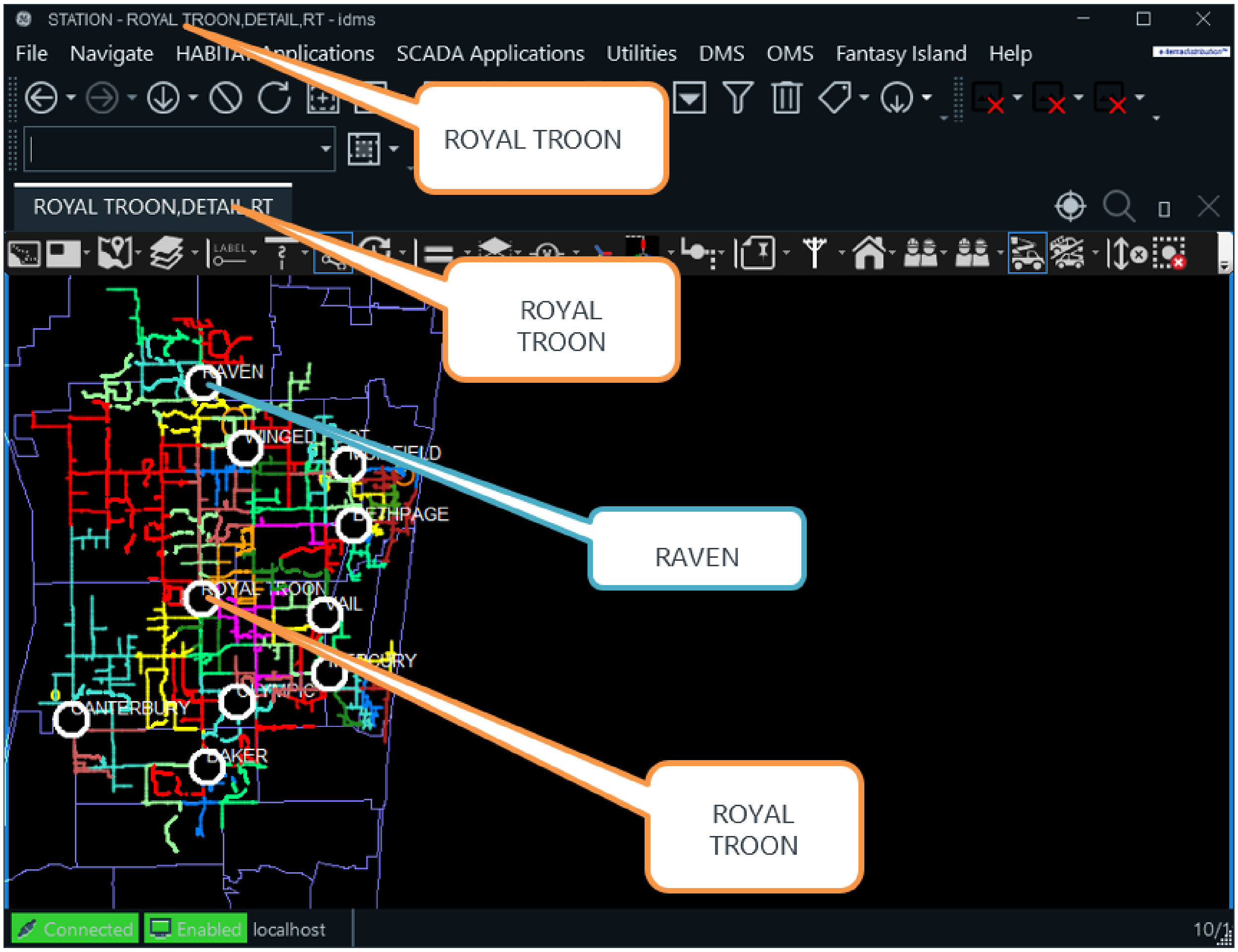


Figure 53. Re-Centered Partial Model View Showing Change of Station Grouping

A more-detailed description of the partial model view is included in Partial Model View - How It Works.

3.2.2. Basic Navigation

The main ADMS display provides the ability to show all parts of the network in a geographic or schematic view. You can navigate the network view using combinations of the keyboard, the mouse, and toolbar buttons.

Table 6. ADMS Main Display Navigation Functions

Function	Action
Pan	Click and drag the cursor in the direction the display is to be panned
Re-Center the Display	Click the scroll wheel; the display is centered on the current cursor position
Zoom In	Press F4, or scroll up with the mouse wheel
Zoom Out	Press F5, or scroll down with the mouse wheel

Function	Action
Return to Original Zoom Level	Press F3
Rubber-Band Zoom	Hold down the Shift key and the scroll wheel, and select the zoom area
Select a Device	Click the left mouse button
Deselect a Device	Click the left mouse button on a blank space, or close the device-specific toolbar

The full list of key combinations is given in Quick Guide.

3.2.2.1. Panning and Zooming

The ADMS geographic and schematic displays can be magnified or reduced by using the scrolling the mouse wheel.

- Scroll Up causes the display to zoom in.
- Scroll Down causes the display to zoom out.

Zoom levels can also be controlled using the keyboard function keys:

- F3: Returns the display to its original zoom level
- F4: Zooms in
- F5: Zooms out

Rubber-band zoom allows the operator to select an area of the screen and then fill the display viewport with the defined area.

1. Select a starting point on the viewport.
2. Hold down the Shift key and the mouse scroll wheel, and drag the mouse diagonally.

   A shaded rectangle is drawn on the screen.
3. When the rectangle covers the required part of the display, release the mouse scroll wheel.

   The display is updated with the area shaded by the rectangle now filling the viewport.

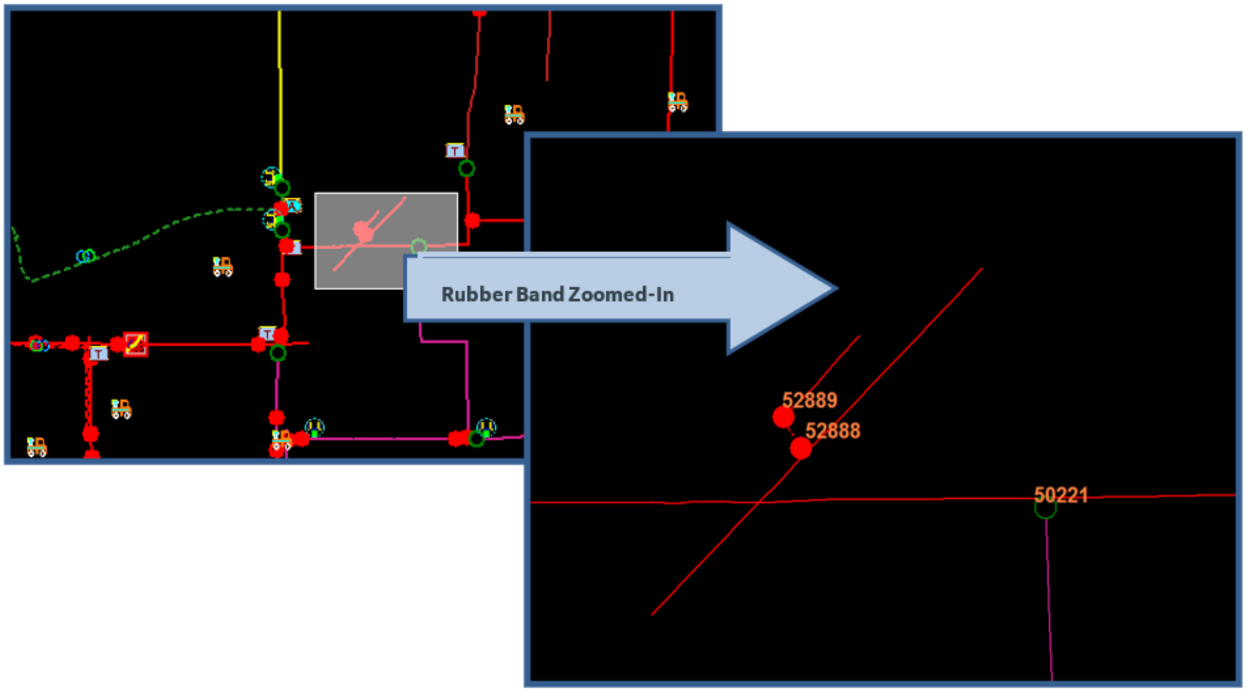


Figure 54. Rubber-Band Zoomed-In View

To re-position the display (that is, pan) in any direction, left-click the display background, hold down the left mouse button, and drag the cursor in the direction you want to pan. The cursor changes to a four-way arrowhead while panning the display. When the mouse button is released, the display is repositioned.

Note that the display can only be panned to the limit of the stations that are currently included in the Partial Model View (PMV). Once the limit of the current PMV has been reached, it is necessary to re-load/re-center the PMV by clicking the Load Surrounding Substations (Update Partial Model View) button on the common toolbar.

The network view can also be scrolled horizontally and vertically by dragging the viewport scroll bars with the mouse pointer.

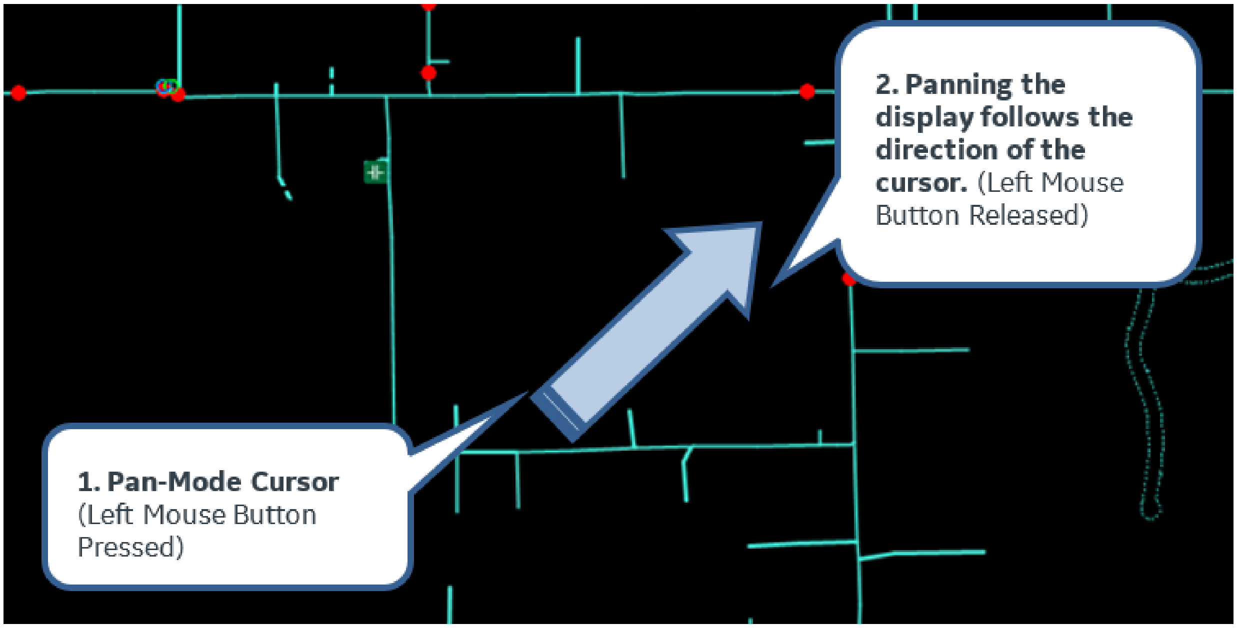


Figure 55. Panning the Display

3.2.2.2. Re-Centering the Display

To shift the view so that the display is centered on the current position of the cursor, click the mouse scroll wheel.

> **Tip**
>
> This can be a very effective method of quickly moving around the display in half-screen-size steps.

3.2.2.3. Navigating by Geographic Coordinates

The geographic view can be centered on a given state-plane coordinate by entering the two values into the text field on the Browser toolbar and selecting "Go to Coordinates on Geographic" on the RTDMS menu.

The coordinates are entered as:

`X=xxxx&Y=yyyy`

Where *xxxx* is the value of the X coordinate, and *yyyy* is the value of the Y coordinate.

Coordinates can be entered to re-center the display to any part of the model. It is not restricted to the stations currently loaded in the partial model view.

> **Note**
>
> When entering coordinates, the values must always be entered using decimal points (for

> example, X=155.22&Y=255.49). Alternate characters such as commas (for example, X=155,22&Y=255,49) must not be used.

3.2.3. Navigating to a Station

3.2.3.1. Station Overview

An easy way to navigate to a station is to use the Station Overview display. Click the Picture in Picture button on the common toolbar and select the Station Overview option to show the station overview as a picture-in-picture in the current viewport, or select Station Overview from the RTDMS menu to show it in a separate viewport.

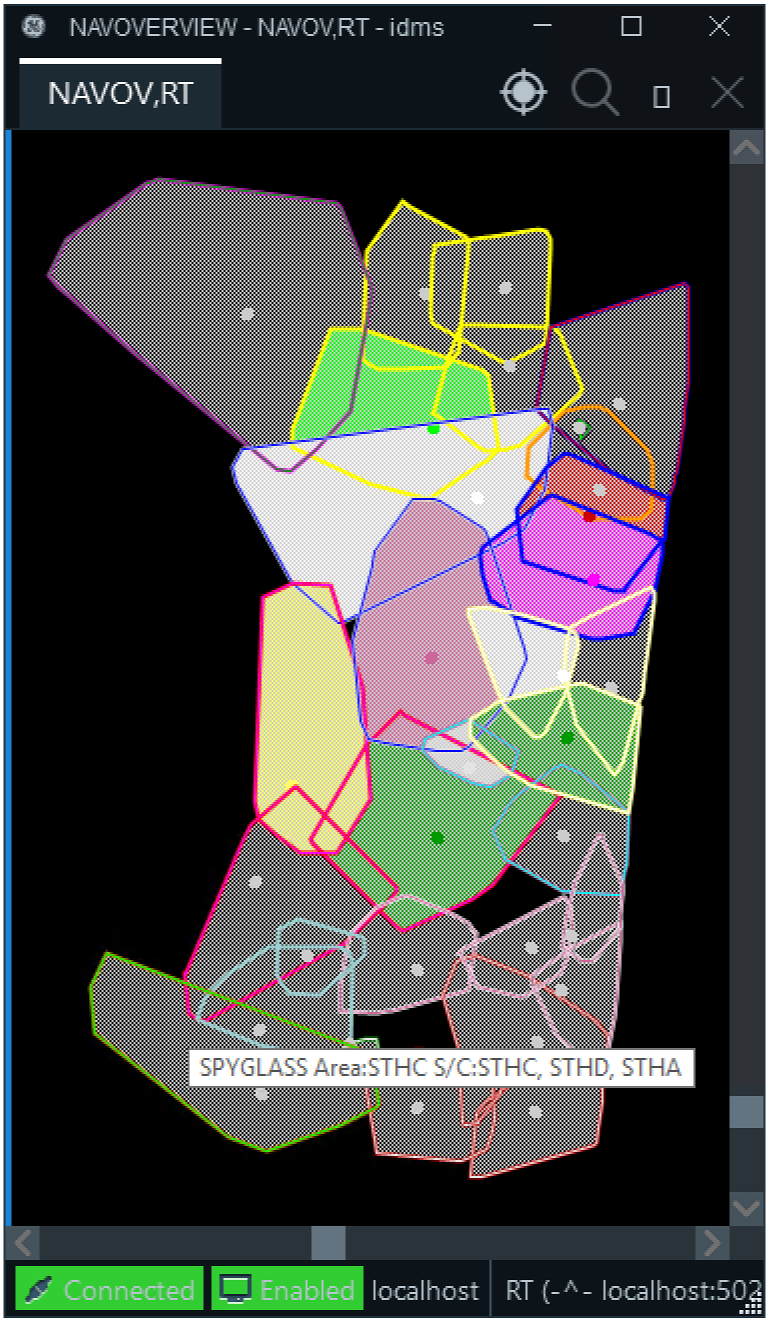


Figure 56. Station Overview Display

The station overview displays a set of polygons that represent the footprint of each substation (that is, the geographic extent of each feeder that is normally supplied from the substation). In most cases, the polygons overlap, reflecting the interleaving of the feeders in the network.

The small circles represent the actual substations. Tooltips are used to identify the substations and other information such as area and service center. Boundary colors indicate different service centers. The highlighted areas are those that are currently shown in viewports.

Tip

> If the tooltips do not appear when the mouse pointer is hovering over the station overview, click in the black portion of the display. This gives the viewport "focus" and tells Windows which particular viewport is currently being viewed.

Clicking a point on the station overview opens a geographic network view centered on that point at a nominal zoom level. Clicking another station footprint in the station overview opens another viewport centered on that station.

The station footprints are grouped with colored outlines. The colors represent the service centers.

To enlarge or reduce the amount of the network being displayed, scroll the mouse wheel; scrolling up zooms in, and scrolling down zooms out.

The display can be repositioned vertically or horizontally by clicking on the display and dragging the mouse pointer. The display can also be repositioned vertically by pressing the Shift button and using the scroll wheel.

When a network view viewport is opened, notice that the relevant station footprints on the network overview are colored to indicate that stations are included in the display. To provide an effective network view, the neighbors of the selected station are also loaded into the display. For a more-detailed explanation of the partial model view process, refer to Study Case Setup and Partial Model View - How It Works.

3.2.3.2. Area Overview

The area overview display can be accessed by navigating to the area from the Area List tabular display that can be accessed from the RTDMS menu. This display behaves the same as the main station overview but is limited to a single area (that is, South, North, and so on).

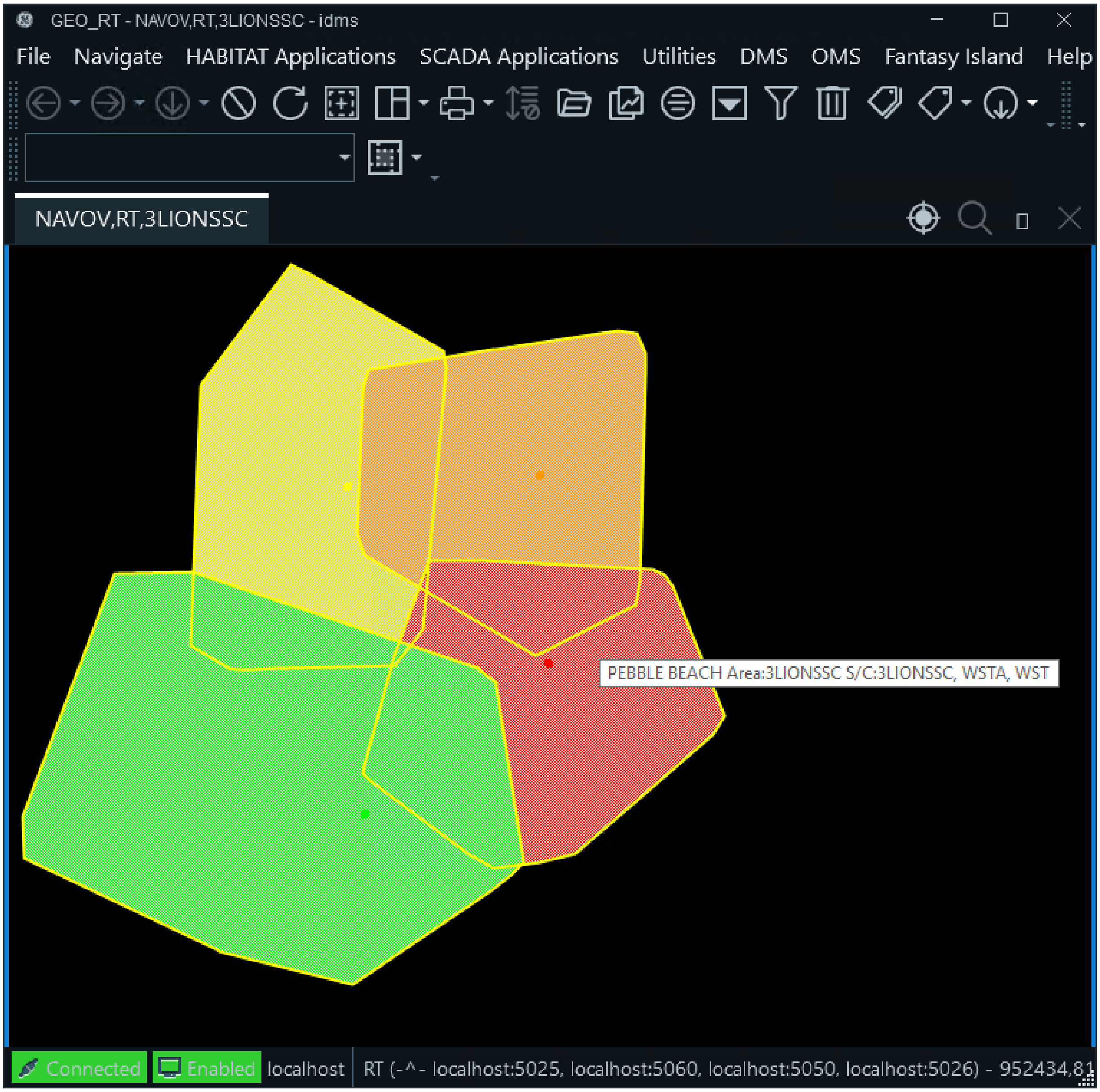


Figure 57. Area Overview Display

3.2.3.3. Stations List

To show the Stations List display, click the RTDMS menu and select Station List, or press Ctrl+U on the keyboard from the geographic view. This display lists all the available stations with icons for the types of view: geographic, SCADA one-line, station internals, and so on.

To navigate to the substation view, click the icon for the desired station and display type.

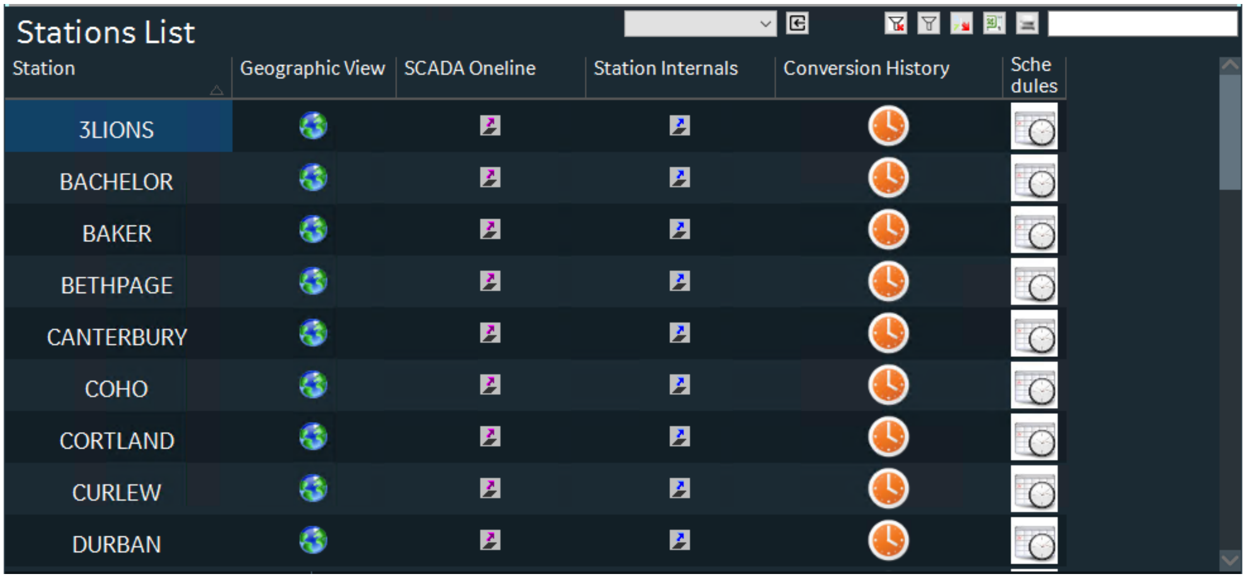


Figure 58. Stations List Display

3.2.3.4. Navigating to a Station by Name

Type the station name into the text entry field on the Browser toolbar, and click the RT DMS button. The requested station is displayed in the current viewport.

3.2.3.5. Navigating to a Station from the Browser Menu

From the Browser menu, click the company name (for example, Fantasy Island), point to the station name, and then select the desired station view from the list. The requested station is shown in the current viewport.

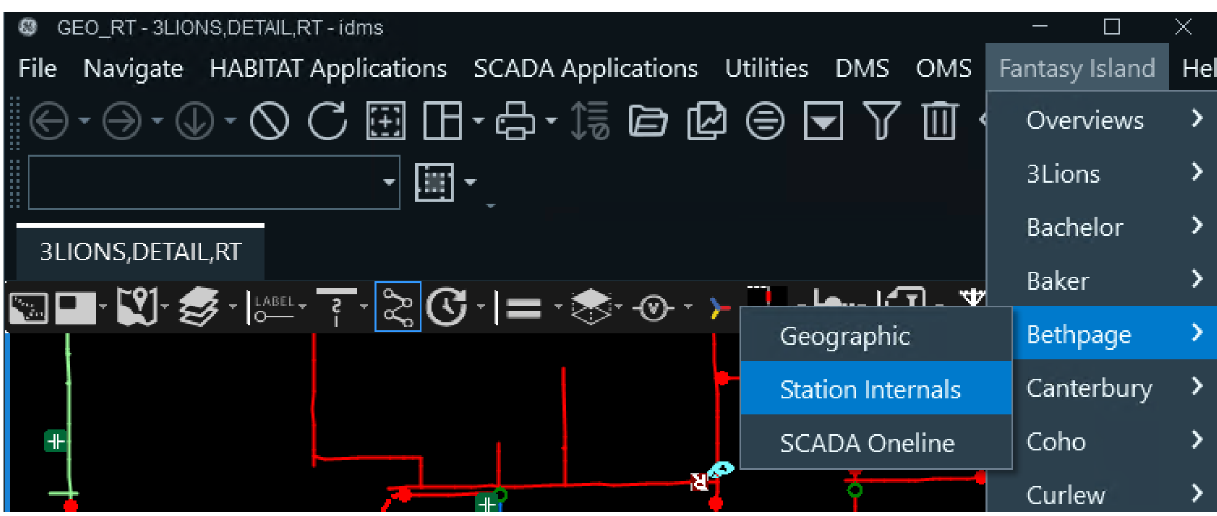


Figure 59. Navigating to a Station View from the Browser

3.2.3.6. Navigating to a Station Schematic

From the geographic network view, select the substation, which displays the device toolbar for stations. The toolbar has buttons to navigate to the SCADA one-line and the Substation Internals display. SCADA one-lines are opened in a new viewport, and substation internals are opened in the existing viewport.

Schematic substation displays can also be selected directly from the Stations List display.

3.2.4. Navigating with the Picture-in-Picture View

The picture-in-picture view opens a small window in one corner of the main viewport that can be panned and zoomed independently of the main view. The picture-in-picture window is also fully operational (that is, devices can be selected and operated in exactly the same way as the main viewport).

You can access the picture-in-picture view by clicking the Picture in Picture icon on the common toolbar and selecting a picture-in-picture option, such as Picture in Picture or Station Overview.

The picture-in-picture window has two modes of operation, follow and fixed:

- In follow mode, the picture-in-picture window displays whatever is in the central area of the main screen, following the main screen at a different zoom level.
- In fixed mode, the picture-in-picture window centers on the current selection on the main screen. The main screen can then be panned or zoomed as required without affecting the picture-in-picture window.

The operating mode of the picture-in-picture window is toggled by clicking the button on the toolbar at the top of the window.

In addition to the standard view, the picture-in-picture window can be used to display the station overview or the navigational overview (see image below).

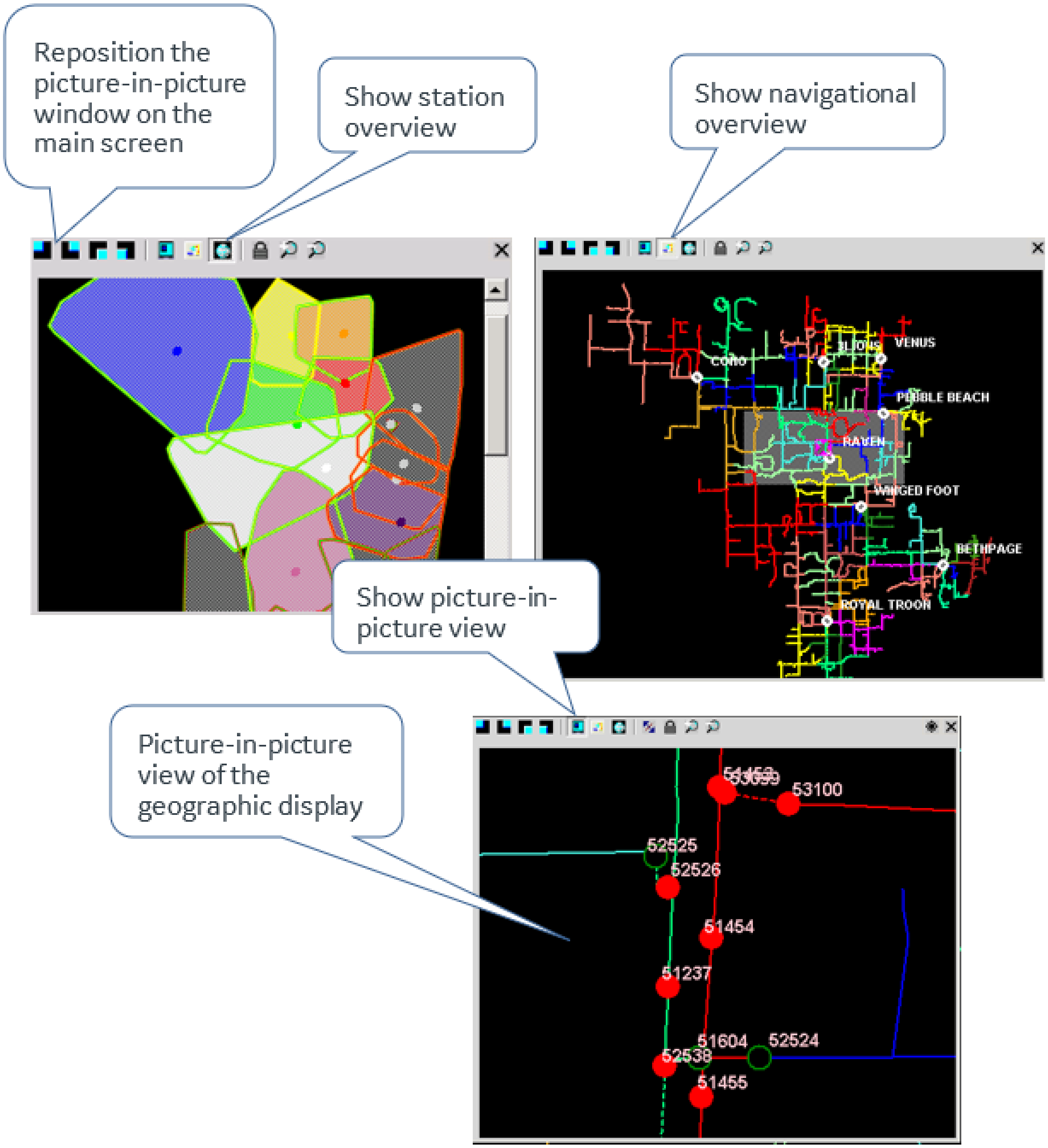


Figure 60. Picture-in-Picture Window Showing All Three Types of Views

An important feature of the picture-in-picture window is that it uses the display attributes of the main viewport at the time it was opened. This allows two different views of the network to be shown simultaneously. The picture-in-picture window can be opened and zoomed out to show an entire feeder view, and the main viewport can be set to show a small section of the network with all the details enabled.

The location of the picture-in-picture window in the main viewport can be changed to any corner by clicking the buttons on the picture-in-picture window toolbar.

To dismiss the picture-in-picture window, click the Close button (**X**) on the picture-in-picture window toolbar. When the picture-in-picture window is closed, all settings are lost.

3.2.5. Navigational Overview

The navigational overview is a small, separate pop-up window that provides an overview of the entire model view that is currently loaded.

The navigational overview can be opened by pressing the F6 key, pressing Ctrl+N, or opening the picture-in-picture window and selecting the navigational overview.

The navigational overview shows all of the stations currently loaded in the partial model view overlaid with a shaded gray rectangle, which indicates the portion of the network that is currently displayed in the main viewport. The magenta cursor shows the current center of the main display.

If there are multiple network view viewports, opening the navigational overview relates to the one that is currently selected.

To reposition the main geographic display, click the required point on the navigational overview. The gray rectangle is then re-centered to that point and the main display is re-positioned accordingly.

To dismiss the navigational overview, click the Close button (**X**) in the upper right corner.

> **Tip**
>
> If the main display is deeply zoomed in, the shaded rectangle on the navigational overview can be difficult to see because it is very small; only the magenta cursor is visible.

> **Tip**
>
> The navigational overview can also be called up for schematic displays, which can be helpful when navigating around a large and complex loop or feeder.

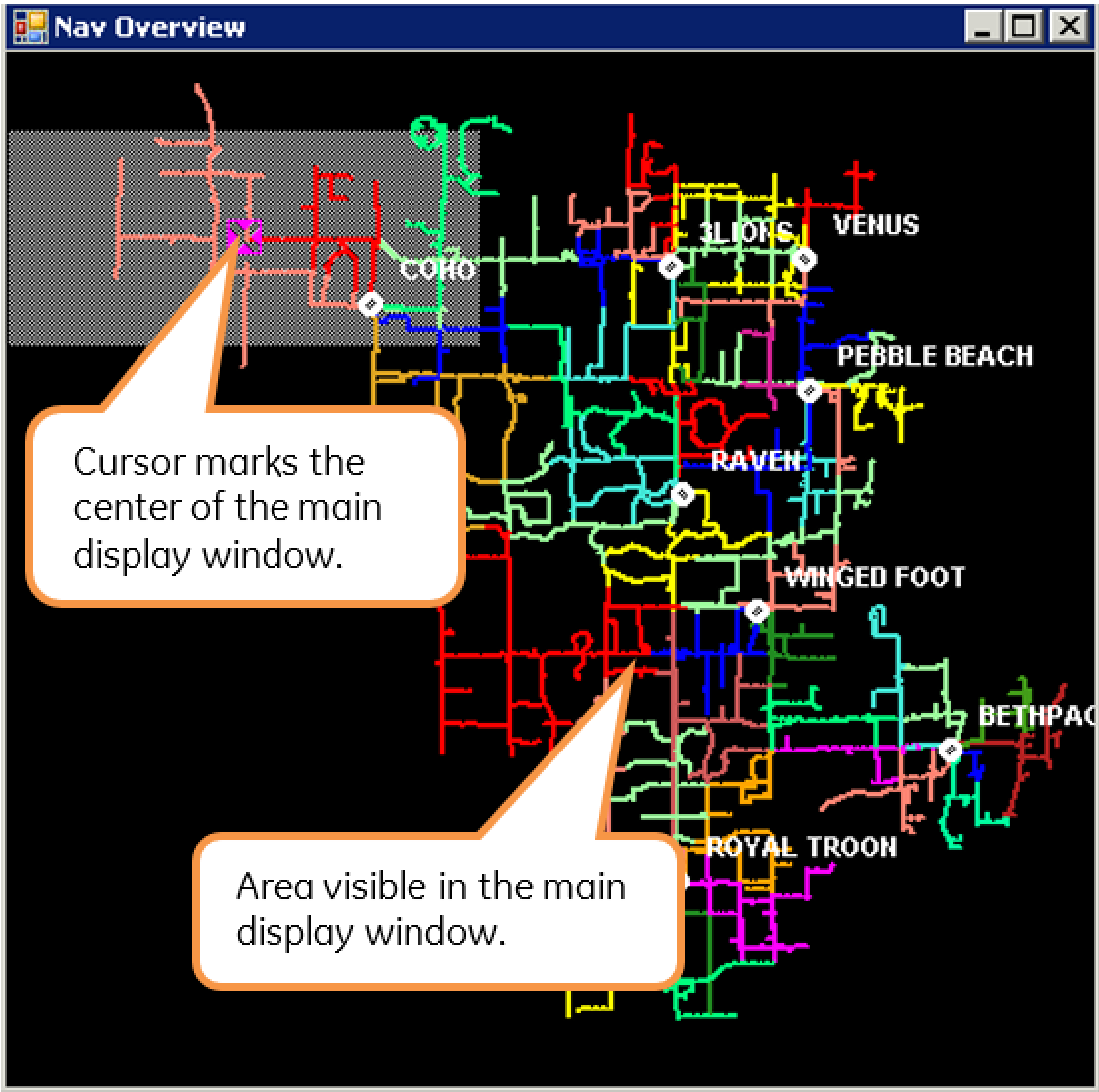


Figure 61. Navigational Overview

3.2.6. Selecting a Device

The selected device is a critical part of the operation of the ADMS user interface. Just as only one viewport can have "focus" at any one time, only one device can be selected.

To select a device, move the cursor over it and click the device. The system then acknowledges the selection of the device by covering it with a magenta check box. The ADMS toolbar is also updated with the device controls, if relevant.

In many parts of the network, there are a number of devices closely grouped together, making selection difficult. In some cases, there may be devices that are directly on top of each other. The selection process assists in this case by selecting in turn each of the devices that are in close proximity to the cursor position. This means that the operator only needs to position the cursor close to the desired device and click the mouse button multiple times, until the required device is shown in the device-specific toolbar and the symbol is shown with the magenta border. The selection process rotates through all the devices within the zone of the cursor.

> **Tip**
>
> If you are attempting to select one device in a closely packed group and you click one too many times, keep clicking. The selection rotates through all the devices and comes back to the one you want. Also, the picture-in-picture window can be used to zoom in on a device without changing the view in the main viewport.

3.2.7. Finding a Component

A system component (such as a device, customer, landbase object, structure, or crew) can be located using the Find tool.

Figure 62. Find Tool Icon

3.2.7.1. Finding a Network Device

ADMS provides two methods of finding devices and equipment within the network. The device search function finds devices by name and should cover most cases. The analyst search allows for searching by device IDs and is aimed for database verification rather than normal operations.

The device search utilizes the Ternary Search Tree (TST) technology to provide a high-performance search function that minimizes space requirements. The ternary search tree is built as part of the model creation process, allowing the required device types and identifier fields to be easily specified. By default, the device search only allows searching by device name. The analyst search does not use the TST and is, therefore, slower — but it allows for searches by both names and device IDs.

The search tool preserves the search history. To show a list of searched items, click the down arrow for the search input field.

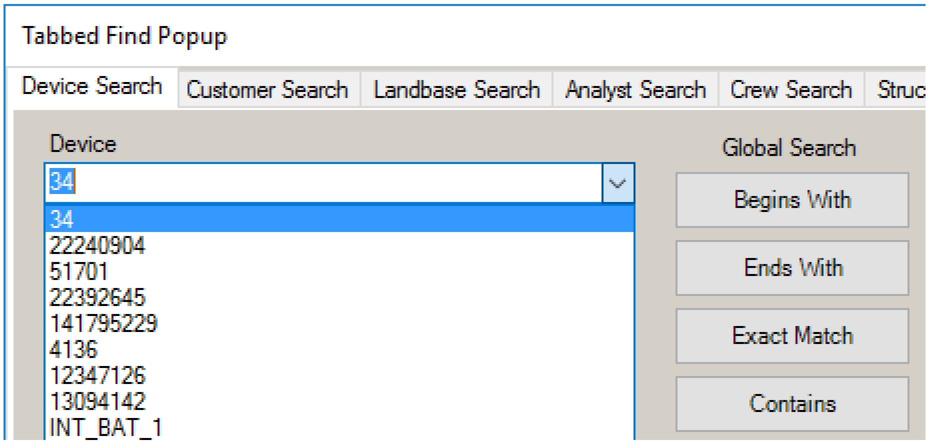


Figure 63. Tabbed Find Popup Window

3.2.7.2. Device Search

Click the Find button on the common toolbar (or press Ctrl+F) to open the Tabbed Find Pop-up window. By default, the Device Search tab is selected, as shown in image below.

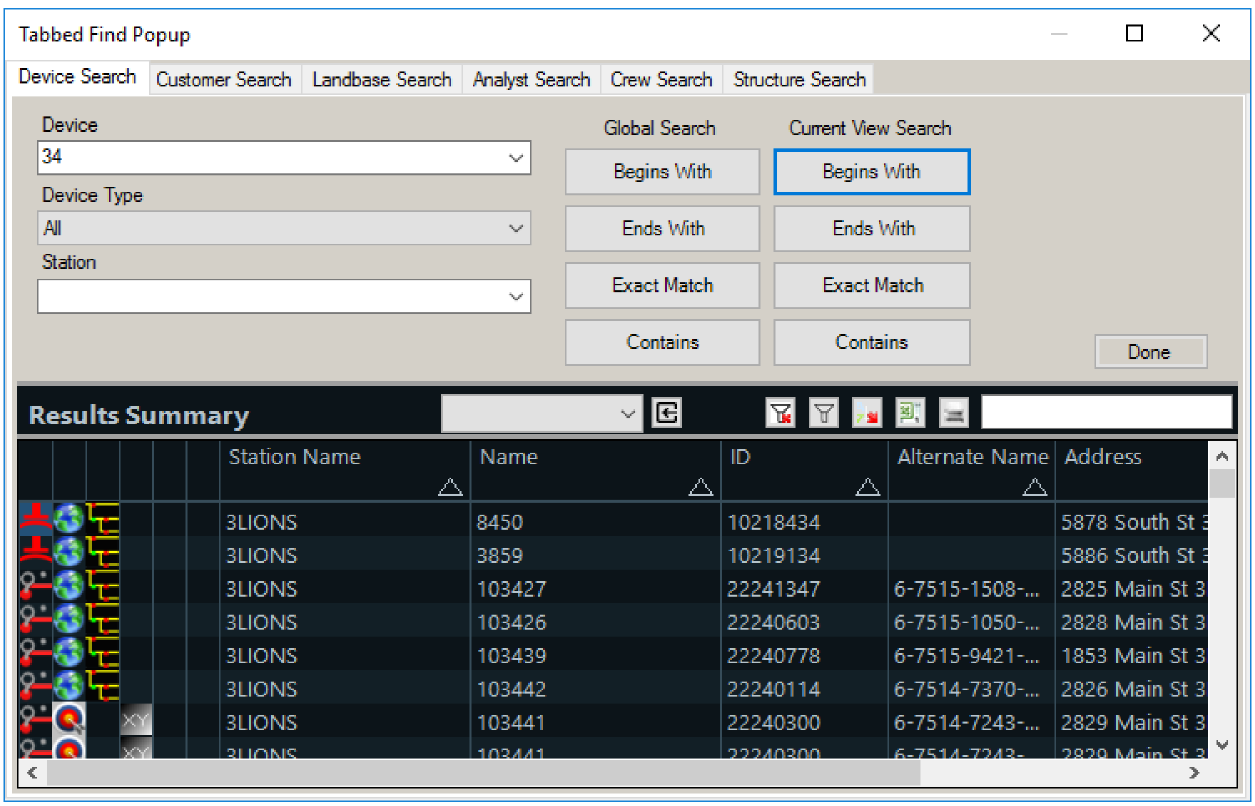


Figure 64. Device Search Tab

Enter all or part of the device name in the Device field, and select the required type of search. If the search is to be restricted to the current loaded stations (that is, partial model view), click the appropriate Current View Search button. If the search is to include the entire network, click the appropriate Global Search button. The results of the search are displayed in the summary table at the bottom of the Find pop-up.

The search can be narrowed by selecting a device type and station name from the pull-down menus.

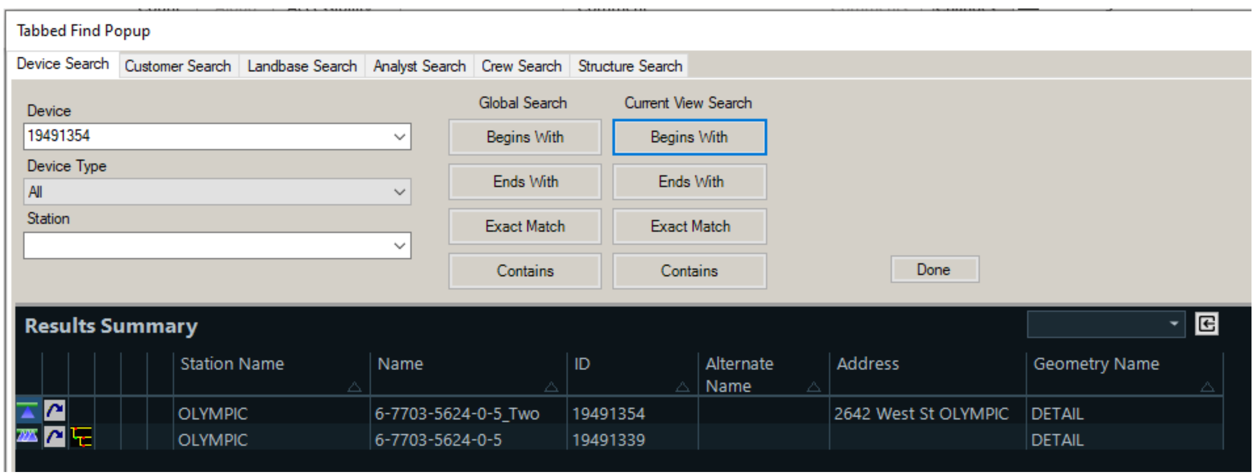
The results can be filtered using the global or per-column filtering options at the top of the results table. The table is filtered to show only those devices that contain the filter text or numbers somewhere in any of the displayed fields. In this way, the results can be filtered by part of a device name, station name, or ID code. Filtering can be removed by clearing the filter text entry field.

The results can also be sorted by clicking the column headers in a similar manner to other tabular displays.

When the required device has been located in the results table, the display can be navigated to the device by the use of the Locate buttons at the left side of the results table.

The left-most icon indicates the type of device, and a tooltip is available for this field. The other icons are Locate buttons, which navigate to the appropriate display for the device and its situation in the network. Devices can display one or more Locate buttons.

If a distribution transformer is found by any search type (for example: Begins With, Exact Match, and so on), the associated transformer bank is also displayed in the results grid along with the distribution transformer.



The meanings of the device Locate buttons are as follows:

- **Globe:** Navigates to the Geographic Network View centered on the device, with the device selected.
- **Bull's-Eye:** Navigates to the Relevant Internal View, centered on the device, with the device selected.
- **XY:** For devices that are part of an internal view, this button navigates to the Geographic Network View centered on the location of the device.
- **Curved Arrow:** Navigates to the Geographic Network View with laterals turned on. The device is centered and selected.

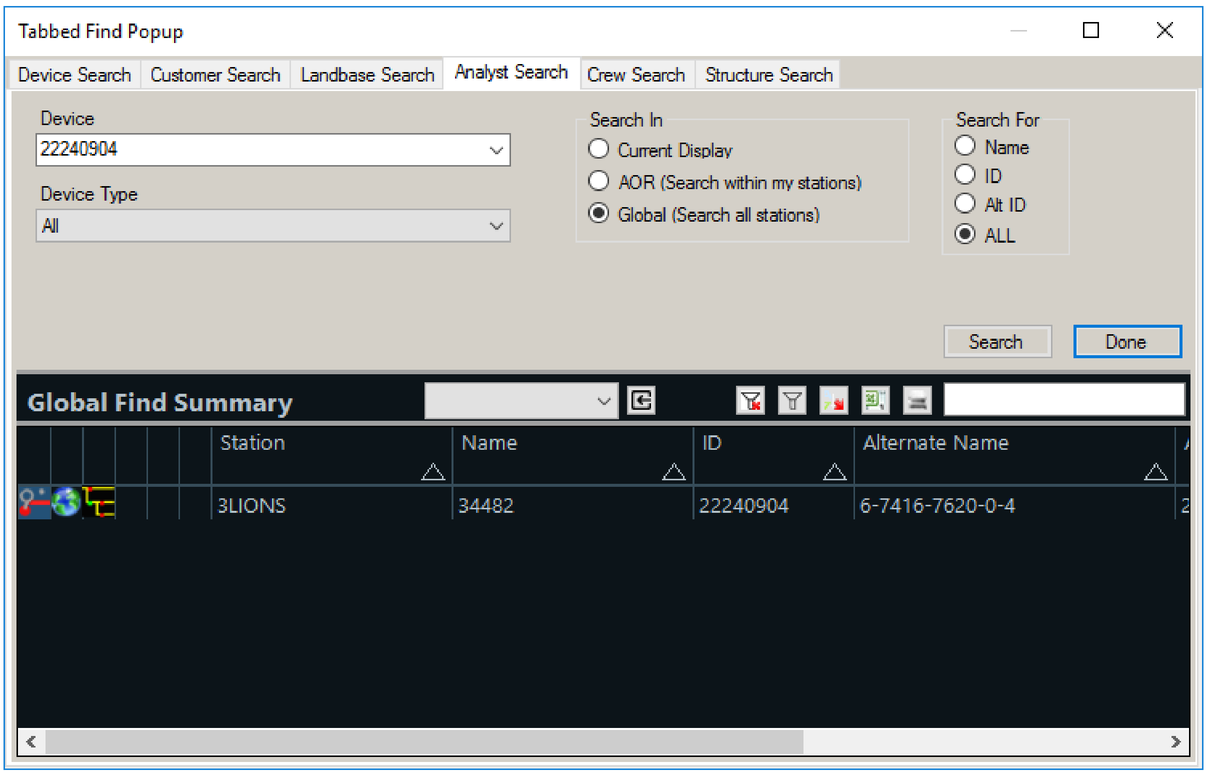
- **Navigate to Station Internals:** Navigates to the Substation Internals View.
- **Navigate to SCADA:** Navigates to the Substation SCADA View.

To dismiss the Find pop-up, click the Done button.

3.2.7.3. Analyst Search

The analyst search allows the network model to be searched by other data types such as the device ID code:

1. Click the Find button on the common toolbar (or press Ctrl+F), and then select the Analyst Search tab on the Tabbed Find Popup window.

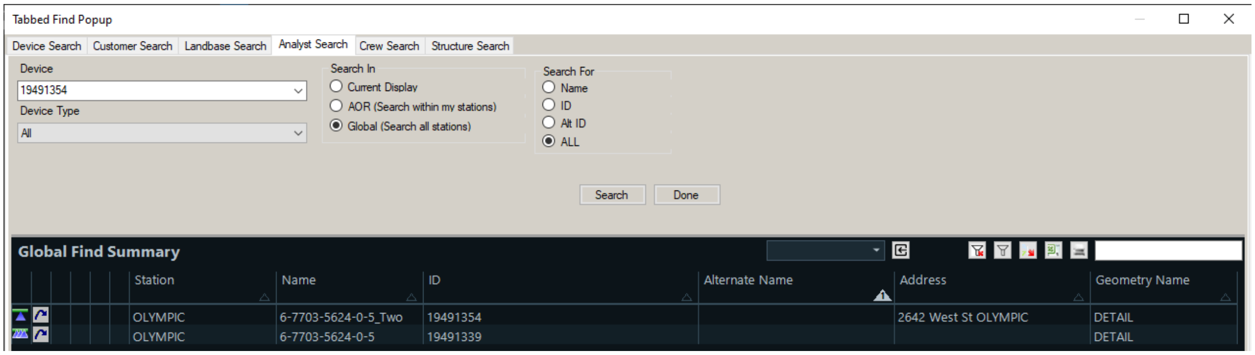


2. Enter the device name or ID in the Device field, and click the buttons for the range of the search (current display (PMV), the current Area of Responsibility, or the entire network) and the data type to be searched for. Then click the Search button.

   The search can be narrowed by selecting a device type from the Device Type pull-down menu.

   The results of the search are displayed in the results table at the bottom of the Tabbed Find Popup window. They can be filtered and sorted in the same way as other tabular displays.

If a distribution transformer is found by this search operation, the associated transformer bank is also displayed in the results grid along with the distribution transformer.



The display can navigate the required device by clicking the Locate buttons at the left of the results table. The meanings of the Locate buttons are described in Device Search.

3.2.7.4. Finding a Landbase Object

On the Landbase Search tab, you can search for geographic background objects, such as roads, buildings, boundaries, and water objects.

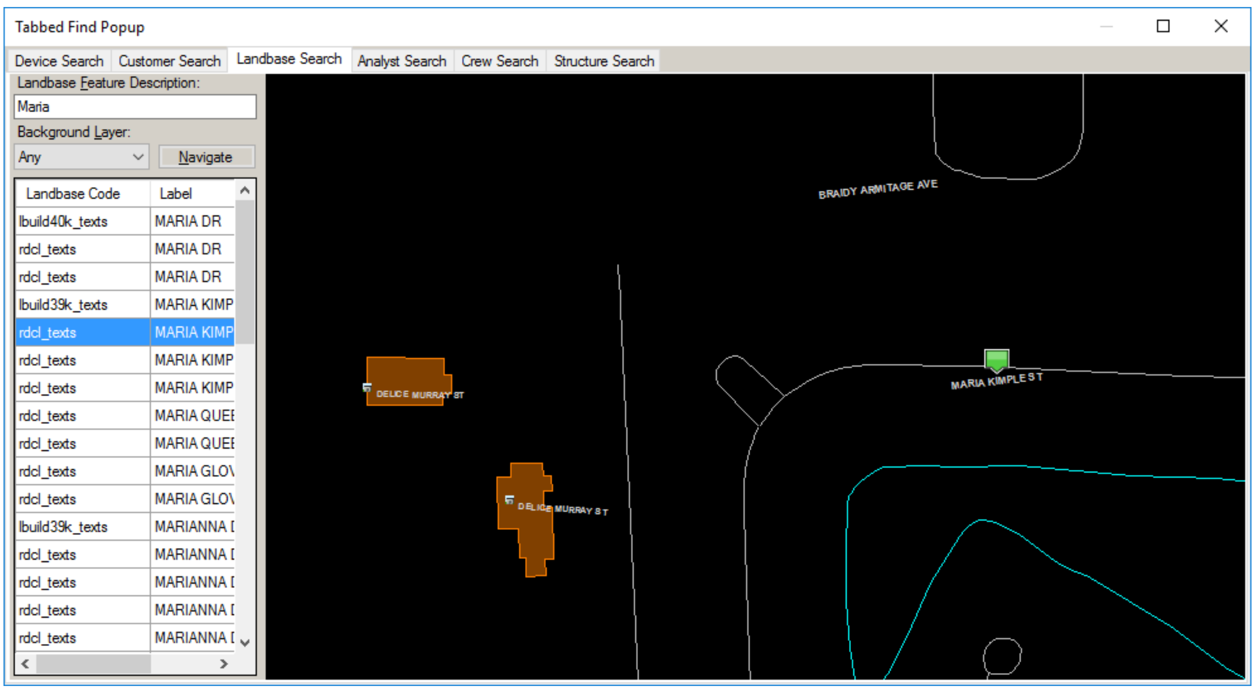


Figure 65. Landbase Search Tab

3.2.7.5. Finding a Customer

> **Note**
>
> The customer search functionality is part of the ADMS Outage Management System. This feature is only available with the OMS system installation or as a custom installation option. For

more information, contact GE.

Customers can be found by using flexible search feature. Select the Customer Search tab, as shown in image below.

This search uses a single-line text field and supports searches by name, account number, address, service point, meter number, customer key, or phone number.

Note

The search starts after three characters are entered. Partial entries are supported, but they must begin with the first characters of the name, street, or other search value. To refine the search, combine part of the name with an address or phone number. For meter numbers, the entire number must be entered.

The search results are shown in the lower part of the window. They can be filtered and sorted in the same way as other tabular displays.

To navigate to the geographic location of a specific customer, click the globe icon on the left side of the search result.

Note

The search starts after three characters are entered. No history is saved, and the search refreshes whenever the search criteria change.

Note that the Customer Search tab only appears on the Tabbed Find Popup window where customers are included as part of the optional Outage Management functionality.

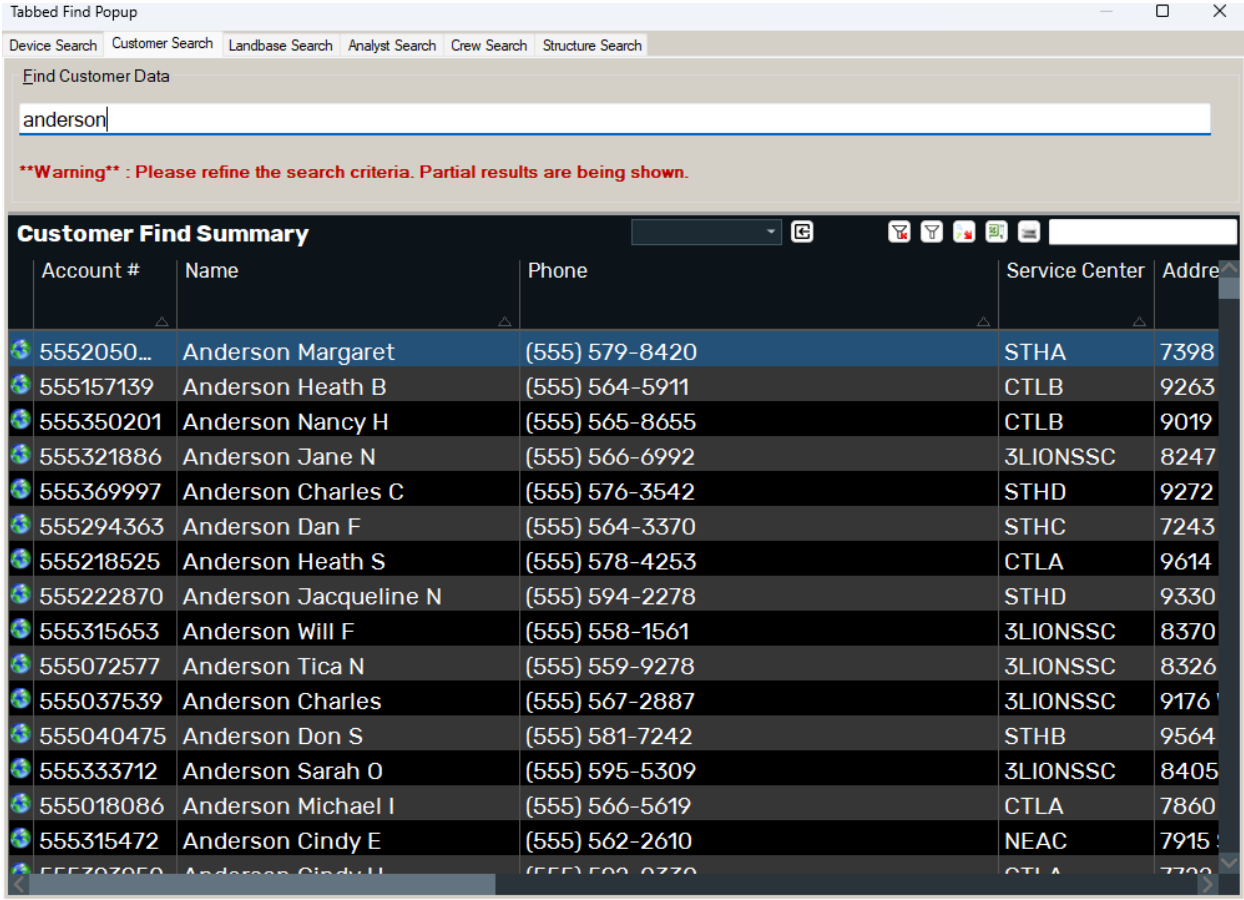


Figure 66. Customer Search Tab

3.2.7.6. Finding a Crew

> **Note**
>
> The crew search functionality is part of the ADMS Outage Management System. This feature is only available with the OMS system installation or as a custom installation option. For more information, contact GE.

The location of a crew can be found by selecting the Crew Search tab of the Tabbed Find Popup window, as shown in image below. Enter the crew name, the primary employee's name (usually the crew leader), or the radio number in the relevant fields and click the Search button.

The search results are shown in the summary table in the lower part of the window. They can be filtered and sorted in the same way as other tabular displays.

To navigate to the geographic location of a specific crew, click the globe icon at the left side of the search result.

Note that the Crew Search tab only appears on the Tabbed Find Popup window where crews are included as part of the optional Outage Management functionality.

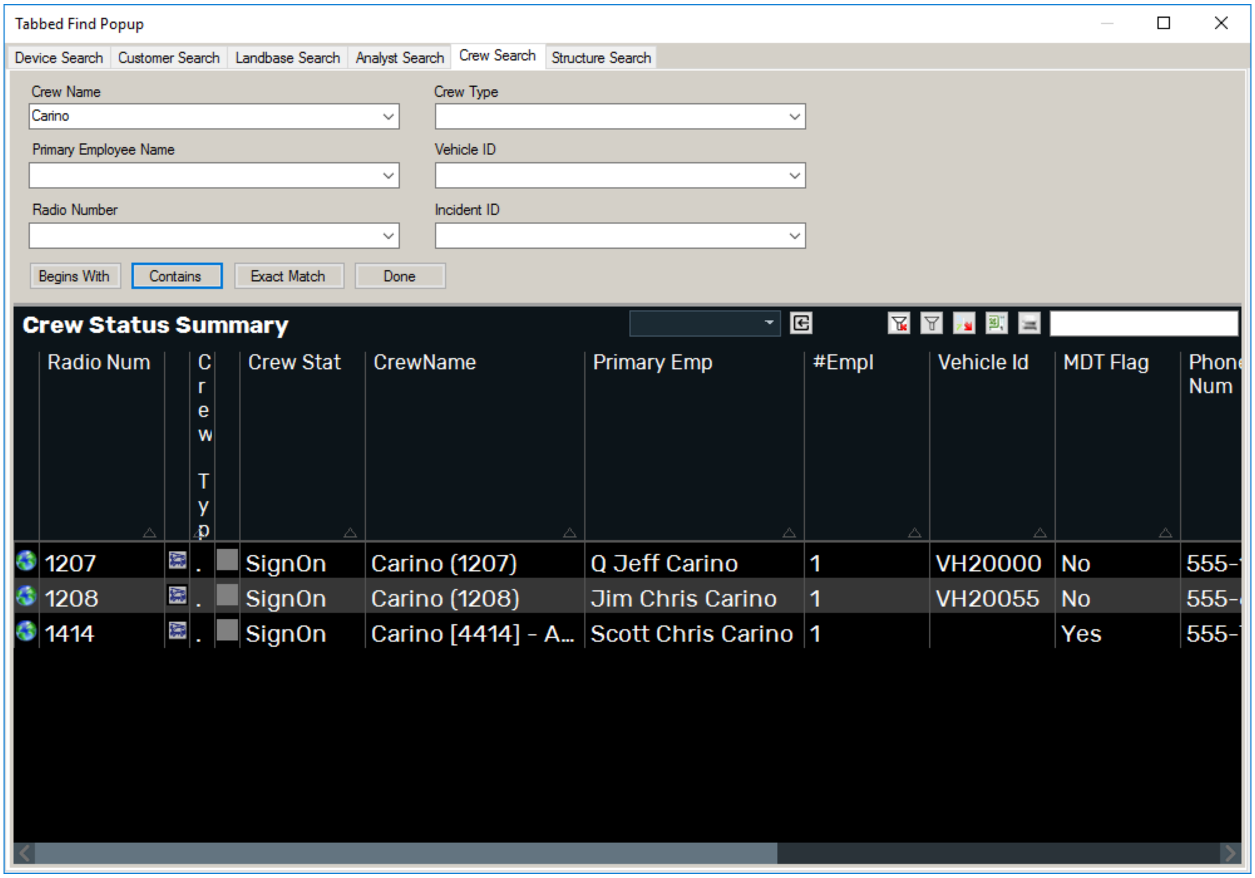


Figure 67. Crew Search Tab

3.2.7.7. Finding a Structure

You can find structures by selecting the Structure Search tab on the Tabbed Find Popup window, as shown in image below. Enter the structure name or ID and click Search.

The search results are shown in the summary table in the lower part of the window. They can be filtered and sorted in the same way as other tabular displays.

To navigate to the geographic location of a specific structure, click the globe icon at the left side of the search result.

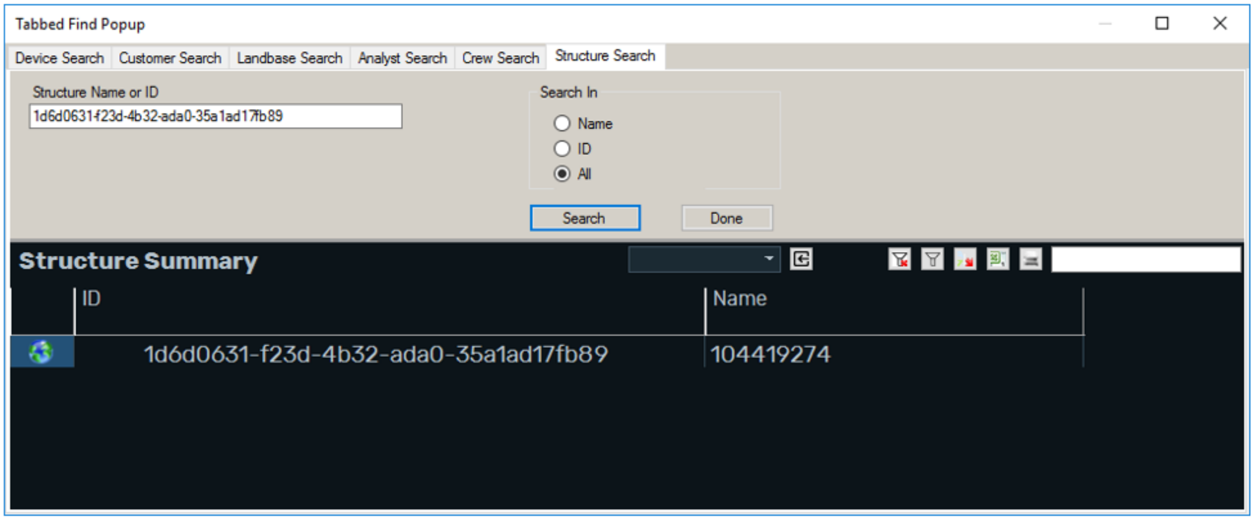


Figure 68. Structure Search Tab

3.2.8. Managing the Detail on the Display

To minimize the amount of unnecessary information being displayed, many of the available layers of information are normally hidden. The display of these additional layers is controlled by the toolbar buttons and the right-click menu. Controls are also provided to further reduce the amount of information displayed when required.

Tip

Displaying the laterals while the display is zoomed out adversely affects display performance because of the large number of objects that have to be displayed and moved on the screen. It is recommended that laterals only be shown when a small portion of the feeder backbone is visible on the screen.

Note

Filtering only applies to the currently selected network view. Other viewports are not affected.

3.2.8.1. Reconfigurable Line Section View

The network display can be concentrated to only show the main elements of the network that are capable of changing the feeder backbone configuration. All line sections, switches, and other devices that are not able to affect the configuration of feeders (typically spurs off the main feeder, lateral lines, and devices) are removed from the display to give the clearest possible view for feeder switching operations, and so on.

Use the Show/Hide Non-Reconfigurable Sections button on the common toolbar to toggle the reconfigurable section view on and off.

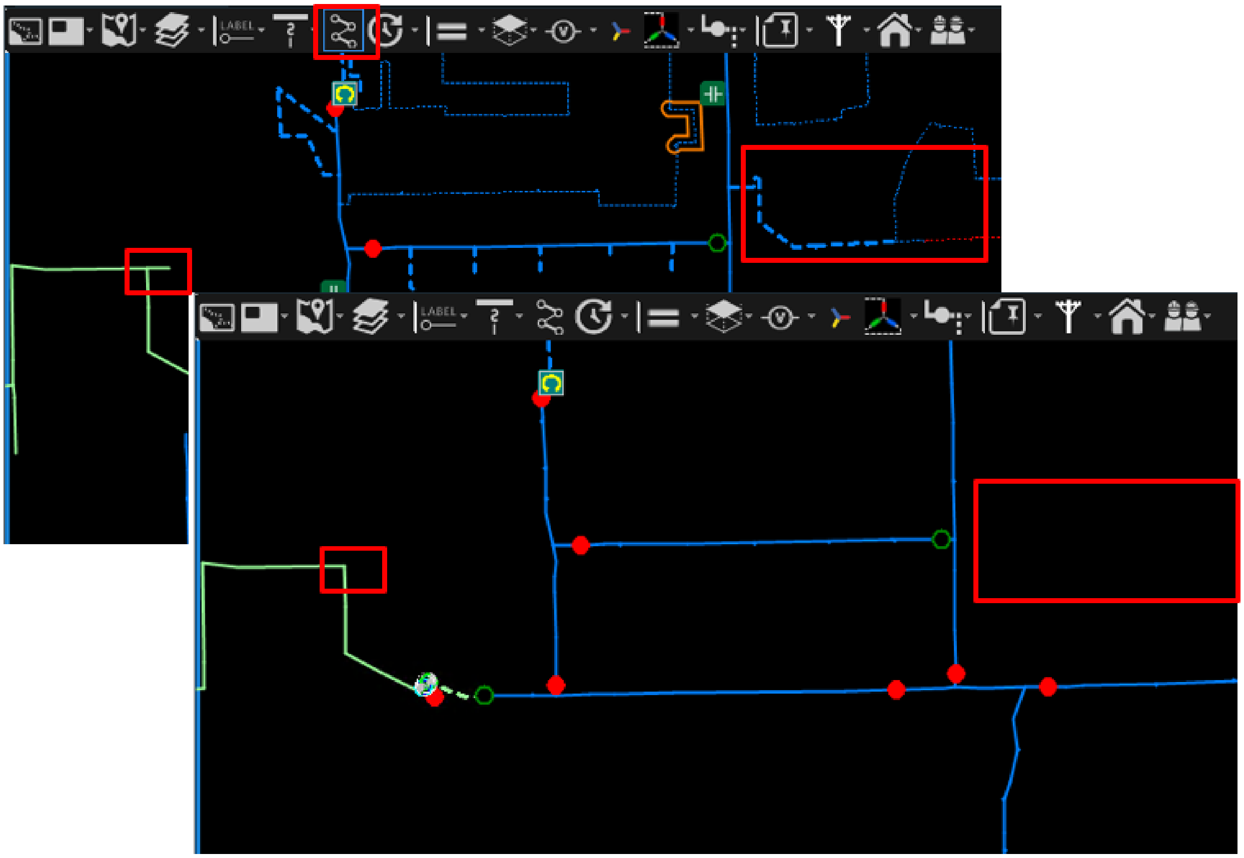


Figure 69. Show Non-Reconfigurable and Hide Non-Reconfigurable (Lower) Views

Enabling the reconfigurable section view is an easy way to remove unnecessary equipment from the current display (for example, for switching operations, and so on) without having to re-arrange the viewport.

3.2.8.2. Show/Hide Laterals

Laterals and their associated fuses are normally displayed at a specified zoom level. The display of laterals can be forced ON or OFF from the menu on the common toolbar or from the right-click menu.

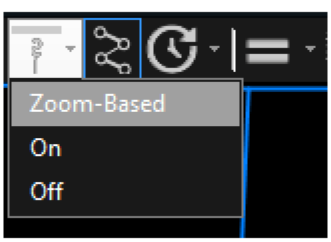


Figure 70. Show/Hide Laterals Icon

The display of laterals can be returned to zoom-based control by selecting the option from the common toolbar or via the right-click menu.

3.2.8.3. Show/Hide Labels

Device labels are normally displayed at a specified zoom level. The display of labels can be forced ON or OFF from the menu on the common toolbar or from the right-click menu.

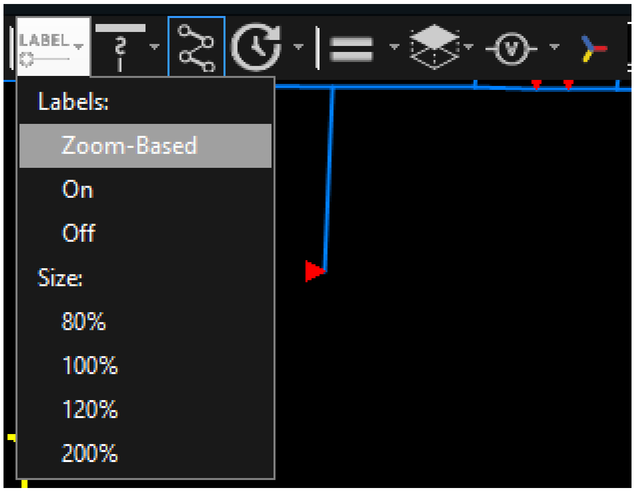


Figure 71. Show/Hide Label Icon

This can be useful for reducing clutter on the display. The display of labels can be returned to zoom-based control by selecting the option from the common toolbar or via the right-click menu. Labels are colored to reflect the voltage rating of the switch.

> **Tip**
>
> A more effective means of identifying switches is to move the cursor over the switch and allow the tooltip to appear.

3.2.8.4. Label Font Size Menu

To adjust the size of the label font on the network displays, click the Show/Hide Labels pull-down menu on the common toolbar and select the required value. You can also change the label font size by pressing Ctrl+Shift and scrolling the mouse wheel.

The default size is 100%. The size control is applied to all the label text displayed on the network displays, including the land-based labels.

> **Tip**
>
> Reducing the size of the label font is a very effective method of de-cluttering displays when a number of devices such as switches are very close together.

3.2.8.5. Showing Land-Based Information

In addition to the electrical distribution network, the geographic display is capable of showing the corresponding land-based information: roads, streets, street names, waterways, major building footprints, and municipal boundaries. Each type of land-based information can be displayed or hidden independently by selecting the check boxes on the Background Layers pull-down menu, as shown in the image below.

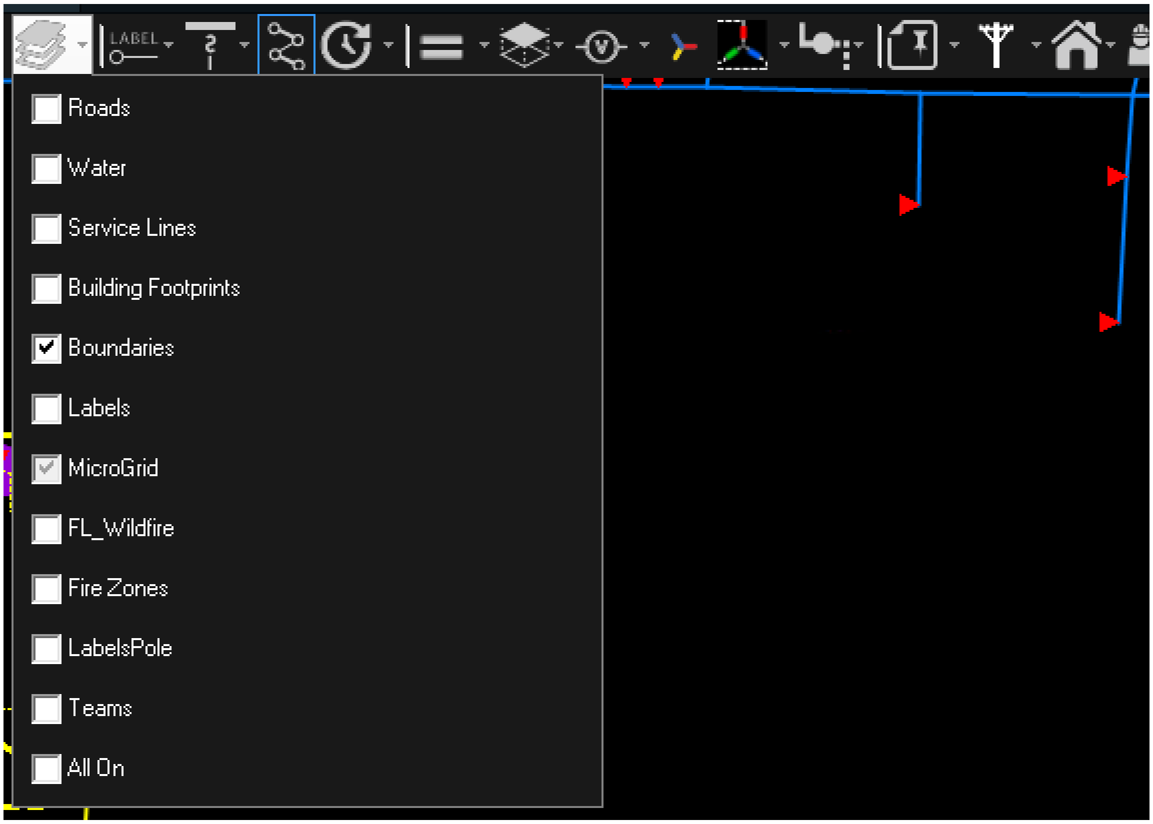


Figure 72. Background Layers Pull-Down Menu

The elements of land-based information that are displayed are dependent on the current zoom level. For example, minor roads are only displayed at high zoom levels (fully zoomed in). If the display is zoomed out, only major roads and freeways are visible.

Figure 73. Geographic Display with Streets, Water, and Buildings

3.2.8.6. Showing Equipment Under Commissioning

Components that are modeled in the distribution electrical network but not active (for example, components that are planned to be put in service, planned to be removed, or recently removed) are placed in a special display layer so they can be shown or hidden in the current view by clicking the Show/Hide Future button on the common toolbar.

Equipment under commissioning can be located by opening the Station Commissioning and Decommissioning display, navigating to the Equipment Commissioning and Decommissioning display for the required station, and navigating directly to the required device using the Locate button.

When it is time to put all or part of the future construction into service, the equipment is made Active, then switching orders are used to authorize closing the isolating switches to energize it. Once the equipment is in service, the GIS is updated and the next model update cycle shows the equipment as normally connected. A similar workflow is utilized when equipment moves from the future remove state to the removed state.

> **Note**
>
> Future equipment that is modelled as part of a commissioning project cannot be energized until the operator manually sets the Active status (for example, from the Equipment Commissioning/Decommissioning display). Future equipment that is modelled without a

commissioning project (legacy approach) can be set to Active and energized by closing the upstream future switchable device attached to an active part of the network.

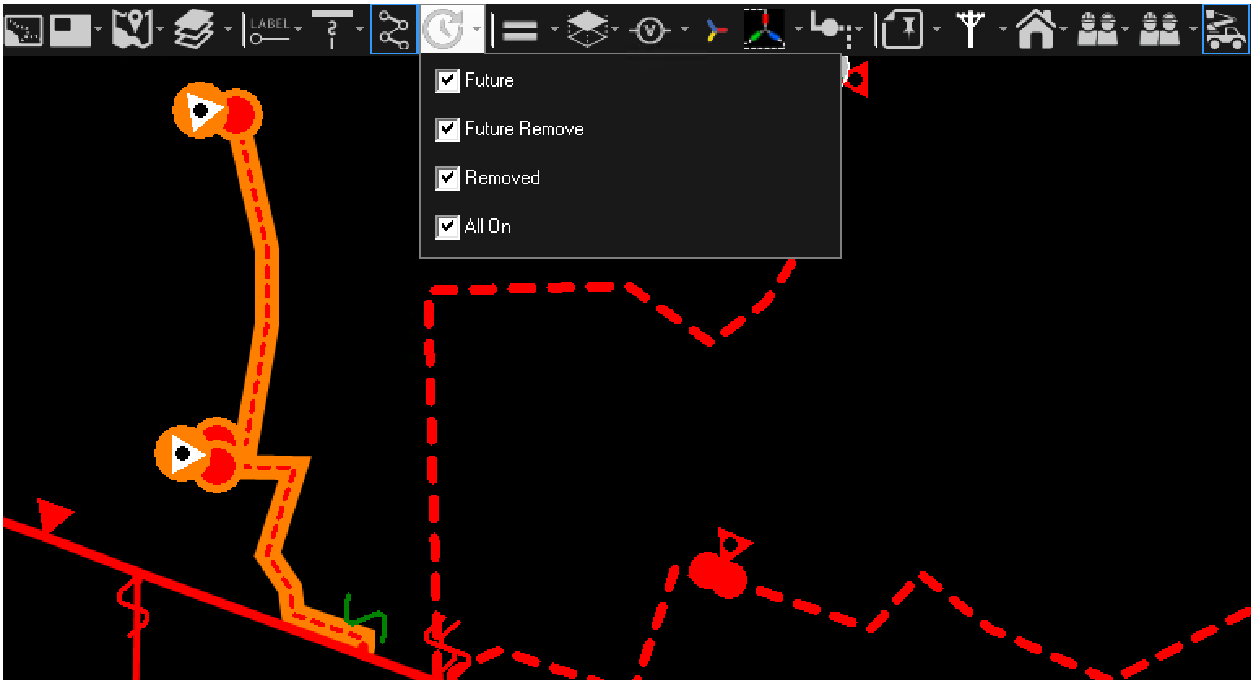


Figure 74. Displaying Equipment Under Commissioning

You can update the device commissioning state from the network view or from the Equipment Commissioning/Decommissioning display.

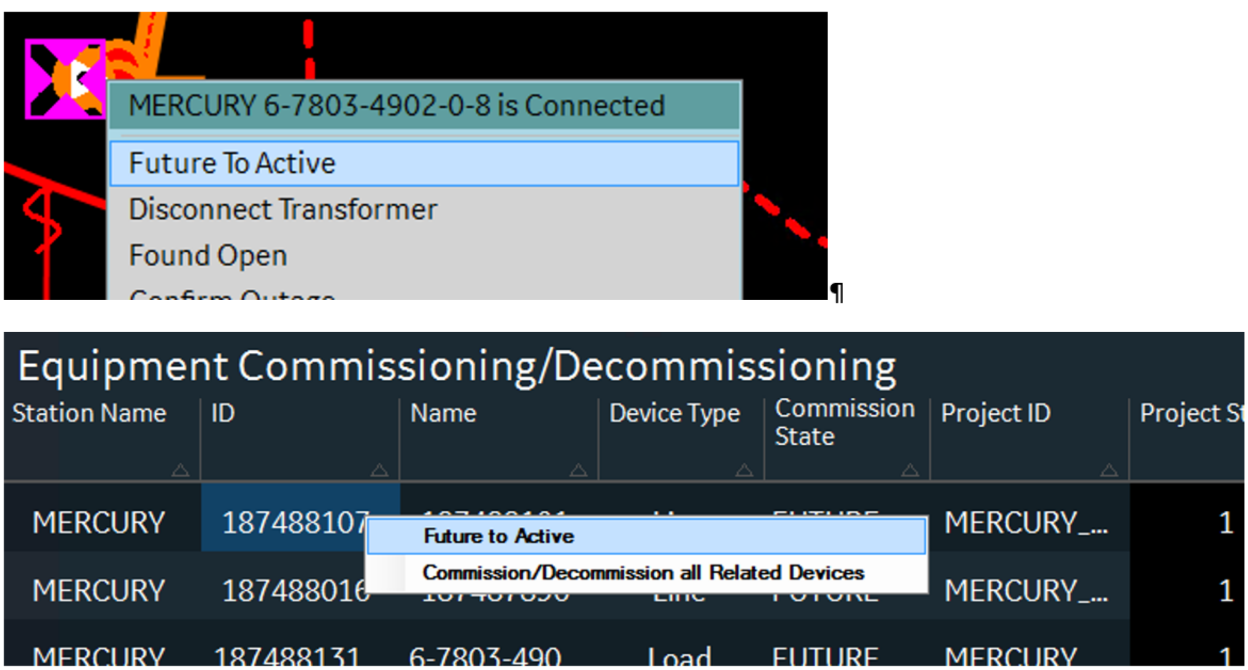


Figure 75. Updating the Equipment Commissioning State

3.2.8.7. Show/Hide Voltage Violations

Voltage violations are calculated as part of the Distribution Power Flow application execution. They indicate elements of the network where the voltage exceeds pre-defined maximum or minimum voltage levels.

Clicking the Show Halos button on the common toolbar shows the menu options to show voltage violations as colored halos.

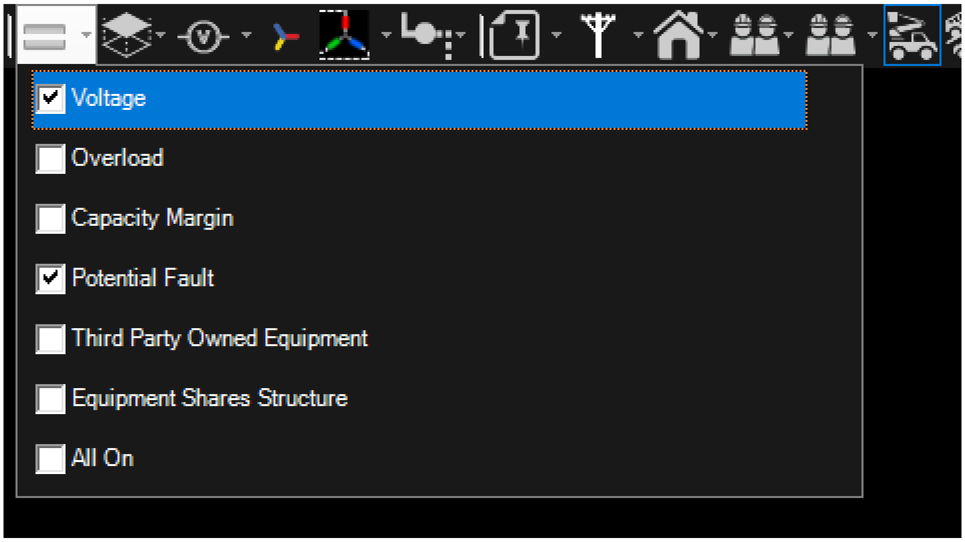


Figure 76. Voltage Check Box in Show Halos Pull-down Menu

As shown in image below, dark blue indicates a violation of the maximum voltage limit. Purple indicates a violation of the minimum voltage limit. At normal zoom levels (whole feeder view), the violation halo is applied to the entire switchable section that contains the device that is in violation. At high zoom levels where all devices such as distribution transformers are visible, the halos are applied to the individual elements.

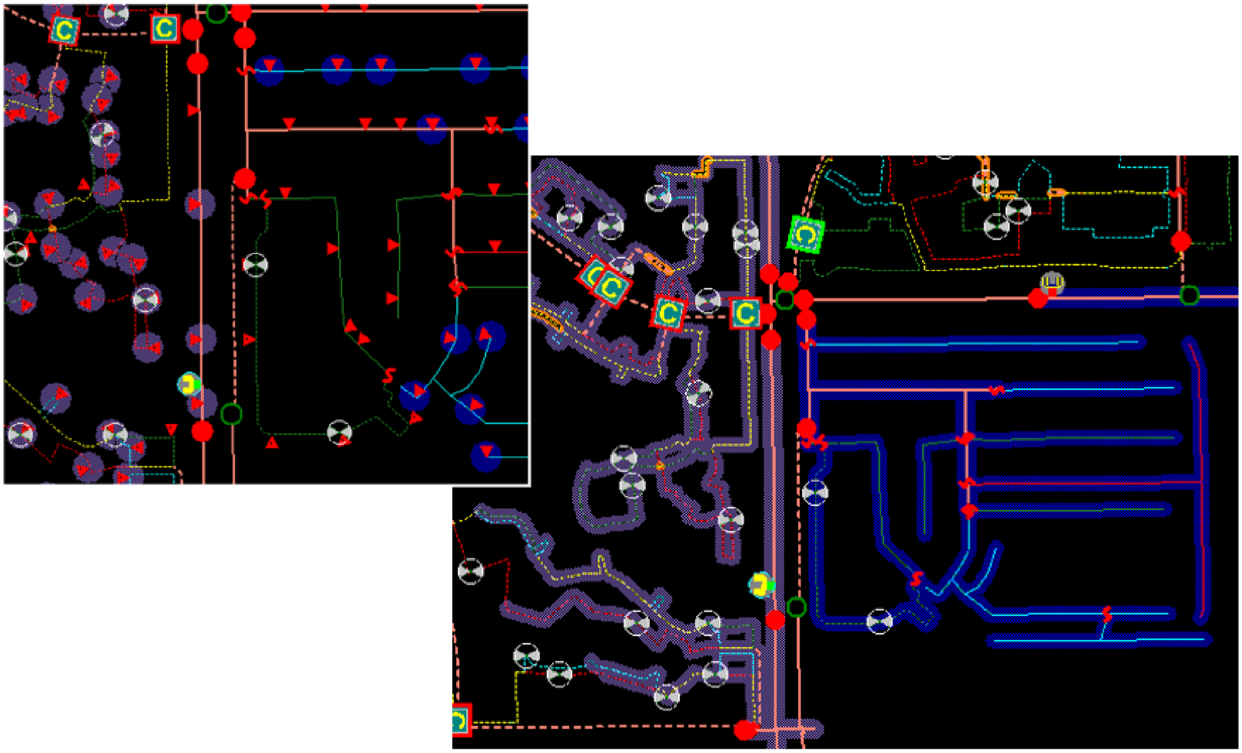


Figure 77. Voltage Violation Halos

3.2.8.8. Show/Hide Overloads

Overload conditions are calculated as part of the distribution power flow. They indicate elements of the network where the current exceeds pre-defined levels.

Clicking the Show Halos button on the common toolbar shows the menu options to show overload conditions as colored halos.

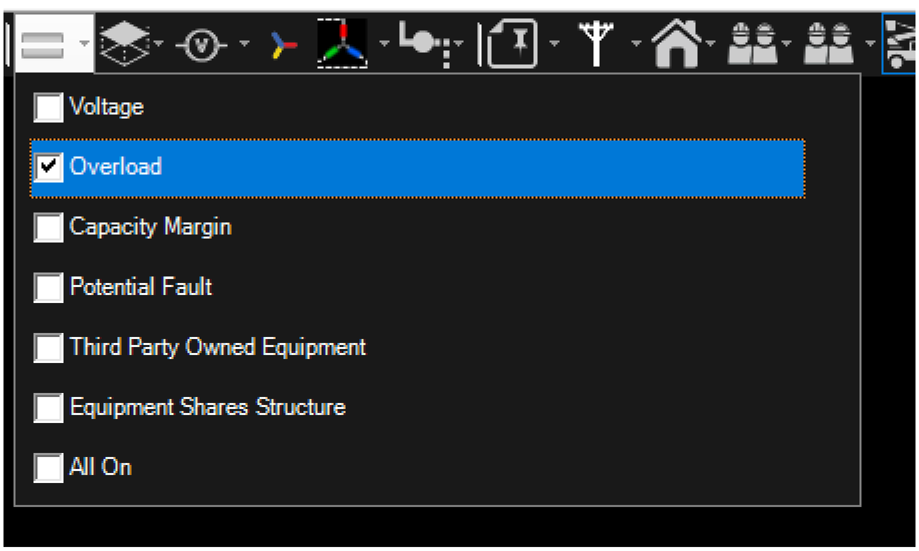


Figure 78. Overload Check Box in Show Halos Pull-Down Menu

A yellow halo indicates a Level 1 overload. A dark yellow halo indicates a Level 2 overload. An orange halo indicates a Level 3 overload.

At normal zoom levels (whole feeder view), the overload halo is applied to the entire switchable section that contains the device that is in violation. At high zoom levels, where all devices such as distribution transformers are visible, the halos are applied to the individual elements.

3.2.8.9. Show/Hide Capacity Margin Halos

Clicking the Show Halos button on the common toolbar shows the menu options to show capacity margin conditions as colored halos.

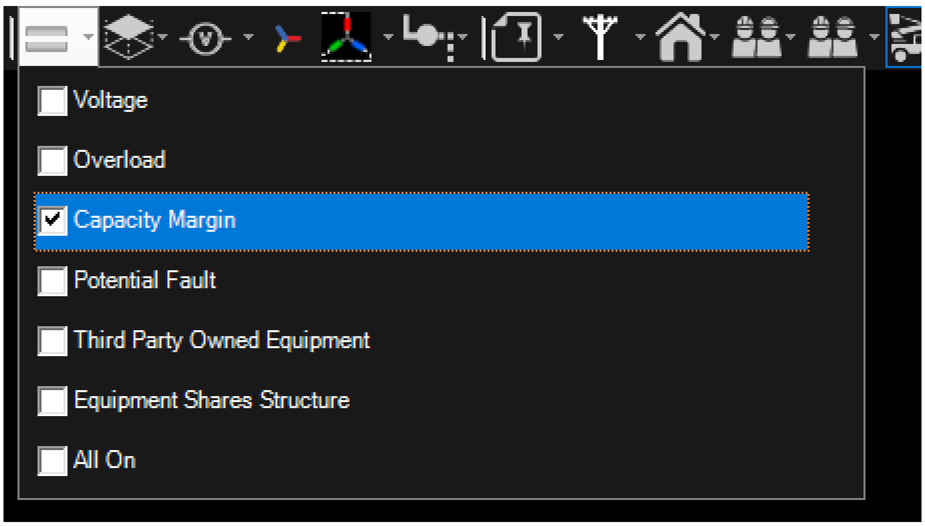


Figure 79. Capacity Margin Check Box in Show Halos Pull-Down Menu

3.2.8.10. Show/Hide Potential Fault Halos

Clicking the Show Halos button on the common toolbar shows the menu options to display the likely locations of a fault based on the fault current seen by the feeder protection relay.

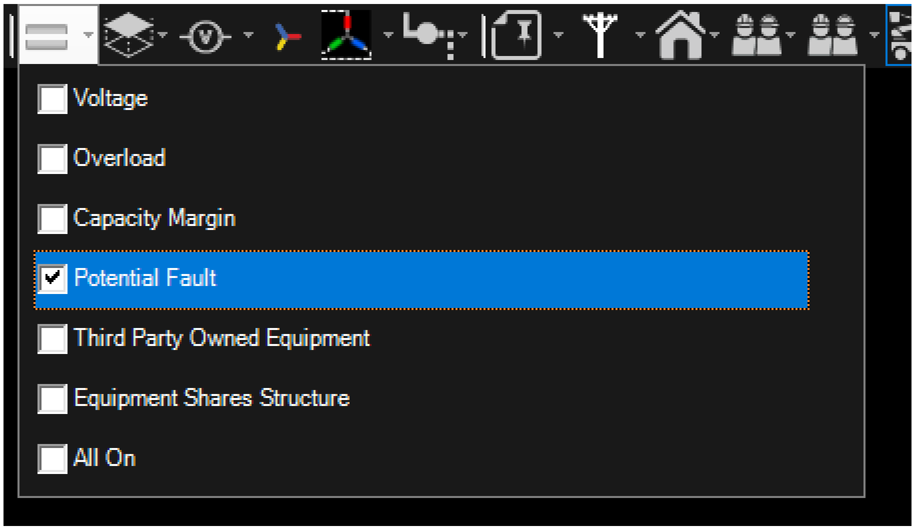


Figure 80. Potential Fault Check Box in Show Halos Pull-Down Menu

When a feeder trip occurs, the fault current information (where available) is passed to the network analysis functions. Short-circuit calculations are performed on each segment of the feeder to determine the fault locations that match the observed fault current.

The halos are three colors, indicating the levels of likelihood that the fault is located on that segment of the feeder. The green halo (Level 1) indicates the most likely location. The display of the potential fault halos is automatically reset after a number of minutes.

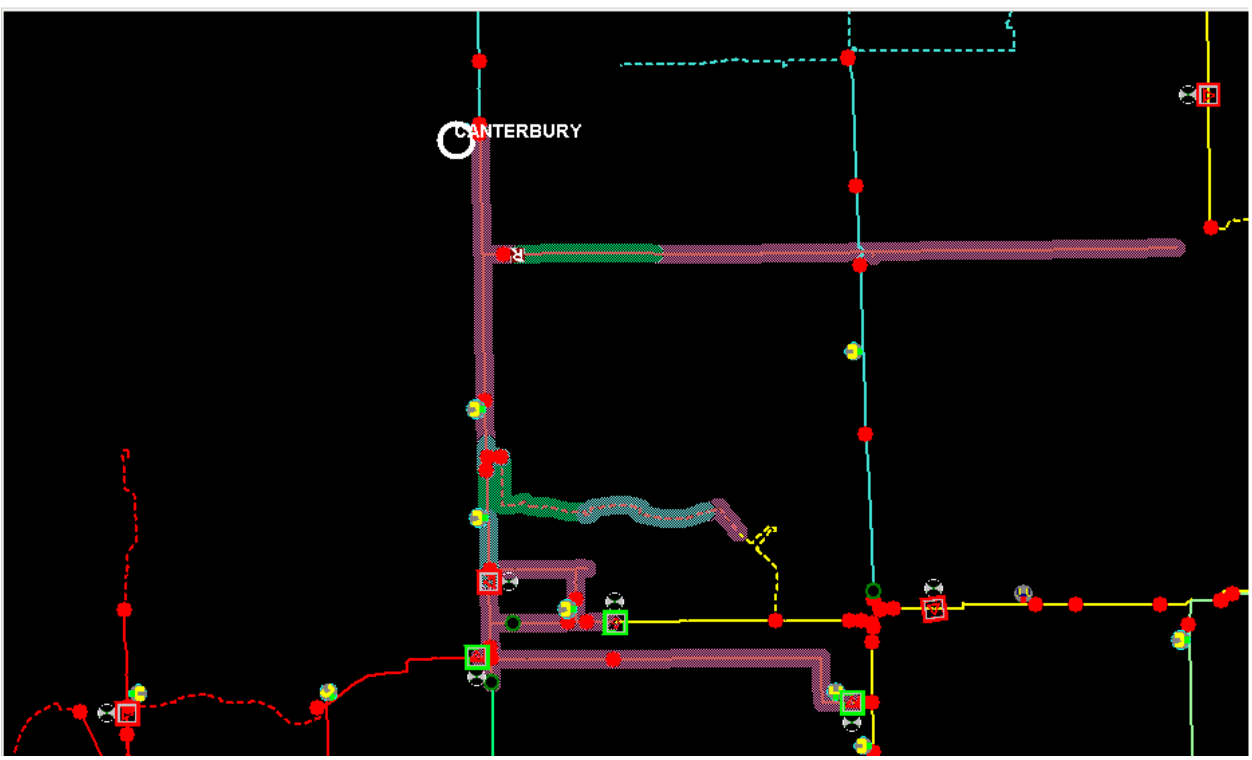


Figure 81. Potential Fault and Fault Indicator Halos

3.2.8.11. Show/Hide Third-Party Owned Equipment Halos

Clicking the Show Halos button on the common toolbar shows the menu option to highlight the third-party owned equipment.

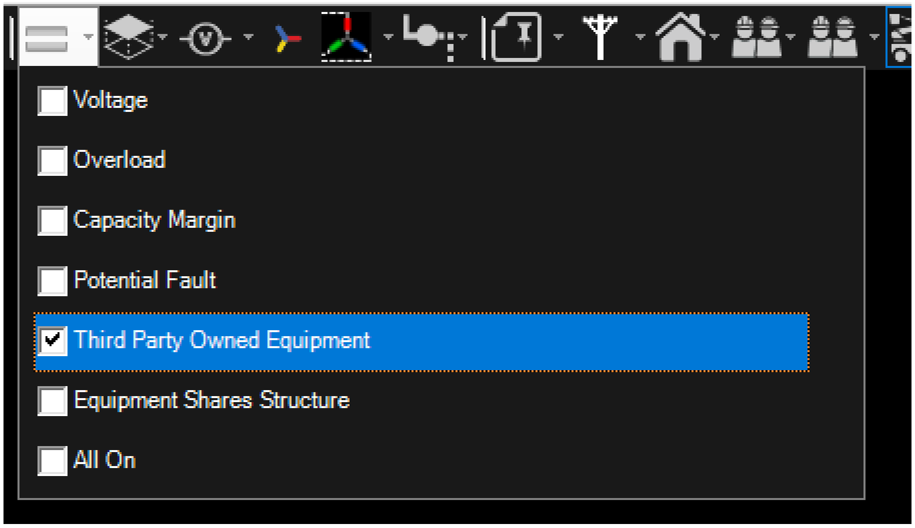


Figure 82. Potential Fault Check Box in Show Halos Pull-Down Menu

Components that are owned by a third-party company are shown with the purple halo. To locate a third-party equipment:

1. Open the Stations Containing Third-Party Equipment display.
2. Navigate to the Third-Party Equipment for the required station.
3. Navigate to the required device using the Locate button.

When an operator attempts to change status of a third-party equipment or when a switch status change affects a third-party owned equipment downstream, a check-before-operate message appears (configurable).

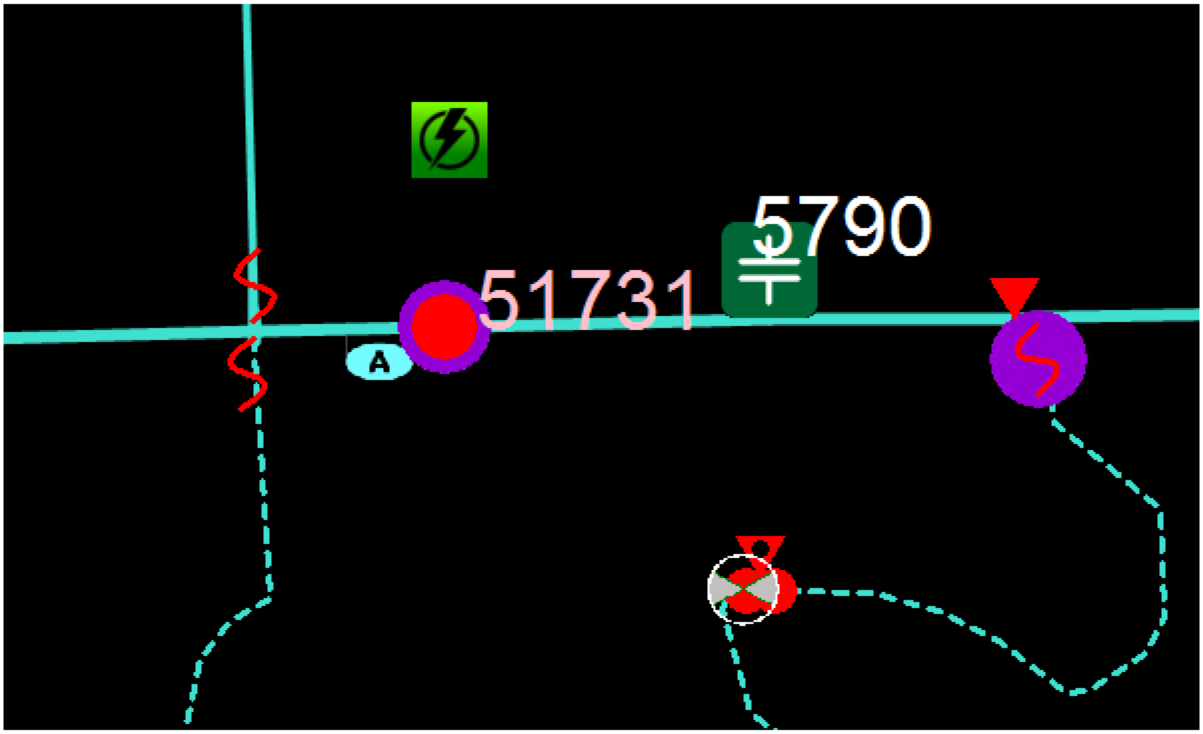


Figure 83. Third-Party Owned Equipment Halos

Third Party Equipment Summary

Station Name	Locate	Feeder Name	ID	Name	Device Type	External Company N
RAVEN		F8863	334881671	13360866	LN	Alpha ltd.
RAVEN		F8863	329340089	12591916	LN	Alpha ltd.
RAVEN		F8861	341118991	22467292	LN	Alpha ltd.
RAVEN		F8864	22239714	51731	SWT	Beta inc.
RAVEN		F8864	22239906	52217	SWT	Beta inc.
RAVEN		F8861	22240018	31852	SWT	Beta inc.
RAVEN		F8864	17541666	11944-12	SWT	Alpha ltd.

Figure 84. Third-Party Equipment Summary Display

3.2.8.12. Show/Hide Equipment Shares Structure Halos

Clicking the Show Halos button on the common toolbar shows the menu option to highlight the equipment that shares structures with equipment from other feeders.

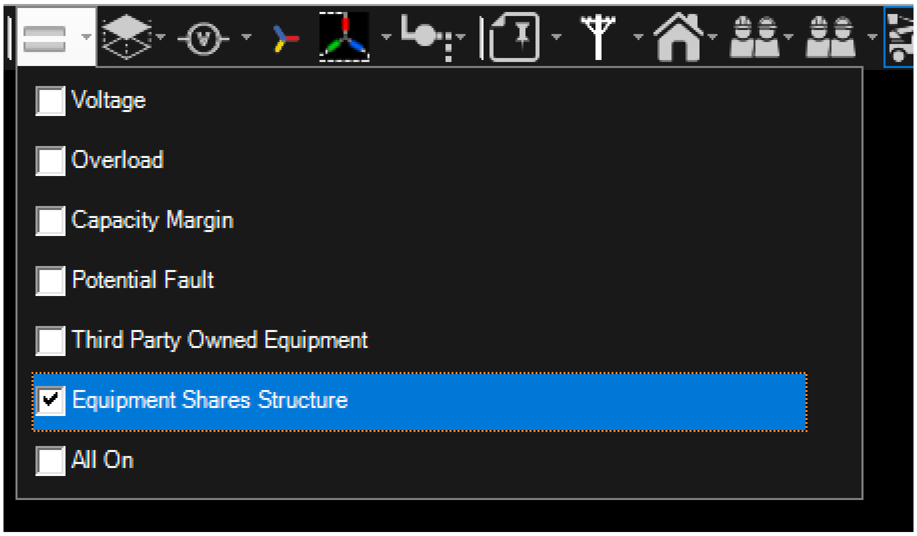


Figure 85. Equipment Shares Structure Option in the Show Halos Menu

Equipment that shares a structure with equipment from a different feeder is shown with a halo. This halo can be used by field crews who are going to perform work on these structures to give them situational awareness and follow any additional safety steps based on the devices on that structure.

3.2.8.13. Show/Hide Fault Indicator Halos

Clicking the Fault Indicator Halos button on the common toolbar indicates the likely segments of a feeder that may be the location of a fault, based on information from manual and telemetered (SCADA) fault indicators.

Figure 86. Fault Indicator Halos Pull-Down Menu

The Fault Indicator Halos button has 4 options:

- **Hide All Halos:** Select this option to hide all fault indicator halos on the geographic view.
- **Show Manual Halos Only:** Shows only manual fault indicator halos.

  Only the manual fault indicators are considered in the fault indicator halo analysis. Telemetered fault indicators (both set and unset) are ignored.

- **Show Telemetered Halos Only:** Shows only telemetered fault indicator halos.

Only the telemetered fault indicators are considered in the fault indicator halo analysis. Manual fault indicators (set, unset or unknown) are ignored.

- **Show Combined Halos:** Both telemetered and manual fault indicator settings are used to determine the placement of the fault indicator halos.

The faulted sections are highlighted with a pink halo. The display of the fault indicator halo is automatically reset after a configured number of minutes. In addition, fault indicator halos may be preserved for a configured number of minutes after the fault indicator status has been reset if the upstream breaker is capable of holding the fault indication.

For manual fault indicators, the state of a fault indicator is reported by the crew that arrived on-site. By default, manual fault indicators in the real time system have an "Unknown" status until changed to either "Set" or "Unset" by the operator. For telemetered fault indicators, when a feeder trip occurs, information is sent via SCADA from fault indicators and automated feeder switches indicating whether they have observed a fault current condition. Analysis of the fault current indications is then used to determine the zone of the feeder where the fault is located. Typically, the fault halo is shown between the furthest downstream set fault indicator and the next set of fault indicators that are unset, or the end point of the network.

Telemetered fault indicators can be modeled either as having a battery backup system, or a super capacitor backup system. Super capacitor backed fault indicators can only provide valid SCADA measurements for a limited time after being de-energized.

> **Note**
>
> Fault indicators with a battery backup that receive SCADA measurements with Suspect or Bad data quality are ignored in fault halo calculations. This prevents the system from displaying halos based on unreliable measurements.
>
> Fault indicators with a super capacitor backup that are de-energized and that receive a suspect SCADA measurement use the prior, non-suspect fault indicator state. This allows the system to remember the fault state of the indicator before the super capacitor lost its charge.

If the super capacitor backed fault indicator receives a suspect SCADA measurement while still energized, the indicator is treated the same as a battery backed up indicator and the state is ignored.

All telemetered fault indicator types show the suspect decoration if a suspect SCADA measurement is received.

The fault indication halo is preserved after the fault indicator status has been reset if Hold Fault Indication is enabled on an upstream breaker object. In this case, the fault indication halos will be maintained on the user interface until the conditions for one of the Hold Fault Indication timers in the server configuration file have been met or until the server is restarted.

3.2.8.14. Show/Hide Voltage Levels

To filter the network view by voltage, click the Show/Hide Voltage Levels pull-down menu on the common toolbar, and select or clear the check boxes as required.

Figure 87. Show/Hide Voltage Levels Pull-Down Menu

Click the network view to apply the changes. This filters the display to show only those network components that are defined with that nominal operating voltage. The default setting is "All On".

3.2.8.15. Show/Hide Phase Colors

By default, a line on the geographic network view is displayed with the color of a feeder it belongs to. To toggle between coloring by feeders and by phases, click the Show/Hide Phase Colors icon.

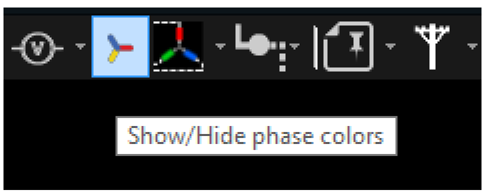


Figure 88. Show/Hide Phase Colors Icon

- When the Show/Hide Phase Colors option is enabled, the A phase is colored red, the B phase is colored green, the C phase is colored cyan, a two phase line is colored yellow, and a three-phase line is colored light-orange.
- When the option is disabled, lines are colored with colors of feeders they belong to.

Feeder colors can be configured using the Viewer Configuration Editor tool. For more information, refer to *Viewer Configuration Editor User Guide*.

> **Tip**
>
> An alternate method of identifying particular phases within the network is to use the phase trace function.

Figure 89. Lines Colored by Feeders (Left), and by Phases (Right)

3.2.8.16. Show/Hide Phases

To filter the network view by phase, click the Show/Hide Phases (Nominal) pull-down menu on the common toolbar, and select or clear the check boxes as required.

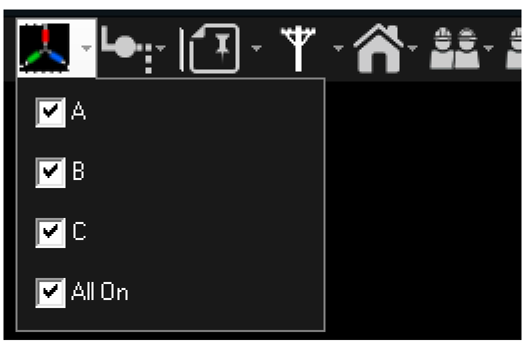


Figure 90. Show/Hide Phases (Nominal) Pull-Down Menu

Click the network view to apply the changes. This filters the display to show only those network components that include the specified phases. The default setting is "All On". This filtering can be very useful to identify specific phases on laterals or URD loops.

> **Tip**
>
> An alternate method of identifying particular phases within the network is to use the phase trace function.

3.2.8.17. Show/Hide Annotations

To filter the display of annotations on the network view, click the Annotations pull-down menu on the common toolbar, and select or clear the check boxes as required.

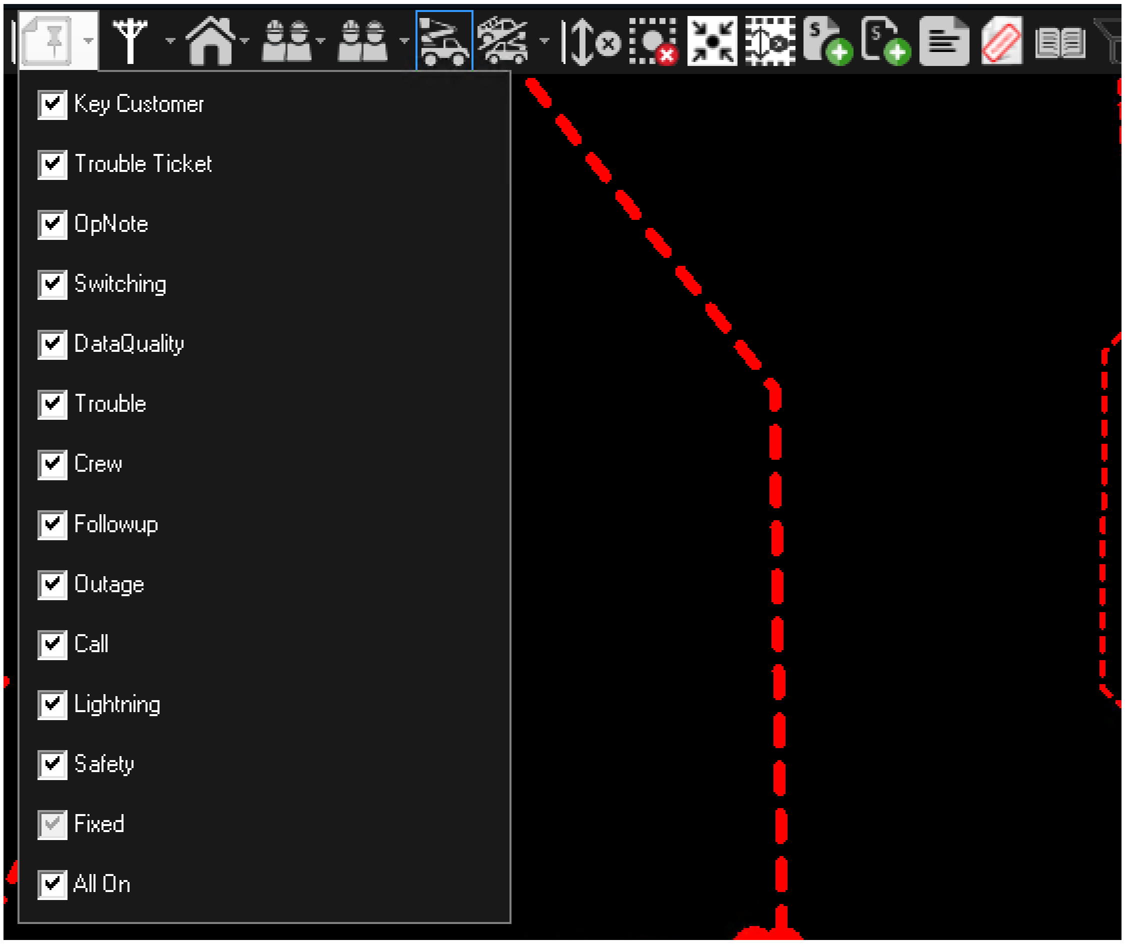


Figure 91. Show/Hide Annotations Pull-Down Menu

Click the network view to apply the changes. This filters the display to show only the selected annotation types. The default setting is "All On".

3.2.8.18. Show/Hide Structures

The geographic display can show the structure information: poles, street lights, call boxes, restricted areas, fire sensitive areas, work boundaries, and hazard areas. Each structure type can be displayed or hidden independently. To filter the display of structures on the network view, click the Show/Hide Structures pull-down menu on the common toolbar and select or clear the check boxes as required.

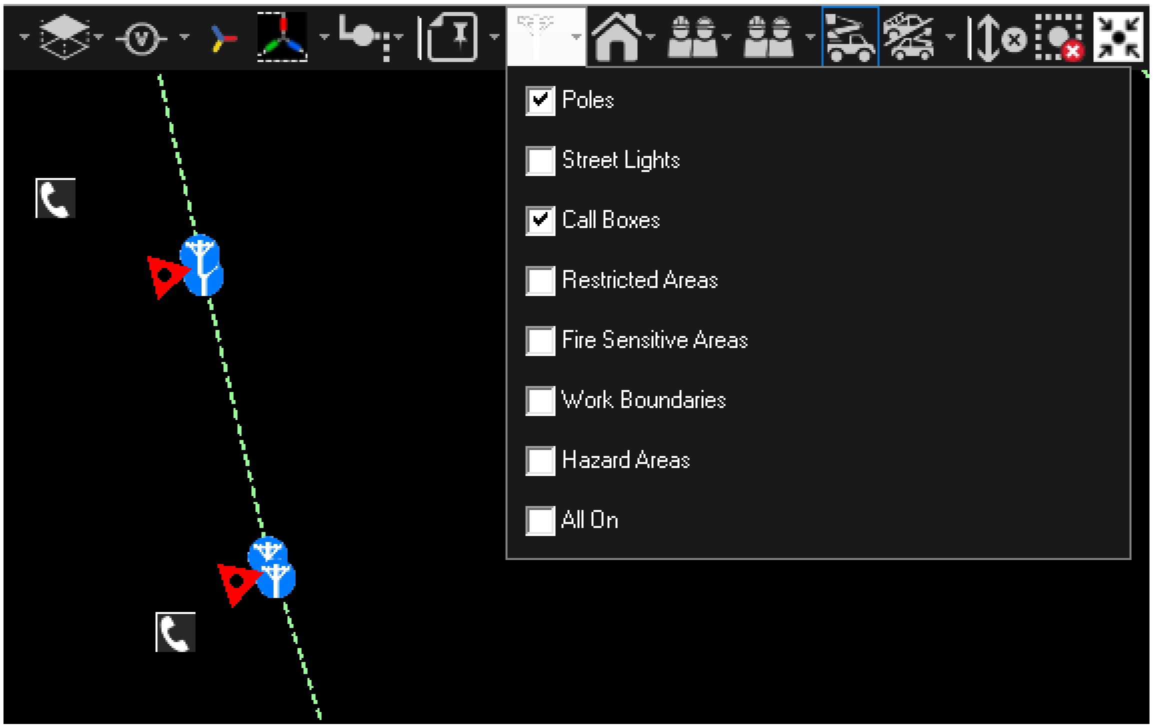


Figure 92. Show/Hide Structures Pull-Down Menu

Click the network view to apply the changes. This filters the display to show only the selected structure types.

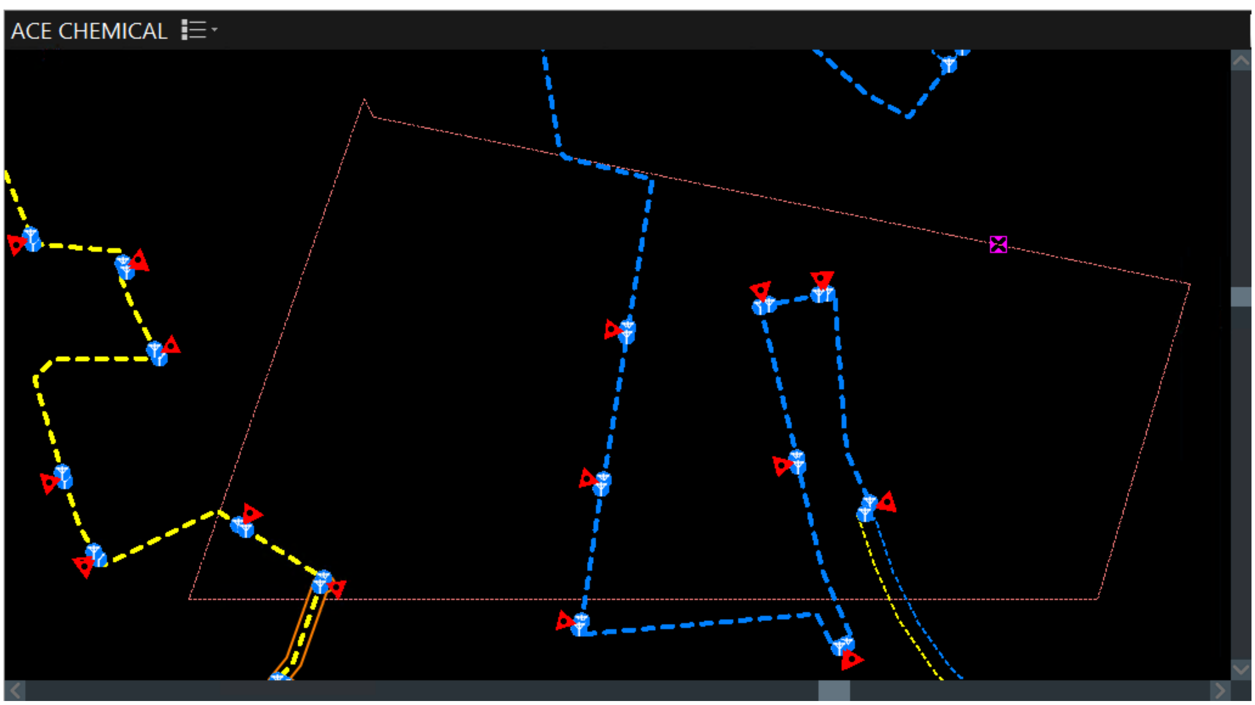


Figure 93. Restricted Area Structure Type

Structures that are assigned to specific feeders follow the feeder filtering function. For more information about filtering the network view by certain feeders, see Feeder Filtered Displays.

Note

The structure filtering supports only nominal feeders. A structure is not re-assigned to a dynamic feeder when the structure area is energized from an abnormal feeder.

3.2.8.19. Show/Hide Web Maps

To filter the display of web maps on the network view, click the Show/Hide Web Maps pull-down menu on the common toolbar and select one of the map types.

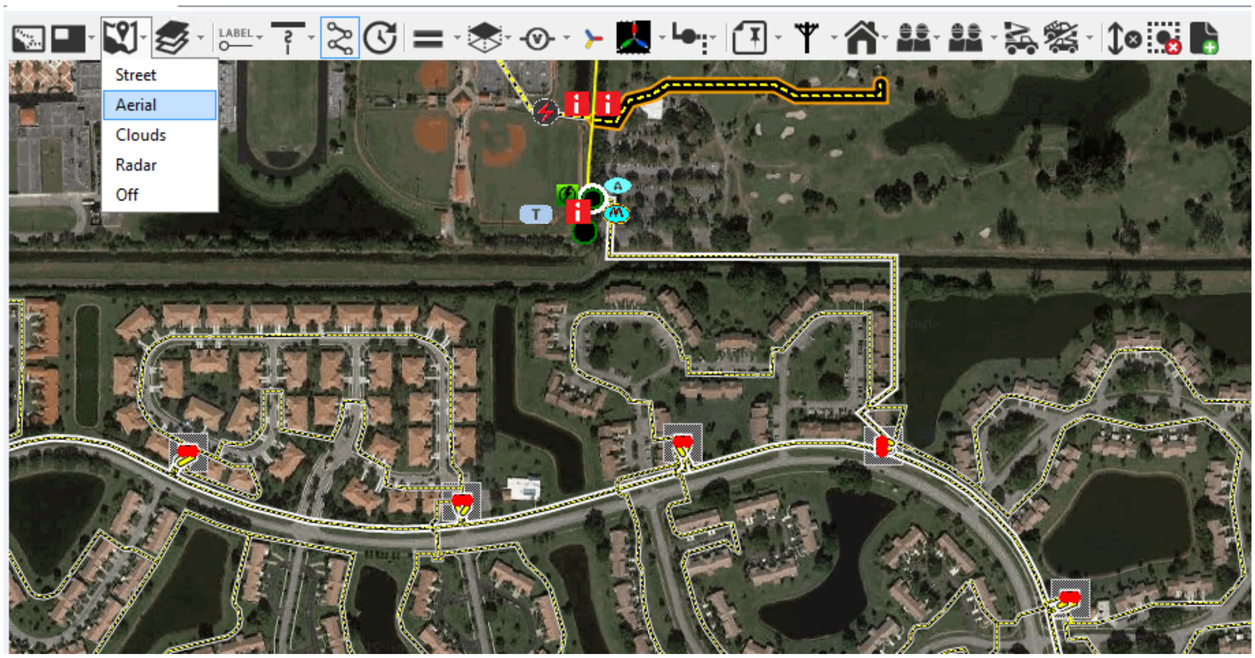


Figure 94. Show/Hide Web Maps Pull-Down Menu

Selecting an option shows only the selected web map type. The default setting is "Off".

3.2.8.20. Show/Hide Overhead/Underground

To filter the network to show only overhead or underground components, click the Show/Hide Overhead/Underground pull-down menu on the common toolbar, and then select the required display mode.

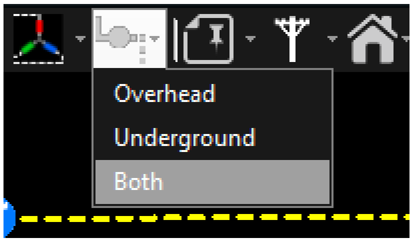


Figure 95. Show/Hide Overhead/Underground Pull-Down Menu

The default setting is "Both".

3.2.8.21. Showing Crews

To filter crews on the network view, use the Show/Hide Crews Pull-down menu or the Filter Crews by Type pull-down menu on the common toolbar.

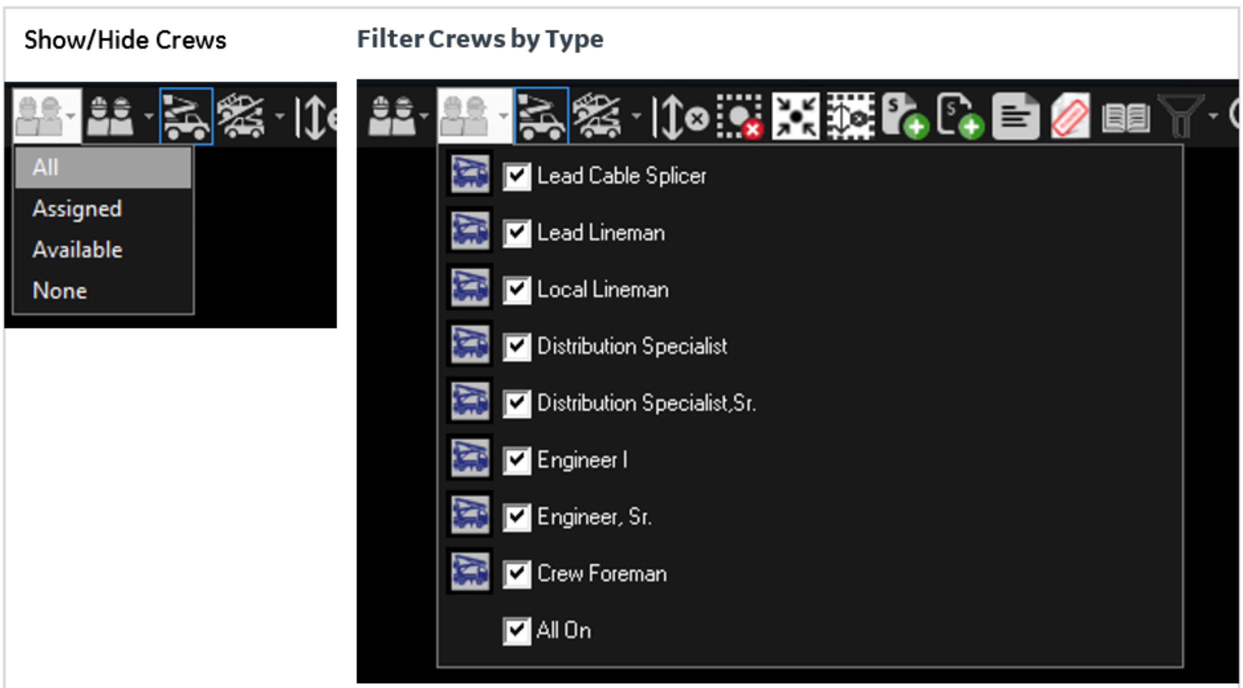


Figure 96. Show/Hide Crews and Filter Crews By Type Pull-Down Menus

Select any options to apply the changes. This filters the display to show only the selected crew types. The default setting is "All On".

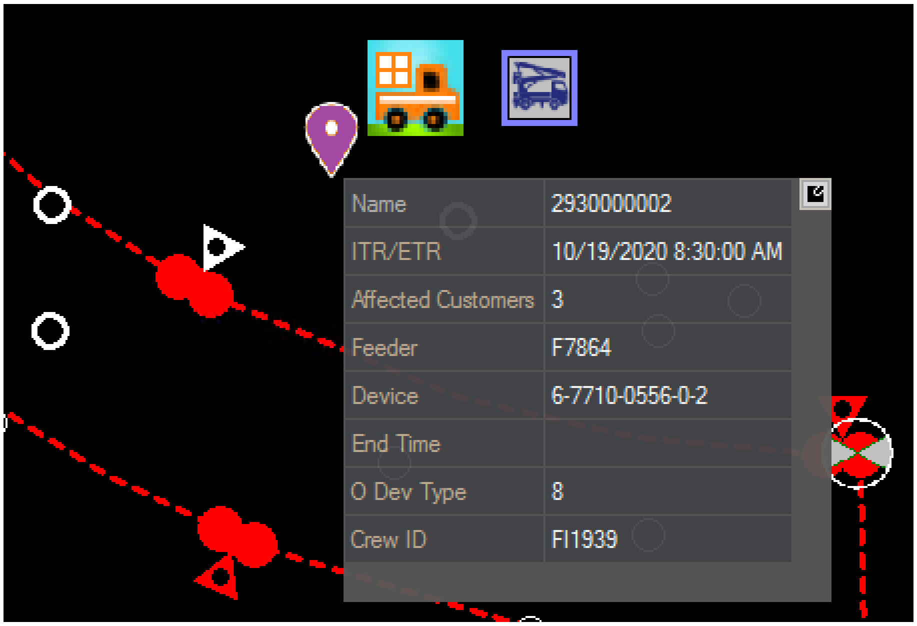


Figure 97. Incident, Crew, and Vehicle on the Network View

3.2.8.22. Showing Vehicles

To filter vehicles on the network view, use the Show/Hide Vehicles pull-down menu or the Filter Vehicles by Type pull-down menu on the common toolbar.

Figure 98. Show/Hide Vehicles and Filter Vehicles By Type Pull-Down Menus

Select any options to apply the changes. This filters the display to show only the selected vehicle types. The default setting is "All On".

3.2.8.23. Feeder Filtered Displays

The feeder filter tree allows selecting feeders to be displayed on the geographic view; you can select from all feeders in the model to feeders per station and single feeders. When filtering is enabled, the Feeder Filter pull-down menu icon changes to the color blue.

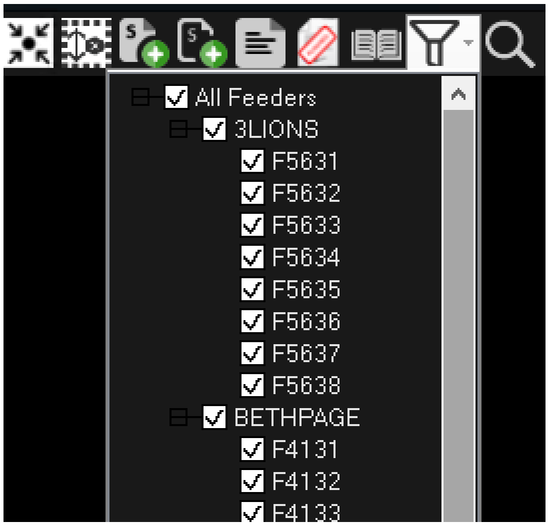


Figure 99. Feeder Filter Pull-Down Menu

To display the feeder filter tree and select the feeders to view:

1. Click the feeder filter button  on the right side of the common toolbar.

   This displays a list of each of the substations and feeders currently loaded in the partial model view.

2. Uncheck the All Feeders checkbox at the top of the tree.
3. Select the desired stations or feeders.
4. Click the Filter button or anywhere on the geographic view to hide the feeder filter tree.

   The geographic display is then restricted to show only the desired feeder(s).

   > **Note**
   >
   > This is the dynamic view of the feeder, so it also shows any other line sections that are currently being supplied by this feeder, as well as any parts of the feeder that are currently being supplied by a neighboring feeder. This is because the topology processor temporarily assigns ownership of all the network elements to one of the feeders under these conditions.

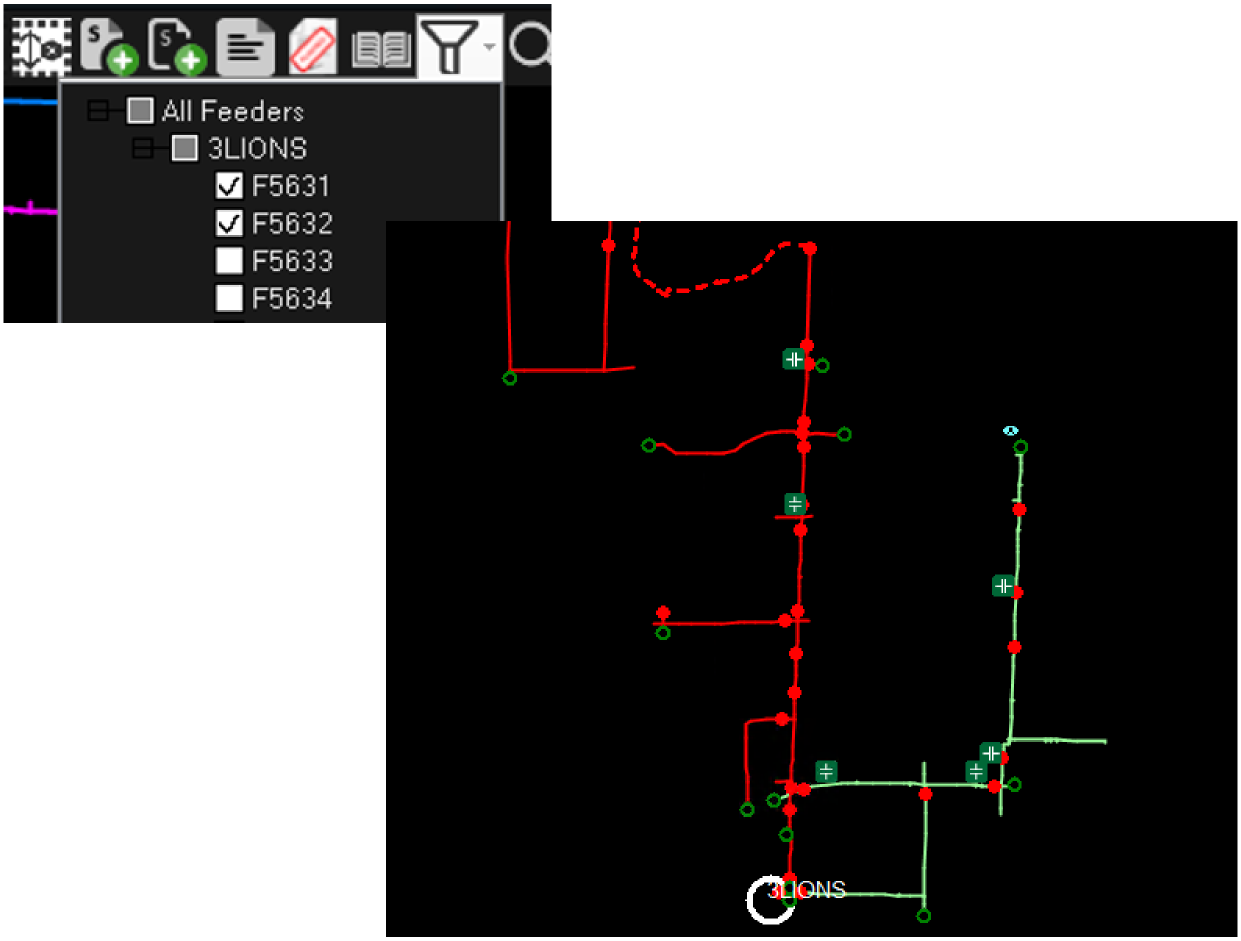


Figure 100. Filtering Feeders on the Geographic View

3.3. Working with Geographic Views

The geographic views of the network correctly place each device according to the provided geographic coordinates. This means that the display can also include land-based information such as roads, buildings, and waterways. These displays also allow for the placement of temporary modification objects such as cuts, jumpers, and grounds.

The key issue in working with geographic network views is managing the amount of information shown in a viewport to ensure that the network can be operated effectively. This is done by controlling the layers that are displayed and setting the appropriate zoom level.

Users are encouraged to make use of tools such as the picture-in-picture view, which allows a small overview in the same viewport as the close-up view of the section of the network being operated. Alternately, the picture-in-picture window can be used to zoom in on a cluster of switches without the need to change the view in the main window.

When switching is being performed on the feeder backbone, the reconfigurable section view is a very effective means of de-cluttering the display.

When operating on the geographic network views, particularly with large monitors, the right-click menus are very effective in reducing the need to drag the cursor back and forth across a large area of the screen.

3.4. Working with Schematic Views

While geographic views of the network show the correct relationships between network devices and other objects (roads, buildings, so on) and are essential for guiding work crews, there are many times when it is more efficient to work with a schematic view of the network. Schematic views provide a clear view of crowded switching arrangements, compress long lines, and present details that are normally inside UI containers, vaults, and switch cabinets.

Schematic displays show any temporary network modifications such as cuts and jumpers that have been applied on the geographic display. However, temporary objects cannot be applied or removed on schematic displays because the line lengths have no relationship with the physical arrangement.

The ADMS system has the capability to automatically generate a schematic view of a selected portion of the network, such as an Underground Residential Distribution (URD) loop.

3.4.1. Geo Schematic View

The geo schematic view provides the clearest view for network switching operations. It shows schematic connection of all switching devices. The switching devices on the geo schematic view are shown in their X and Y coordinate locations, and the lines are drawn orthogonally. The view does not include laterals and most of the network elements.

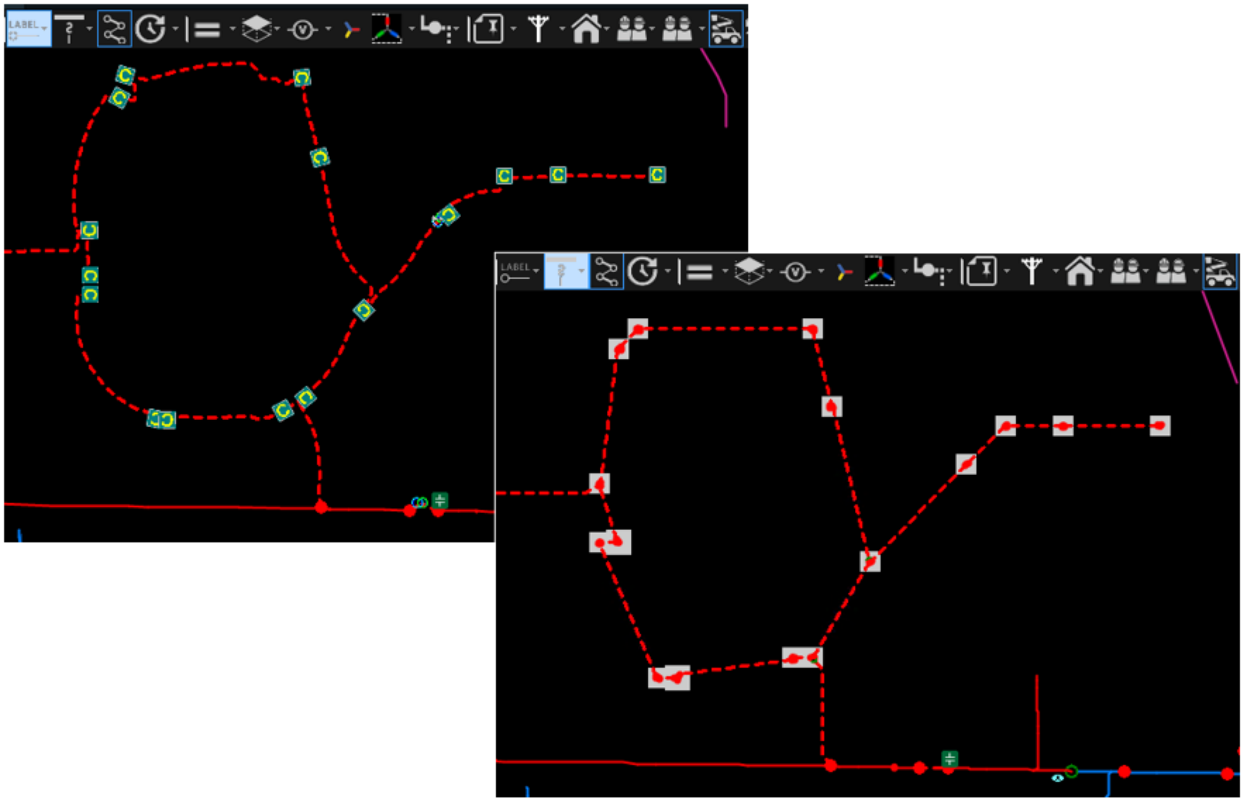


Figure 101. Geo Schematic View of the Network (Right)

To display the geo schematic view, click Draw a Geo Schematic on the line toolbar or press Ctrl+H. To restore the geographic view, press Ctrl+G.

3.4.2. Feeder Schematic View

The feeder schematic view shows the schematic view for the feeder main (medium voltage) part. On-the-fly schematics can be generated for any single feeder. The schematic view automatically creates a rectilinear representation of the feeder main and associated devices. The schematic is generated using the nominal feeder and is automatically updated to show the operated status of all lines and devices including current state, halos, and tracing.

To show the feeder schematic view for a single feeder, select the Automatic Feeder Schematic option from the line right-click menu.

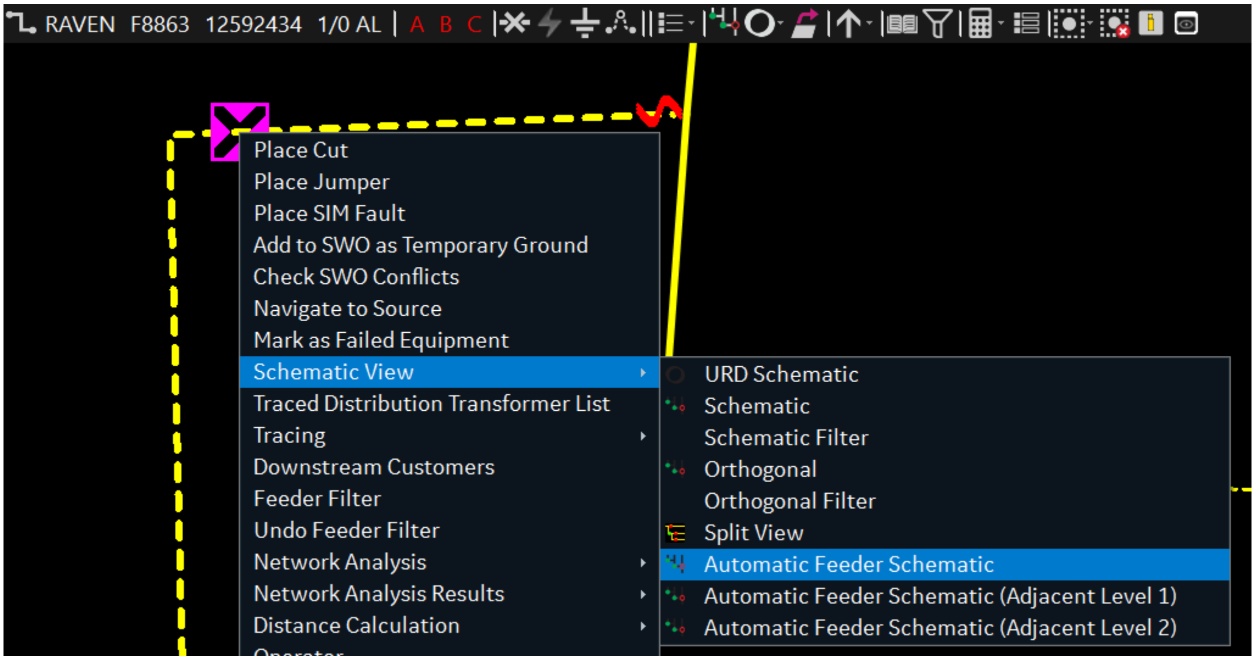


Figure 102. Showing the Feeder Schematic View

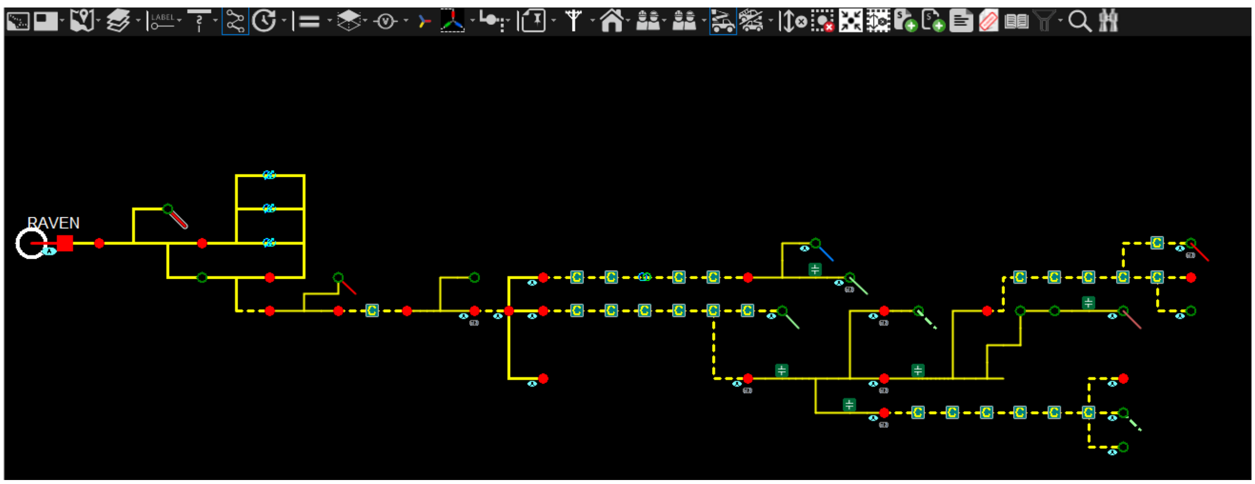


Figure 103. Feeder Schematic View

The schematic view starts from the substation, followed by the feeder head breaker and then extends to the right. Each section extent ends with the last feeder main branch or a normally-open boundary switch and the first branch of the neighboring feeder. In the schematic view, each of the objects are shown in the order they physically exist.

All feeders that are not part of the current schematic but are connected to the feeder that is shown in the schematic are shown as small diagonal lines going to the right and down. These lines indicate the location where a new feeders starts. You can right-click on these diagonal lines and request a new feeder schematic for the selected feeder.

You can also view Automatic Feed Schematic for the following two levels:

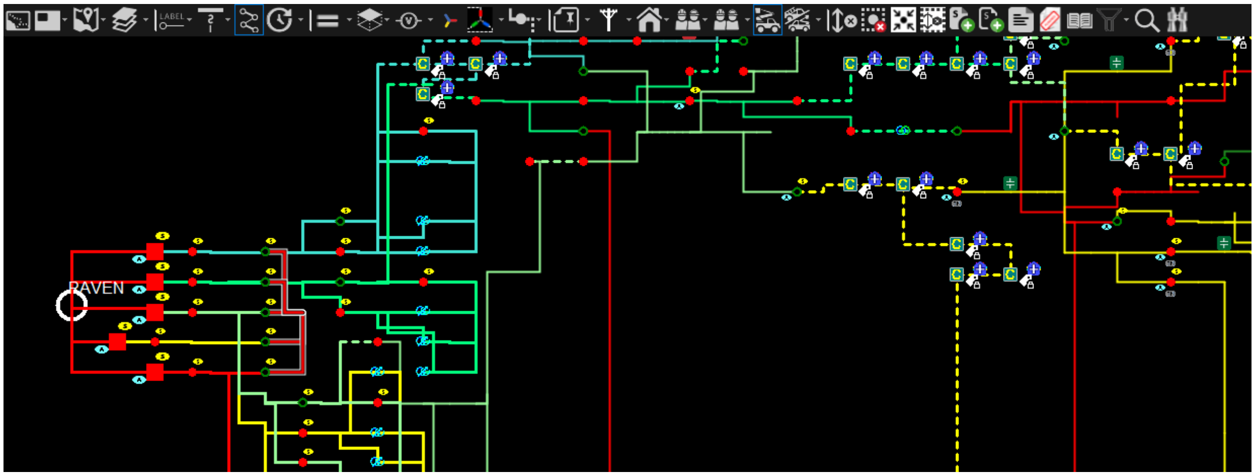
- Automatic Feeder Schematic (Adjacent Level 1): Generates a feeder schematic view of the selected feeder as well as:

    - All the feeders that are connected to the selected feeder.

- Automatic Feeder Schematic (Adjacent Level 2): Generates a feeder schematic view of the selected feeder as well as:

    - All the feeders connected to the selected feeder.
    - All feeders connected to the feeders which are connected to the selected feeder.

> **Note**
>
> When you select a feeder to generate its schematic, the station of the feeder will always be the leftmost station in the schematic. The feeder that was selected to generate the schematic will be leftmost feeder in the schematic.



The schematic includes station internals and externals for the selected feeder. Temporary devices, such as cuts and jumpers, are also displayed on the schematic. However, temporary devices may not be placed on the schematic view due to safety concerns and lack of geographic awareness.

> **Note**
>
> The feeder schematic view does not support rendering low voltage devices, such as distribution transformers.

Operations can be performed on the devices within a schematic display. Devices can be selected in the same way they are selected on the geographic diagram, and the relevant device-specific toolbars that appear have the same functionality as the toolbars in the geographic network view. Device attributes and labels are also available from the schematic view.

3.4.3. Underground Residential Distribution Loop

Underground Residential Distribution loops (URD) contain more equipment than it is possible to effectively display on a geographic network view. To provide a clearer view of URD loops, a special function has been provided to automatically generate a schematic diagram of the selected loop that includes all the information contained in the database. This includes the isolating "elbow" switches on either side of each distribution transformer and the length of cable between each distribution termination point (that is, transformer, cabinet, pole, and so on).

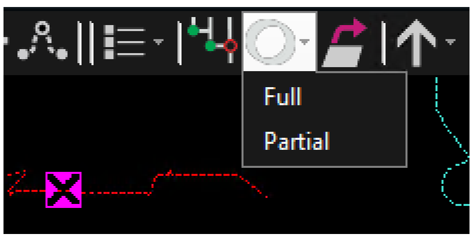
The lines in the loop diagram are colored to represent the three phases. Phase A is presented in red, B is presented in green, and C is presented in yellow.

To display the schematic:

1. Select an underground line by clicking a line segment.

   If the system detects that the selected line section is a URD loop, the Loop Schematic button appears on the line toolbar.

2. Click the Draw a Multi Loop Schematic toolbar button to show the loop schematic diagram options.

   The schematic for the loop is displayed in a new viewport, as shown in image below.

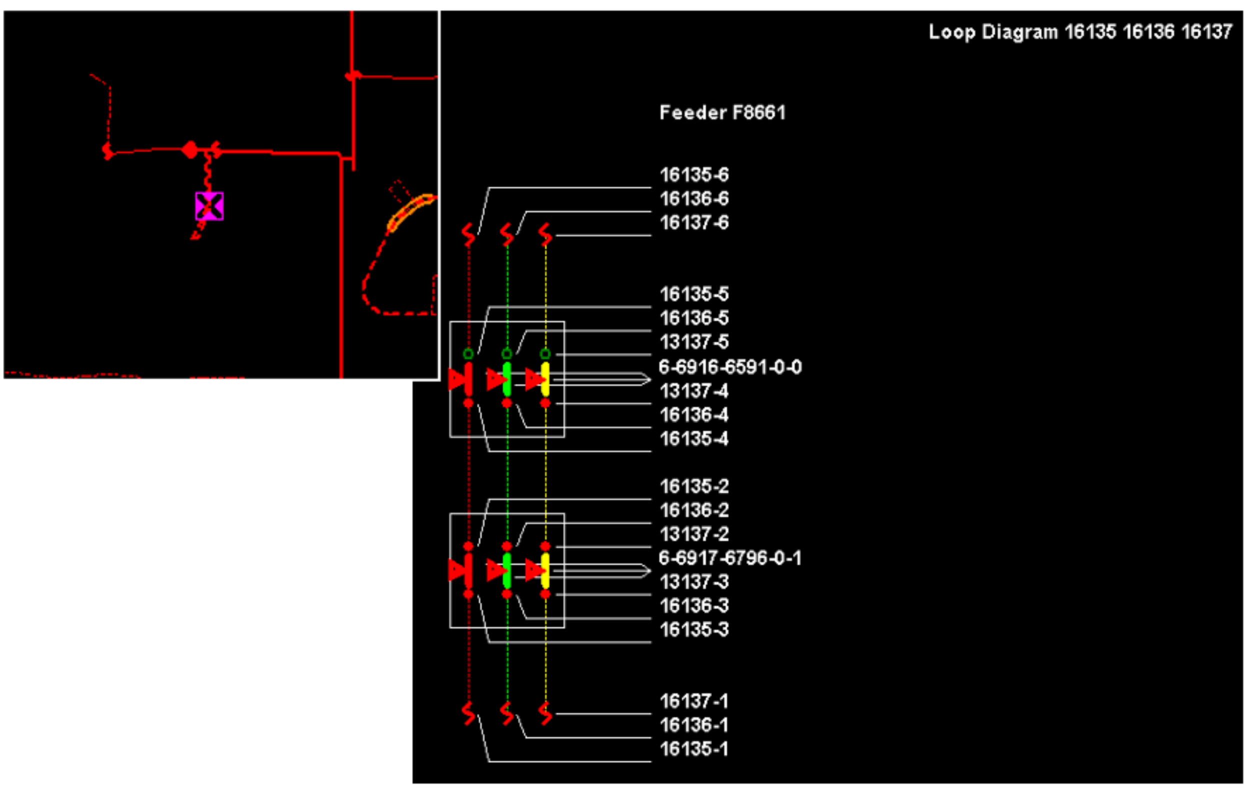


Figure 104. Geographic and Schematic Views of a URD Loop

The relevant feeder numbers are displayed at each end of the loop. In this way, loops that also form feeder ties are clearly indicated.

The fuse names, which are effectively the loop number, are shown in the title of the URD Loop Schematic display.

Device names are shown for each transformer and elbow switch.

Each segment of cable is shown with a device identifier and the cable length.

Loops can be either single-phase or multi-phase. The schematic automatically displays all the equipment that is determined to be part of the loop by the topology processor.

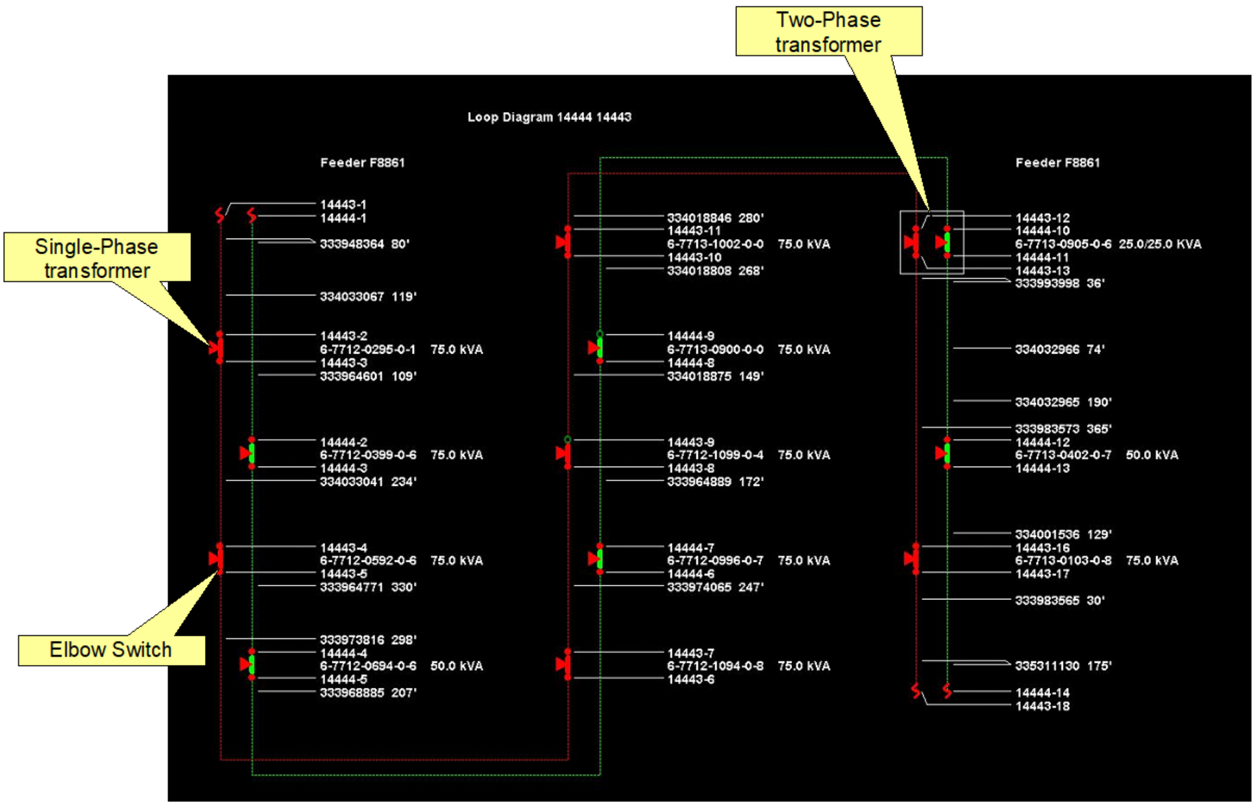


Figure 105. Schematic Display of a Two-Phase URD Loop

Operations can be performed on the devices within a loop. The devices displayed on the schematic can be selected in the same way they are on the geographic diagram. Selecting a device causes the relevant device-specific toolbar to be displayed with the same functionality as the geographic network view. In the schematic view, each of the switches or transformers is shown as they physically exist (that is, a multi-phase loop can include banks of single-phase devices and can be operated accordingly).

3.4.4. Internal World View

It is not always possible to present all the devices contained within a UI container, vault, or switch cabinet in a single geographic view.

Internal world views are a means of displaying the contents of a UI container, vault, or switch cabinet in a schematic form that is still connected to the main diagram.

To display the internal world view of a container object, select the object, and then click the Navigate to Related Display button on the device-specific toolbar. A separate viewport opens showing the internals of the selected object.

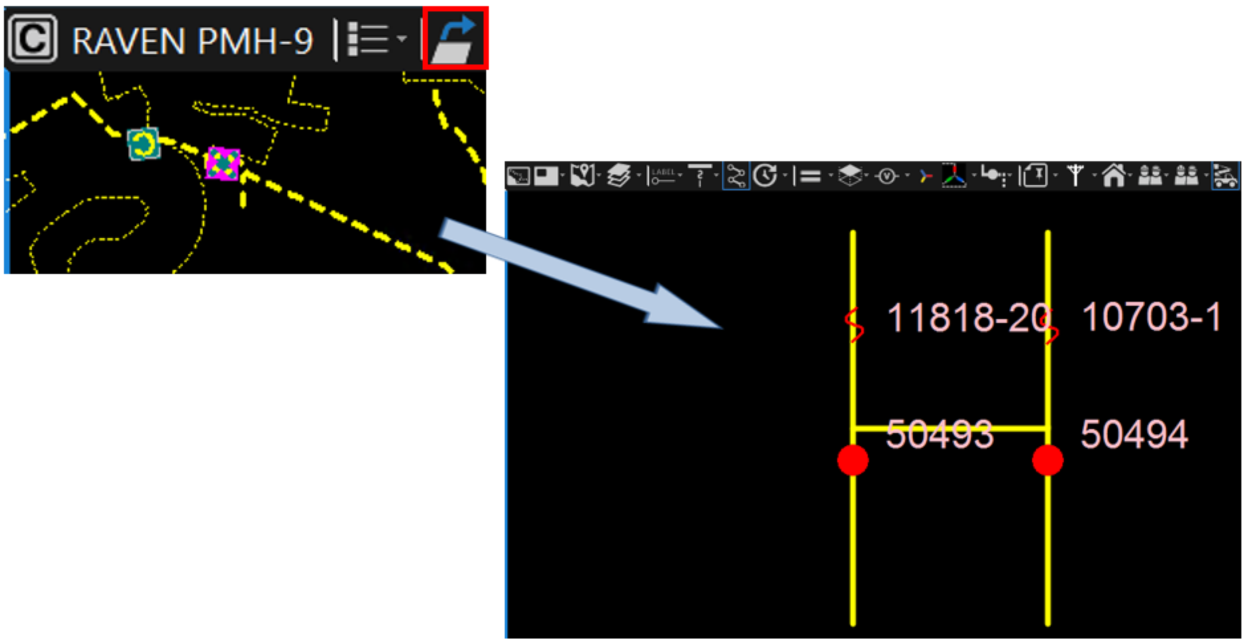


Figure 106. Internal World View of a Switch Cabinet

The internal world view of a container object is usually a schematic representation, but it supports the same operational functionality as the main geographic view. Devices can be selected, traces can be run, and switches can be operated. The effect of these changes is correctly reflected on the main geographic view.

In many cases with vaults, there are vaults within vaults. In these cases, the same procedures apply. The vaults that appear on an internal world view can be selected and another internal world view opened. This can be repeated as many times as required.

Tip

Windows that display internal world views are independent of the main geographic display and should be closed manually as soon as they are no longer required to reduce screen clutter and confusion. Internal world view schematics all look very similar.

3.4.5. Station Internal View

The equipment inside the substation is normally presented in a schematic view. Two types of schematic internal substation views are available:

The SCADA One-Line: A schematic Browser display is driven directly from the SCADA system and operates independently of the ADMS functionality. This display shows measured values on the screen. The feeder numbers at the bottom of the screen are linked to the geographic network view. Click a feeder number to navigate to a filtered geographic view of the selected feeder.

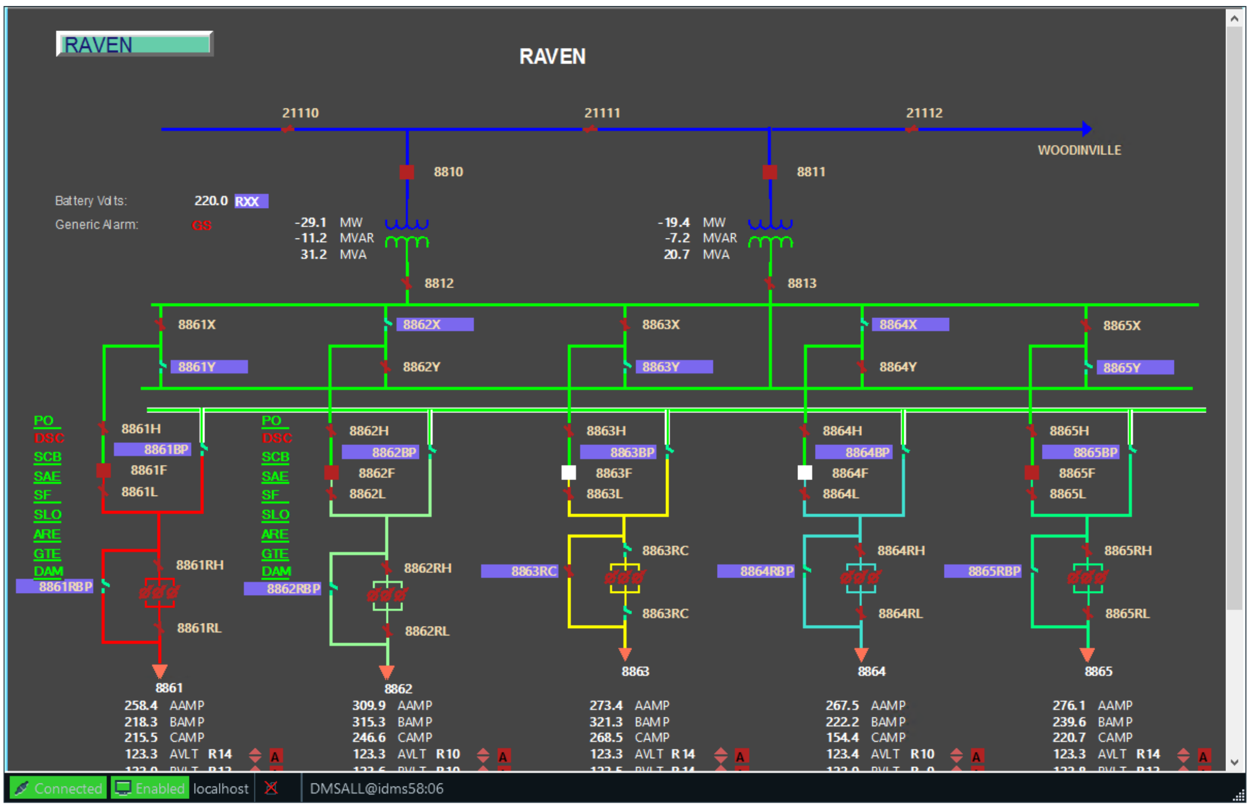


Figure 107. SCADA One-Line Display

Substation Internals: A schematic substation display that has been created within the ADMS model and behaves in a similar manner to the geographic network view. This display is linked to the SCADA system to provide full control of the substation switches.

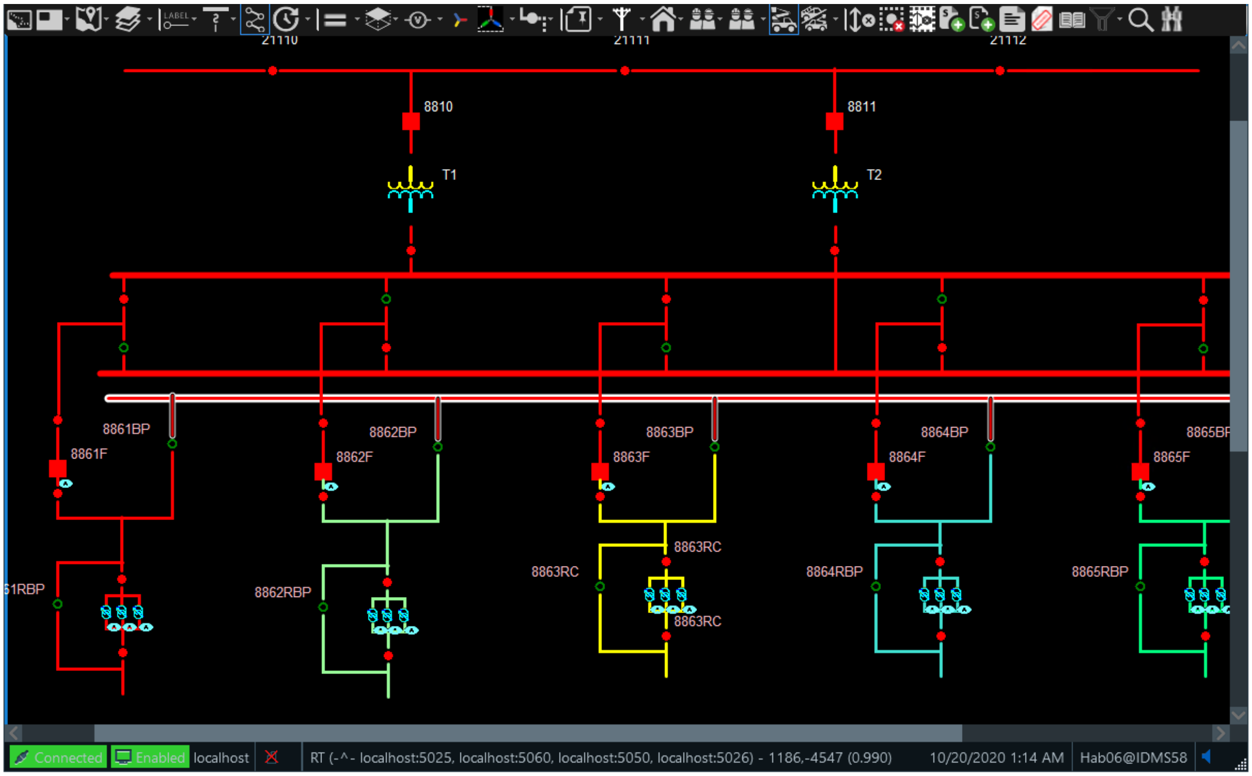


Figure 108. Substation Internals Display

3.4.6. Schematic View Graph Layout Settings

When the adjacency level is 0 the Schematic Engine will use Layered layout and the edge routing settings will be taken from LayoutSettings > EdgeRoutingSettings. Otherwise, the edge routing settings will be taken from FastIncrementalLayoutSettings > EdgeRoutingSettings

Graph offsets and scale for both layered and incremental layout will be taken from LayoutSettings > OutputSettings

Setting	Description	Possible Values
GraphSettings		
ContainerView	Defines whether to show container internals. i.e: vaults, cabinets, ganged switches, and so on.	true/false
LogEnabled	Defines whether to output Schematic Engine debug logging	true/false
GraphSettings/LayoutSettings		
GraphSettings/LayoutSettings/OutputSettings		
LayoutSettings/OutputSettings/Scale	Scale the graph layout	Any positive number
LayoutSettings/OutputSettings/OffsetX	Offset the graph in the X axis	Any real number

Setting	Description	Possible Values
LayoutSettings/OutputSettings/OffsetY	Offset the graph in the Y axis	Any real number
GraphSettings/LayoutSettings/EdgeRoutingSettings		
EdgeRoutingSettings/EdgeRoutingMode	Defines the way graph edges are routed	• Spline • SplineBundling • StraightLine • SugiyamaSplines • Rectilinear • RectilinearToCenter • None
EdgeRoutingSettings/ConeAngle	The angle in degrees of the cones in the routing	From 0 to 180 degrees expressed in radians
EdgeRoutingSettings/Padding	Amount of space to leave around nodes	Any positive number
EdgeRoutingSettings/PolyLinePadding	Additional amount of padding to leave around nodes when routing with polylines	Any positive number
EdgeRoutingSettings/CornerRadius	For rectilinear edge routing, the degree to round the corners	From 0 to 180 degrees expressed in radians
EdgeRoutingSettings/BendPenalty	For rectilinear, the penalty for a bend, as a percentage of the Manhattan distance between the source and target ports	Any value from 0 to 100
EdgeRoutingSettings/RoutingToParentConeAngle	Cone angle to find a relatively close point on the parent boundary	From 0 to 180 degrees expressed in radians

Setting	Description	Possible Values
EdgeRoutingSettings/SimpleSelfLoops ForParentEdgesThreshold	If the number of the nodes participating in the routing of the parent edges is less than the threshold then the parent edges are routed avoiding the nodes	Any positive value
IncrementalRoutingThreshold	Defines the size of the changed graph that could be routed fast with the standard spline routing when dragging	Any positive value
EdgeRoutingSettings/KeepOriginalSpline	If set to true the original spline is kept under the corresponding EdgeGeometry	true/false
EdgeRoutingSettings/RouteMulti EdgesAsBundles	If set to true routes multi edges as ordered bundles, when routing in a spline mode	true/false
LayoutSettings/MinimalHeight	Layered: The resulting layout should at least as high as this value	Any positive value
LayoutSettings/MinimalWidth	Layered: The resulting layout should be not more narrow than this value	Any positive value
LayoutSettings/Straighten	Straighten Edges	true/false
GraphSettings/FastIncrementalLayoutSettings		
FastIncrementalLayoutSettings/NodeSize	Graph node size	Any positive number
FastIncrementalLayoutSettings/ContainerSize	Graph node size for containers	Any positive number
GraphSettings/GraphDirection	Graph direction	• TB • LR • BT • RL • None
GraphSettings/CollisionMoveBehavior	Behavior when detecting collisions between lines and nodes on different feeders	• Unchanged • TopRight • BottomLeft
GraphSettings/DrawNextFeederDiagonal	Defines whether to draw lines to the next feeder on the AFS diagonally	true/false

3.5. Network and Model Indications

The indications and tools that can be used to operate the distribution network are described in the following sections.

3.5.1. Phasing Indication

For all network operations, it is important to know which phases are involved for any given device. This is particularly important when working with fuses and laterals that can be ganged, operating all phases at once, or separate, requiring each phase to be operated individually.

The phase indicator appears on the toolbar for each device type and shows the nominal phases that are connected for this device, as shown in image below. By default, the phases are shown as A, B, and C. The colors show if the phase is energized (red), de-energized (green), or not connected to the device (gray).

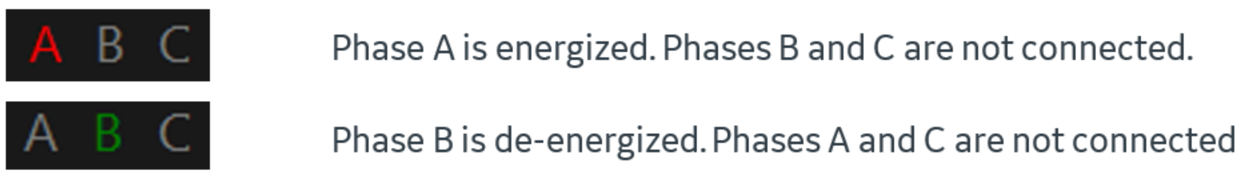


Figure 109. Phase Indicator

The actual phases may not correspond to the nominal phases if dynamic switching changes or jumpers have been placed in the model. Dynamic phase information can be obtained by looking at the analyst attributes for a given object. The dynamic phase for each nominal phase is shown in parenthesis.

Connectivity Data (Real Time)	
Station 1 and Feeder 1	RAVEN F8864
Phase(s) Energized	ABC
Phase(s) De-energized	
Phase(s) Abnormally Energized	ABC
Phase(s) Looped	
Phase(s) Removed	
Dynamic Phase(s)	A(C) B(A) C(B)
Dynamic Feeder Is Looped	No

Figure 110. Dynamic Phase Information

3.5.2. Attribute and Output Displays

Details for any device on the network are shown on attribute pop-up displays. The real-time power flow results for any device on the network are shown in the output pop-up displays.

These pop-up displays are opened by selecting the device, clicking the Attributes pull-down menu on the device toolbar, and then selecting the required option. When selecting another device, the open attribute pop-up display refreshes with new device data.

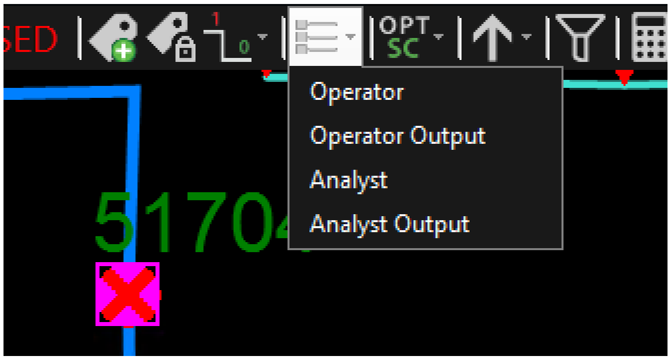


Figure 111. Attributes Pull-Down Menu

There are two types of attribute displays:

- The Operator Attributes display gives a concise list of the information that is relevant to normal operations.
- The Analyst Attributes display gives a more-detailed list of information about the selected device.

There are two types of output displays:

- The Operator Output display gives a summary of the main operating values for the selected device.
- The Analyst Output display gives a more-detailed list of the power flow output values for the selected device. The Analyst Output display also shows power flow direction indicators for the device.

For more information about the Distribution Power Flow, refer to Network Analysis and Optimization Functions.

The attribute and output displays appear as a separate pane at the left edge of the display. To remove a display, click the Close button (**X**) at the top-right corner of the pane.

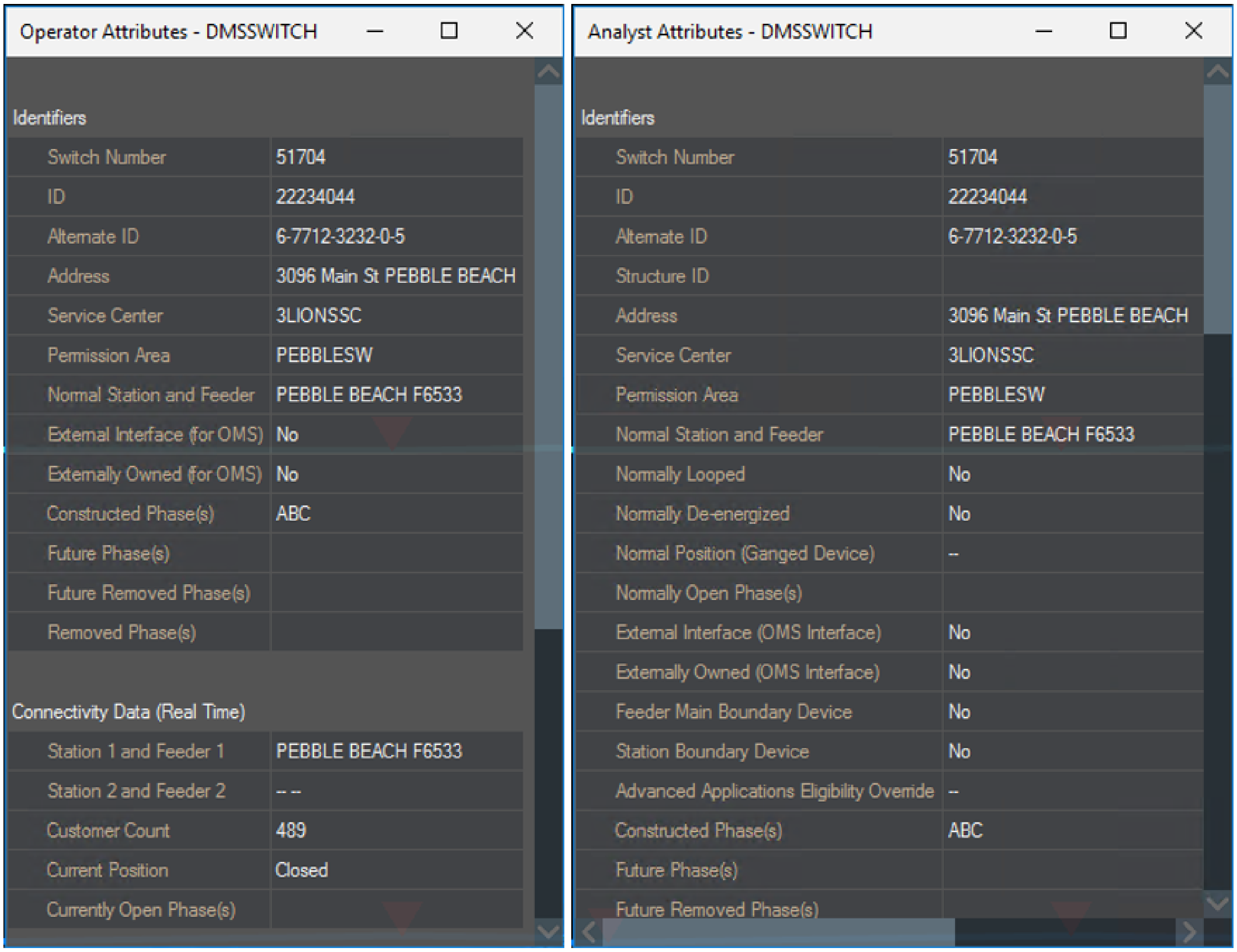


Figure 112. Operator Attributes Window and Analyst Attributes Window

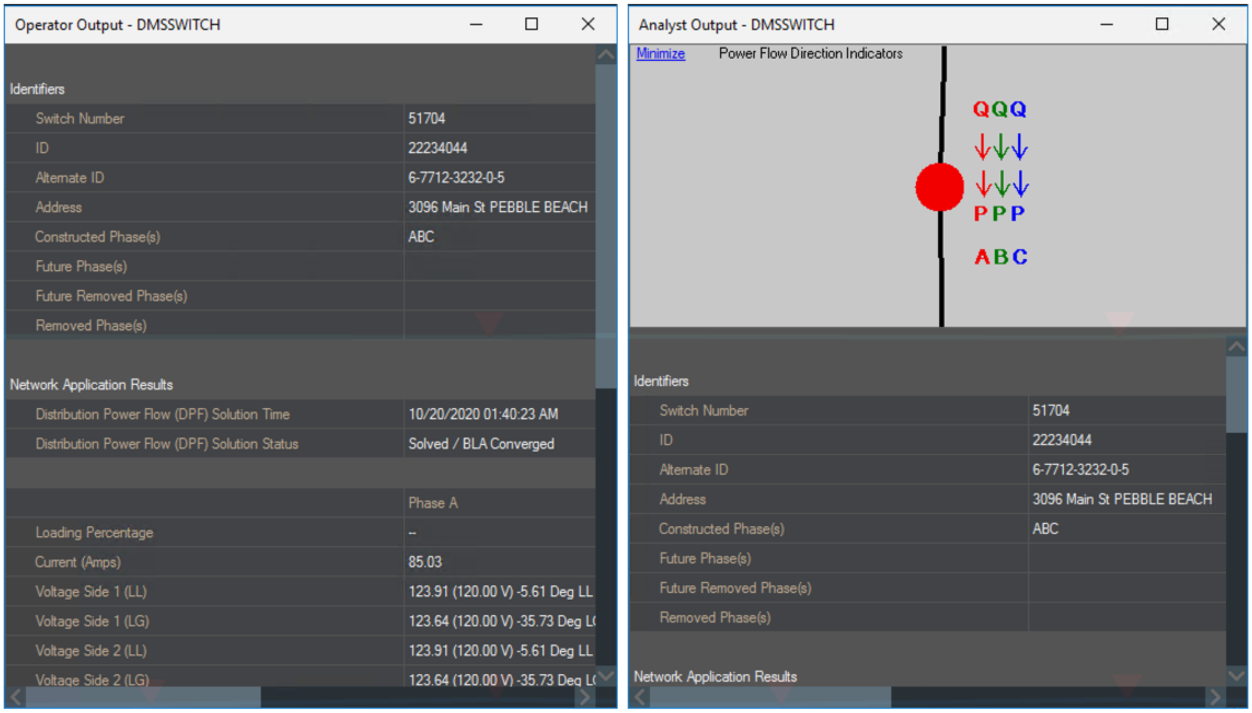


Figure 113. Operator Output Window and Analyst Output Window for a Switch

> **Note**
>
> Attribute and output windows do not automatically refresh. If there has been a change to the network, it is important to recall the display in order to get the latest values. You can re-select the device or re-select the attribute option on the toolbar to update the pop-up display.
>
> In an attribute or output display, you can copy the contents of a cell by pressing Ctrl+C.

Power Flow Direction Indicators

The Power Flow Direction Indicators pane is available on Analyst Output windows for certain devices, such as switching devices, lines, and transformers. This pane provides a visual indication of the direction that real (P) and reactive (Q) power flows through a given device.

The flow directions are determined by the value of power flow results that can be observed in the Analyst Output pop-up via the Real Power and Reactive Power fields. Therefore, to show the power flow direction, the Distribution Power Flow analysis must be performed for the station the device belongs to.

3.5.3. Topology Processing Results

A change of switch status automatically invokes the Network Topology Processor (NTP) so that the energization status of the network is updated to reflect the effect of the switching.

Normally Energized Lines

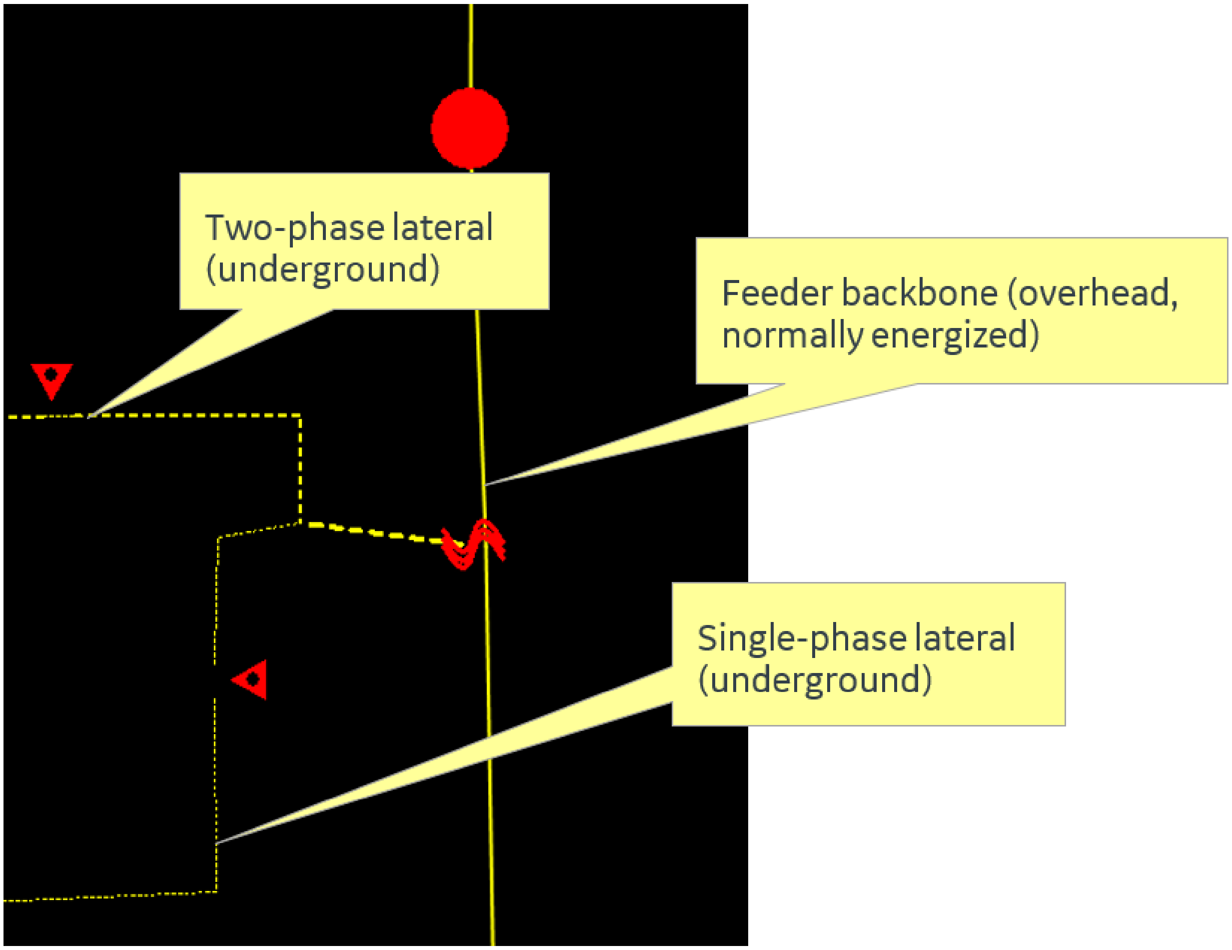


Figure 114. Normally Energized Lines

De-Energized Lines

If a switch is opened, the downstream portion of the feeder is de-energized and the feeder is shown with a white halo.

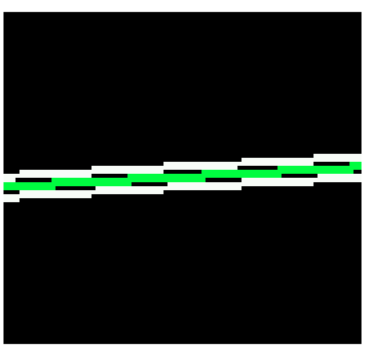


Figure 115. De-Energized Line (White Halo)

Abnormally Energized Lines

Closing another switch or placing a jumper can cause the de-energized lines to become energized from an abnormal source. The color of the lines- reflects the color of the feeder that the lines are now being fed from. The image below shows a feeder that is abnormally energized from a yellow feeder. The abnormal halo (while cross-hatching) shows that the line is energized from an abnormal source.

> **Note**
>
> The line is shown as abnormally energized if any of its phases are energized from an abnormal source.

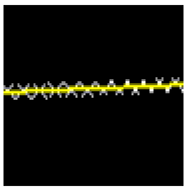


Figure 116. Abnormally Energized Line (Fed by Yellow Feeder)

Partially Energized Lines

If there is a partial energization on a feeder (for example, one phase is de-energized and the others are energized), this condition is shown with a light-orange halo, as shown in image below.

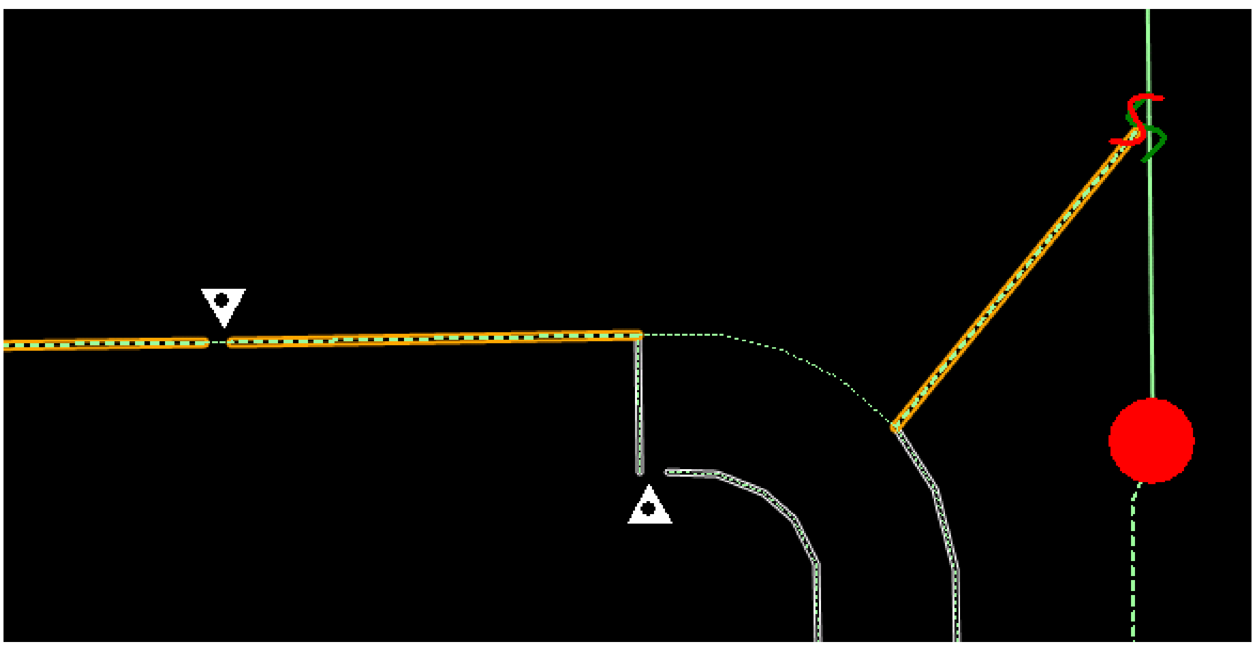


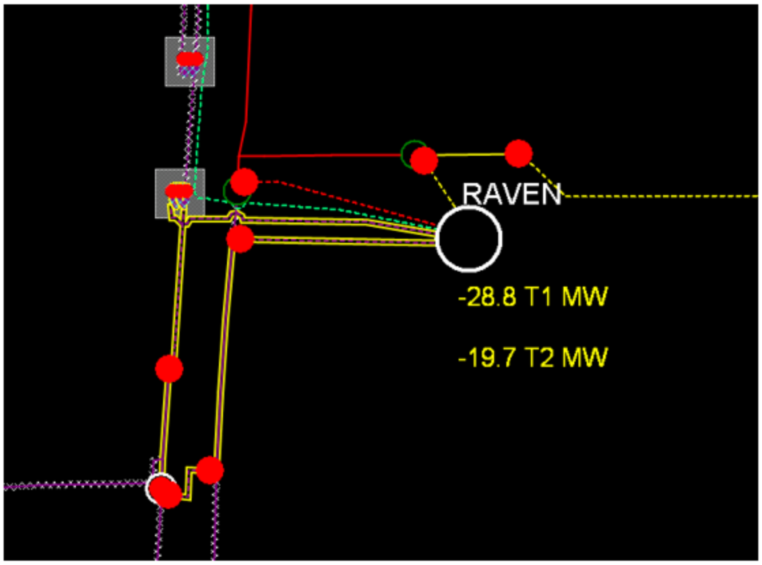
Figure 117. Partial Energization (Light-Orange Halo)

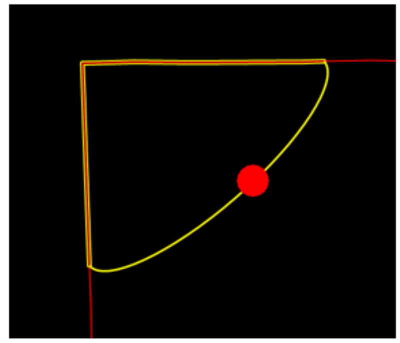
Looped Lines

Switching can also cause a loop condition where the line segments are energized from two or more points. The energization points can be on the same or neighboring feeders that can belong to the same or different energization sources (reference bus interfaces).

The following two types of loop conditions are distinguished:

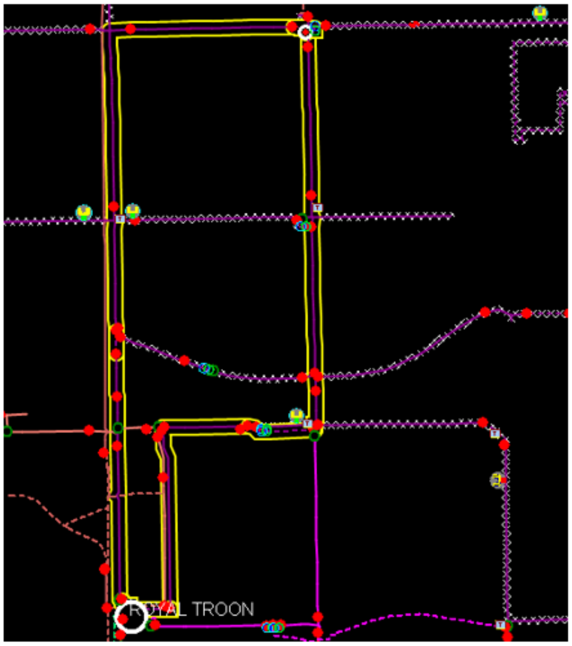
- **Network Loop:** A loop with a single energization source (reference bus interface) is identified on the network view with a default narrow, yellow halo on the affected conductors.

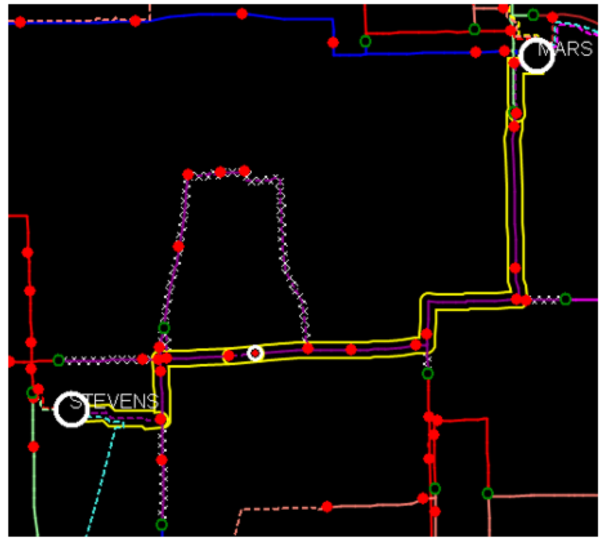




> **Note**
>
> A network loop that is connected to a parallel loop is shown with a Parallel Loop halo.
>
> The Network Loop halo does not appear for lines with the NormallyLooped or NormallyParallelLooped flag set to True.

- **Parallel Loop:** A loop energized from more than one energization source is shown with a wide, yellow halo on the actual loop path between the energization points.





> **Note**
>
> The Parallel Loop halo does not appear for lines with the NormallyLooped or NormallyParallelLooped flag set to True.

- **Fed from Multiple Feeders:** In Fantasy Island default configurations, when a loop (either parallel or network) exists between two or more feeders, lines are colored magenta to indicate being fed by multiple feeders and appear with a yellow outline halo to signify the looped conductor. However, the abnormally energizedline color when fed from multiple feeders is configurable in the Viewer Configuration Editor on the Lines tab. Additionally, the halo effect can be removed by setting the "Width (px)" in the Viewer Configuration Editor to 0. The specific line colors and halo behavior can be configured via the Viewer Configuration Editor line tab.

> **Note**
>
> The Fed from Multiple Feeders halo does not appear for lines with the NormallyLooped or NormallyParallelLooped flag set to True.

Mixed Flow Lines

The opposing flow condition (for example, when one phase is being supplied from the opposite end of the line, or from another feeder) is shown with an orange halo, as shown in image below. The color of the line reflects the color of the feeder that supplies the opposing-flow phase.

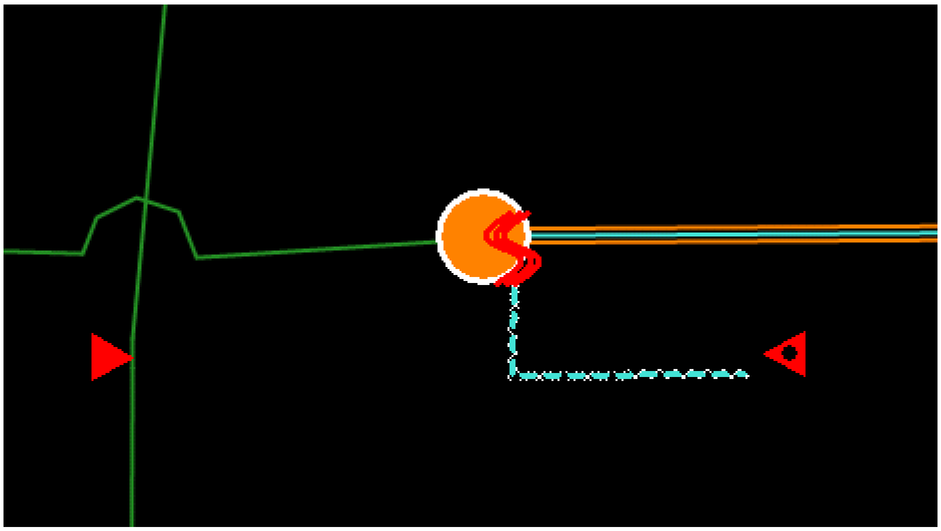


Figure 118. Mixed Flow Line (Orange Halo)

The status of the switch on the network view also reflects both its current status and the normal state. A white halo appears around the switch symbol to indicate that it is not in its normal state. The color of the switch within the halo indicates the current state of the switch.

Grounded Lines

When a temporary ground is applied to a de-energized line, the status of the line is updated to grounded. This status is shown by a yellow and green halo, as shown in image below.

If a line is in a grounded state, adjacent, open switches are prevented from being closed, and cuts on this line are prevented from being removed. Any attempt to remove a cut or close a switch between an energized section and a grounded section generates a warning until the ground is removed.

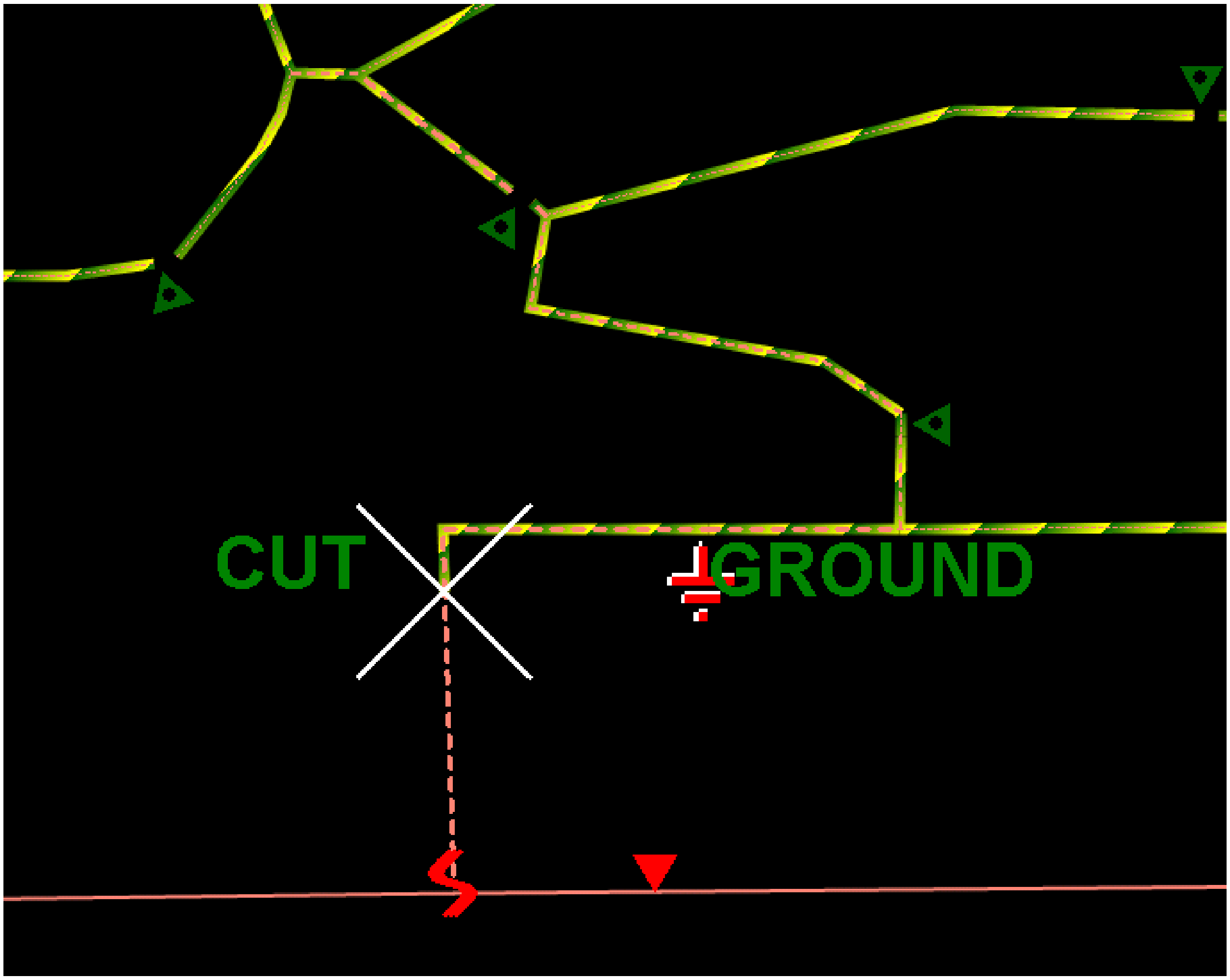


Figure 119. Grounded Line

3.5.4. Power Flow Animation

The power flow animation visualizes the real and reactive power flow on the network view by making conductor lines circulate in the direction of the flow. The function can be performed on a three-phase or per-phase basis so that the connectivity and power flow of multi-phase laterals and URD loops can be identified.

To enable the power flow animation:

1. Right-click a point on the network view.
2. Point at the Power Flow Animation option.
3. Select the animation option.

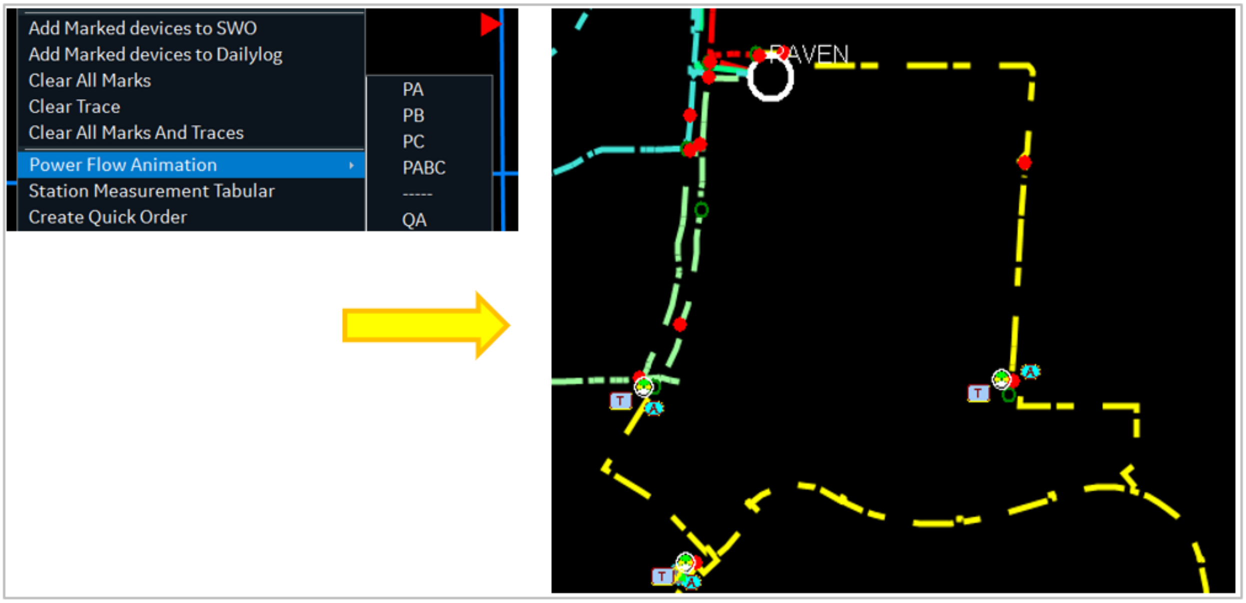


Figure 120. Power Flow Animation Options and View

The animation can be enabled or disabled from the network view context menu. By default, the animation stops after 10 minutes.

Note that flow direction data is available after running the DPF solution for a station.

> **Note**
>
> This feature is available for demonstration purposes only and is not intended for production use.

3.5.5. Outage Prediction Indication

If the Outage Management functions of ADMS are enabled, the system is able to show sections of the network that are predicted to be de-energized due to an outage incident, as shown in image below.

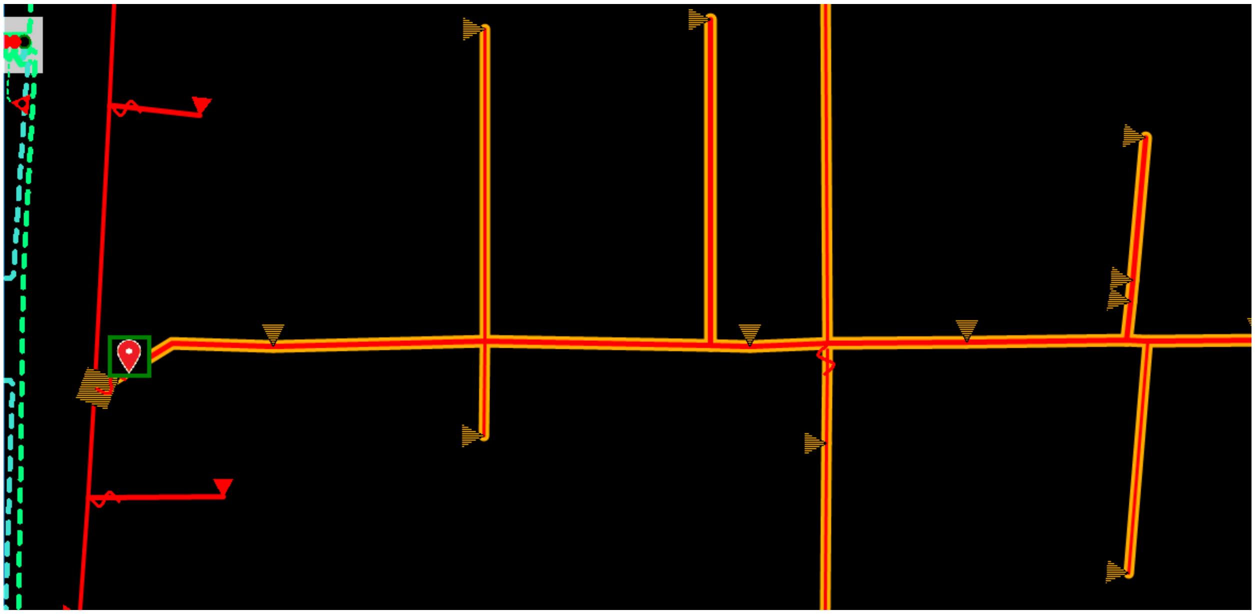


Figure 121. Laterals, Fuse, and Transformers Predicted To Be De-Energized

3.5.6. Specific Device State Decorations

Decorations appear next to switching devices and fault indicators to show a specific device condition, measurement status, or control type.

Scada Decorations

Scada-related decorations show the device control type (automated or telemetered) or a specific status. For example, several common Scada decorations are as follows:

- SCADA Controllable (A)
- Telemetered (T)
- Manually Replaced (M)
- Suspect Quality (S)
- Reclosing Disabled ®

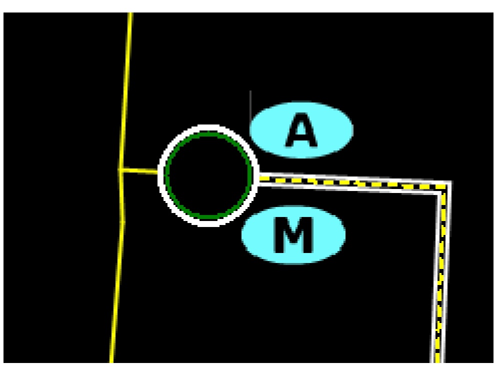


Figure 122. Scada Controllable (Automated) Switch in Manually Replaced State

Pending Permanent State

The Pending Permanent State decoration appears for switching devices that currently have "pending permanent" status. This means that the normal state of the switch has been changed, but the change has not yet been confirmed by an update of the GIS/AMS data. The network topology processing continues to use the existing normal state of the switch until the change is confirmed.

Dangerous Switching Condition

The Dangerous Switching Condition decoration appears on open switches whenever closing a switch could result in a dangerous network condition. Currently, this decoration is placed wherever the following conditions occur:

- **Dynamic Phase Mismatch:** Dynamic phases on either side of the switch do not match.
- **Phase Angle Mismatch:** The phase angles (as calculated by DPF) on either side of the switch exceed the configured maximum difference.
- **Nominal Voltage Mismatch:** The nominal voltages on either side of the open switch are different. Note that this check uses only the nominal voltage, not the currently calculated voltage.

The switch Analyst Attributes panel shows the results of the above three cases so that operators can determine why the dangerous condition icon appears on a given switch. Additionally, if configured, the Check Before Operate (CBO) function notifies the operators of the situation before closing the switch.

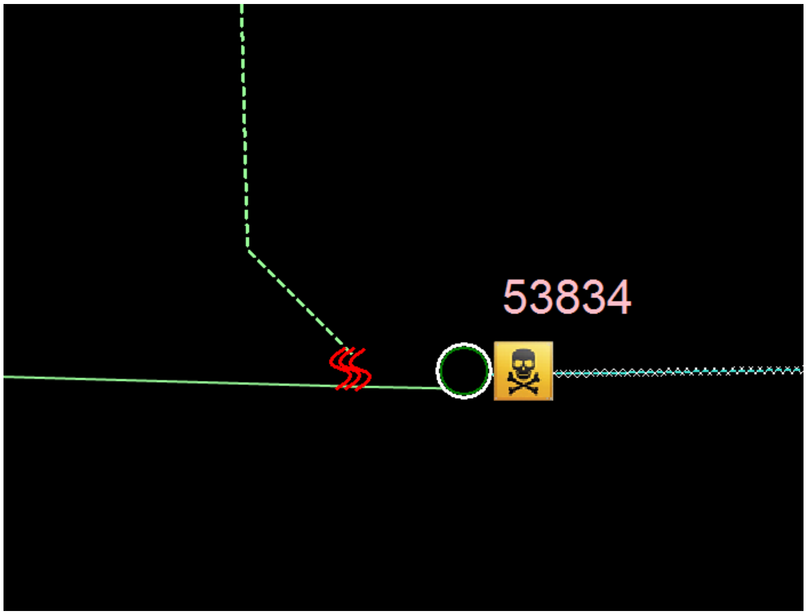


Figure 123. Dangerous Switching Condition

3.6. Alarms and Events

Alarms are part of the Scada system. They are managed using the standard SCADA alarm displays and menus.

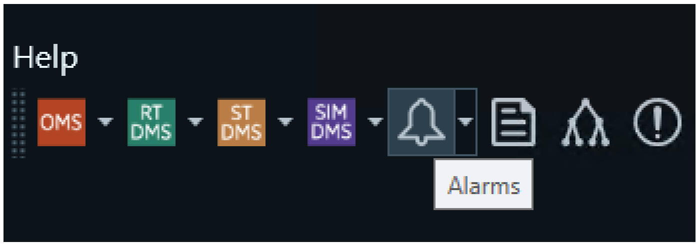


Figure 124. Alarms Pull-Down Menu

Alarms and events that are generated within the distribution network functions, such as operation of manual distribution switches, are included in the standard alarm event logs.

3.7. Working with Tabular Displays

There are a large number of distribution-specific tabular displays. They are all managed in a similar fashion.

For example, consider the Station Exception display, which is a primary tabular display that is used for normal operations. This display is arranged such that the most important outages (customers and abnormal states) are listed at the top. Further details are available by clicking the relevant fields; this drills down into subsequent displays. On the lists of individual devices, such as switches in abnormal states, Locate buttons are provided that navigate directly to the device on the geographic network view.

The station exception summary lists only those stations for which the operator currently has responsibility (that is, operational control).

The order in which the information is shown on the tabular displays can be manipulated as required (see image below).

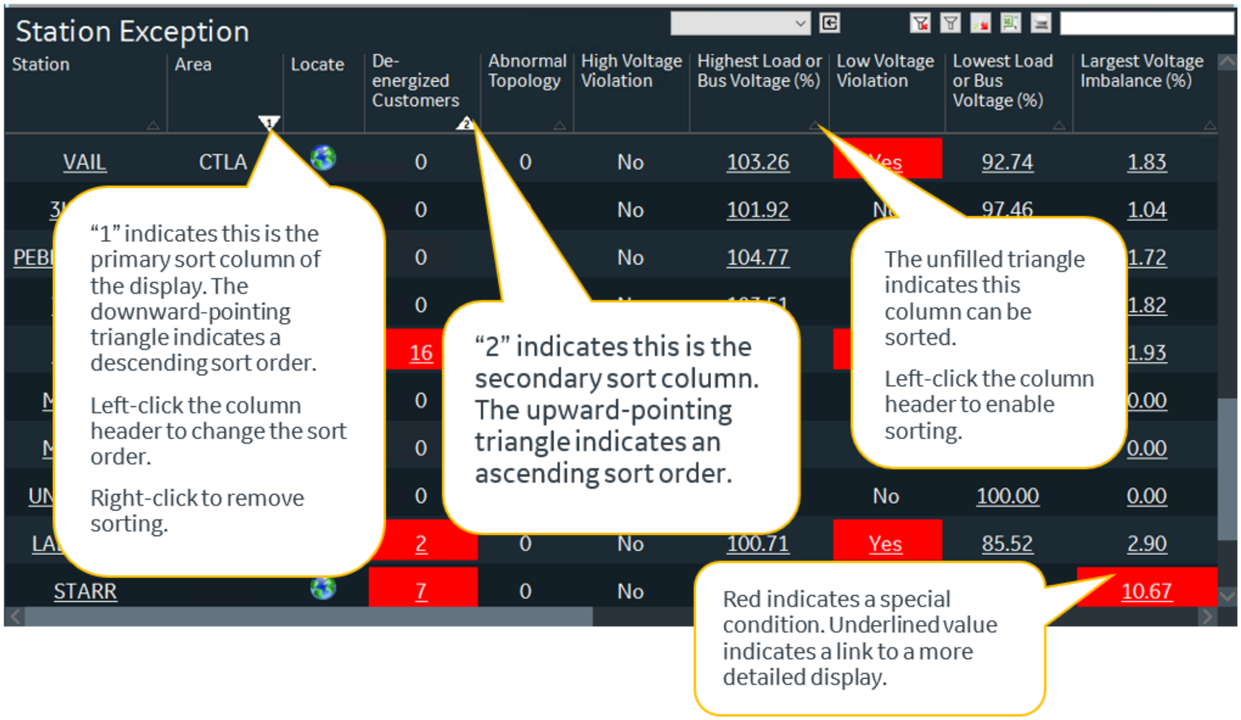


Figure 125. Tabular Display

Each column with an unfilled triangle (Δ) on the header row can be used to sort data on the display by the data in that column. Initially, clicking a column header sorts the data in ascending order, which is indicated by a filled upward-pointing triangle. Clicking a column header a second time re-sorts the display in descending order, indicated by a filled downward-pointing triangle.

A number 1 inside a triangle on the header row indicates the primary sort column. Clicking the header row for another column causes a secondary sort of the display by the data in that column, and a number 2 appears inside the triangle.

To remove the sorting of a particular column, right-click the column's header. This removes the sorting for that column and re-organizes the sorting of the other columns. If, for example, a primary and a secondary sort have been applied and the primary sort is removed, the secondary sort is promoted to become the primary sort. To remove all custom column sorting on the tabular display, click the Reset Sorting Order button.

You can change row height, column width, and reorder columns as needed. To hide a column (size a column to its minimum width), double-click the column's right border. To unhide a column (restore a column to its default width), double-click the column's right border again.

> **Note**
>
> The display's last used column order and sort settings are saved for the current Windows user and applied each time the display is opened.

Cells with a colored background indicate a special condition. More details are available by clicking cells with an underlined value, which navigates to the corresponding linked display. For example, selecting a substation name navigates to an exception summary for each feeder of the substation.

Information shown on a tabular display can be filtered using the Global Filter entry field on the title bar, using the filtering panel, or by using column-specific filters that are enabled by clicking the Show/Hide Column Filtering Fields icon . Per-column filtering also enables the Filter On Cell Value context menu item; enhanced filtering allows you to save filtering panel configurations.

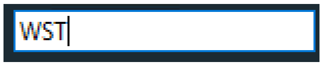


Figure 126. Global Filter Entry Field

Figure 127. Filtering Panel Configuration

> **Note**
>
> The per-column filtering function supports logical conditions ("=", "!=", ">", "<", ">=", and "=<").

When filtered, only rows of data that contain the text (or numbers) entered in the filter fields are shown in the table. To return the tabular display to its normal state, clear the Filter field and click the Clear Filters button to clear all column-specific filters.

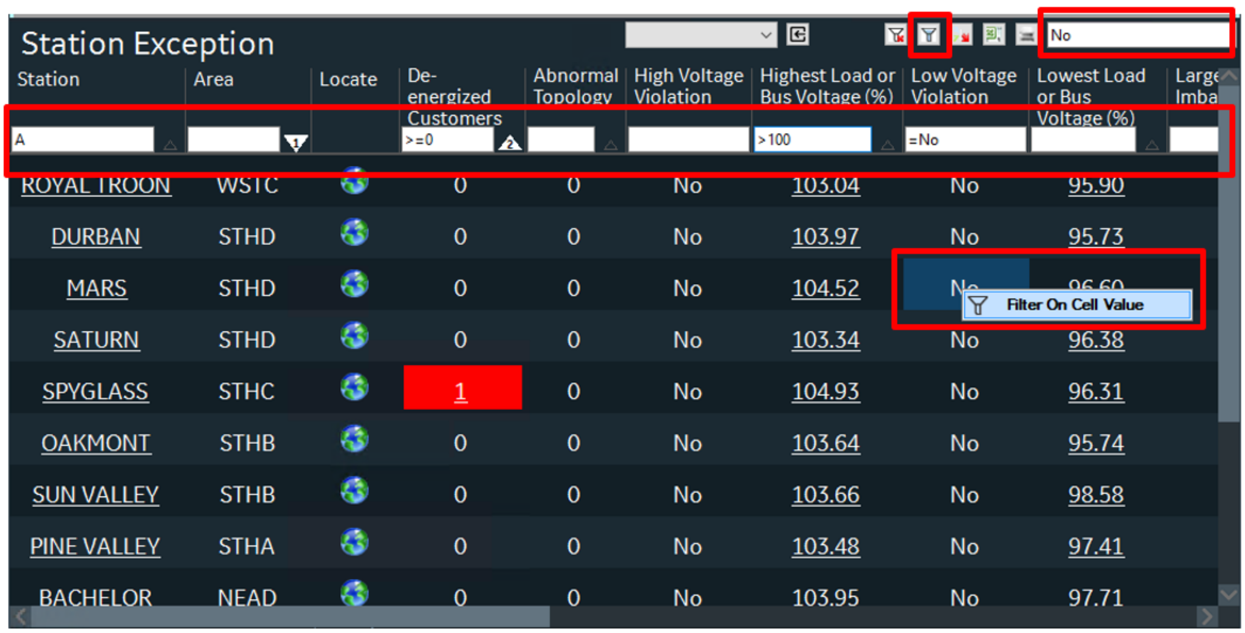


Figure 128. Global and Column-Specific Filtering on a Tabular Display

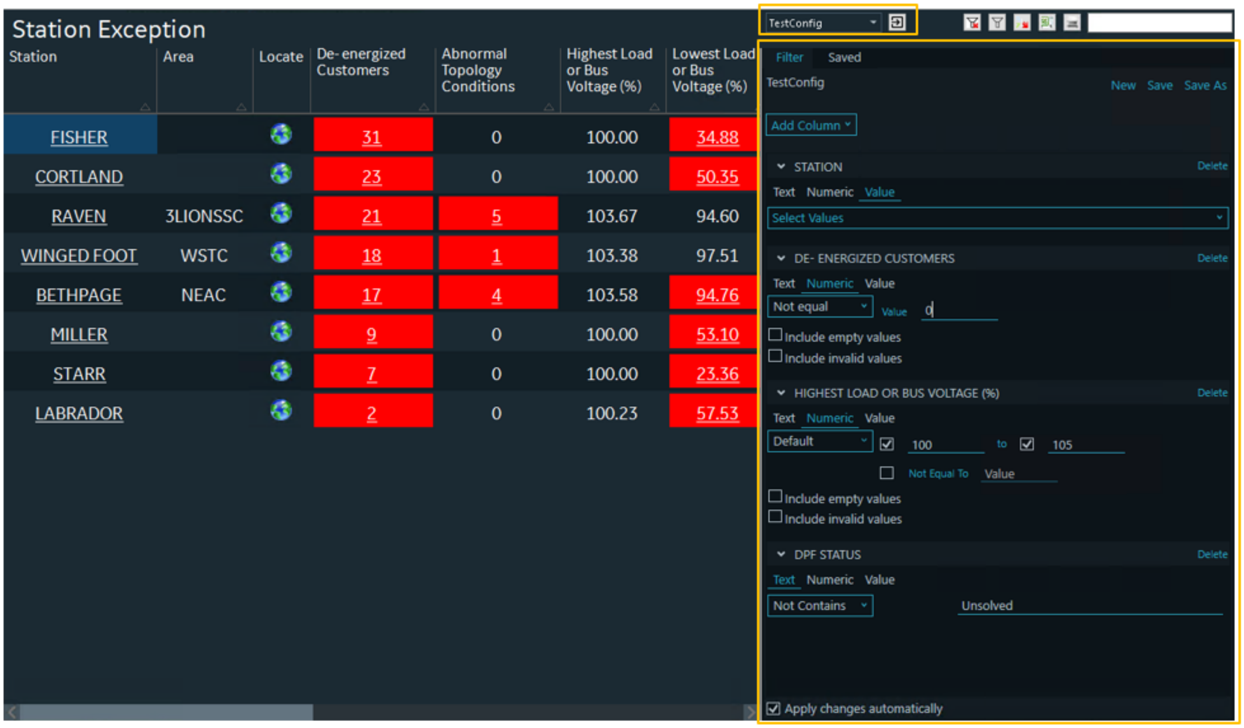


Figure 129. Filtering Panel on a Tabular Display

To export table data to a CSV file that can be opened by Excel, click the Export button. To print the grid, click the Print button.

Note

You can copy the contents of a selected cell or range of cells by pressing Ctrl+C.

3.8. Printing SCADA and Geographic Displays

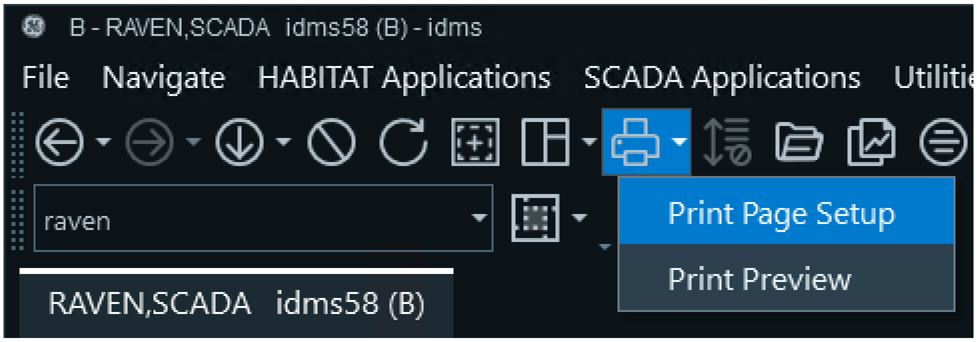
Before printing, each workstation must be configured to use the correct printer, paper size, and page layout. The best results are obtained by using a color printer and A3 or 11" x 17" paper in landscape mode.

To print SCADA and geographic displays:

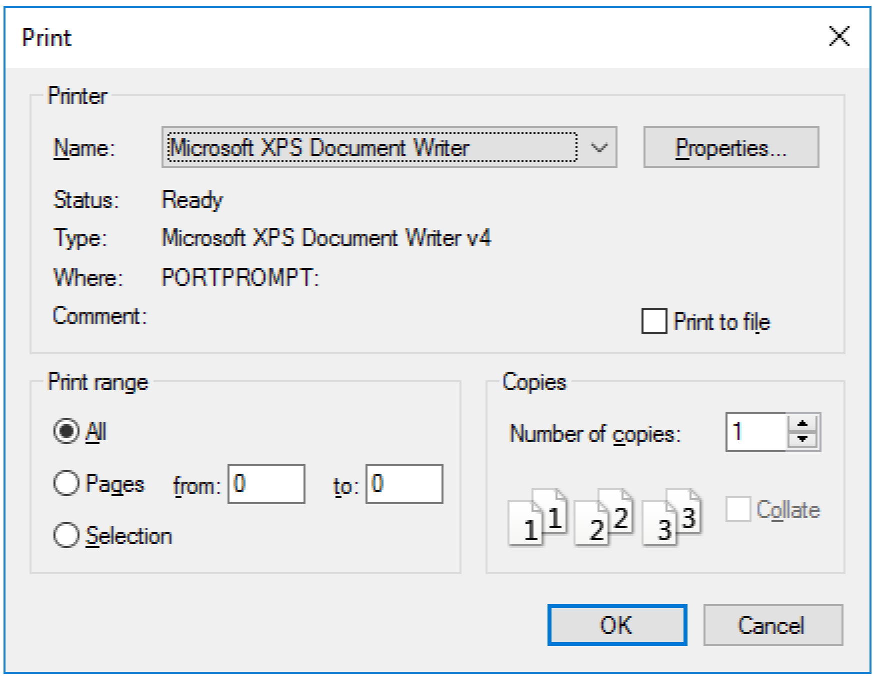
1. Re-size the viewport so it is slightly smaller than the chosen paper size. The display does not print correctly if the viewport is at the full size of the monitor.
2. Position the display as needed within the viewport.
3. Set the display to fit.

   For SCADA one-line displays, you can click the "Turn On Scale to Fit" button to show the entire screen in the re-sized viewport.

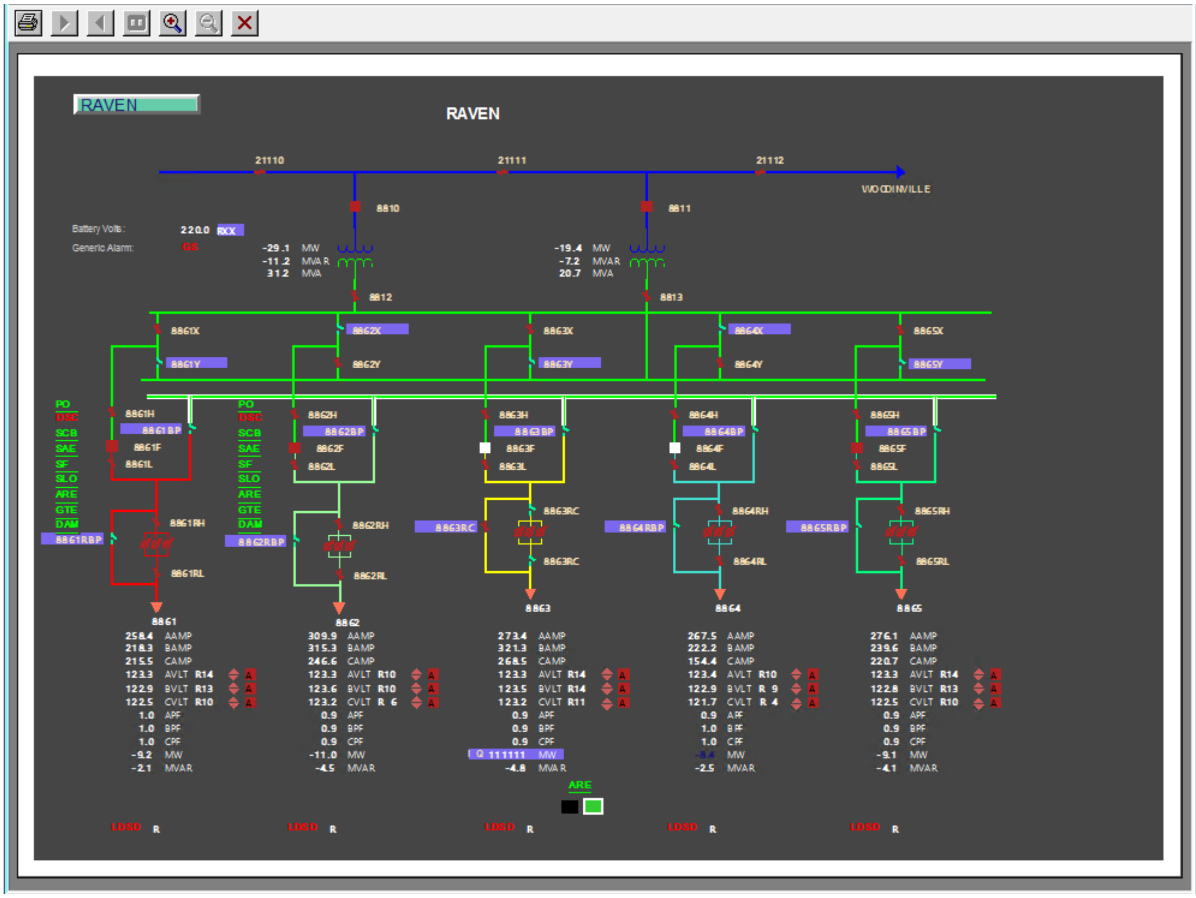
4. On the Print icon menu, click Print Page Setup.



5. Enter the required settings and click OK.



6. Navigate to the Print Preview dialog box.



7. Click the Print button.

4. Performing Network Operations

Network operations applied to the real-time views update the central network model and are broadcast to all of the operator workstations that are connected to the system. The section of the network that is able to be operated by an operator is known as the "Area of Responsibility" (AOR). An AOR is a station, or a group of stations, for which the operator has operational control. The operator can make changes to the real-time network model, operate switches, and apply tags.

Actions that do not alter the real-time model, such as tracing, can be performed regardless of the current AOR status. Attempts to make changes on a station for which the operator does not have responsibility are prevented by the system, and warning messages are displayed.

The operation of switches and other devices on the network is normally managed as a formal process involving Switching Orders or other procedural instructions. The details of the Switching Order functions are described in the *Switching Operations User Guide*.

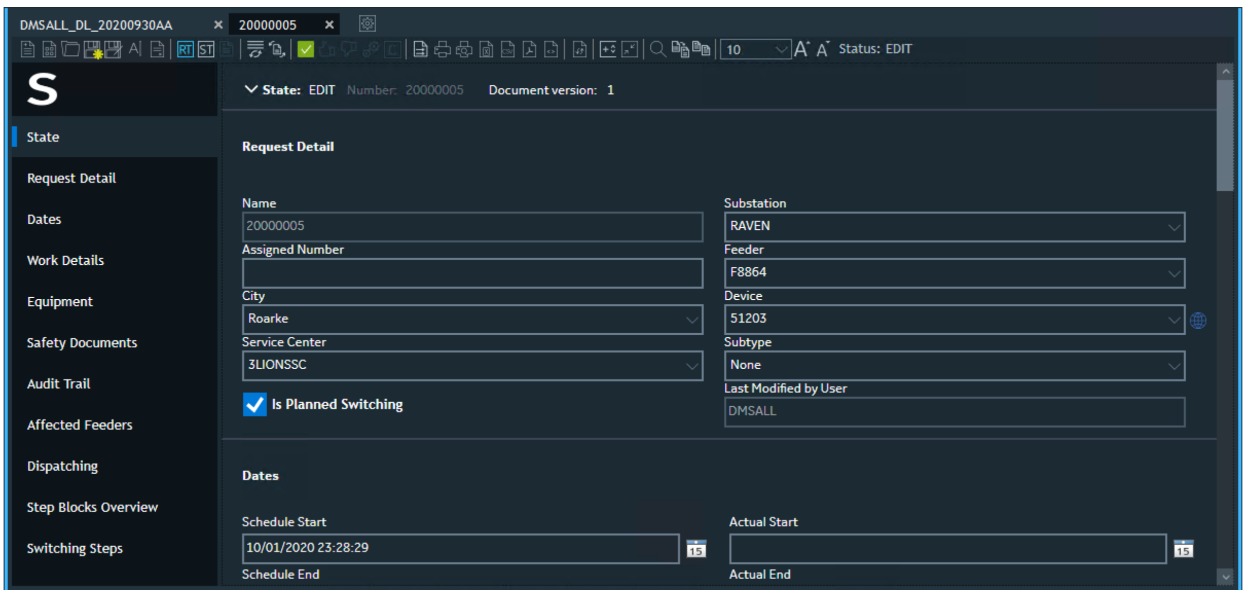


Figure 130. Switching Order Application

4.1. Working with Switching Devices

4.1.1. Switch Operations

A switch can be selected by clicking the center of the symbol. When the switch is selected, it becomes highlighted in magenta and the lower toolbar changes to provide operating functions for that switch. For the selected switch, the phase indicator symbol shows the phasing of the selected device.

The majority of switching devices in a distribution network are manually operated devices (for example, disconnect switches and fuse switches). This means switches are actually operated by field crews under the instruction of control room operators. In each of these cases, it is the job of the operator to update the switch states within the ADMS model in synch with the operations being performed in the field.

Manually operating a switch is performed by selecting the desired switch, clicking Status on the device-specific toolbar, and then either selecting the Change Switch Status option to toggle the switch state or selecting the Toggle Phases option to toggle individual switch phases.

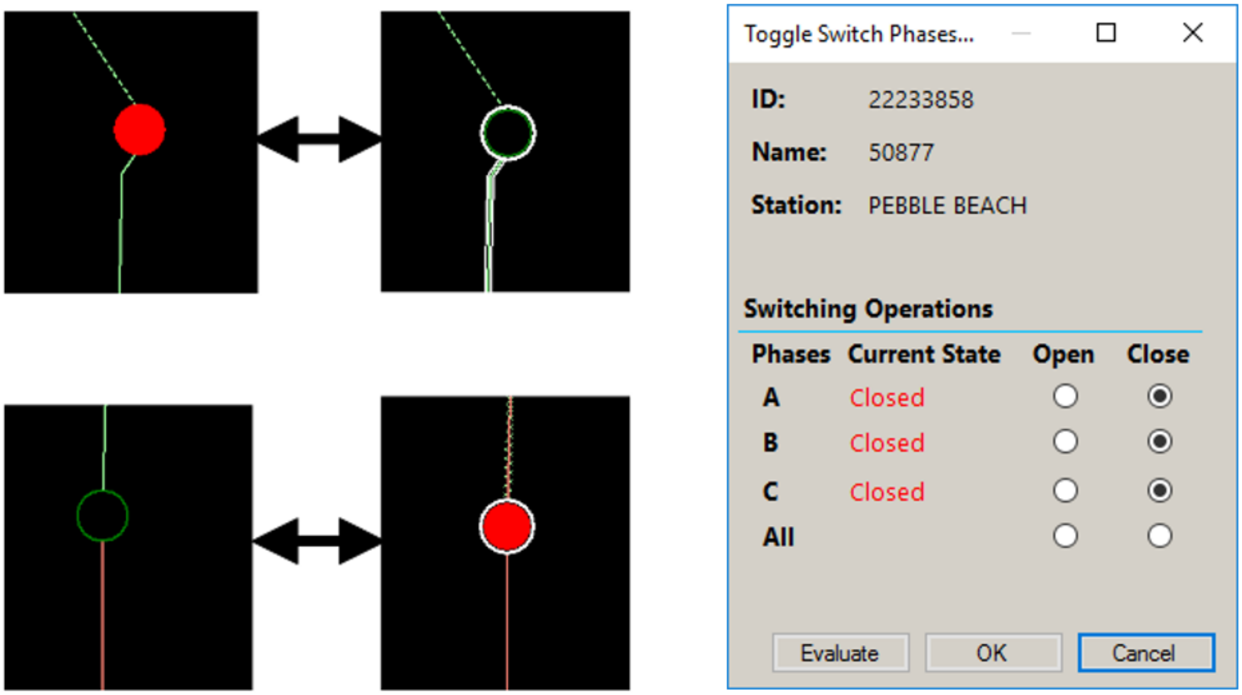


Figure 131. Changing Switch Status

As shown in image above, a switch is shown with a white halo to indicate that it is in an abnormal state. The normal state of a switch is loaded into the model from the source data (usually the GIS), and it is the source data that must be updated to change the normal state. Clicking the Change Permanent Status button on the device toolbar marks a switch as pending a normal state change. This is flagged by a special symbol that is displayed next to the switch (see image below), which indicates that the normal state has been changed and the system is awaiting an update of the model from the GIS.

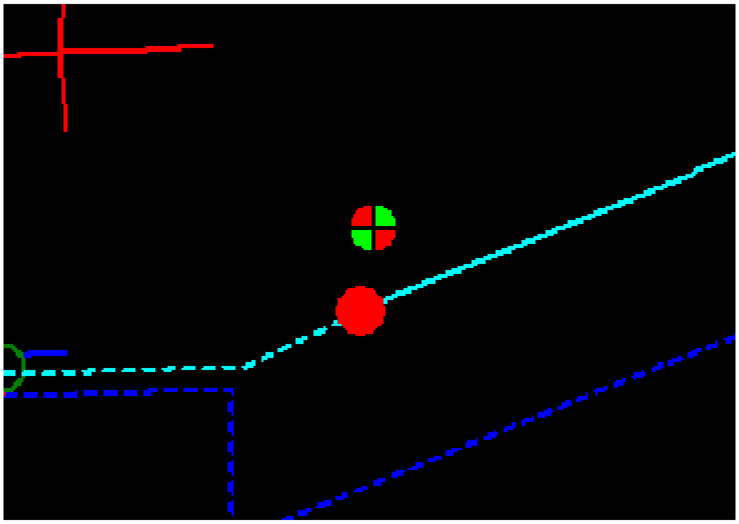


Figure 132. Switch Showing Symbol for Pending Change of Normal State

You can set a specific open state classification (Normal Open, Temporary Open, or Bad Open) using the Change Switch Classification drop-down list on the device toolbar, as shown in image below.

- The Normal Open classification indicates that the switch is normally open.
- The Temporary Open classification indicates that the open requirement is of temporary nature.
- The Bad Open classification indicates that the switch must not be closed under normal condition. The Bad Open classification provides a visual indication and a Check Before Operate notification to alert an operator that closing this point leads to bad parallel condition between two sources and can lead to undesired interruption and damage to equipment or personnel.

The Global Designated Switches and Designated Switches grid tabular displays show switching devices with the Normal Open, Temporary Open, or Bad Open Point classification for all substations or a specific substation, respectively.

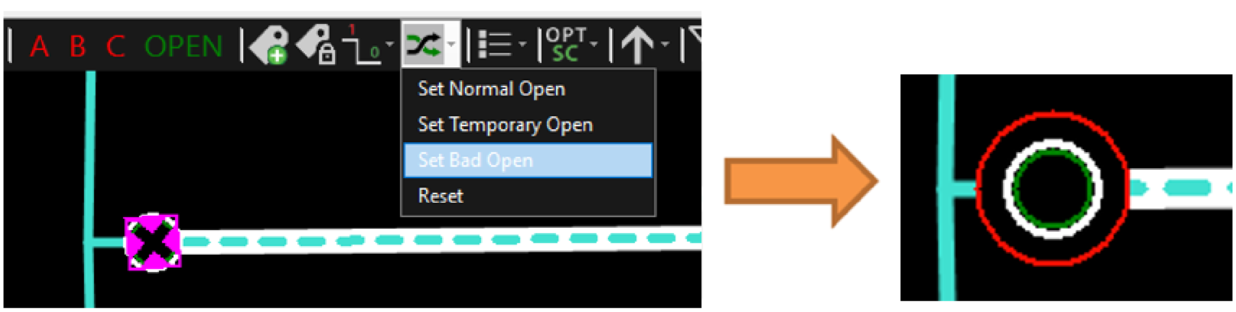
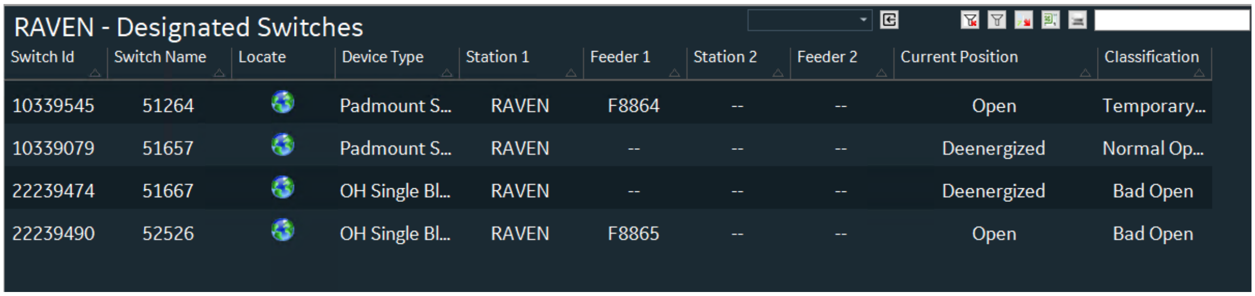


Figure 133. Setting the Switch Classification to Temporary Open


RAVEN - Designated Switches

Switch Id	Switch Name	Locate	Device Type	Station 1	Feeder 1	Station 2	Feeder 2	Current Position	Classification
10339545	51264		Padmount S...	RAVEN	F8864	--	--	Open	Temporary...
10339079	51657		Padmount S...	RAVEN	--	--	--	Deenergized	Normal Op...
22239474	51667		OH Single Bl...	RAVEN	--	--	--	Deenergized	Bad Open
22239490	52526		OH Single Bl...	RAVEN	F8865	--	--	Open	Bad Open

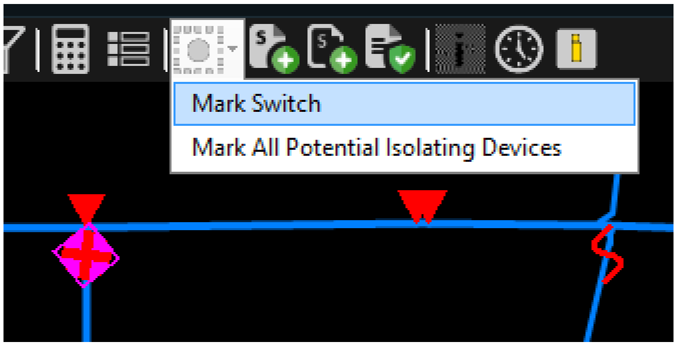
Figure 134. Designated Switches Display

There are a number of additional switch operations available from the device toolbar, such as those related to distribution system optimization functions. For more information about device toolbar buttons, refer to the *ADMS Display Reference Manual*.

4.1.2. Marking Switches

Switches can be temporarily marked with a highlighting symbol. This allows the operator to identify a switch that may be required for switching, or some other operational procedure that can easily become "lost" while panning and zooming. The Mark symbol is only visible on the operator’s workstation. Multiple switches can be marked, and they can be cleared individually or as a group.

- To mark a switch, select the switch, click the Mark button on the toolbar, and then select the desired mark option.



- To show all marked devices in a single view, click the Show Marked Devices button on the common toolbar.
- To clear mark from an individual switch, select the switch, and click the Mark Switch button.
- To clear all marks on the display, click the Clear Marks or Clear All Marks and Traces button on the common toolbar.

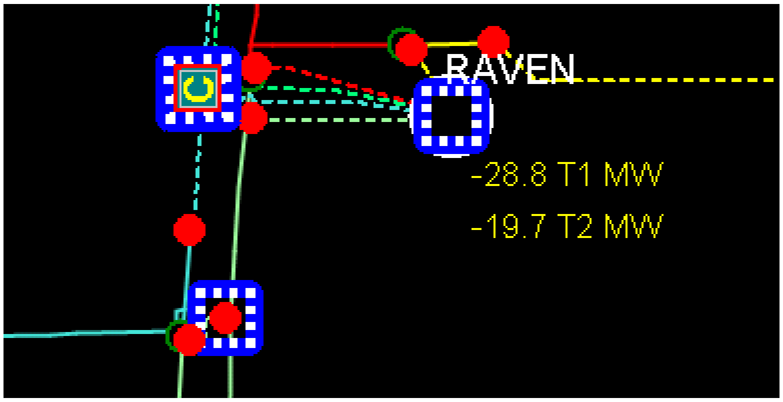


Figure 135. Marked Switches

When a switch is marked on an internal view, such as the station internals or vault internals view, the container object is highlighted with a mark symbol on the geographic network view.

4.1.2.1. Marking Switches to Isolate a Switch or a Line Segment

To mark the switches that need to be tripped to isolate a given switch or line section, select the component. From the toolbar, click the Mark pull-down menu and select the Mark All Potential Isolating Devices button or the Mark and Trace All Potential Isolating Devices button.

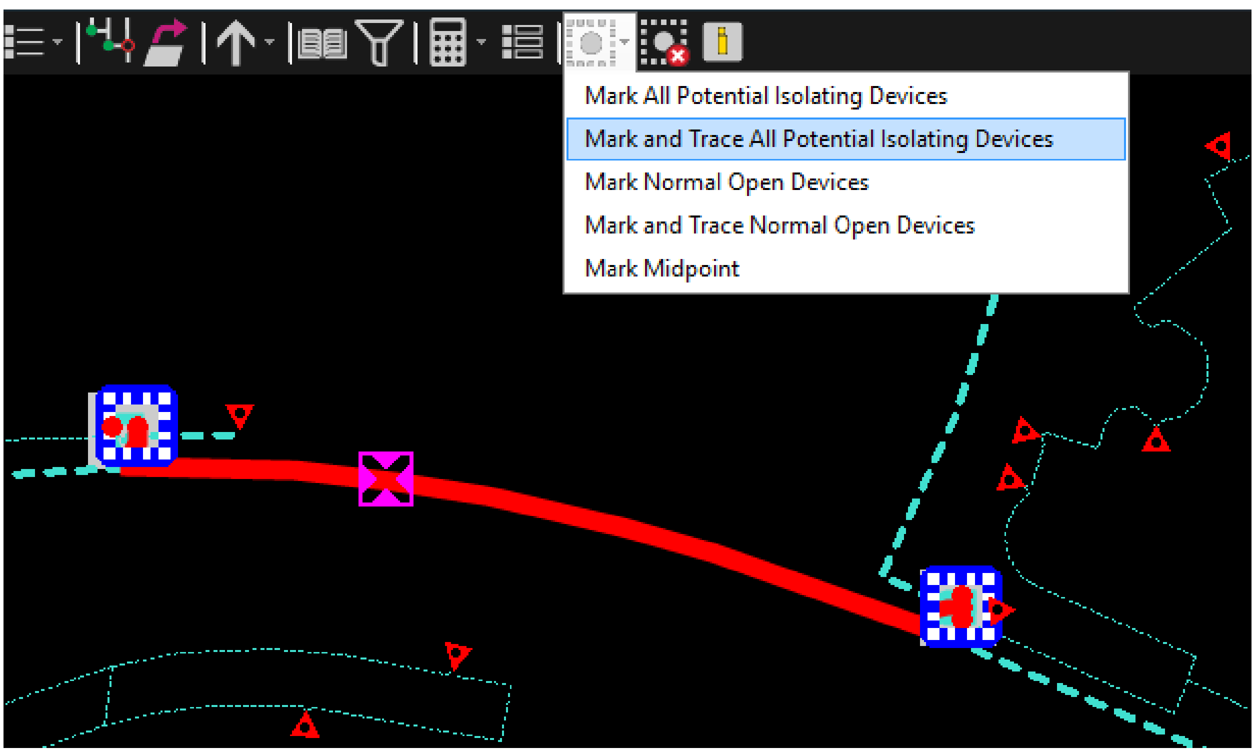


Figure 136. Marking and Tracing All Potential Isolating Devices

All of the switches (open and closed) that need to be open to fully isolate the selected switch or line segment are highlighted with the Mark symbol. When using the Mark and Trace All Potential

Isolating Devices option, the system also draws the trace between the switches.

4.1.2.2. Marking Normally Open Switches on a Feeder

To mark all normally open switches on a feeder:

1. Select a point on the line.
2. From the toolbar, click the Mark pull-down menu and select Mark Normal Open Devices or Mark and Trace All Normal Open Devices.

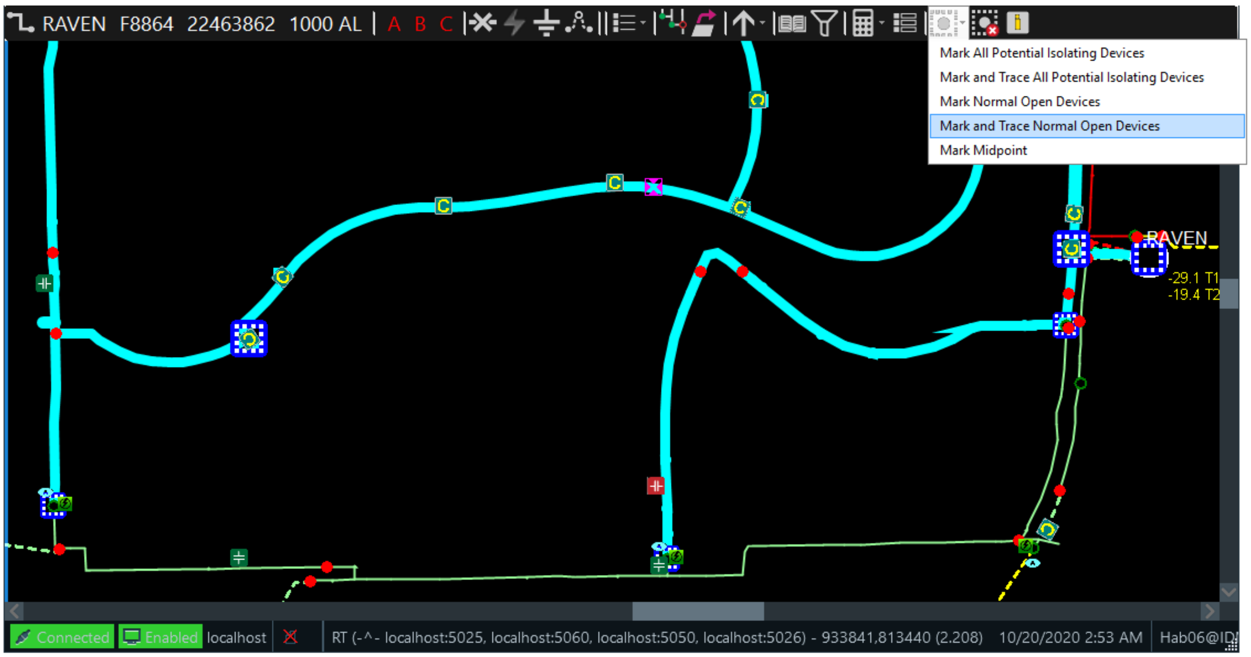


Figure 137. Marking and Tracing All Normally Open Devices

All normally open switches on the feeder are highlighted with the Mark symbol. When using the Mark and Trace All Normal Open Devices option, the system also draws trace between the switches.

4.1.2.3. Marking a Feeder Midpoint Switch

To mark the switch that is closest to the geographical midpoint of a feeder, select a line segment on the feeder. From the toolbar, click the Mark Switches pull-down menu and select Mark Midpoint.

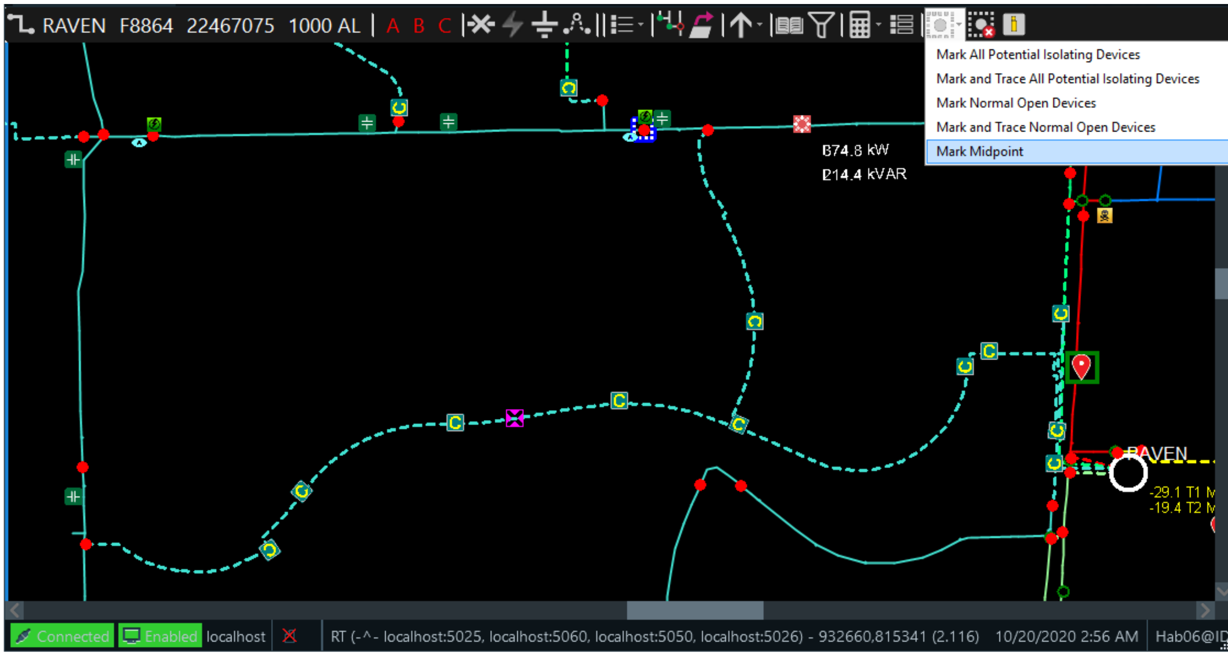


Figure 138. Mark Midpoint Option

4.1.3. Fuse Operations

Fuses are used to both protect and isolate the network. They behave in the same way as manually controlled switches, but there is no indication of abnormal status. Fuse switches are always considered to have a "closed" normal state.

The image below shows how a closed fuse and an open fuse are displayed.

Figure 139. Closed and Open Fuses

4.1.4. Regulators and Bypass Switches

There are no operations performed on regulators — only the bypass switches. This is done using the switch toolbar.

4.1.5. Reclosers and Automated Feeder Switches

Reclosers and other automated feeder switches normally consist of a main switch and a bypass, and they are normally SCADA controlled. On the network view, reclosers with bypass switches are shown as recloser UI containers.

Reclosers and switches can be operated by selecting the main device from the network view or from the UI container internal (schematic) view. The individual switches are operated as standard SCADA (or manual) switches.

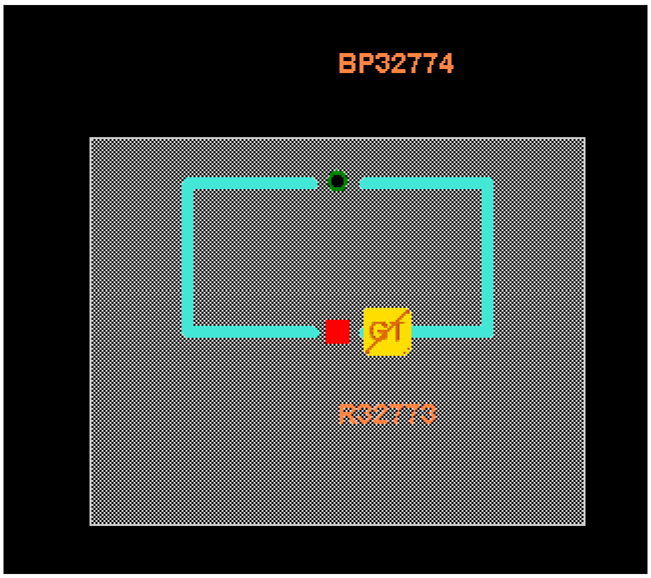


Figure 140. Recloser Internal View

4.1.6. Composite Switches

Composite switches or multi-position switches can operate independently according to local controllers. For example, throwovers and transfer switches automatically transfer the load to an alternate source of supply if the preferred supply fails. This automatic logic is modeled within the switch, and the model behaves in the same way as the real network. These switches can also be operated manually to force them to a particular state or turn off supply to the load completely.

The states of the composite switches are predefined in the static model. The switch device-specific toolbar contains the Status button with options to manage all multi-position switch states. The image below shows composite switch states for several types of composite switches.

The automatic/manual operation mode state is important. When the switch operates in the automatic mode, the button to manually control the states is disabled.

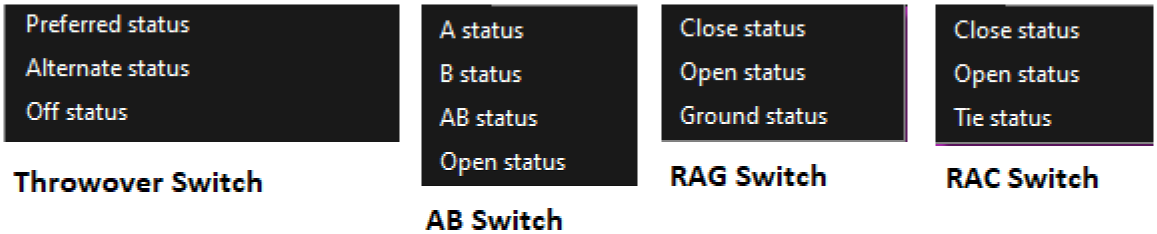


Figure 141. Examples of Composite Switch States

Figure 142. Status Button for a SCADA Composite Switch

In real-time mode, the Status button for the SCADA-controlled composite switches, does not display all composite switch states. Instead there is an option "Go to SCADA Control (EMP)", which indicates that the SCADA control is available. Clicking this option invokes the standard SCADA control pop-up for the selected composite switch. In the simulator and study modes, the Status button shows composite switch states for SCADA controlled and manually operated composite switches.

On the network view, multi-position switches are typically located inside the vaults.

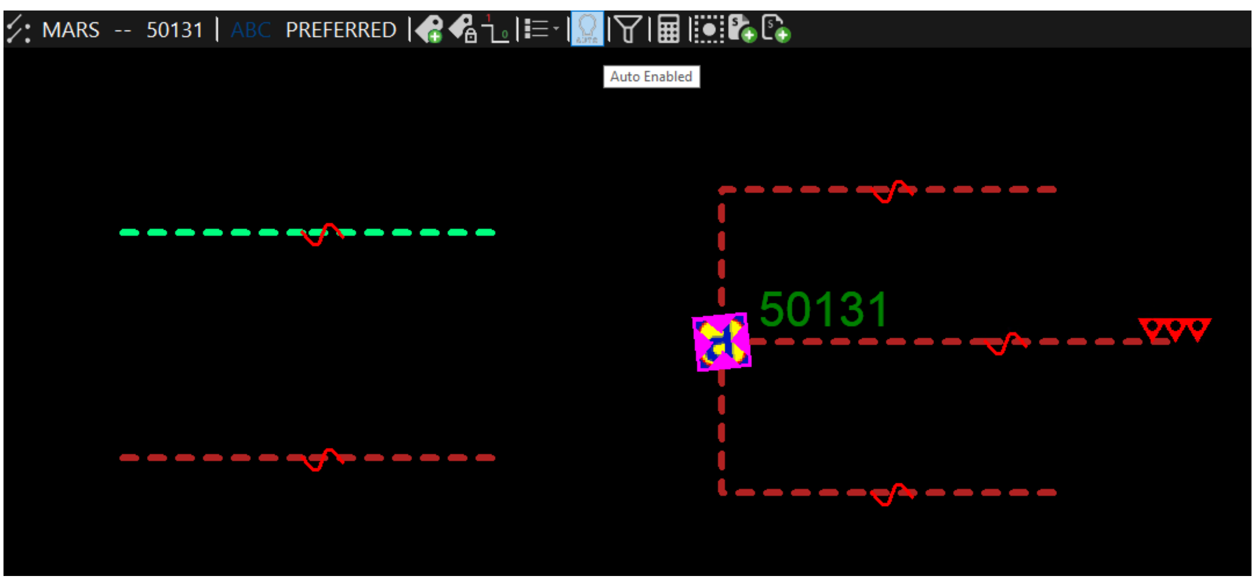


Figure 143. Throwover Switch - Auto Mode

4.1.7. SCADA Switching Operations

Switches that are remotely controlled and monitored can be directly operated from the control room. The integration of Scada with ADMS allows these actions to be performed directly from the distribution network view in a similar manner to manual switching.

When a SCADA-controlled switch is selected, the Status button on the device-specific toolbar includes the Go to SCADA Control (EMP) option indicating that SCADA control is available.

Clicking this option invokes the standard SCADA control pop-up for the selected switch. The standard SCADA two-step operational sequence can then be performed, as shown in image below. Similarly, the status of SCADA control switches is automatically updated on the network views.

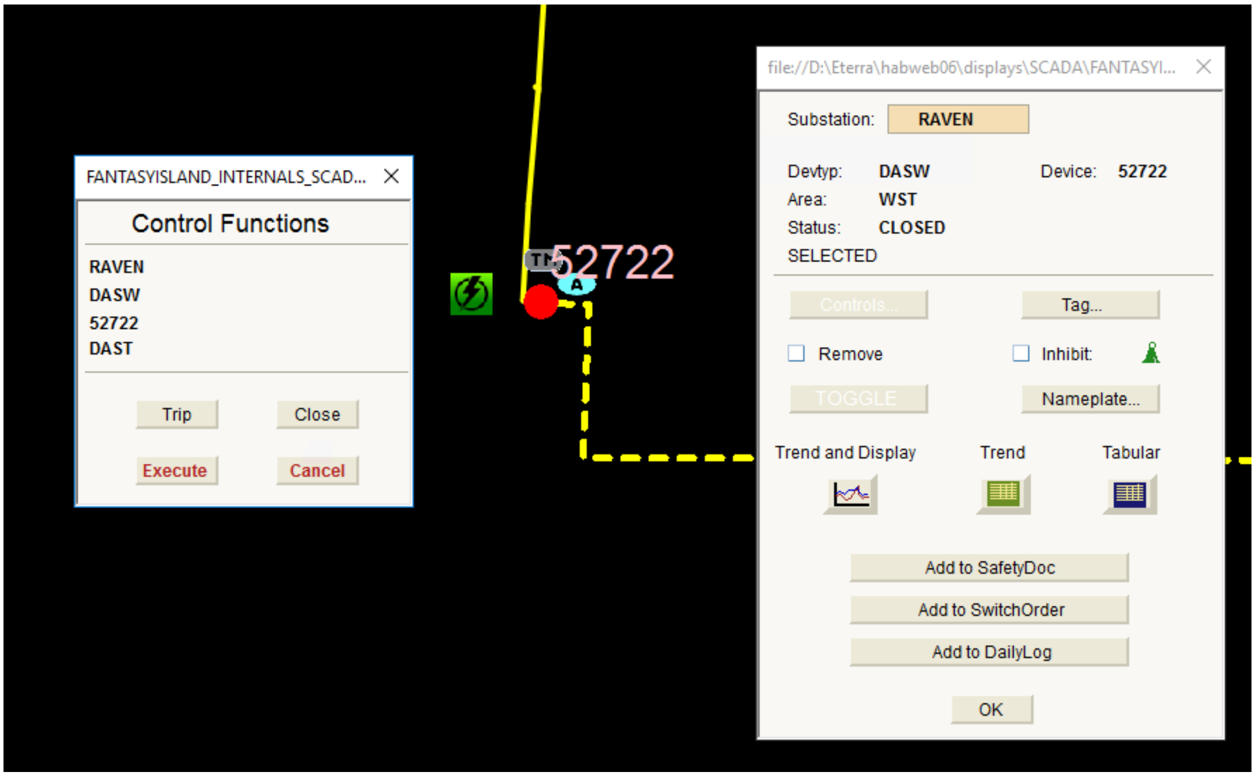


Figure 144. SCADA Switch Operation

4.1.8. Check Before Operate

The Check Before Operate (CBO) functionality issues notifications when the operator is attempting to issue a switching step or perform a switching operation that affects customers, crews, specific equipment, or network topology.

For example, notification messages are issued when:

- Energizing a grounded line
- Energizing future equipment
- De-energizing a number of customers
- Closing a boundary switch (normally open switch between two substations)
- Placing a jumper between lines with different voltages, crossed phases, and not connected

nodes

In addition, the Check Before Operate functionality can show results from a Distribution Power Flow (DPF) analysis of the post operation conditions for switching operations, tap changing transformer tap position changes, and capacitor status changes. For more information, refer to the *Distribution Network Analysis Functions (DNAF) User Guide.* To run this check, click Evaluate on the device state change dialog.

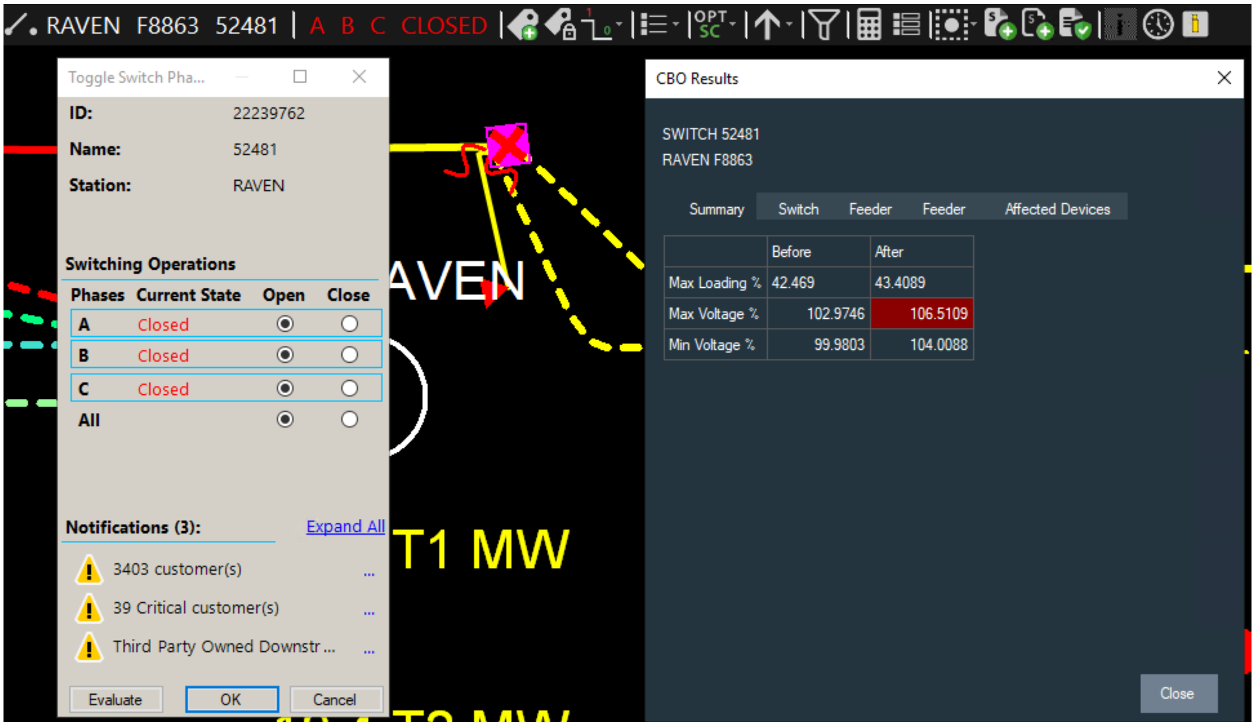


Figure 145. Check Before Operate Notifications

The Check Before Operate functionality can also show results from a Protection Validation (PRV) analysis of the post operation conditions for switching operations. The Protection Validation check notifies the operator if a relay setting violation could occur because of a switching action. For more information, refer to the *Distribution Network Analysis Functions (DNAF) User Guide.* To run this check, click Evaluate on the device state change dialog.

Operators can consider Check Before Operate notifications in their decision of whether or not to complete the switching action. There are three notifications severity types: informative, warning, and prohibitive. The prohibitive notifications cannot be ignored by the operator.

The Check Before Operate messages issuing and severity are configurable via the Viewer Configuration Editor tool. If notifications are configured to never show, the Check Before Operate function is disabled. For more details, refer to the *Viewer Configuration Editor User Guide*.

For composite switches, the functionality of issuing the notifications is similar to regular two-state switching devices. When another composite switch state is selected using the Status toolbar button , the Check Before Operate rules are evaluated and a dialog with notifications appears. For manual switches, the dialog shows the selected composite switch details, including the previous state, selected state, and the notifications, as shown in image below. For SCADA (automated) switches, selecting the Go to SCADA Control (EMP) option from the Status menu shows a dialog with notifications and the Continue and Cancel buttons. Clicking the "Continue" button shows the respective SCADA control popup.

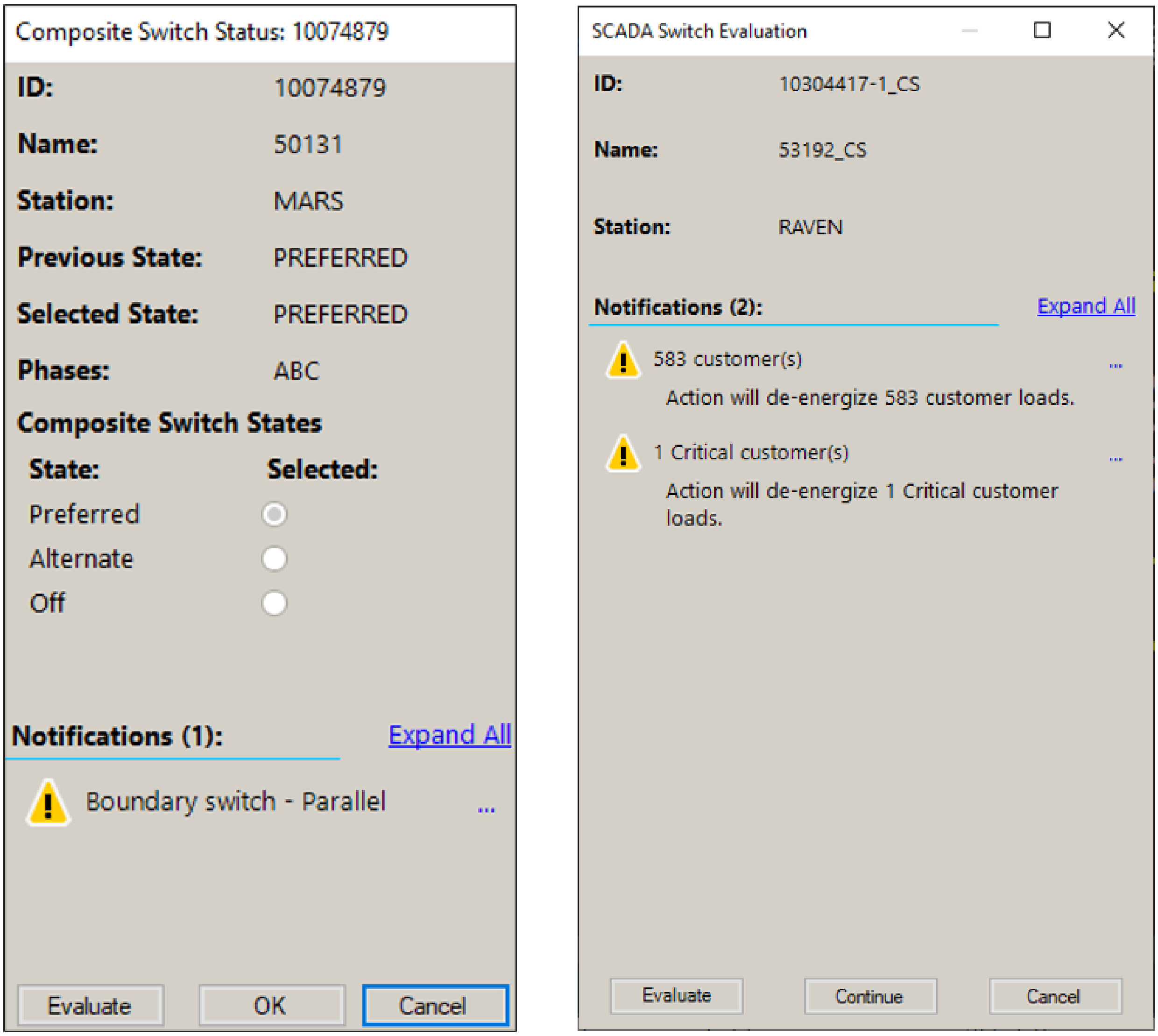


Figure 146. Check Before Operate Notifications for Manual and SCADA Composite Switches

4.2. Working with Vaults, Switch Cabinets, and UI Containers

Vaults, switch cabinets, and UI containers represent physical objects that exist in the network. These are container objects that house one or more switches. These switches can be part of the feeder backbone or they can be associated with laterals. It is also quite common that vaults can contain other vaults or cabinets.

To view the internals of a container, select the object and open the related display by clicking the relevant button on the device-specific toolbar. The internal view is opened in a separate viewport. In most cases, this is a schematic view of the internals.

To give some indication of the state of the switches within a vault or cabinet, an outer border of the symbol is colored to indicate whether the internal switches are open, closed, or in an abnormal state.

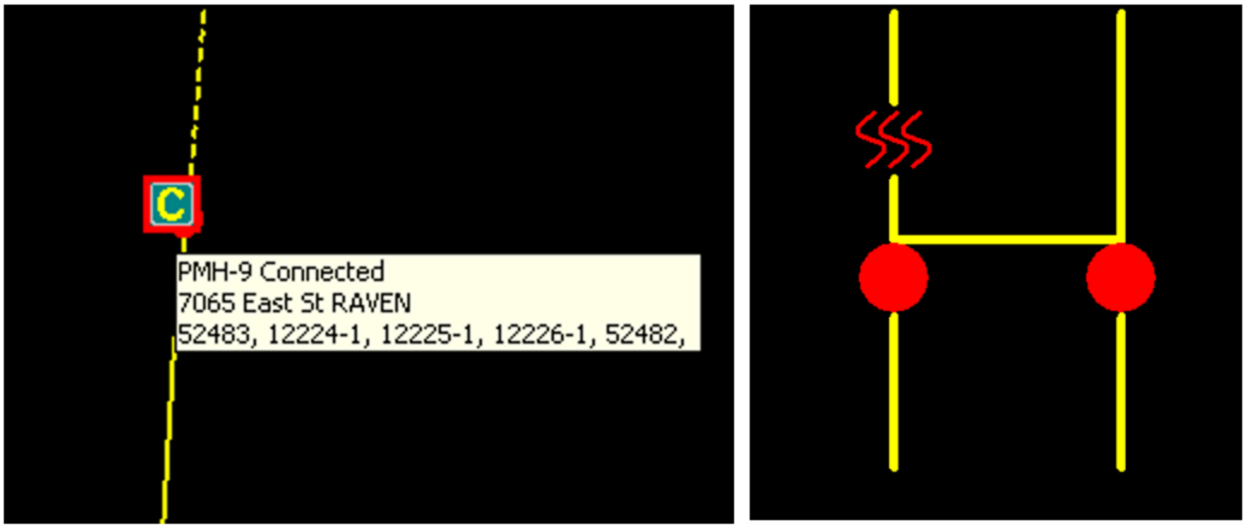


Figure 147. Red Border Indicates That the Feeder Circuit Is Closed

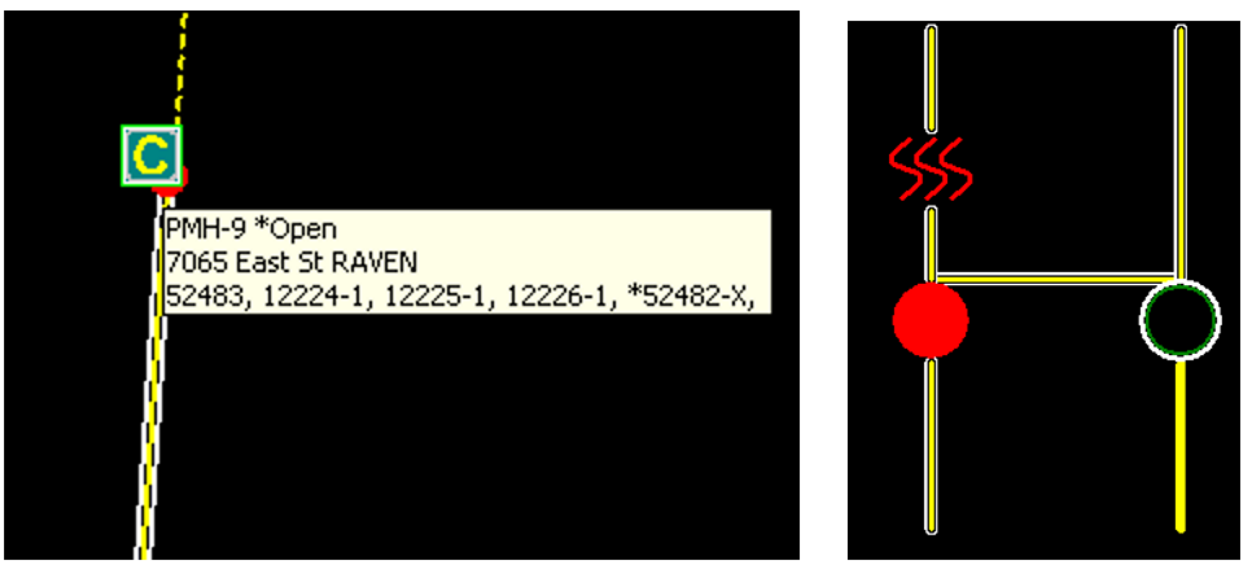


Figure 148. Green and White Borders Indicate Open Switch in an Abnormal State

For containers that contain switches that are effectively part of the feeder backbone and can change the feeder configuration, a simplified internal view is available directly from the geographic network view. As the display is zoomed in on a container, it transforms to an "exploded" view, showing the main switches in a schematic view. All of the devices in the exploded view are operational, allowing main feeder switching to be performed without having to open additional windows.

If a switch or fuse within a container is tagged, a tag symbol is also shown beside the container symbol, as shown in image below. When the display is zoomed in and the contents of the container are visible, a tag symbol is also shown next to the relevant switch. In the case where a fuse inside the container is tagged, this is indicated by the container tag symbol.

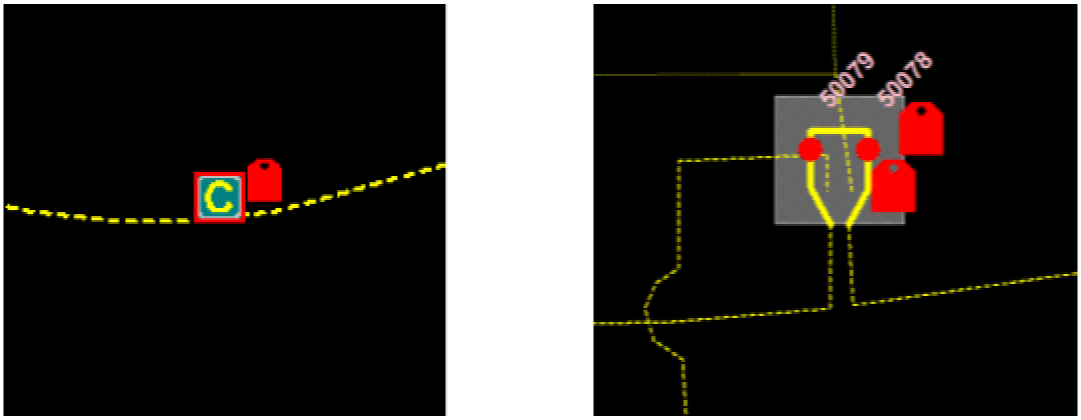


Figure 149. Cabinet Containing a Tagged Switch

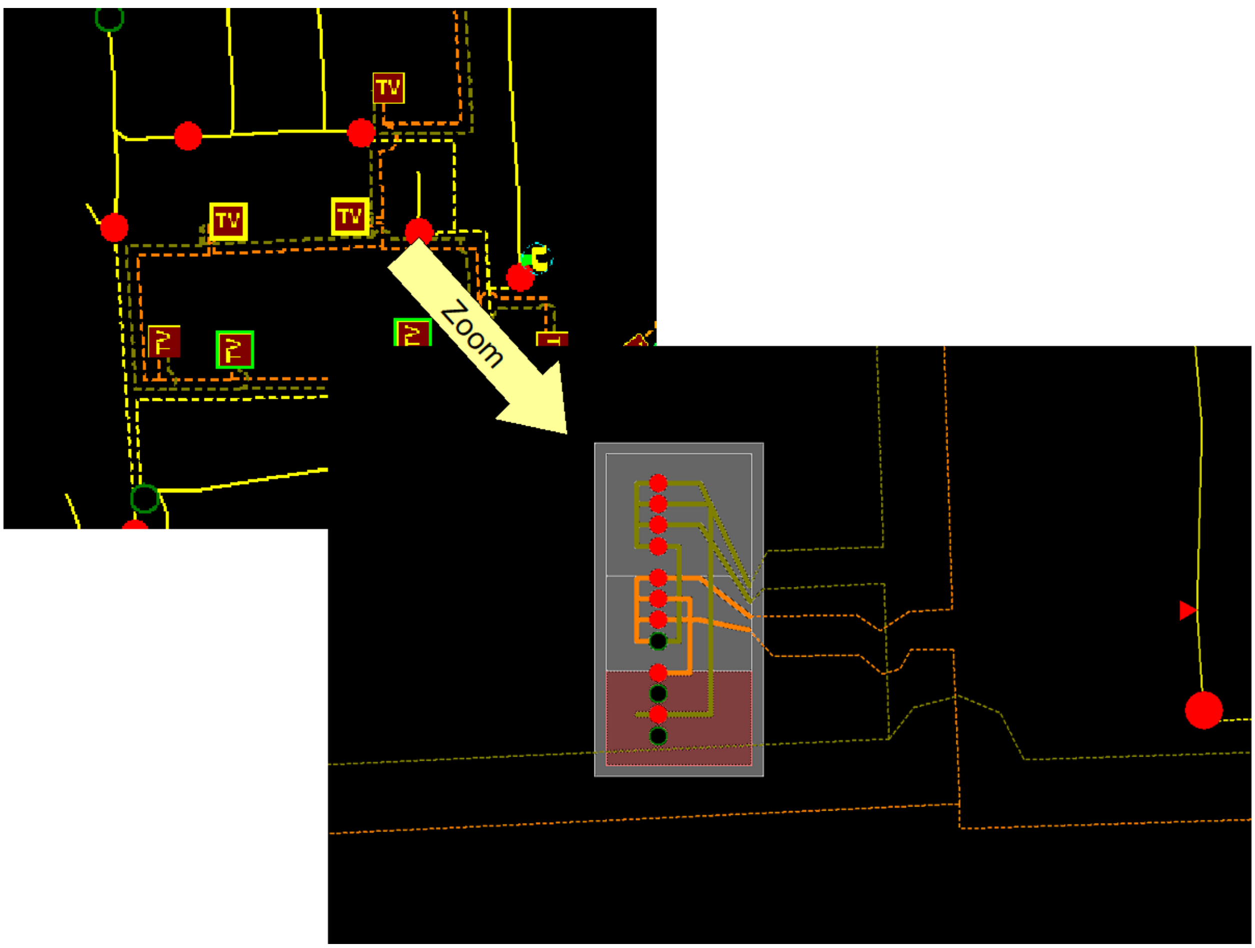


Figure 150. Zoomed-In View of a Vault

4.3. Managing Energy Resources

Distributed Energy Resources (DERs) and Distributed Generators (DGs) are small-scale energy sources that are connected to the distribution grid. DER and DG types include wind, solar, micro turbine, fuel cell, battery storage, gas, biomass, and others.

When the output from DERs or DGs exceeds the aggregate load, power flows back from distribution into transmission, which may cause damage to equipment. The Distribution Power Flow application monitors such backfeed conditions and the operator is notified of them via special alarms.

In case of a backfeed, the operator can curtail the DER or DG generation. The following DERs and DGs can be connected/disconnected:

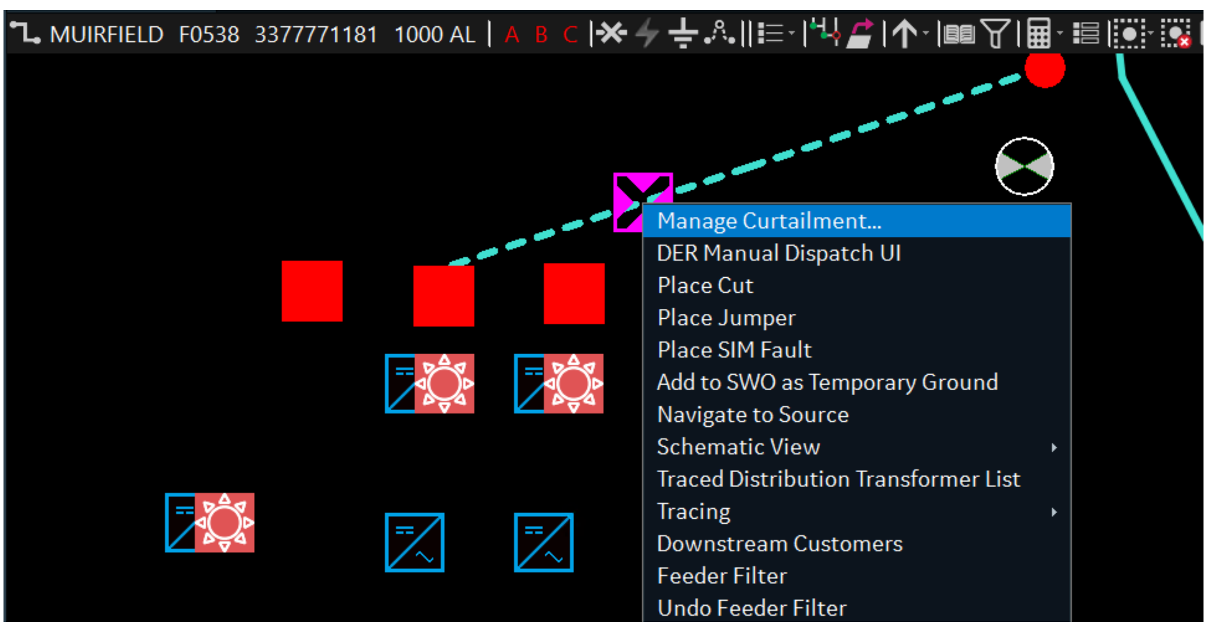
- SCADA-controlled DERs or DGs that are connected to the network through SCADA breakers or reclosers.
- AMI-controlled DERs or DGs that are connected to the network through normal switches.

> **Note**
>
> AMI curtailment is available only as part of the optional Outage Management functionality.

To control DER or DG output:

1. Right-click a feeder with connected DERs or DGs.

   The Manage Curtailment option appears.

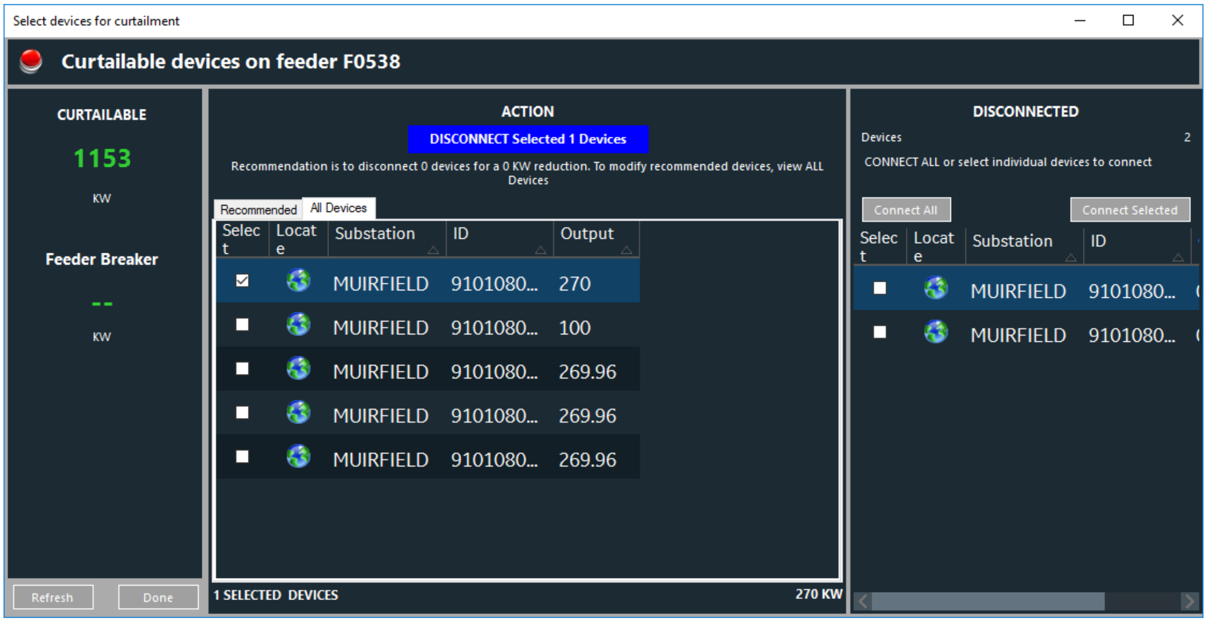
2. Click Manage Curtailment.

   The Curtailment dialog box appears.

   - The Recommended tab shows the devices that Distribution Network Analysis Functions recommend for curtailment.

  ◦ The All Devices tab shows all devices that are available for curtailment.

3. Select DERs to be disconnected.



4. Click the Disconnect Selected Devices button.

   The selected DERs are disconnected from the network.

5. To reconnect the DERs: in the Disconnected pane, select the disconnected DERs and then click Connect Selected. Alternatively, click Connect All.

   The DERs are reconnected.

4.4. Line Segments and Loop Operations

A line segment (single- or multi-phase) can be selected by clicking the desired point on the network diagram. The system acknowledges the selection by displaying a small magenta box at that point on the line. The operations toolbar for the line is also displayed.

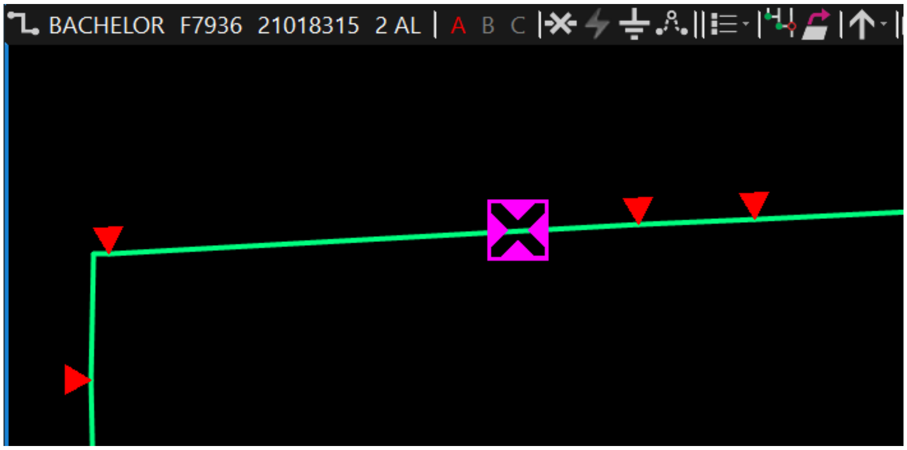


Figure 151. Selecting a Line on the Network View

Using the provided functions, you can place cuts or jumpers in the line and review the results of the real-time load flow or show the selected lateral or URD loop in a schematic view.

For detailed descriptions of line symbology, refer to Topology Processing Results.

4.5. Transformers

With the zoom level of the distribution network view set appropriately (and the display of laterals not forced OFF), the distribution transformers on the feeders and laterals are displayed. The transformers themselves can then be selected for further examination.

To select a transformer, click the lower point of the triangular symbol. The device toolbar shows that a transformer is selected.

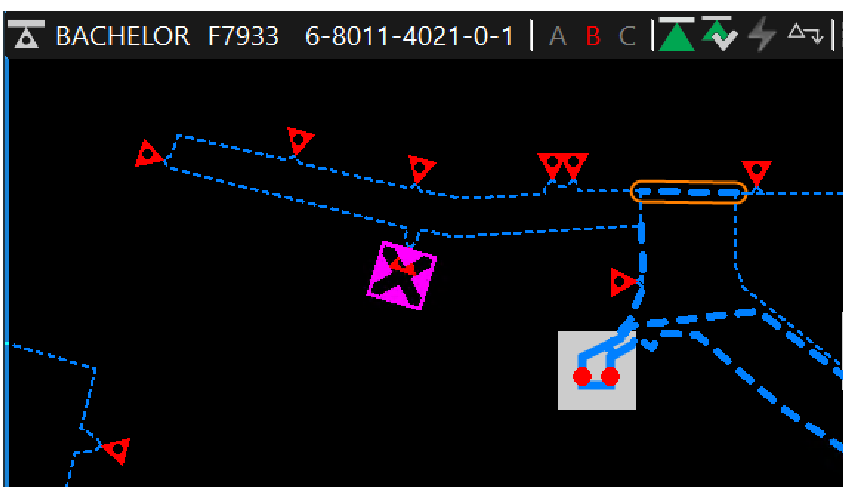


Figure 152. Selecting a Transformer on the Network View

Transformers can be disconnected from the network by removing the high-voltage fuse. This is replicated on the network view by clicking the Disconnect Transformer button on the device toolbar. The transformer symbol is colored white to indicate that it is disconnected.

4.6. Capacitors

Capacitor bank switching is performed manually in the field, and the dispatcher needs to update the status of the network operations model. Capacitor banks can be operated directly from the distribution network view.

The capacitor bank symbol is selected by clicking the short stub that is perpendicular to the main capacitor symbol. The operations toolbar shows that a capacitor is selected.

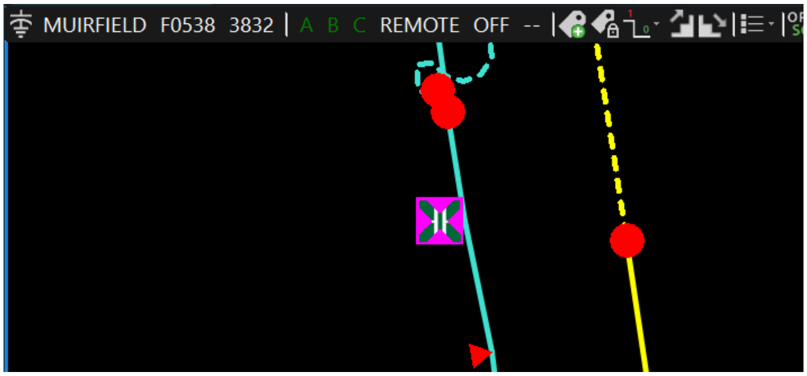


Figure 153. Selecting a Capacitor on the Network View

The color of the capacitor bank symbol is used to identify the current status. The stem of the capacitor bank symbol is shown in red to indicate ON status and green to indicate OFF status. To change the status of a capacitor bank, select the required symbol, and click the Toggle Capacitor Status button on the device toolbar.

There are a number of other distribution system optimization functions available from the device toolbar.

4.7. Tracing the Network

The network trace function is implemented locally within the ADMS client. It has no effect on the operations model or other clients viewing the network.

This allows dispatchers to identify sources, end points, or connectivity on the current view of the network. Tracing can also be performed on a per-phase basis so that the connectivity of multi-phase laterals and URD loops can be readily identified. Tracing is invoked from the device-specific toolbar.

The starting point for the trace is the selected device (such as a switch or a transformer) or a point on the line section. Trace function buttons are presented on the device-specific toolbar depending on the device and its current state. Traces can also be performed from the right-click menu.

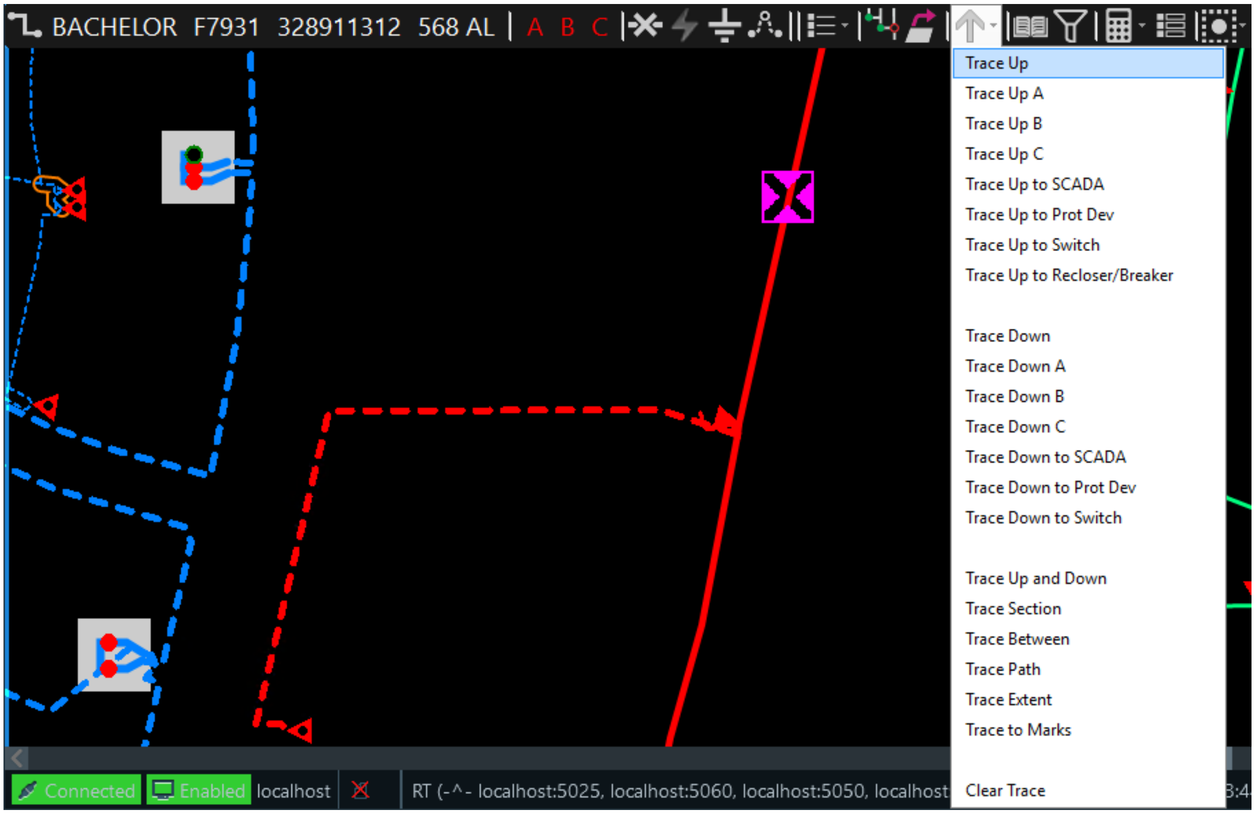


Figure 154. Trace Pull-Down Menu

Table 7. Network Trace Types and Functions

Trace Type	Function
Trace Upstream	Traces upstream from the selected point to the supply source.
Trace Downstream	Traces downstream from the selected point out to the end points, including laterals.
Trace Section	Traces the switchable section. Highlights all the feeder backbones and laterals that are within the bounding switches.
Trace Between	Traces the shortest path between the selected points and highlights that path and all laterals and devices connected to it.
Trace Path	Traces the shortest path between the selected devices or points on the line sections and highlights it.
Trace Up and Down	Performs both the Trace Upstream and the Trace Downstream functions.
Trace Extent	Traces all connected end points of the network that are not energy sources, including branches and laterals.
Trace Up to SCADA Device	Trace upstream from the starting point to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an info message appears and the tracing is canceled.
Trace Up to Protection Device	Trace upstream from the starting point to the next upstream protective device such as a fuse, recloser, breaker, network protector, or sectionalizer. If there is no upstream protective device, an info message appears and the tracing is canceled.

Trace Type	Function
Trace Up to Switching Device	Traces from the selected point to the first upstream switching device, including temporary switches, such as cuts and jumpers. If no device is found, an info message appears and the tracing is canceled.
Trace Up to Recloser/Breaker	Traces from the selected point to the first upstream recloser or breaker. If there is no upstream recloser or breaker, an info message appears and the tracing is canceled.
Trace Down to SCADA Device	Traces downstream to the first SCADA controllable (not only telemetered) device. If no device is found, an info message appears and the tracing is canceled.
Trace Down to Protection Device	Traces downstream to the first protective device such as a fuse, recloser, breaker, network protector, or sectionalizer. If no device is found, an info message appears and the tracing is canceled.
Trace Down to Switching Device	Traces downstream to the first switching device including temporary switches, such as cuts and jumpers. If no device is found, an info message appears and the tracing is canceled.
Trace to Marks	Traces from the starting point to the marked device. If there are no marked devices between the starting point and network end points, traces to all connected end points.
Trace to All Potential Isolating Devices	Marks the switches that need to be tripped to isolate the selected point on the line, and also draws the trace to the switches.
Trace to All Normally Open Switches	Marks all normally open switches that are connected to the selected point on the line, and also draws the trace to the switches.
Clear Trace	Clears all traces in all viewports.
Clear All Marks and Traces	Clears all marks and traces in all viewports.

Selecting one of the trace options invokes the trace. The traced part of the network is highlighted in Trace 0 color (cyan by default). The next trace is shown with another color. A configurable number of traces can simultaneously exist on a geographic display. When the maximum number of line halos is reached, the oldest trace is cleared and the new trace is shown with the color of the cleared trace. This logic applies until all traces are cleared using the Clear Trace or Clear All Marks and Traces option.

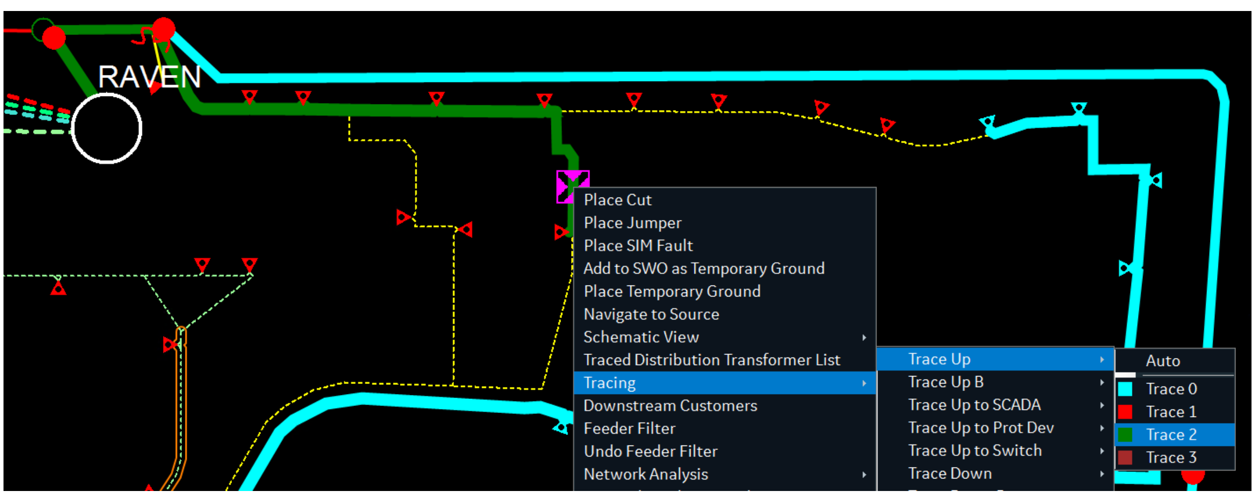


Figure 155. Multiple Traces Displayed on a Network View

In addition, for downstream and point-to-point traces, the number of transformers and customers included in the trace halo is shown on the message line at the bottom of the geographic view. This information is only shown for 10 seconds after the trace has been completed.

To clear the trace, select the Clear Trace or Clear All Marks and Traces option from the geographic toolbar or right-click menu.

A composite switch is not a topology object. So, after the selection of a composite switch, the first identified closed switch within the composite switch is considered as the starting point for tracing.

In the below RAG composite switch example, when "RAG-4" is selected for tracing, the regular switches "10304417-4" and "10304417-4_GSW" are determined to be the associated two state switching devices. The switch "10304417-4" is found closed and considered as a starting point for tracing.

Similarly, when"RAG-3" is selected for tracing, "10304417-3" and "10304417-3_GSW" are identified. The switch "10304417-3" is found closed and used for tracing.

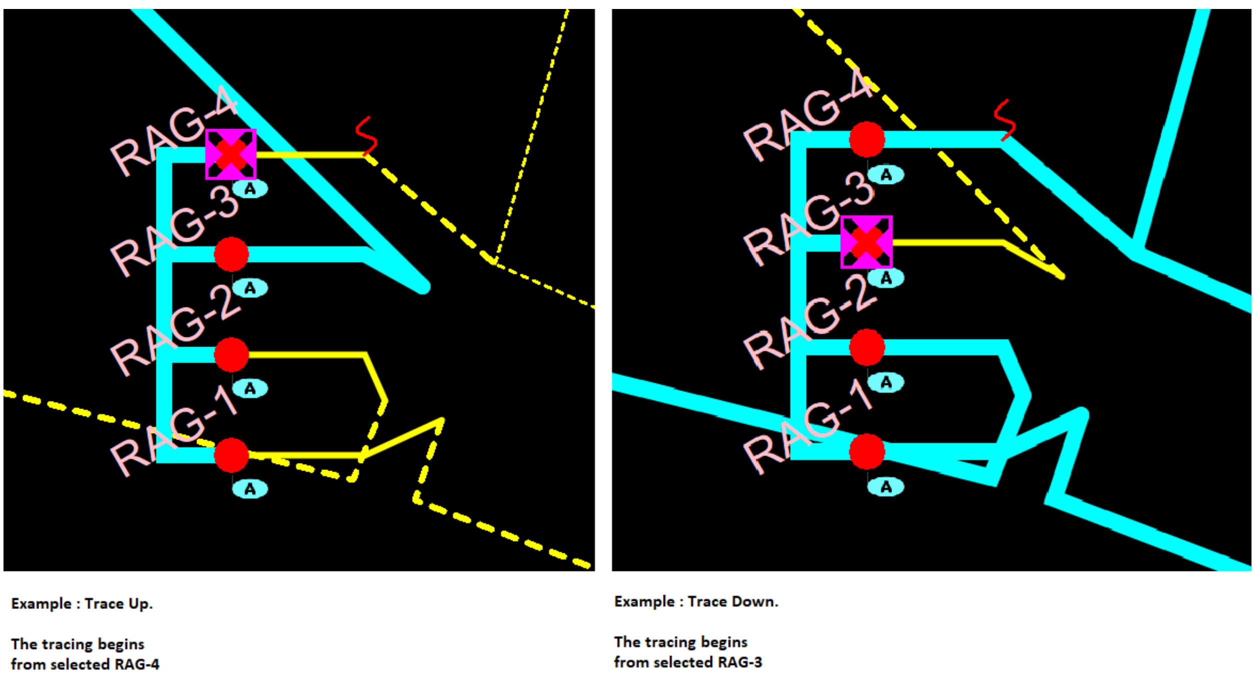


Figure 156. Tracing Examples of a Composite Switch

The multiple "Trace Up" and "Trace Down" tracing options will not be available if the regular switches associated to the selected composite switch are found open. In the below example, when"RAG-3" is selected for tracing, both the associated regular switches "10304417-3" and "10304417-3_GSW" are found open and the upstream and downstream tracing options are not available.

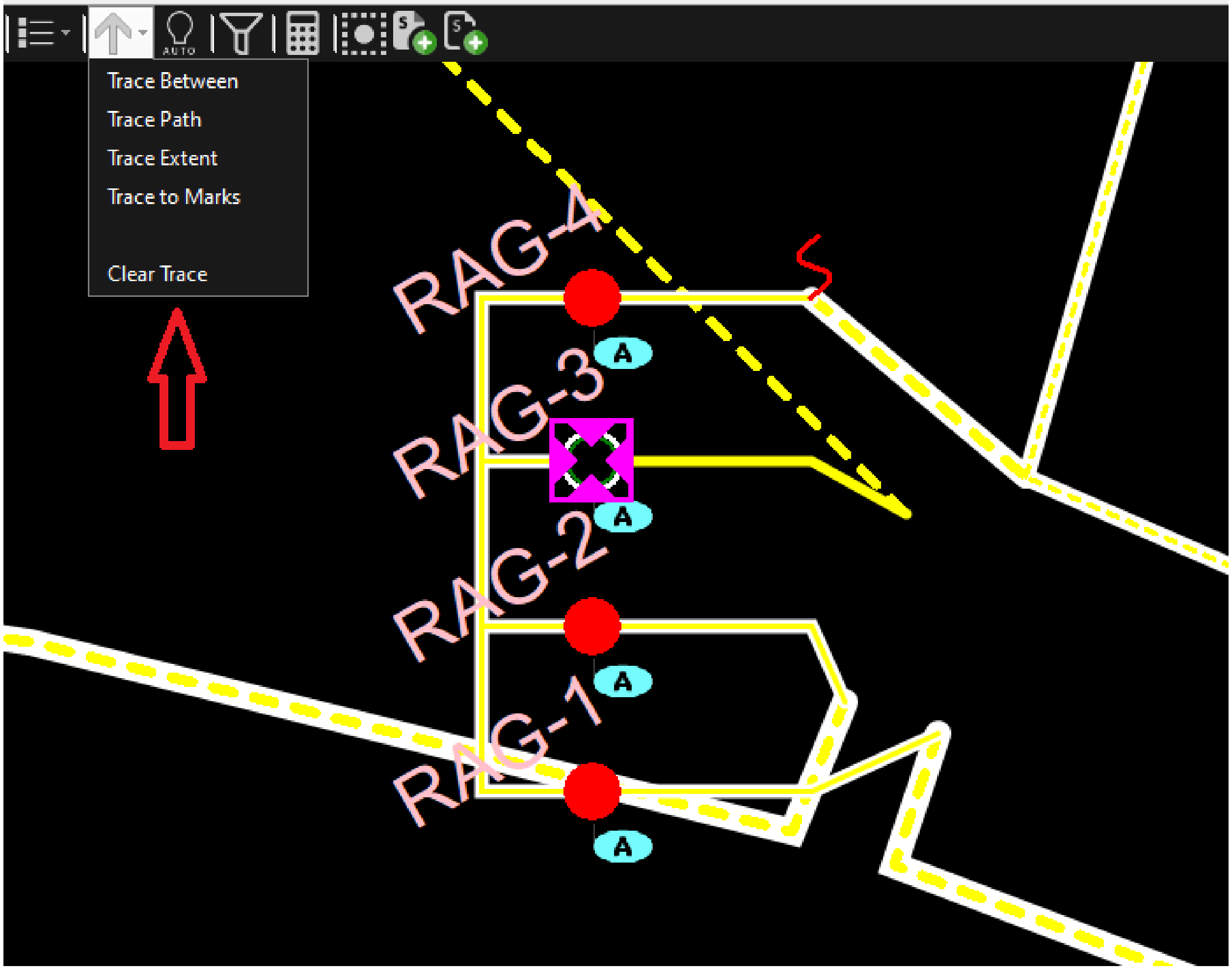


Figure 157. Tracing Options of an Open Composite Switch

4.7.1. Trace Upstream

An upstream trace runs directly from the starting point to the source of energy. It does not highlight other branches of the feeder or laterals that might be connected to the traced path.

It is not possible to perform an upstream trace on a de-energized portion of the network, because there is no energy source to define the upstream direction.

Note

The trace upstream function is disabled for the components on the actual loop path in the looped part of the network. When tracing upstream from branches that are not part of the actual loop path, the trace goes upstream to one of the energization points (not necessarily to the nominal energization source). You can disable the trace upstream function for branches that are not part of the actual loop path by using the ISDYNFEEDERLOOPED property instead of the COMPISLOOPED property in the visibility condition fields on the Viewer Configuration Editor.

To perform an upstream trace, click a starting point on the display, and then click the Trace Up button on the toolbar. The circuit is traced with a cyan halo from the starting point toward the substation.

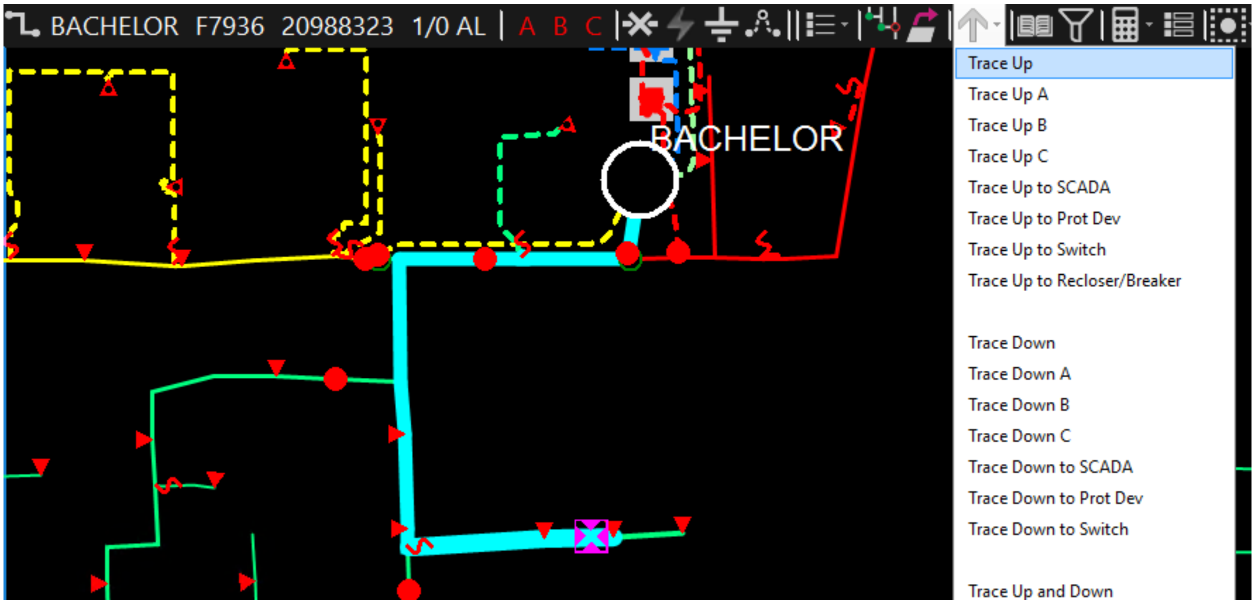


Figure 158. Trace Upstream View

The message line at the bottom of the screen shows the message "Upstream Trace Complete". The number of customers and transformers that are connected to the traced portion of the network is not shown for upstream traces.

There are three specific upstream tracing commands to trace to specific devices:

- **Trace Up to SCADA:** Traces upstream to the first SCADA controllable (not only telemetered) device. If no device is found, an info message appears and the tracing is canceled.
- **Trace Up to Protective Device:** Traces upstream to the first protective device such as a fuse, recloser, breaker, network protector, or sectionalizer. If no protective device is found, an info message appears and the tracing is canceled.
- **Trace Up to Switching Device:** Traces upstream to the first switching device including temporary switches, such as cuts and jumpers. If no device is found, an info message appears and the tracing is canceled.
- **Trace Up to Recloser/Breaker Device:** Traces upstream to the first recloser or breaker. If no device is found, an info message appears and the tracing is canceled.

4.7.2. Trace Downstream

A downstream trace extends from the starting point to all connected end points of the network that are not energy sources, including branches and laterals.

A downstream trace can be performed on a de-energized section of the network.

Note

The trace downstream function is disabled for components on the actual loop path in the looped part of the network.

To perform a downstream trace, click a starting point on the display, and then click the Trace Down button. The circuit is traced by showing a cyan halo from the starting point to the ends of all connected circuits.

Note

When a line is traced downstream, the trace also includes the line segment that the selected point belongs to.

On de-energized lines the downstream trace function will trace up and down stream of the entire outage section.

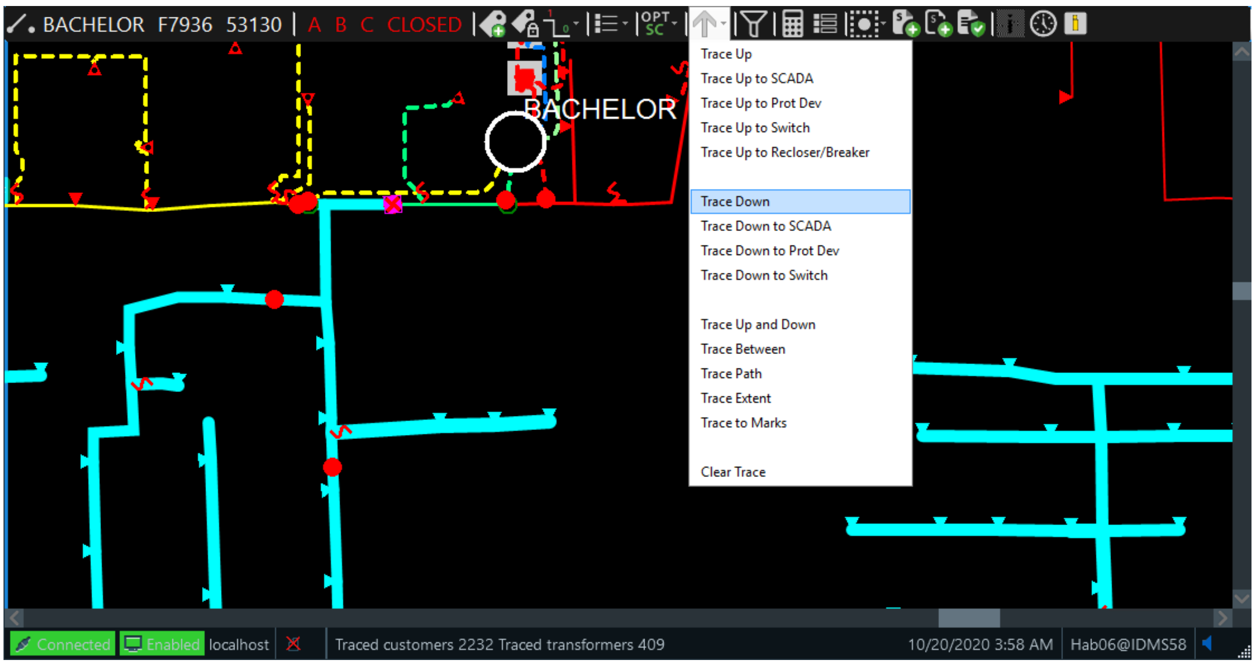


Figure 159. Trace Downstream View

The message line at the bottom of the screen temporarily shows the number of customers and transformers that are connected to the traced portion of the network.

Tip

A downstream trace of a de-energized section of the network is an easy way to get the total number of affected customers and transformers in the entire de-energized section.

There are three specific downstream tracing commands for tracing to specific devices:

- **Trace Down to SCADA:** Traces downstream to the first SCADA controllable (not only telemetered) device. If no device is found, an info message appears and the tracing is canceled.

- **Trace Down to Protective Device:** Traces downstream to the first protective device such as a fuse, recloser, breaker, network protector, or sectionalizer. If no device is found, an info message appears and the tracing is canceled.
- **Trace Down to Switching Device:** Traces downstream to the first switching device including temporary switches, such as cuts and jumpers. If no device is found, an info message appears and the tracing is canceled.

4.7.3. Trace Between

The Trace Between option traces the shortest path between the selected points and highlights that path and all elements connected to it.

To perform a trace between two points in the network:

1. Select the first point on the network.

   The point is highlighted by the magenta cursor.

2. Click the Trace Between button on the toolbar.

   This causes the selection point on the network to turn yellow, which indicates that the system is waiting for a second selection point.

3. Select the second point on the network.
4. Click the Trace Between button a second time.

If a connection exists between the two points, it is traced out with a thick cyan line. All other laterals or lines connected to this direct path are also highlighted. The message line at the bottom of the screen temporarily shows the number of customers and transformers that are highlighted by the trace halo.

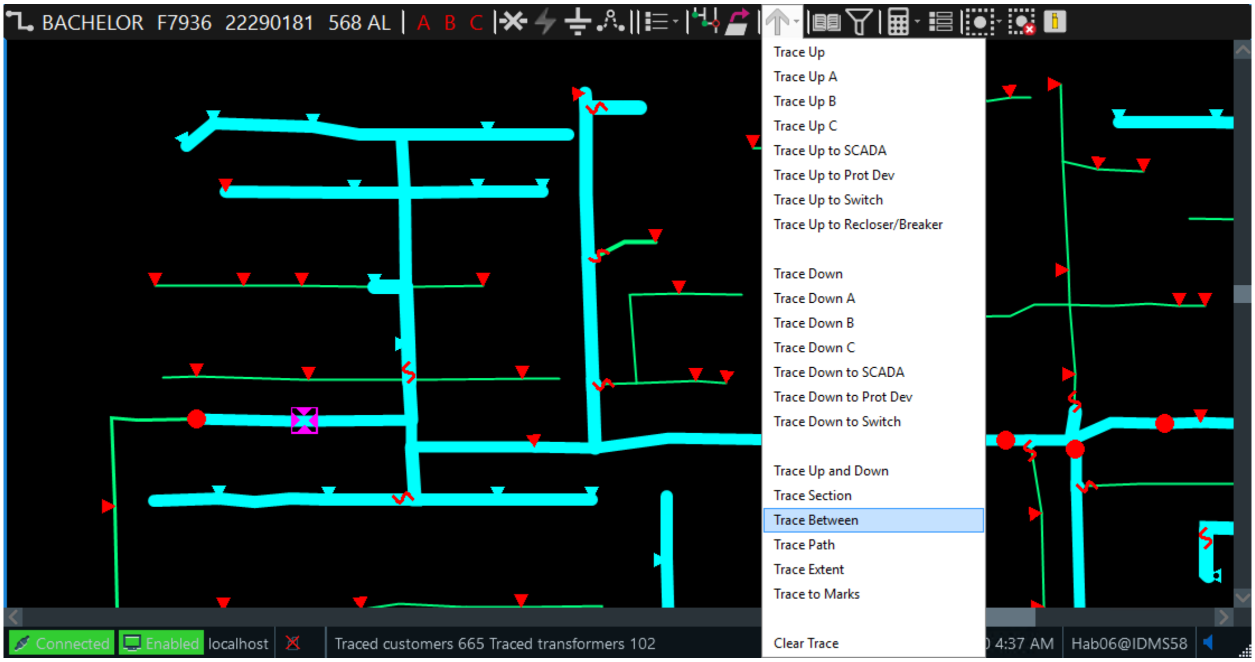


Figure 160. Trace Between Two Points

Tip

This trace is useful for determining the total number of affected customers when a segment of the network needs to have power removed.

Note

The Trace Between function also traces the line segments the selected points belong to. The Trace Between function is limited to one feeder. To trace the path between points on different feeders, use the Trace Path function.

4.7.4. Trace Switchable Section

A useful feature is to be able to quickly identify all the bounding switches of a section of feeder backbone. This is known as a "switchable section".

To display a switchable section, select a point on the feeder of interest and select the Trace Section option from the Tracing menu on the device-specific toolbar.

A cyan trace highlights all feeder backbones and laterals that are within the bounding switches. Bounding switches are devices such as breakers, reclosers, sectionalizers, or regular switches that can be opened or closed. Fuses on the feeder backbone are also considered as bounding switches. (Note that lateral and feeder main boundary fuses are not considered as bounding switches.) This function ignores cuts and jumpers in the identification of new boundaries of the section.

The number of transformers and customers within that section (including those on the laterals as well as those directly on the feeder) is temporarily presented in the message line shown at the bottom of the screen.

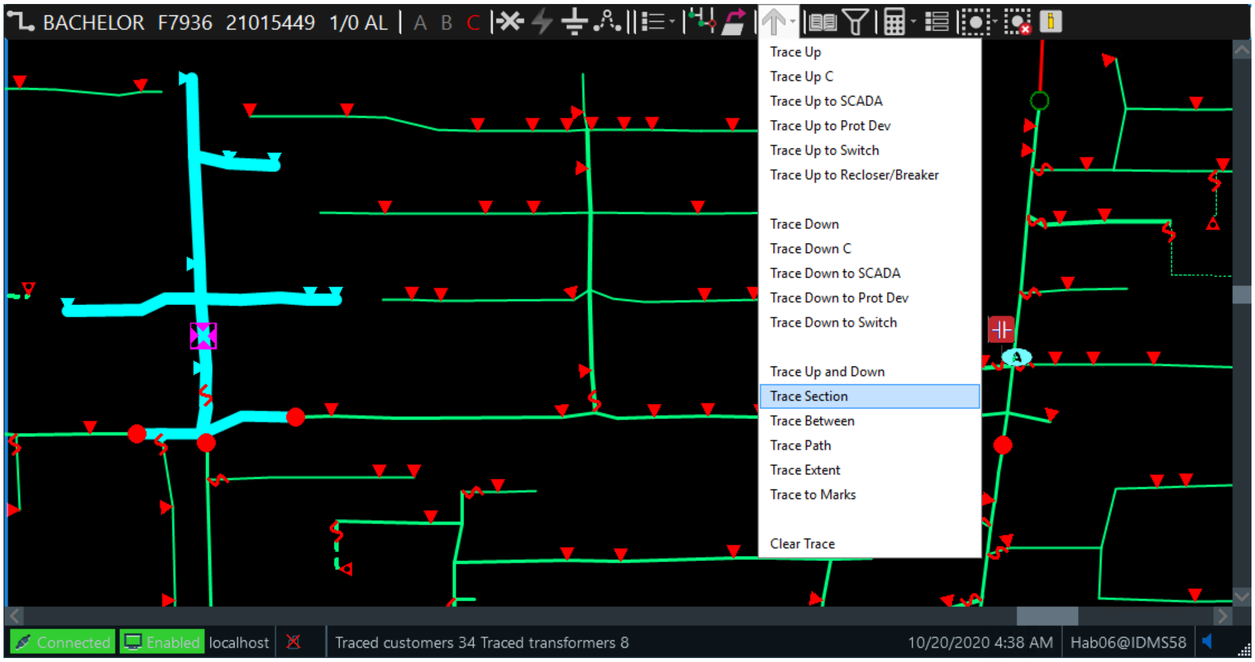


Figure 161. Tracing a Switchable Section

4.7.5. Trace Extent

An extent trace extends from the starting point to all connected end points of the network that are not energy sources, including branches and laterals. For de-energized sections, this traces outward to the first open device found. It can be used to determine the total number of customers without power in a network outage.

To perform an extent trace, click a starting point on the display, and then click the Trace Extent button. The circuit is traced by showing a cyan halo from the starting point to the ends of all connected circuits.

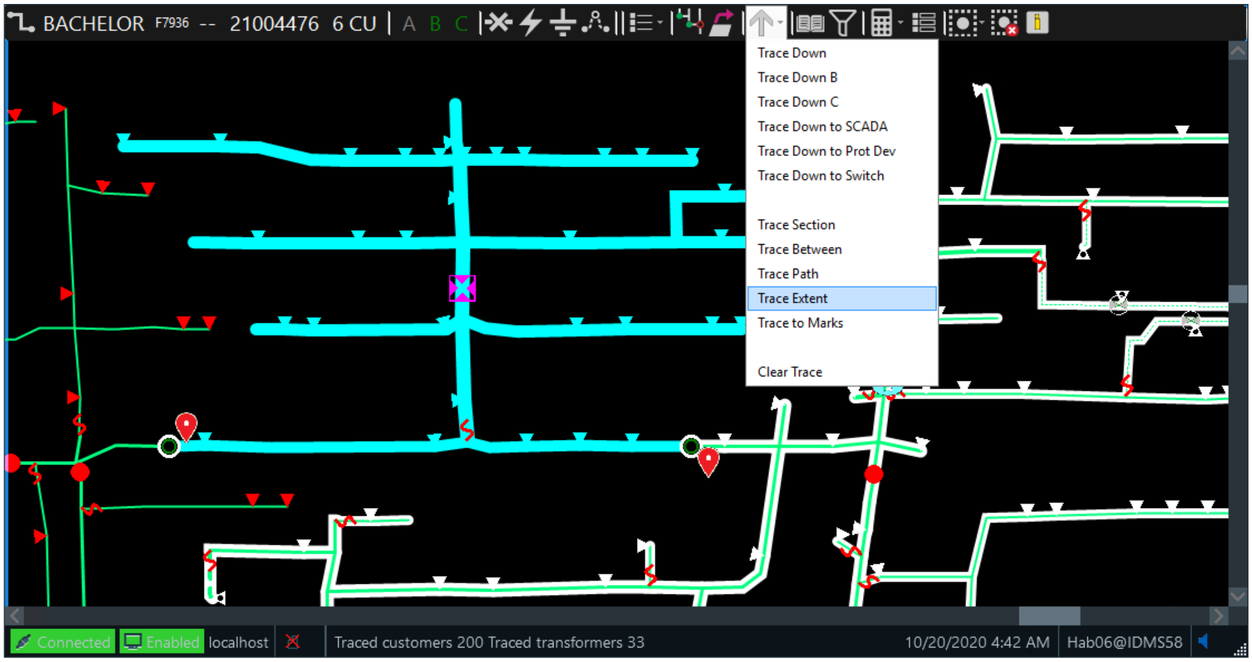


Figure 162. Tracing a Network Extent

Note that in energized sections, a trace extent marks from the feeder head out to the full extent of the feeder (including laterals).

4.7.6. Phase Tracing

For network laterals, and particularly in multi-phase URD loops, it is important to be able to clearly identify the path of individual phases. This can be done using the Trace Phase buttons on the line device toolbar. The phase trace function performs an upstream or downstream trace for one of the available phases in the selected line segment. Only the buttons for the relevant phases are displayed on the toolbar.

As shown in image below, a point on a three-phase URD lateral has been selected and a Phase C trace performed. The separate paths of the phases within the URD loop are visible.

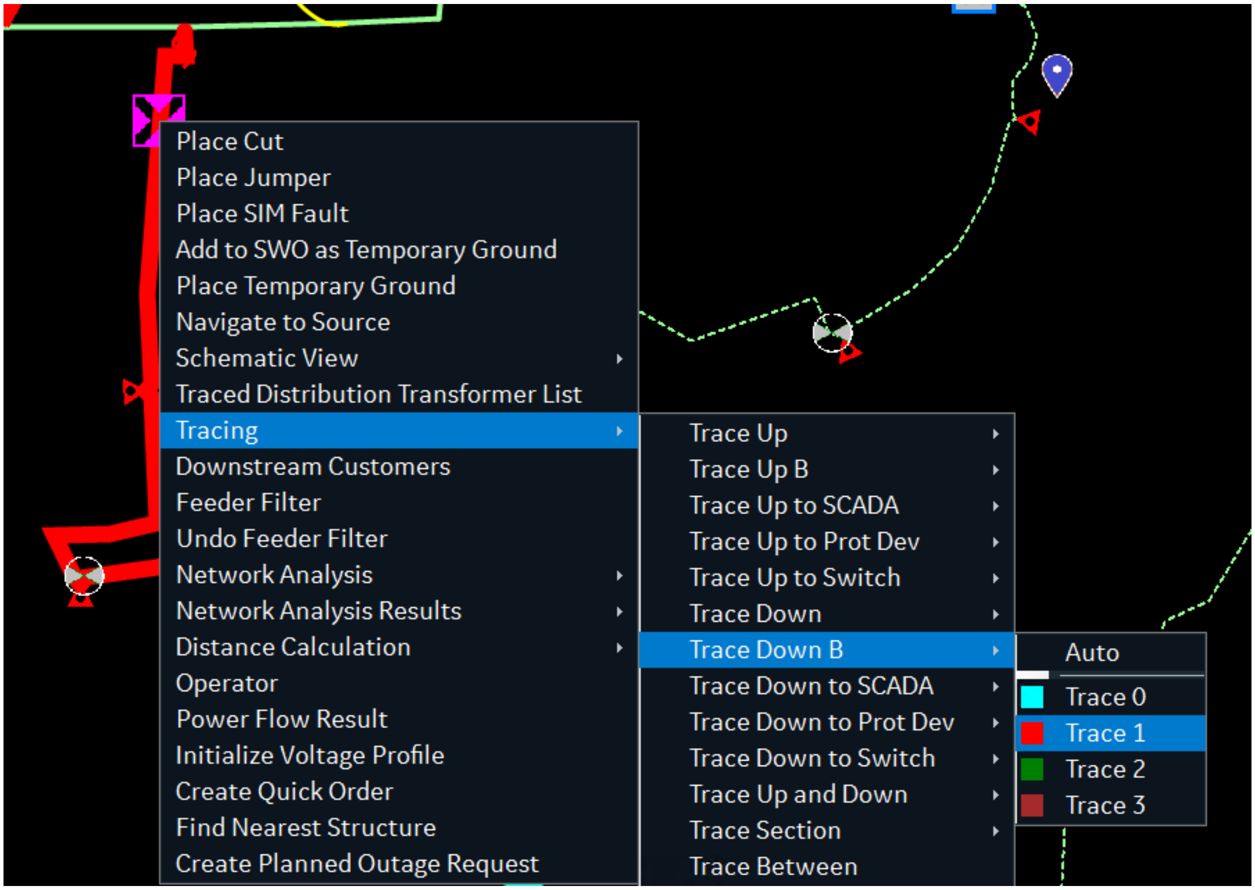


Figure 163. Phase Tracing

4.7.7. Trace Up and Downstream

An upstream and downstream trace functions as if an upstream trace were run immediately after a downstream trace.

4.7.8. Trace Path

The Trace Path option traces the shortest path between the selected points.

To trace path in the network:

1. Select the first point on the network.

   The point is highlighted by the magenta cursor.

2. Click the Trace Path button on the toolbar.

   This causes the selection point on the network to turn yellow, which indicates that the system is waiting for a second selection point.

3. Select the second point on the network.

4. Click the Trace Path button a second time.

If a connection exists between the two points, the most direct path is traced out with a thick cyan line.

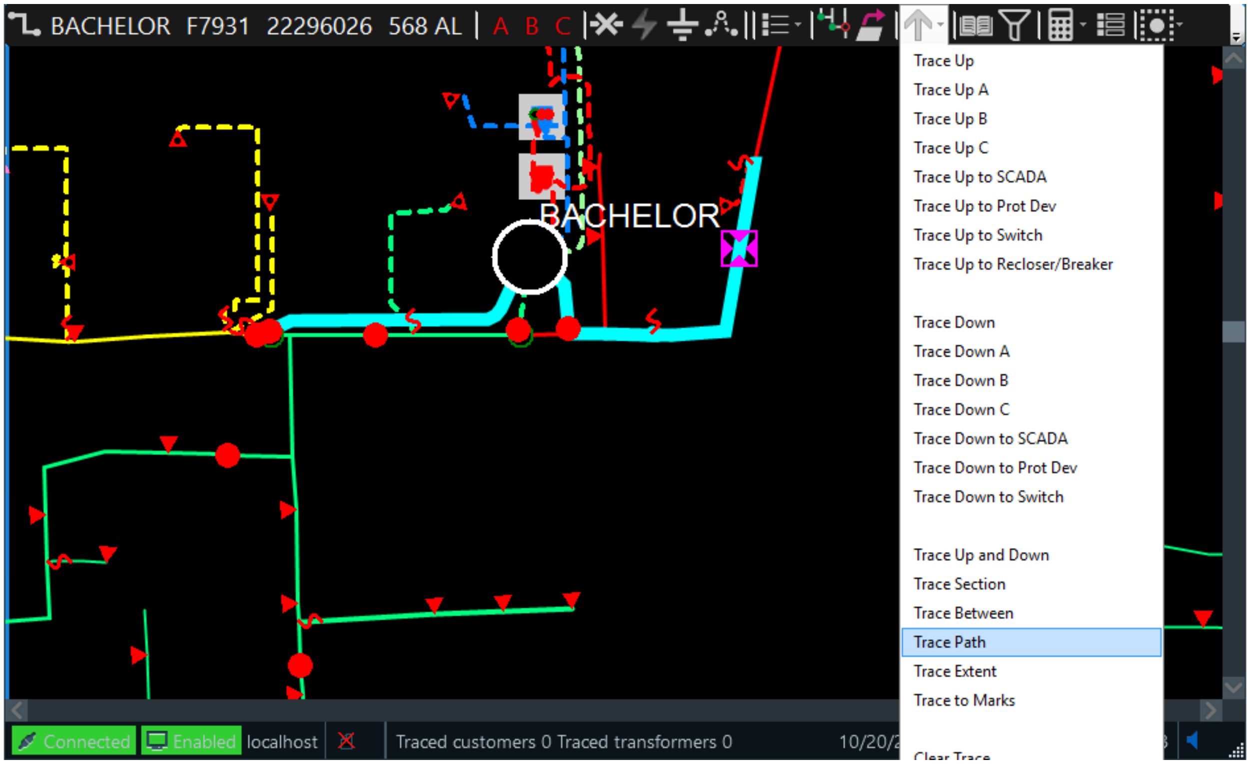


Figure 164. Tracing a Path between Two Points

> **Note**
>
> Only customers and transformers that are on the direct path are highlighted and counted on the display.
>
> The Trace Path function also traces the line segments the selected points belong to. In addition, the Trace Path function is phase-dependent: it solves for points that have common phases.

4.7.9. Trace to Marks

A Trace to Marks is similar to an extent trace. It extends from the starting point to the first marked device. If no marked device is discovered, it traces to all connected end points of the network that are not energy sources, including branches and laterals.

To perform this trace, select a component to be removed from service, then select the Mark Isolating devices button. After the isolating devices are marked, select the Trace Between Marks button. The circuit is traced by showing a cyan halo from the starting point to the ends of all connected circuits or to the marked devices.

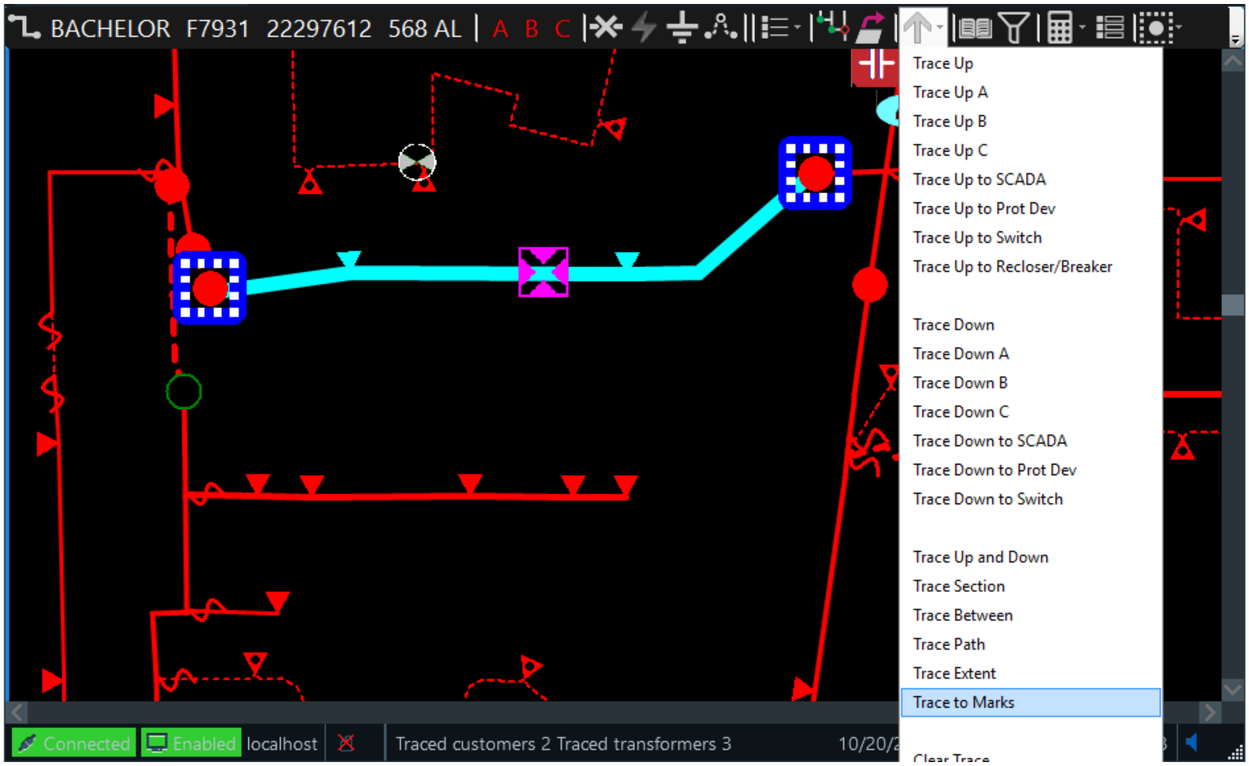


Figure 165. Tracing to Marked Devices

Note that all affected customers within the traced area are reported on the display.

Tip

This trace can be used to determine the total number of customers that will be affected by de-energizing a network segment.

4.7.10. Trace to All Potential Isolating Devices

You can mark the switches that need to be tripped to isolate the selected point on the line, and also draw the trace to the switches.

To mark and trace all potential isolating devices:

1. Select a point on the line.
2. From the line toolbar, click the Mark pull-down menu and select Mark and Trace All Potential Isolating Devices.

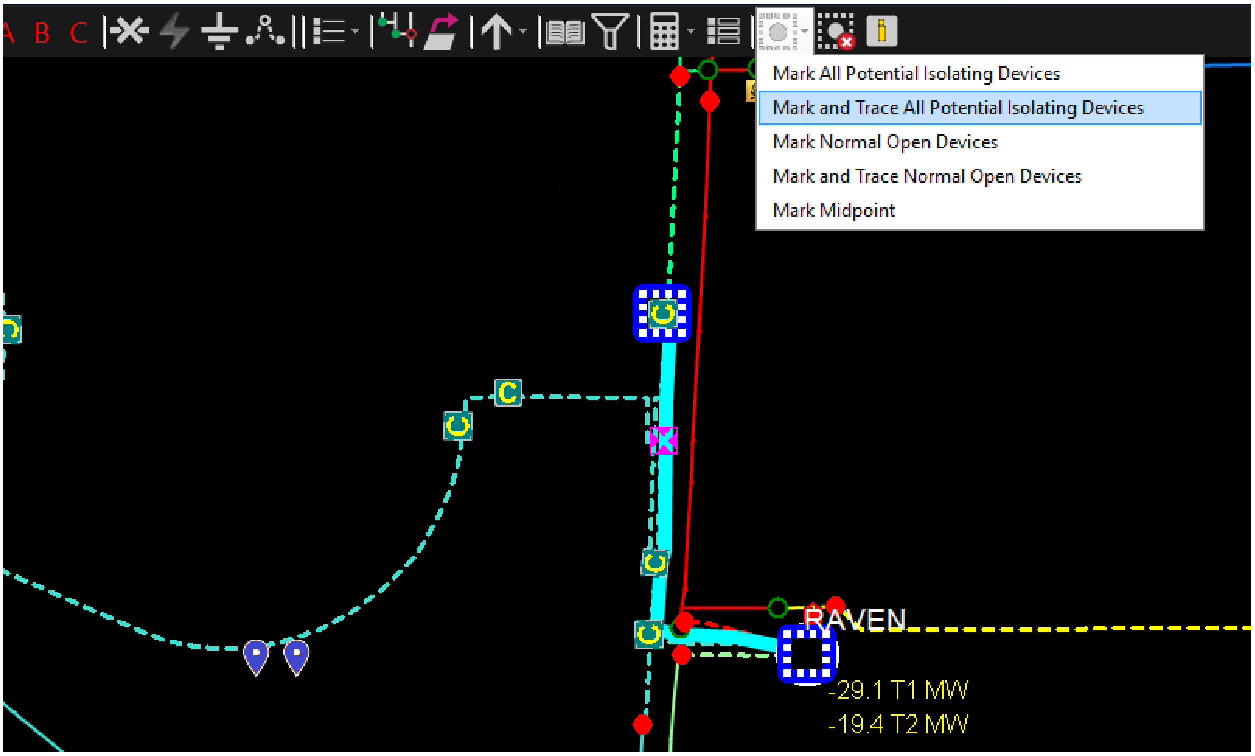


Figure 166. Mark and Trace All Potential Isolating Devices Option

All of the switches (open and closed) that need to be open to fully isolate the selected switch or line segment are highlighted with the Mark symbol and the trace is drawn between the switches.

4.7.11. Trace to All Normally Open Switches

You can mark all normally open switches that are connected to the selected point on the line, and also draw the trace to the switches.

To mark and trace all normally open switches:

1. Select a point on the line.
2. From the line toolbar, click the Mark pull-down menu and select Mark and Trace Normal Open Devices.

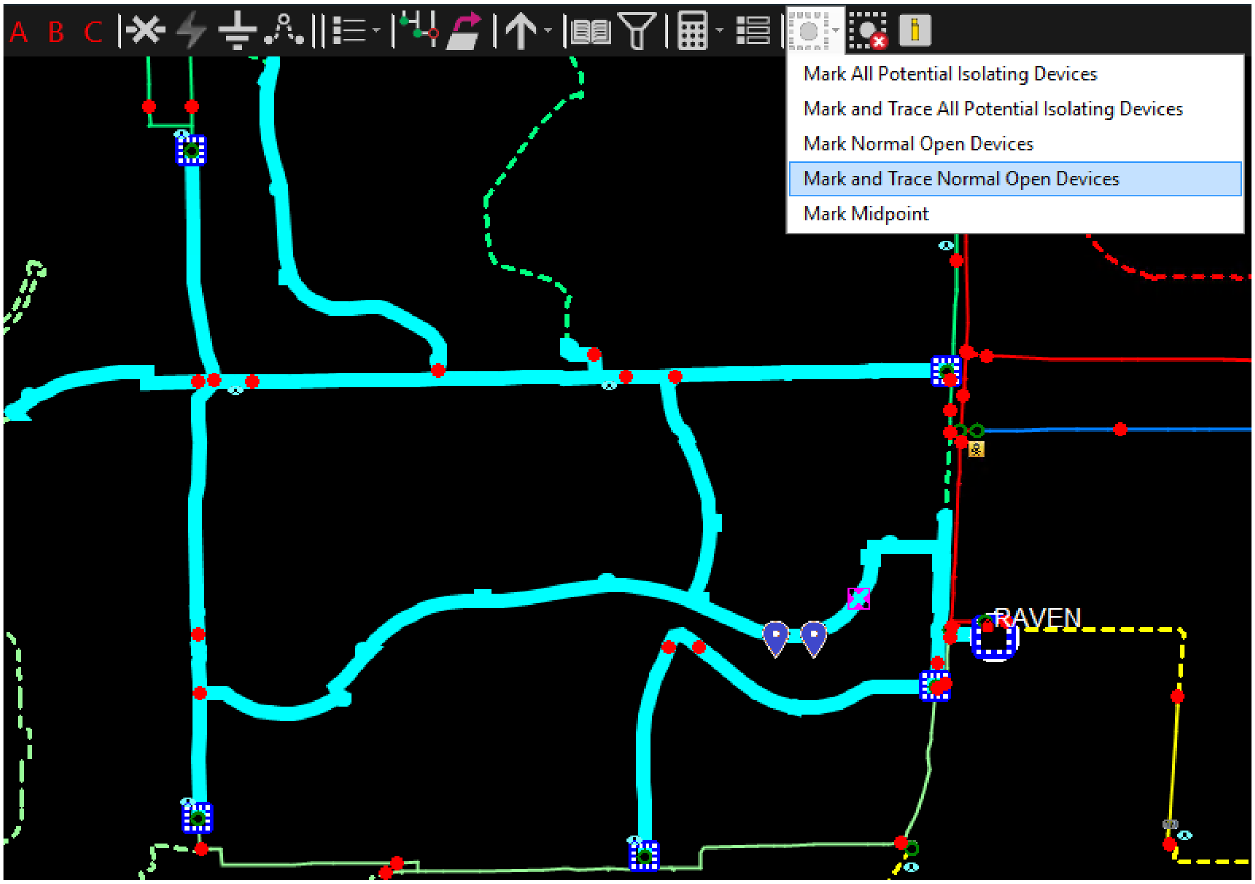


Figure 167. Mark and Trace All Normally Open Devices Option

All of the normally open switches that are connected to the selected point on the line are highlighted with the Mark symbol and the trace is drawn between the switches.

4.8. Temporary Network Modifications

Temporary modifications are made to the distribution network to change the network configuration to isolate faults, restore services, or perform maintenance. These modifications are performed by field crews, and they must be reflected in the network operations model in order to maintain the correct network topology. The system provides tools to allow the dispatcher to apply cuts, jumpers, temporary switches, and grounds at any point in the network.

The Abnormal Summary display lists the jumpers, cuts, and grounds that currently exist. The placement and removal of temporary modifications are also recorded on the system activity log.

4.8.1. Cuts and Temporary Switches

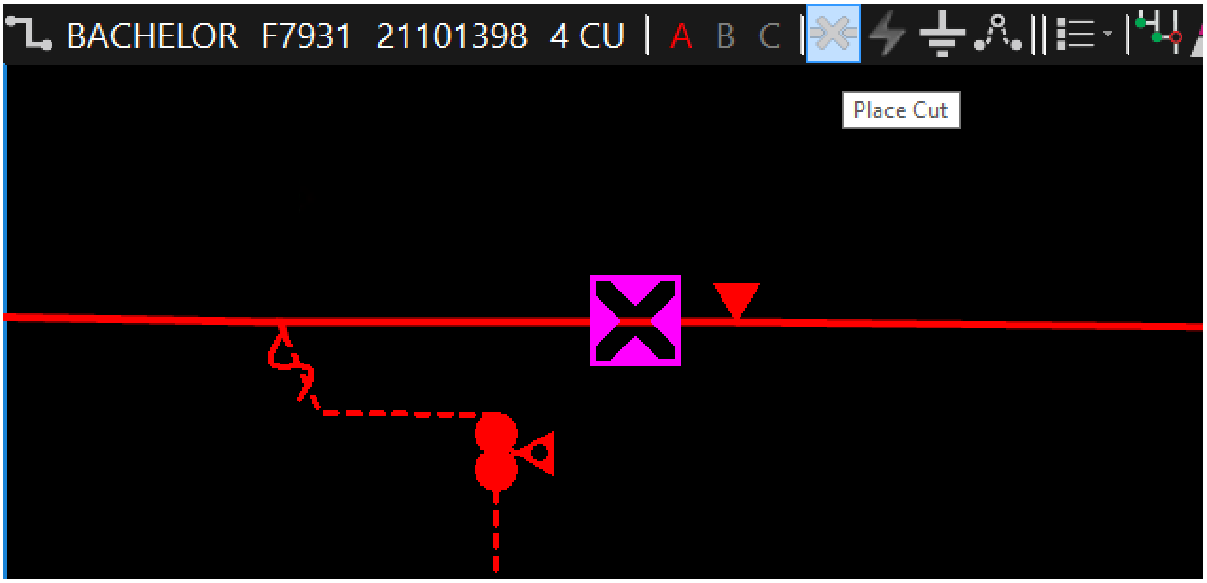
A "cut" is a temporary break of one or more phases. It can be placed on any type of line.

To place a cut:

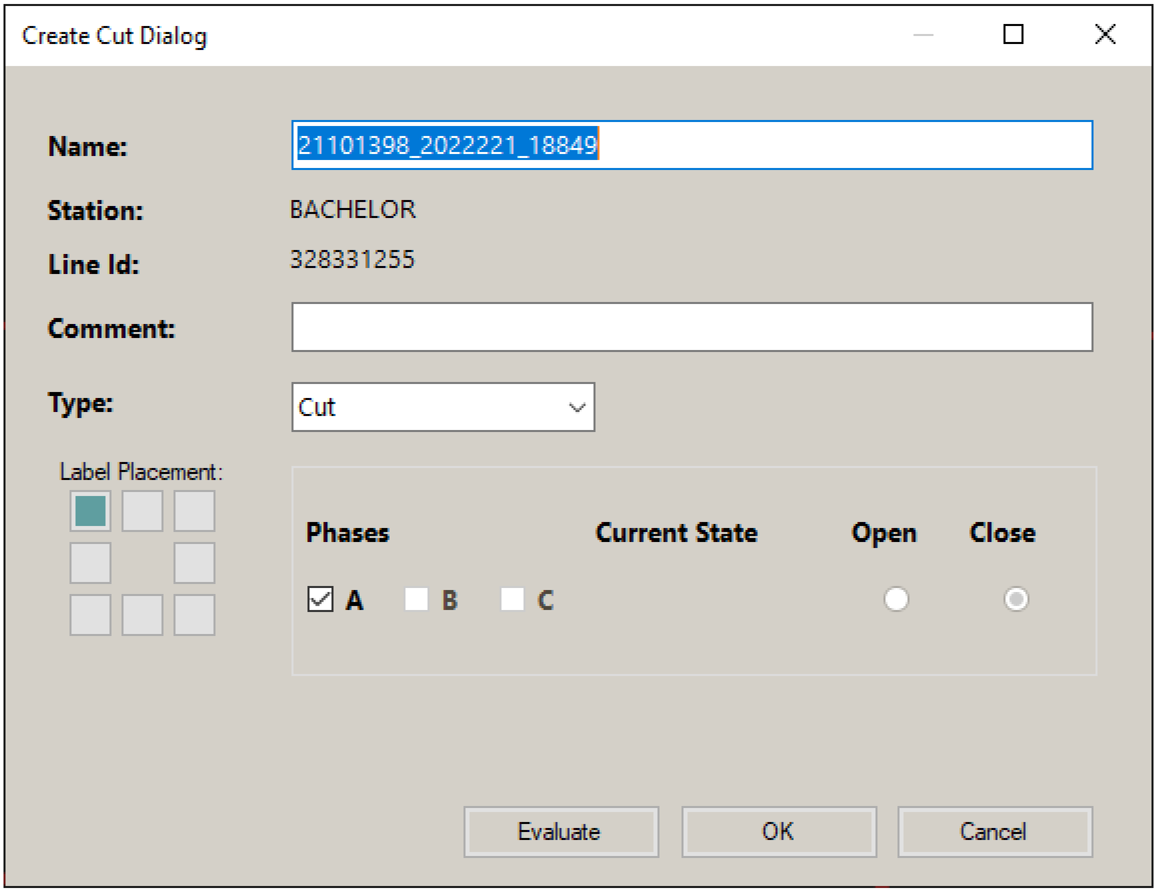
1. From the distribution network view, select the desired point on the line segment by clicking the line.

   The system acknowledges the selection by placing a magenta cursor on the line. The device-specific toolbar shows the station name and line ID.

2. Click the Place Cut button.

   The Create Cut Dialog window opens.



The initial state of the cut is closed by default.

3. Enter a name in the Name field or use the automatically generated name. (By default, the name field is automatically populated with the line name and the current timestamp.)

> **Note**
>
> The Name field is required. The OK button is not enabled until the name of the cut is entered.

4. Enter a description for the cut in the Comment field.
5. Select the respective check boxes for phases that should be cut.

   The three check boxes allow you to select which phases to cut. By default, all of the check boxes are selected, indicating that all the phases in the line are cut.

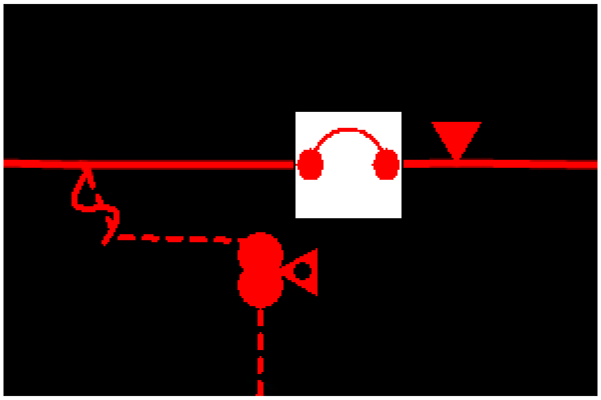
6. If required, select the relative placement of the cut label by clicking one of the Label Placement buttons.

   The default is northwest of the cut symbol.

7. Select the temporary object type from the drop-down list as required (for example, cut, temporary switch, fuse, sectionalizer, or recloser). The default type is cut.

8. To place the cut, click OK.

   By default, a cut is placed in the closed state.

When a cut is opened, the cut symbol on the line changes to a large white "X", and the network topology is updated to show the sections of the network that are now de-energized (shown as white haloes).

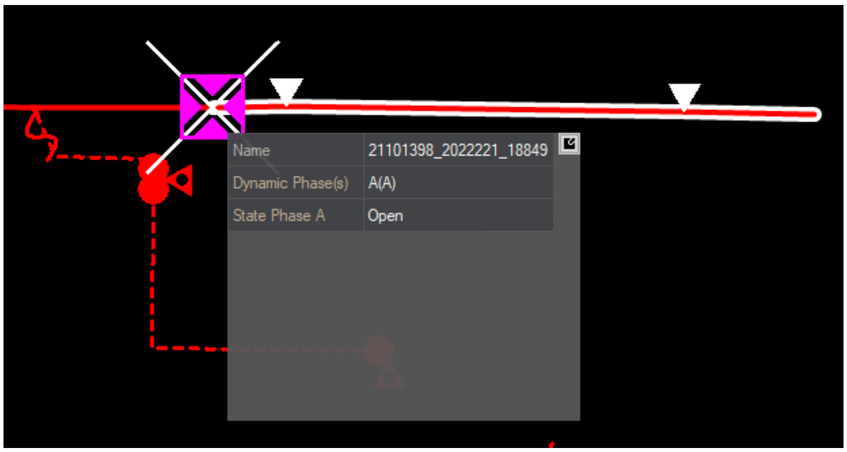


Figure 168. Placing a Cut

Hovering the mouse cursor over a cut shows the cut tooltip panel.

Once a cut has been placed, it acts as a temporary switch. Select the cut to display the device toolbar, which provides buttons for editing the cut, toggling the status of the cut (open/close), placing a tag on the cut, repositioning the cut label, and removing the cut. The phase indicator on the cut toolbar also shows which phases have been cut.

Similarly, you can place other temporary devices, such as a temporary switch, elbow switch, recloser, sectionalizer, or fuse, as shown in image below. This is done by selecting the corresponding option on the cut pop-up.

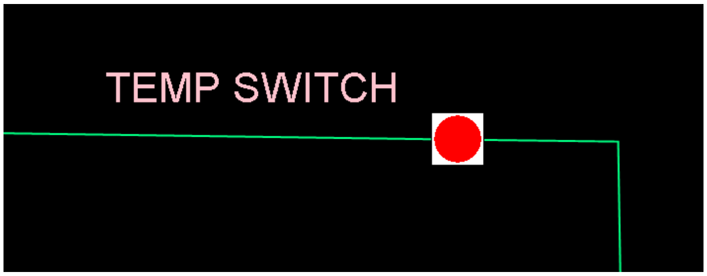
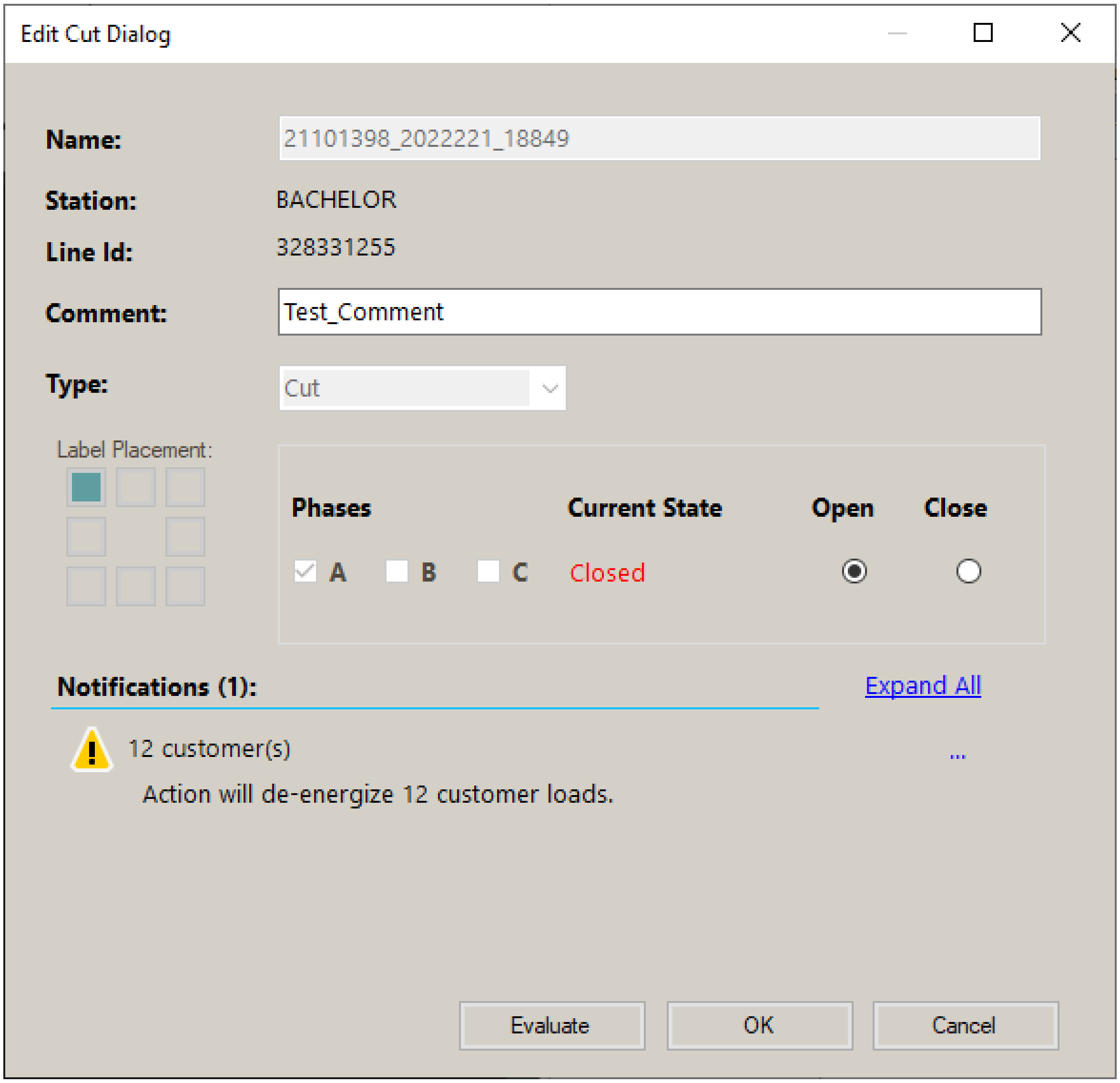


Figure 169. Temporary Switch Inserted in a Line

To edit a cut:

1. Click the cut symbol on the display.
2. On the Cut toolbar, click the Edit Cut button.

   You can update the Comment and Current State fields. In addition, you can evaluate the impact of the cut status change by clicking the Evaluate button.



To remove a cut:

1. Click the cut symbol on the display.
2. Click the Remove Cut button on the Cut toolbar.

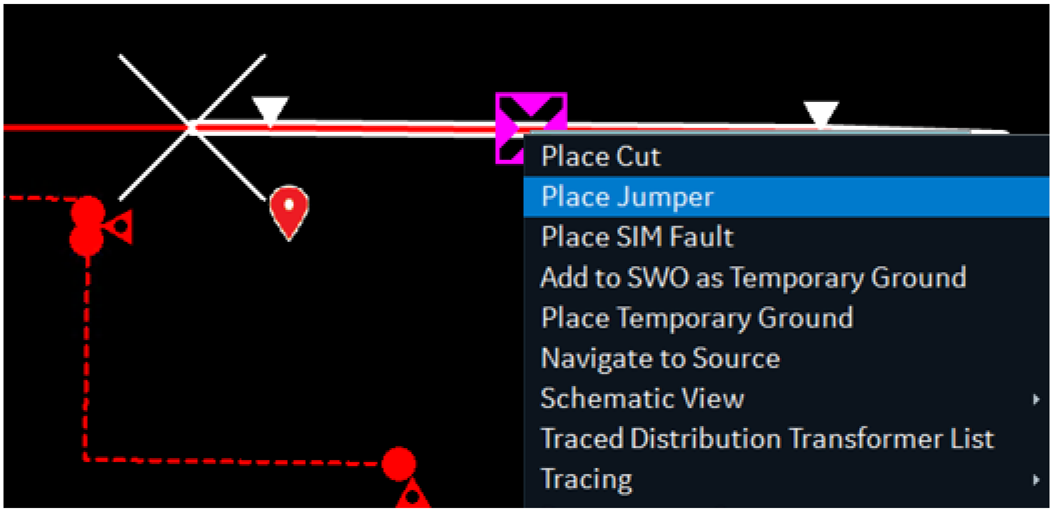
   The cut is removed and network topology is updated to reflect the re-energization of the line segment.

4.8.2. Jumpers

A "jumper" is a temporary switch connection between two points in the network. It is used to provide a means of energizing equipment that has been isolated because of faults or maintenance. The ends of a jumper can be attached to any point on a line, or to a distriibution transformer. When combined with cuts, jumpers allow all types of temporary network modifications to be accurately modeled.

To place a jumper:

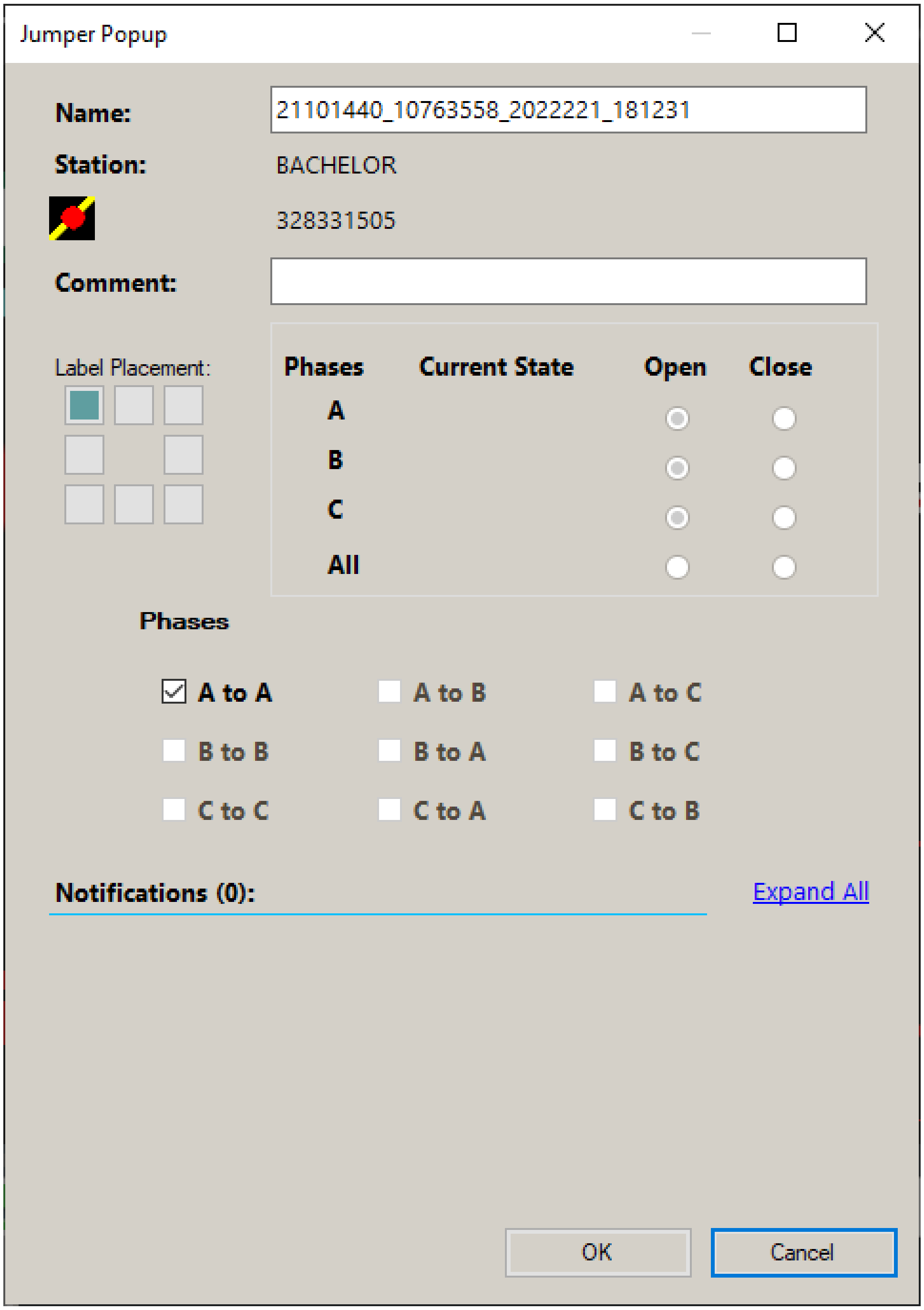
1. Select a point on a line, or distribution transformer, which becomes one end of the jumper.
2. Click the Place Jumper button on the toolbar, or use the right-click menu.



   The first selection point is displayed in yellow.

3. Click the point of the line or transformer that is the other end of the jumper, and then click the Place Jumper button a second time.

   This displays the Jumper pop-up dialog box. The initial state of the jumper is open by default.



4. Enter a name in the Name field or use the automatically generated name. (By default, the name field is automatically populated with the associated line names and the current timestamp.)

   You can enter a name or a reference to the jumper.

   > **Note**
   >
   > The OK button is not enabled unless a name is entered.

5. Select the phase of the jumper using checkboxes. The default connection simply connects the phases A-A, B-B, and C-C. If an alternate arrangement is required, select the other check boxes.

   The check boxes show all possible connections between the two lines. In cases where there is a phase or voltage mismatch between the two ends of the proposed jumper (for example, a B-C cross-phasing is selected), warnings appear in the message box at the bottom of the jumper pop-up. However, the warnings do not prevent the jumper from being placed.

6. Use the optional Comment field to add more information about the jumper.
7. If required, select the relative placement of the jumper label by clicking one of the Label Placement buttons.

   By default, a label is placed at the upper-left (Northwest) of the jumper symbol.

8. Click OK.

   This places the jumper symbol between the two points.

By default, a jumper is placed in the open state. When a jumper is closed, the network topology processor updates to show the resulting energization status of the lines.

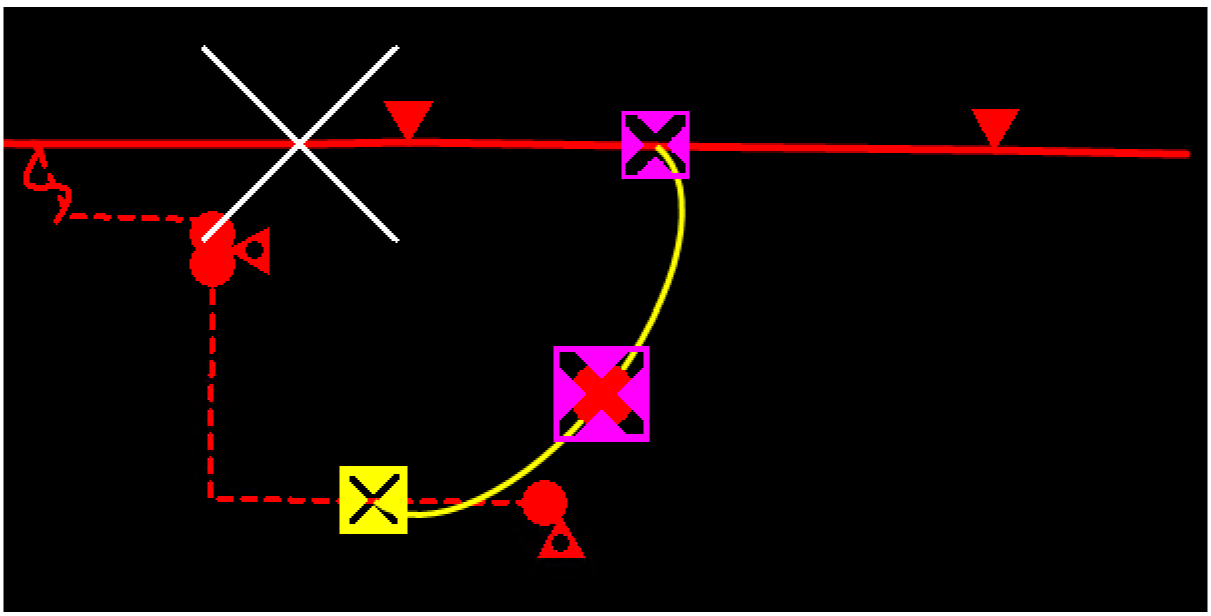


Figure 170. Placing a Temporary Jumper

Hovering the mouse cursor over the jumper switch symbol shows the jumper tooltip panel.

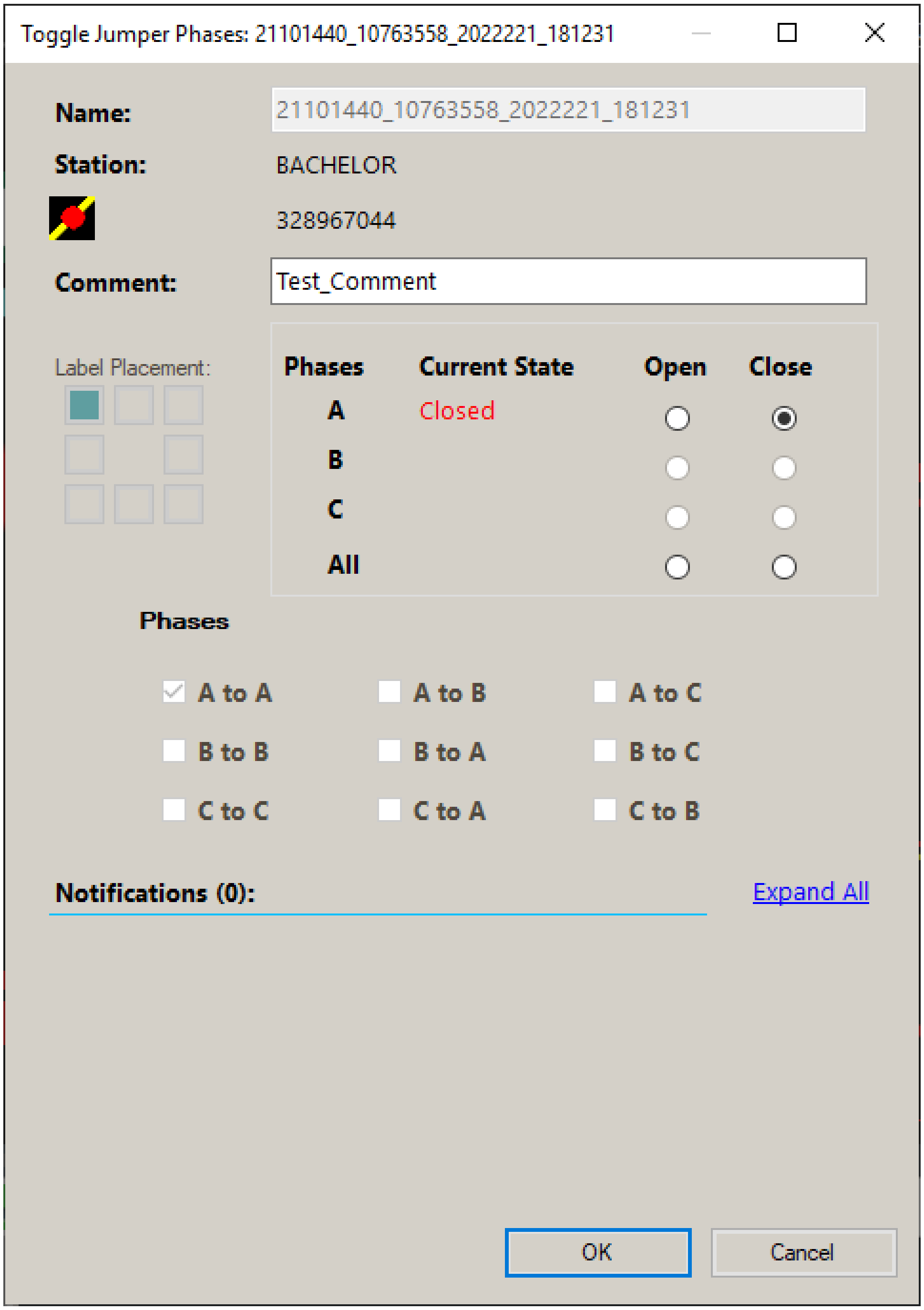
Jumpers are actually temporary switches with a normally Open status. To change the state of a jumper, click the switch symbol (green or red circle) in the middle of the jumper, and then click the Toggle button on the device-specific toolbar. Buttons on the jumper toolbar allow the jumper label to be repositioned if required.

By default, a jumper is drawn as a curved line between the two end points. It is possible to move the center point of a jumper by using the move mode, as described in Moving a Temporary Object. This can be very useful to avoid the jumper obscuring other network devices.

To edit a jumper:

1. Click the jumper symbol on the display.
2. Click the Edit Jumper button on the jumper toolbar.

   You can update the Comment and Current State fields.



To remove a jumper:

1. Click the jumper symbol on the display.

2. Click the Remove Jumper button on the jumper toolbar.

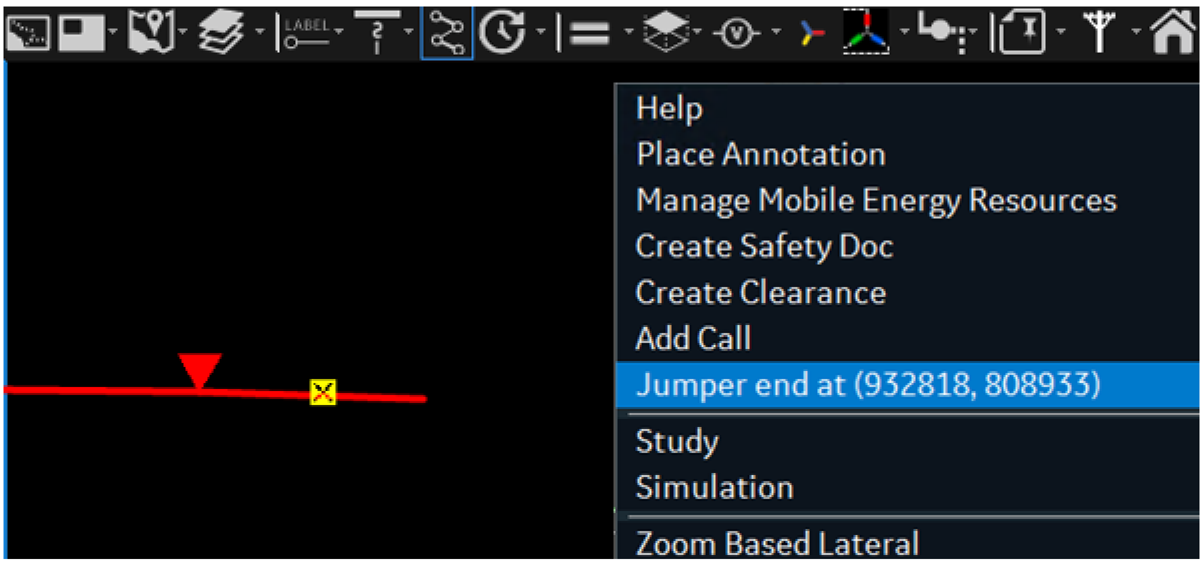
   The jumper is removed and network topology is updated.

4.8.2.1. Jumper to Nowhere

A jumper to nowhere is applied in a case where one end of a jumper must be connected to a line and the other end is connected to nothing.

To place a jumper to nowhere:

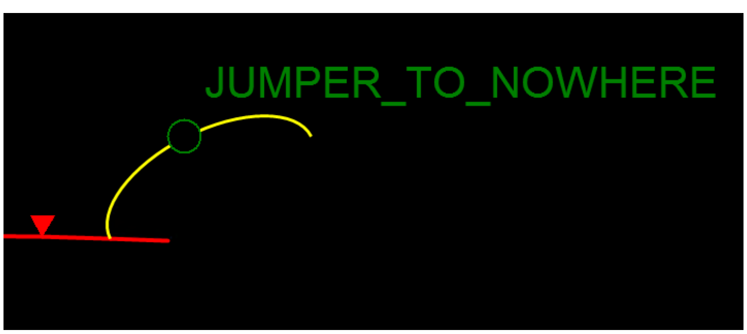
1. Select a point on a line, a feeder backbone, or a lateral, which becomes one end of the jumper.
2. Click the Place Jumper button on the device-specific toolbar.
3. Right-click the location in the black area where the jumper should end, and select the JUMPER END AT (x, y) menu option.



   This displays the Jumper pop-up dialog box.

4. Enter the jumper settings as described in Jumpers.
5. Click OK.

   The jumper is placed on the network view



4.8.3. Temporary Grounds

Temporary grounds can only be placed on de-energized sections of a line.

To place a Temporary Ground:

1. Select a point on the de-energized line segment, and click the Place Temporary Ground button on the line toolbar.

   This displays the Temporary Ground Popup dialog box.

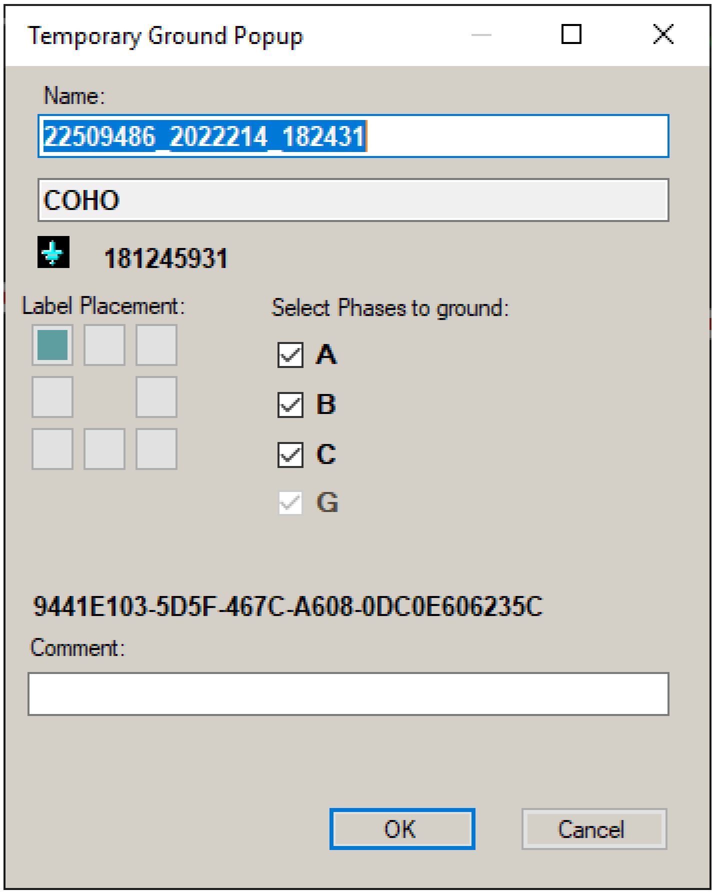
2. Enter a name in the Name field or use the automatically generated name. (By default, the name field is automatically populated with the line name and the current timestamp.)

   For example, you can enter the Temporary Ground name or reference to the Temporary Ground placement such as a switching order number.

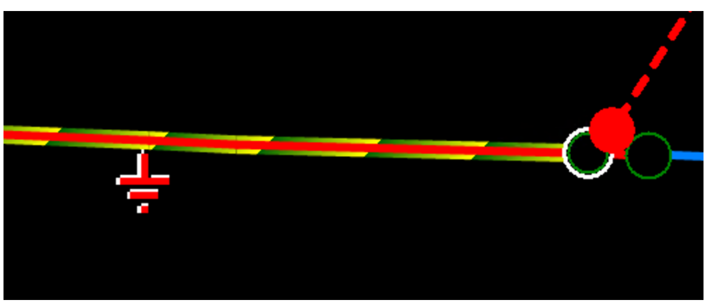
   > **Note**
   >
   > The OK button is not enabled unless a name is entered.

3. Select phases that are to be grounded.

   The check boxes allow the selection of the phases to be grounded. By default, all phases on the line section are grounded.



4. The optional Comment field can be used to add more information about the Temporary Ground.
5. Click OK.



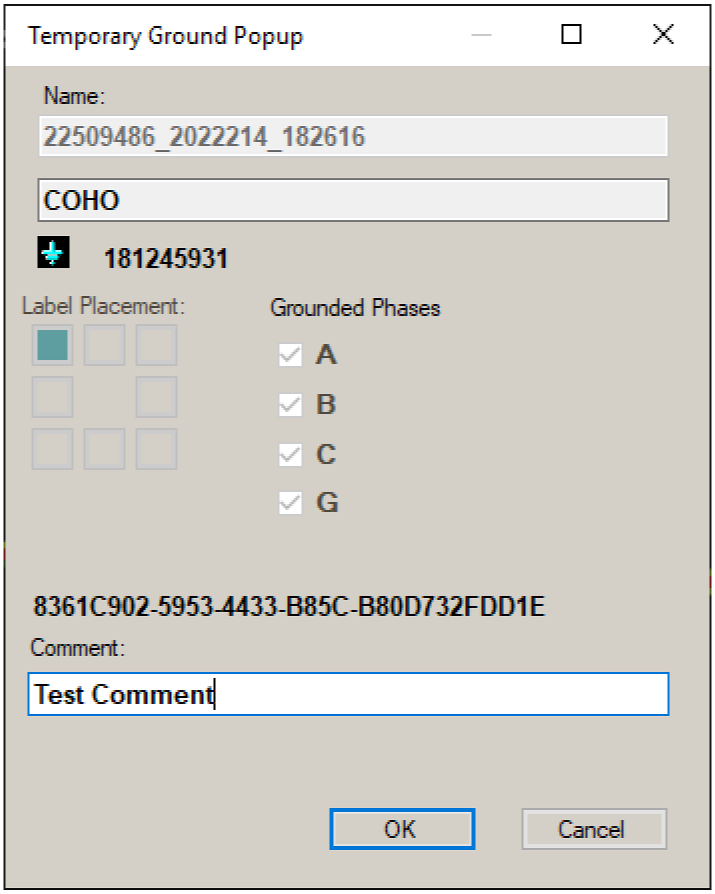
The ground symbol is placed on the line. The grounded section of the line is displayed with a yellow and green halo.

Hovering the mouse cursor over the Temporary Ground shows the ground tooltip panel.

To edit a Temporary Ground:

1. Click the Temporary Ground symbol on the display.
2. Click the Edit Temporary Ground button on the ground toolbar.

You can only update the Comment field.



To remove a Temporary Ground:

1. Click the Temporary Ground symbol on the display.
2. Click the Remove Temporary Ground button on the ground toolbar.

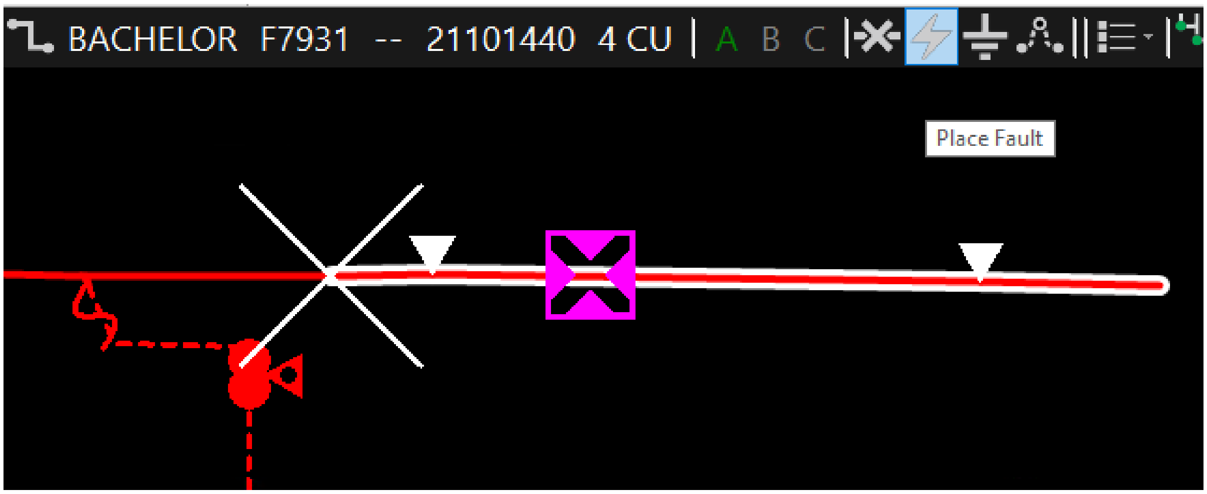
   The Temporary Ground is removed and network topology is updated.

4.8.4. Faults

Faults can only be placed on the de-energized sections of a line.

To place a fault:

1. Select a point on the de-energized line segment, and click the Place Fault button on the device-specific toolbar.

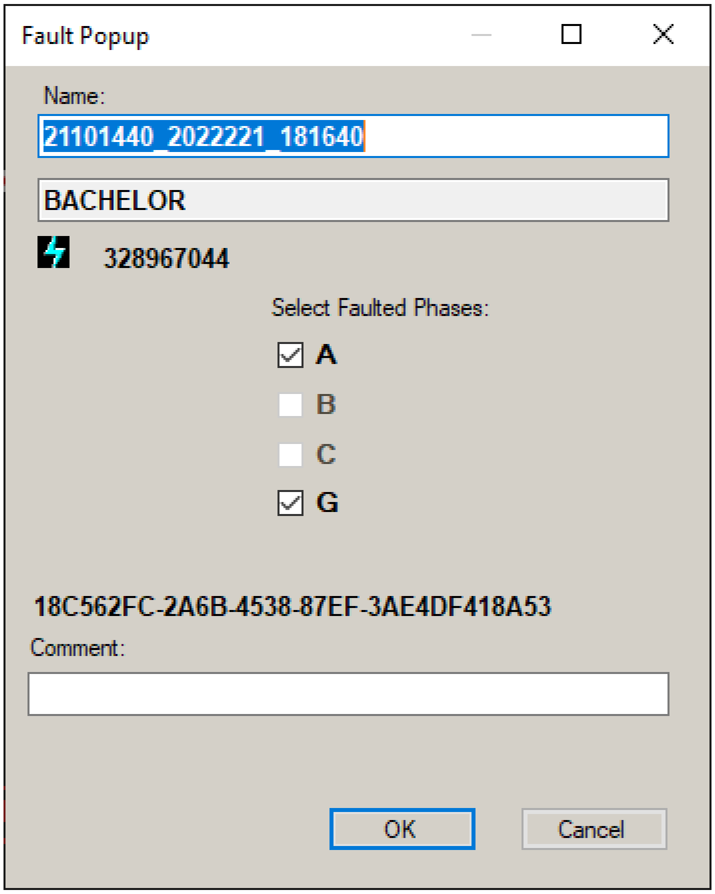


The Fault Popup window is displayed.

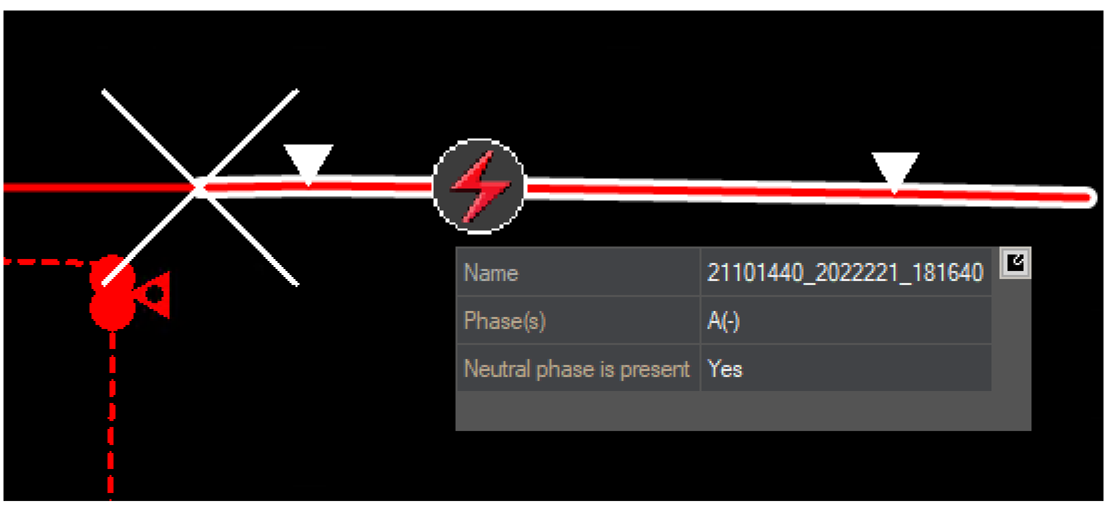
2. Enter a name in the Name field or use the automatically generated name. (By default, the name field is automatically populated with the line name and the current timestamp.)

> **Note**
>
> The OK button is not enabled unless a name is entered.

3. From the Select Faulted Phases options, choose the faulted phases.
4. Optionally, in Comment field, add more information about the fault.



5. Click OK to place the fault symbol on the line.



> **Note**
>
> Hover the mouse cursor over the fault to view the fault tooltip panel.

To edit a fault:

Note

> You can only update the Comment field.

1. Click the fault symbol on the display.
2. Click the Edit Fault button on the fault toolbar.

To remove a fault:

1. Click the fault symbol on the display.
2. Click the Remove Fault button on the fault toolbar.

   The fault is removed and network topology is updated.

4.8.5. Orphaned Temporary Objects

A temporary object such as a cut, ground, or jumper becomes orphaned when its placement on a target line can no longer be determined due to changes in the GIS network topology.

When a new model is loaded, all temporary objects for that station are validated before being reapplied to their target components. Some changes to target line placements are automatically corrected before application. These include:

- Line shifts from one substation to another
- Single phase changes from one phase to another
- To/from node reversal
- Minor geometry changes (distance is configurable)

Under some circumstances, a temporary object cannot be applied to the new network model. This could occur under the following situations:

- Deletion of the target line
- Multiple phase changes in the target line
- Changes in target line placement in the geometry

In these circumstances, the temporary object becomes orphaned and is not applied to the network model. Orphaned objects are shown in the displays with a yellow circle, as shown image below. If the target line is restored in the model, orphaned objects are reapplied and become normal objects again.

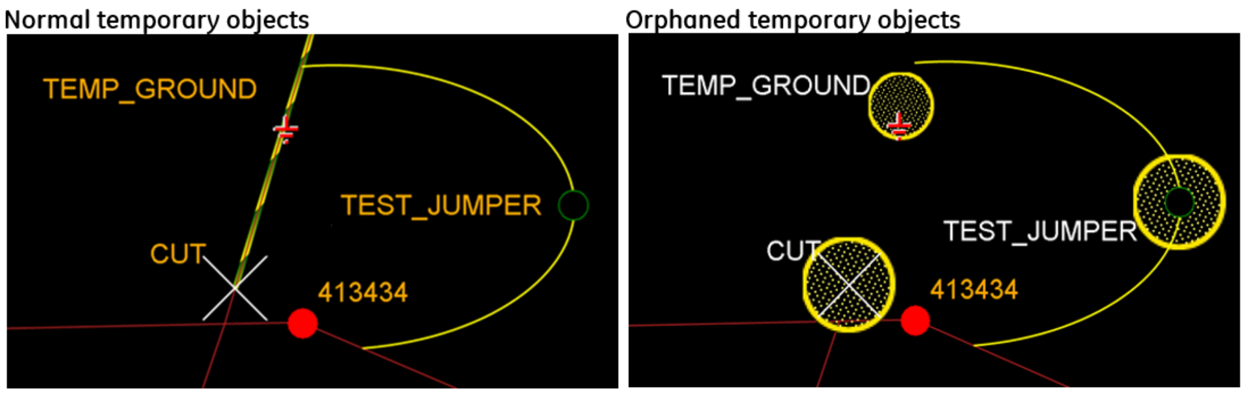


Figure 171. Normal and Orphaned Temporary Objects

Orphaned objects do not change the network model and require operator intervention to correct the underlying issue with their placement. You can manually delete the objects from the network view and then recreate new temporary objects to ensure the network model remains in sync with the real world topology.

> **Note**
>
> Orphaned objects support only "neutral" operations such as closing a cut or opening a jumper. This is to allow deletion of the objects from a neutral state.

Orphaned objects are also cleared after a pre-configured number of days (by default, 1) during midnight dynamic compression.

Use orphaned device summary tabular displays to review all orphaned objects. The displays can be accessed by selecting Orphan Summary from the RT menu.

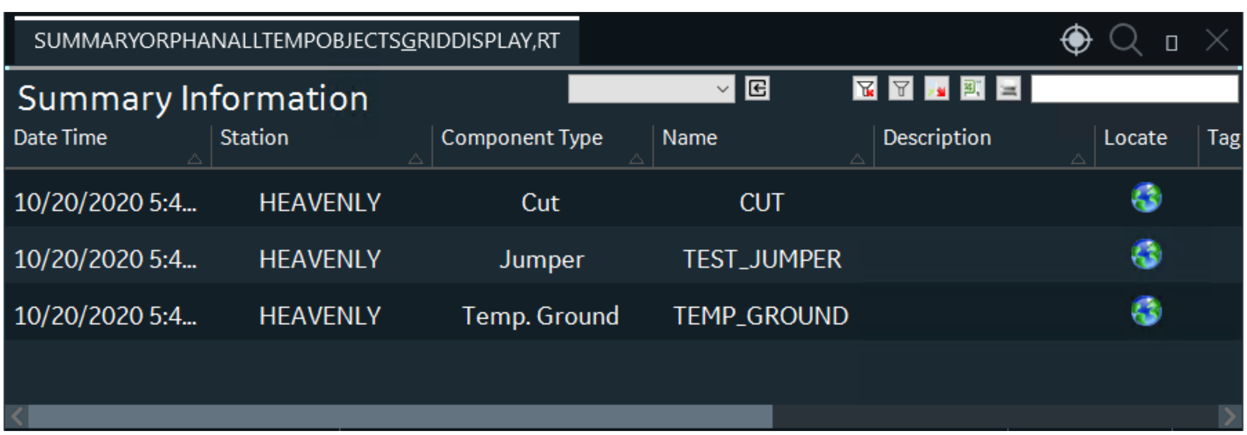


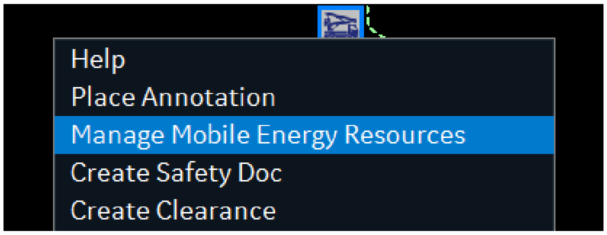
Figure 172. Orphaned Objects Summary

4.9. Mobile Substations

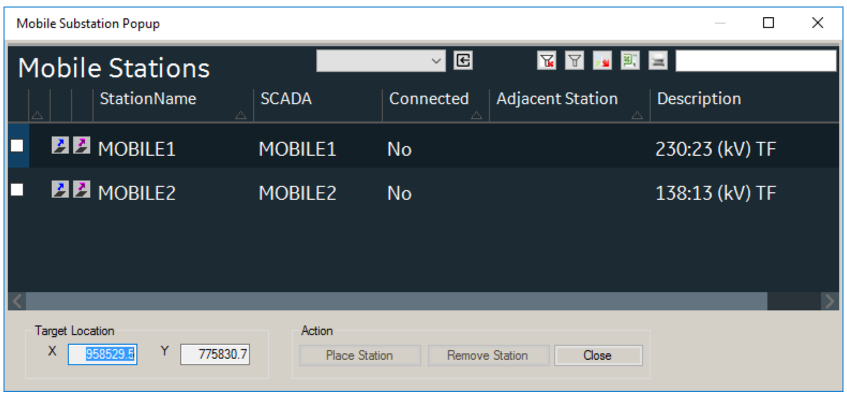
Mobile substations can supply temporary power in case of emergencies or planned outages. A mobile substation can be placed at any point on the network view and connected to the distribution network via jumpers.

To connect a mobile substation:

1. Right-click at a point on the network view and select Manage Mobile Energy Resources.



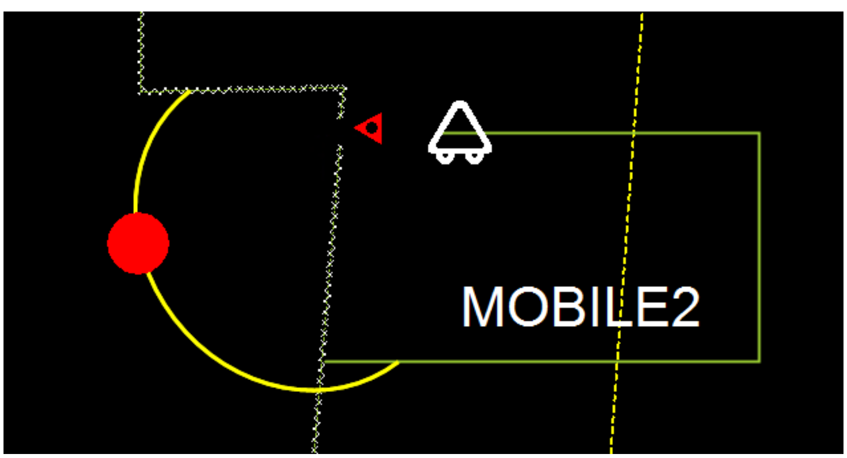
The Mobile Substations display opens.



2. Select a mobile substation from the list and then click Place Station.

   The mobile substation appears on the network view.

3. Connect the mobile substation to the network via jumpers.



The connected network segments are now shown energized.

To remove a mobile substation:

1. Remove jumpers that connect the mobile substation to the network.

2. Right-click the network view and select Manage Mobile Energy Resources.

   The Mobile Substations display opens.

3. Select a mobile substation from the list and then click Remove Station.

   The mobile substation is removed.

> **Note**
>
> Mobile substations can be modeled by modeling tools such as the Substation Editor and are not part of the user interface. For more information about the Substation Editor, refer to the *Substation Editor User Guide*.

4.10. Adding Devices to a Safety Document

In the ADMS client, a **Safety Document** can be created and quickly populated with all associated devices based on the system topology.

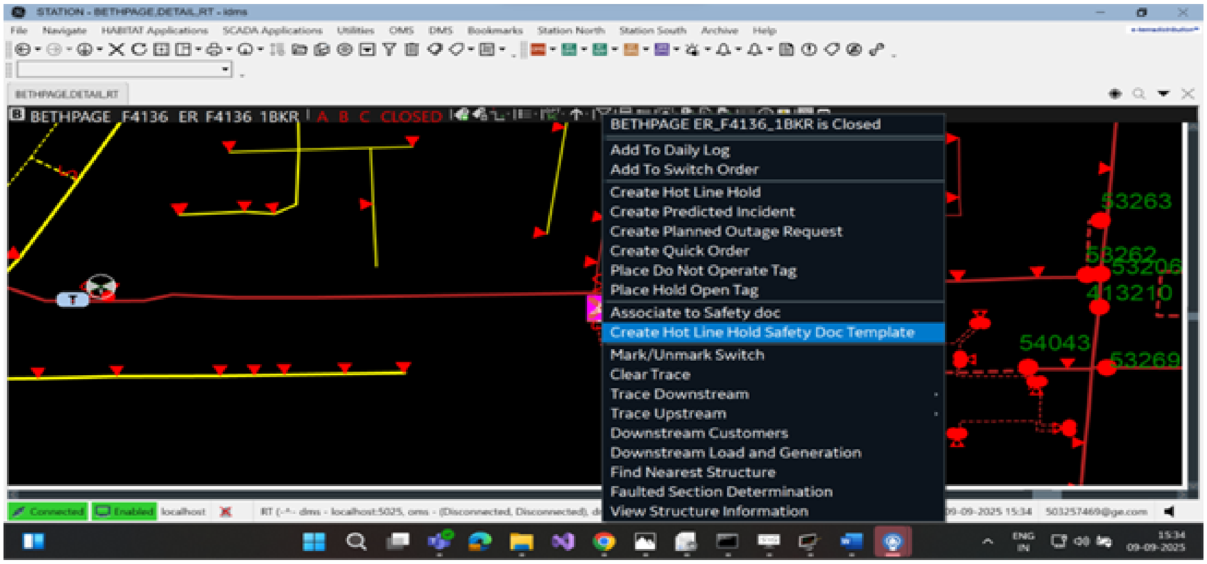
By selecting one or more relevant devices on the map, the system automatically determines which additional devices should be included in the safety document. The included devices are determined by both the topology and the type of safety document being created (**Clearance**, **TOC** or **Hot Line Hold**).

> **Note**
>
> To review the device selection logic used during the creation of HLH, Clearance and TOC safety documents, navigate to *Safety Doc Types* section of *Switch Order Related Tab* in *Switch Order Client Configuration Editor User Guide*.

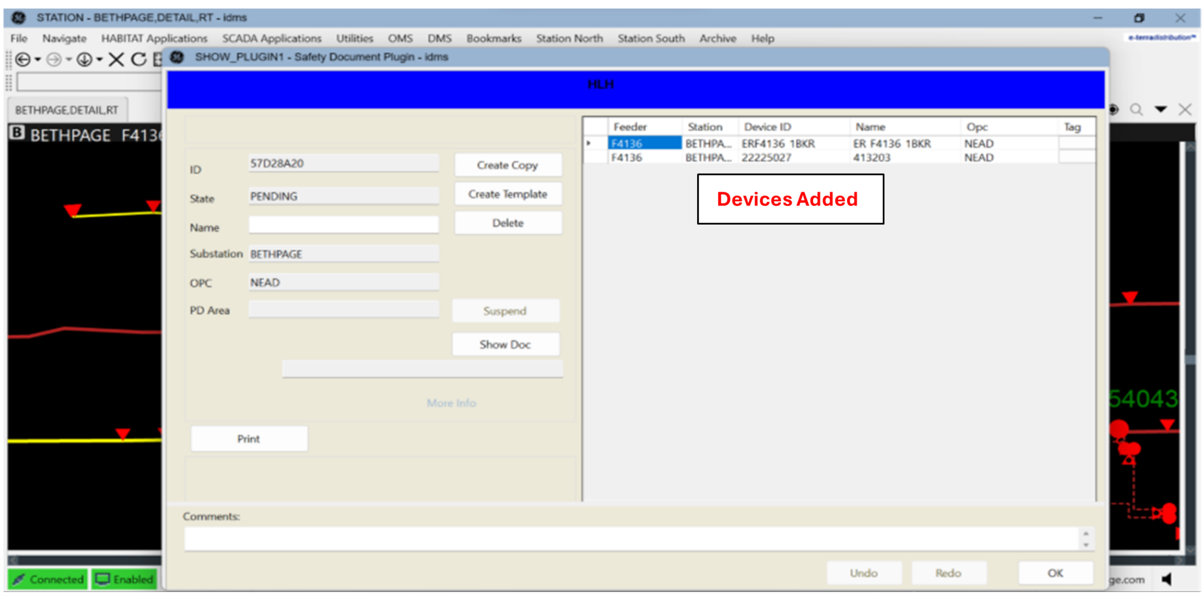
4.10.1. Adding Devices for Hot Line Hold:

To add devices while creating a Hot Line Hold Safety Doc, perform the following actions:

1. Open the geographical view.
2. Select a supported device type, for example `dmsRecloser`, `dmsSectionalizer`, `dmsBreaker`, or `dmsLine`.
3. Right-click the selected device and choose **Create Hot Line Hold Safety Doc Template** from the context menu.
4. The system automatically creates a hot line hold safety doc and adds all required upstream reclosing devices—based on configuration rules, including the selected device if applicable.



5. Device details such as **Feeder**, **Station**, **Device ID**, **Name**, and **OPC** are automatically populated in the safety document.



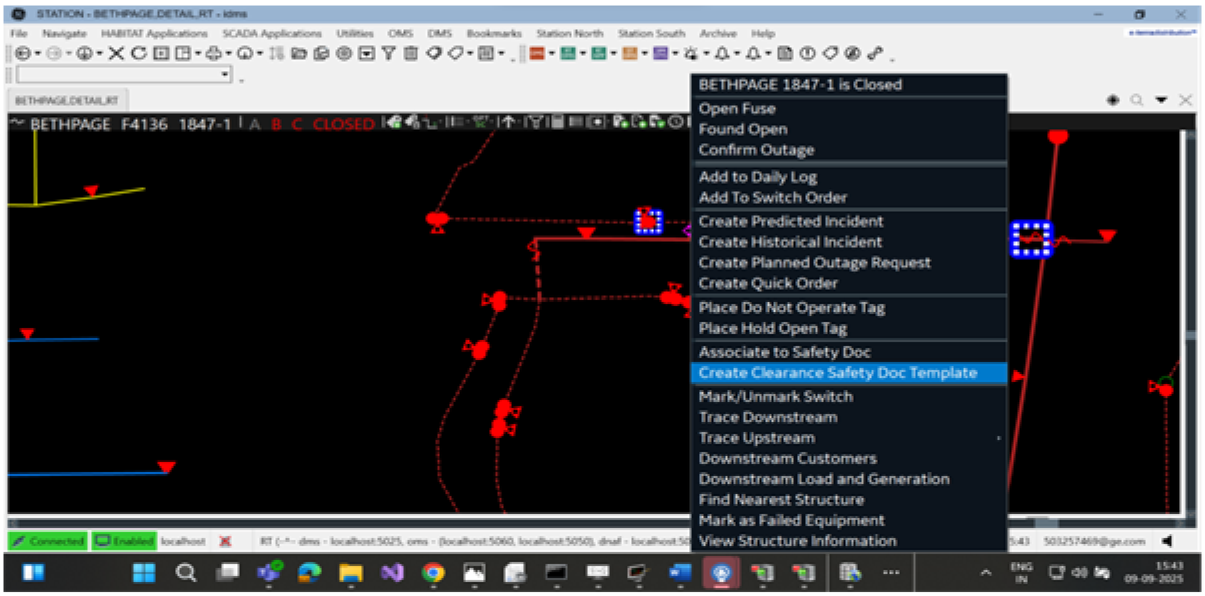
> **Note**
>
> Ensure that the **Tag Type** of the open **Hot Line Hold** safety document matches the **Tag Type** defined in the **Switch Order Client Configuration Editor**.

4.10.2. Adding Devices for Clearance:

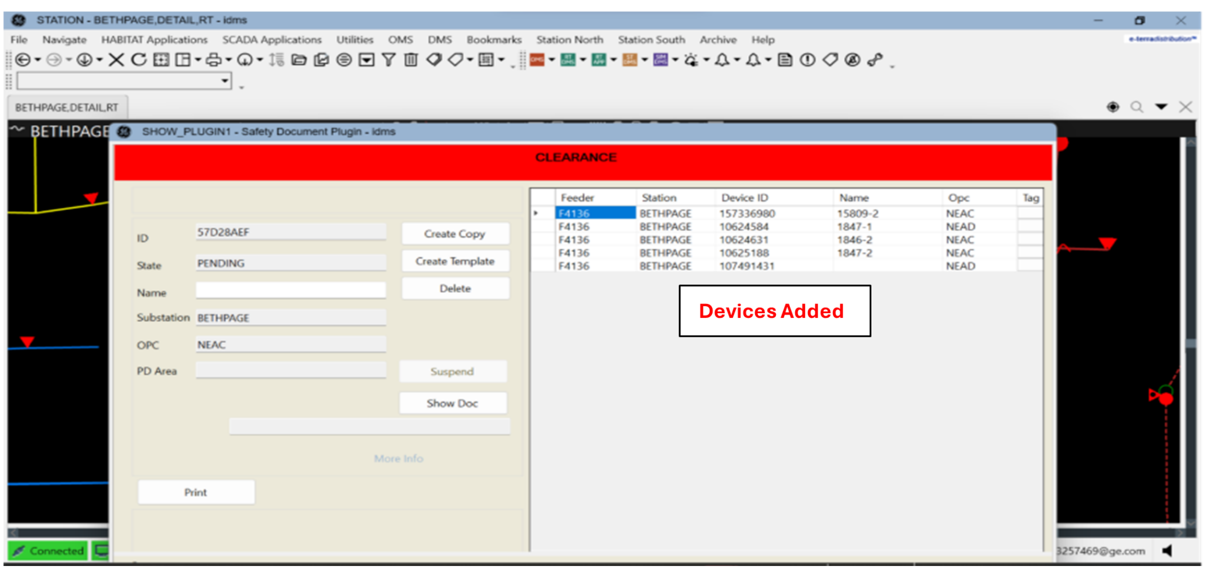
To add devices while creating a Clearance safety document, perform the following actions:

1. Open the geographical view.

2. Select a supported device type, for example `dmsTempSwitch`, `dmsSwitch`, `dmsSectionalizer`, `dmsElbow`, `dmsFuse`, `dmsLine`, `dmsTransformer`, `dmsCapacitor`, `dmsGenerator`, or `dmsEnergyResource`.
3. Right-click the selected device and choose **Create Clearance Safety Doc Template** from the context menu.
4. The system automatically creates a clearance safety doc and adds all required upstream and downstream switching devices, including DERs, according to the configuration rules. The selected device is also included if it qualifies as a switching device.



5. Device details such as **Feeder**, **Station**, **Device ID**, **Name**, and **OPC** are automatically populated in the safety document.



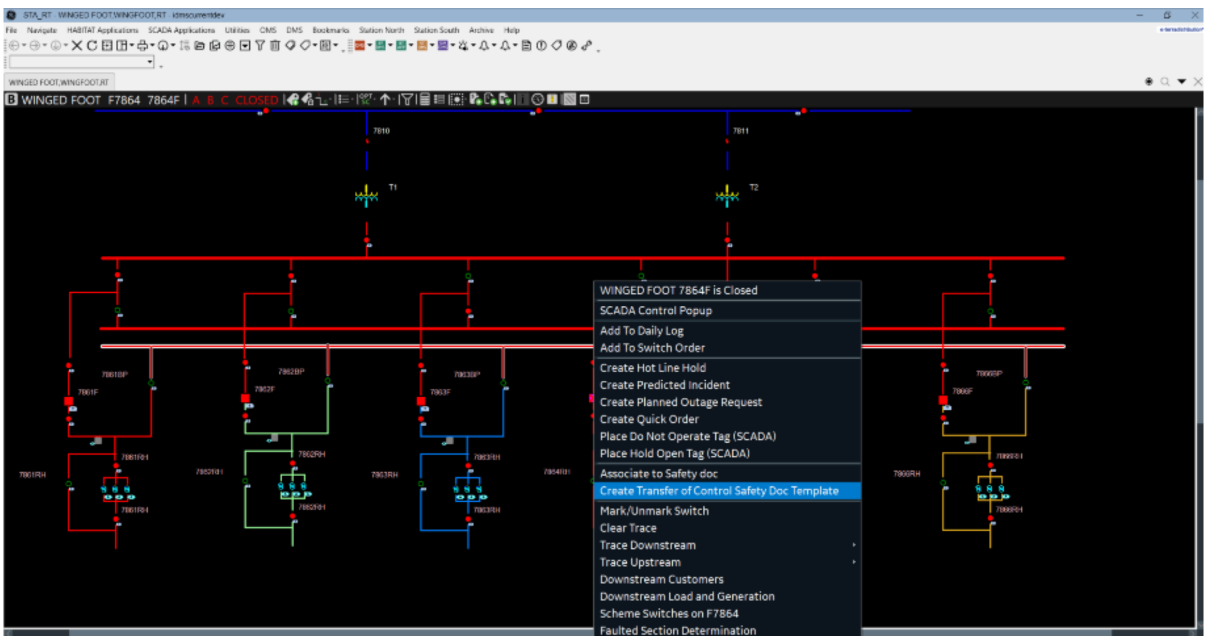
> **Note**
>
> Ensure that the **Tag Type** of the open **Clearance** safety document matches the **Tag Type**

defined in the **Switch Order Client Configuration Editor**.

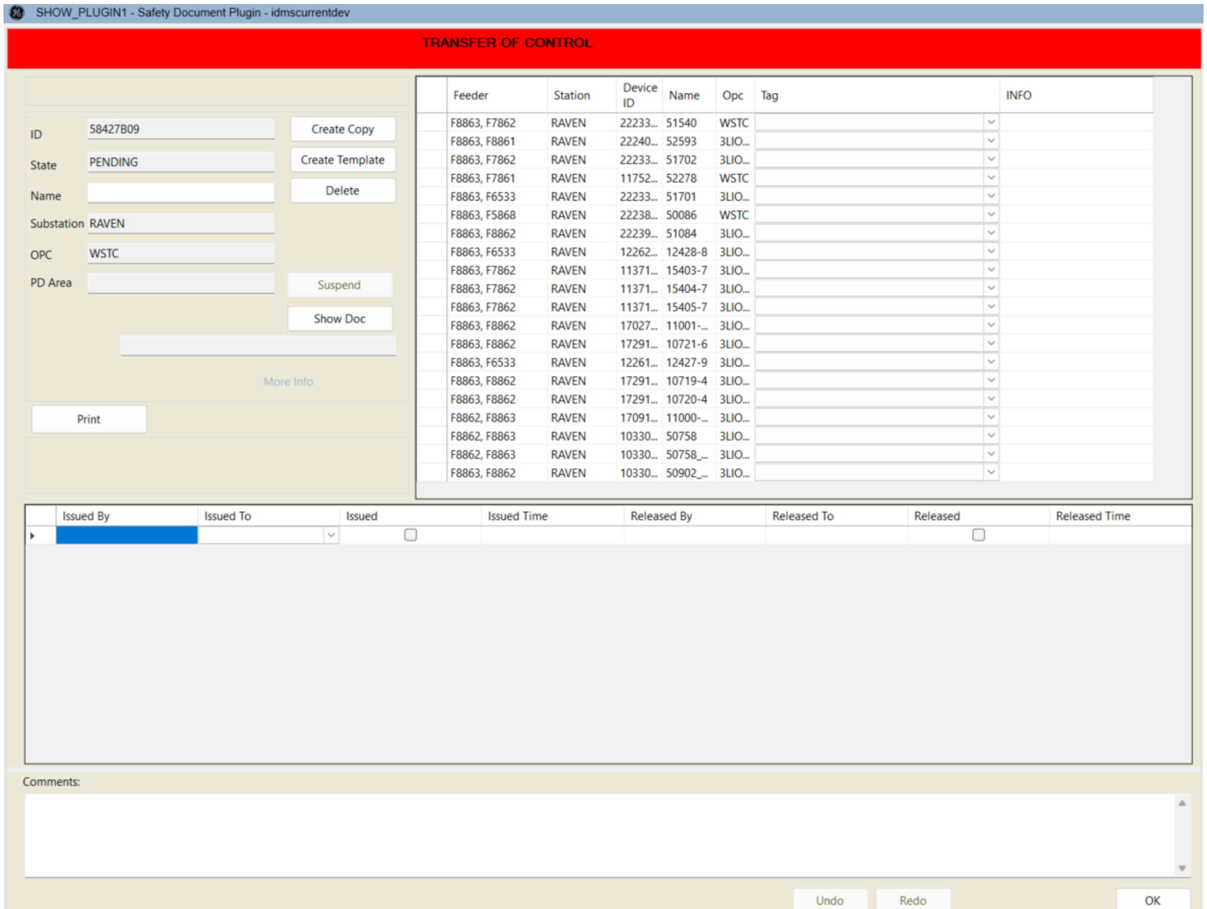
4.10.3. Adding Devices for Transfer of Control:

To add devices while creating a Transfer of Control (TOC) safety document, perform the following actions:

1. Open the internal substation view.
2. Select a supported device type, for example, a feeder breaker or a load side disconnect near feeder head.
3. Right-click the selected device and choose **Create Transfer of Control Safety Doc Template** from the context menu.
4. The system automatically creates a TOC safety document and adds all normally open devices that connect to a different feeder other than the one where the TOC safety document was initiated, based on configuration rules.



5. Device details such as **Feeder**, **Station**, **Device ID**, **Name**, and **OPC** are automatically populated in the safety document.



> **Note**
>
> Ensure that the **Tag Type** of the open **Transfer of Control** safety document matches the **Tag Type** defined in the **Switch Order Client Configuration Editor**.

4.11. Placing and Removing Tags

Tags are placed on devices to indicate that some abnormal condition exists. Usually, tags are applied to indicate that a switch is open to isolate a section of line that is under maintenance clearance. The tag is there to warn operators that the switch must not be operated until maintenance is complete.

Tagging is often associated with Switching Orders and Safety Documents. For more information about these functions, refer to the *Switching Operations User Guide*.

Tags can also be used to add special comments to a device or to indicate a malfunction. Multiple tags can be placed on a single point. If more than one tag is applied to a point, the most important tag (with its associated tag color) is displayed. Tags can be applied to switches and temporary modifications (jumpers, cuts, and grounds).

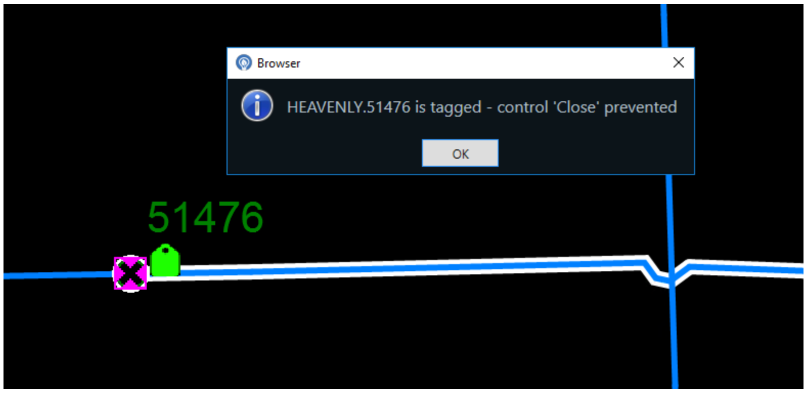


Figure 173. Operation Inhibited Due to Tag

For the purposes of illustration, three types of tag can be applied: Do Not Operate (Red), Hold Open (Green), and Info (White). It is expected that each implementation has tagging colors and types that are tailored to the needs of the utility. The details and usage of these tags are described in the following sections.

4.11.1. Applying a Tag

Tags are applied and managed as part of the Scada system. To apply a tag, select the device, and click the Place Tag button on the device-specific toolbar.

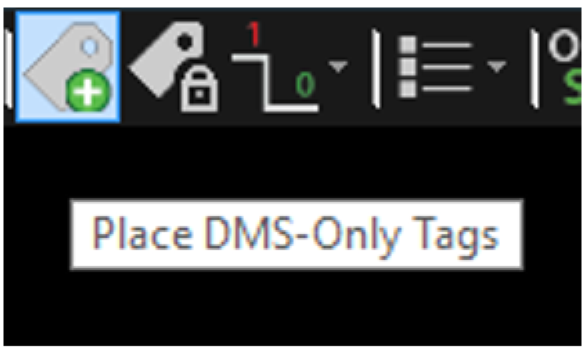


Figure 174. For DMS Devices

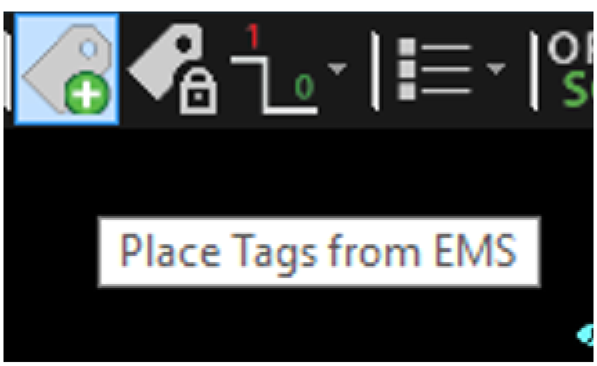


Figure 175. For SCADA Devices

For DMS devices, the Tagging Operations dialog box opens.

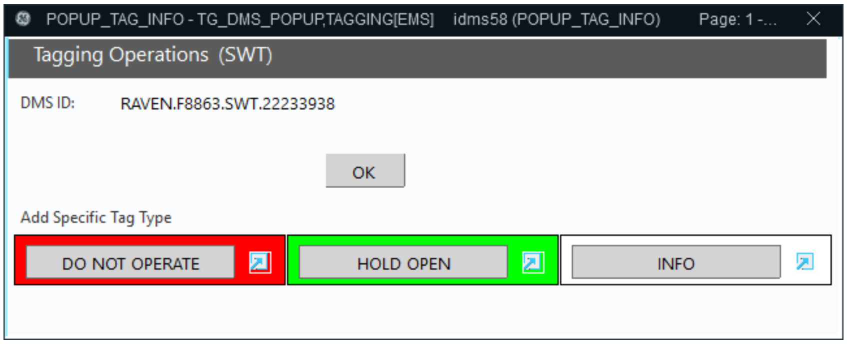


Figure 176. DMS Tagging Operations Dialog Box

For SCADA devices, the following dialog box is shown, where you need to click the Tag button.

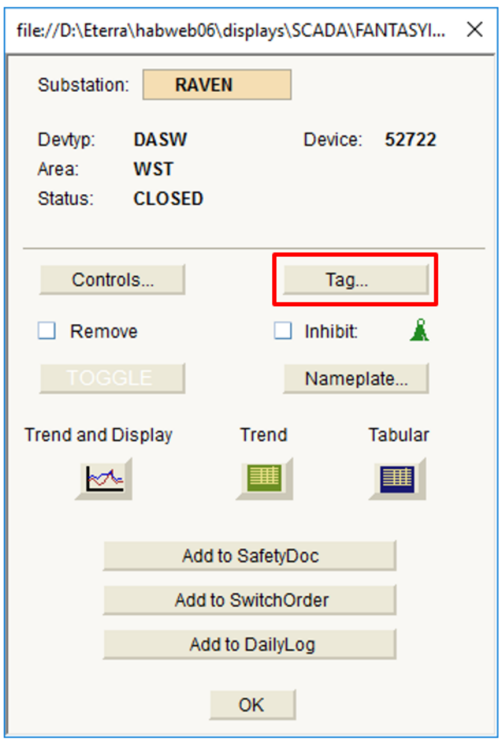


Figure 177. SCADA Control Dialog Box

The SCADA Tagging Operations dialog box opens.

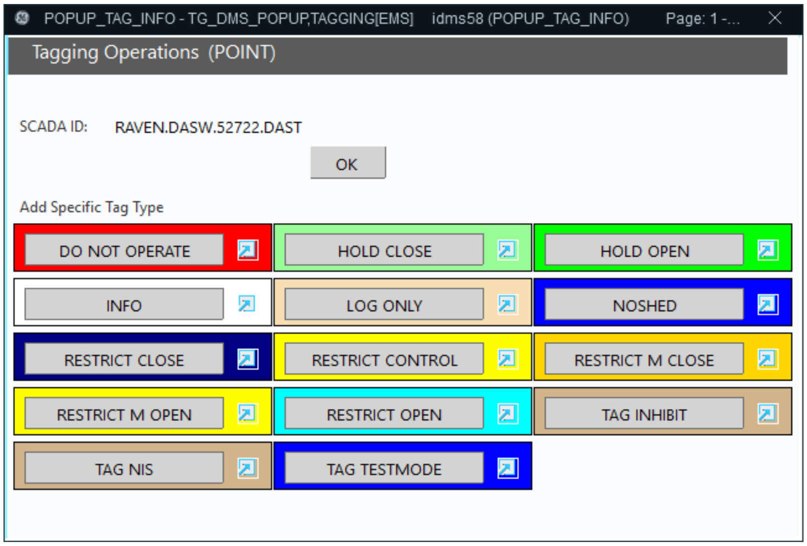


Figure 178. SCADA Tagging Operations Dialog Box

The DMS and SCADA tagging dialog boxes function similarly, however the SCADA dialog box provides more tag types.

To open the tag processing list, click the type of tag to be applied.

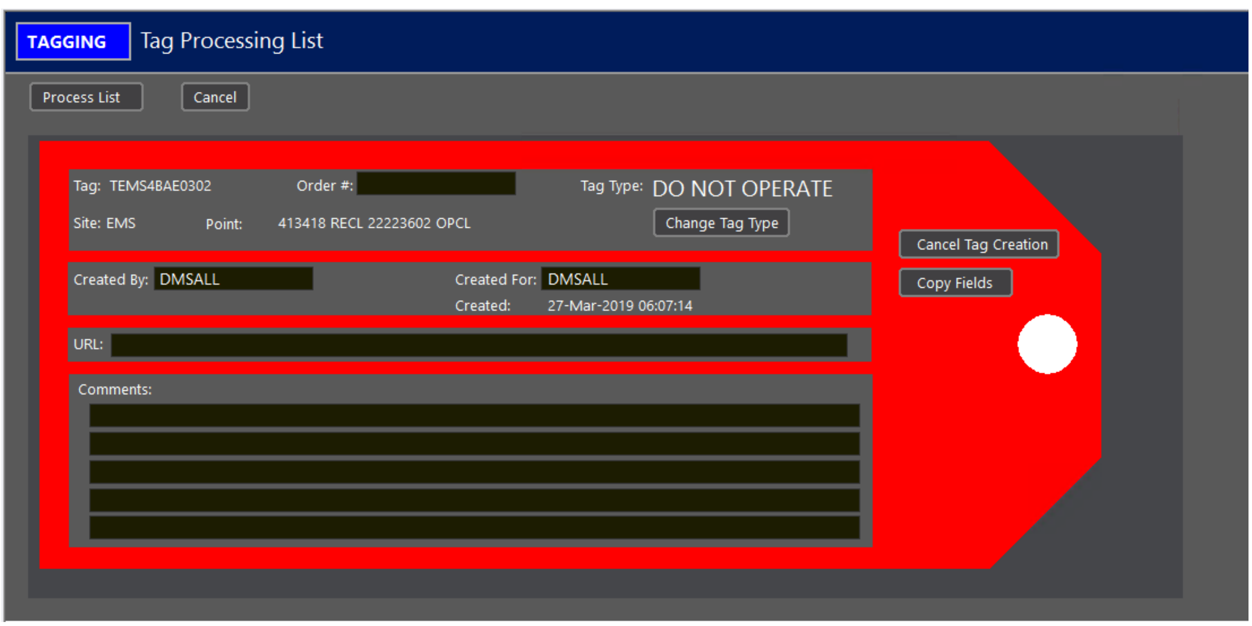


Figure 179. Tag Processing List

When the information on the pop-up form has been completed, including a clearance number where required, click the Process List button to apply the tag. The tag symbol then appears on the geographic network view.

Tags can be reviewed on the tagging summary displays.

In the case of multiple tags, the tag symbol on the network view shows a single tag with a plus ("+")

sign. The color of the tag represents the highest-priority tag that is currently applied.

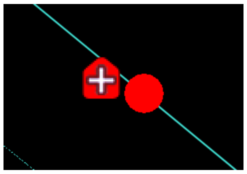


Figure 180. Multiple Tags Applied

In the case where a tag is applied to a switch that is inside a UI container, vault, or switch cabinet, a tag symbol is also displayed next to the container symbol, indicating that there is a tagged device inside. When the display is zoomed in and the internal switches are displayed, a tag symbol is also displayed for any of these switches that can be tagged.

> **Note**
>
> The tag can only be applied or removed by selecting the individual switching device. Container objects do not support tagging operations.

You can move a tag to another position on the network view. To activate the Move mode, use the Move Tag on the tag right-click menu.

> **Note**
>
> When you create a tag on a device, move the tag to the new position, remove the tag and then create a new tag on the same device during a day, tags may not immediately appear in the expected position until the next day because tag decoration dynamics are cleared only once a day, during the midnight compression.

4.11.2. Placing a Single-Click Tag from the Network View

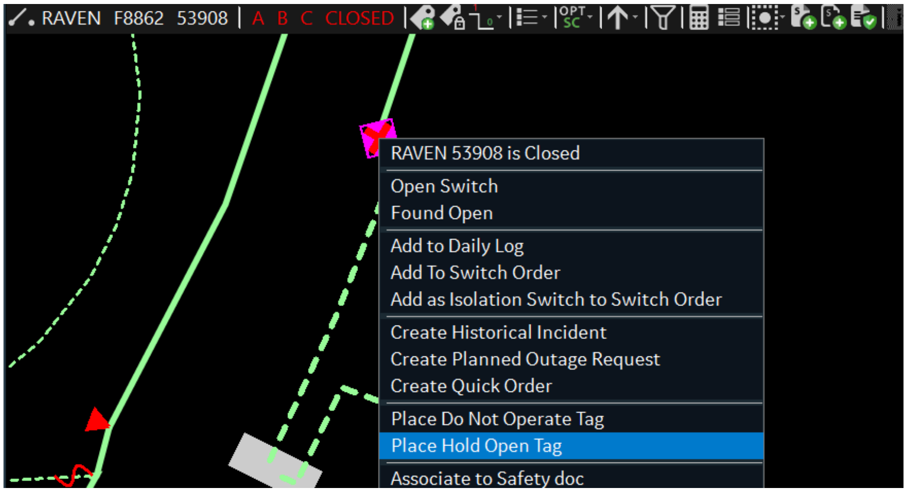
ADMS supports a command syntax to initiate tag creation directly from the network view via the switch's right-click menu.

To place a single-click tag from the network view:

1. Right-click a switch.

   On the right-click menu, find these commands:

   - "Place Do Not Operate Tag"
   - "Place Hold Open Tag"



2. Select the desired tag option.

   The corresponding tag icon appears next to the switch on the geographic display.

3. Select any switch, and click the Place Tags icon on the common toolbar.
4. Select the View button on the Tag pop-up.

   The Tag List display appears with the right-click applied tags listed.

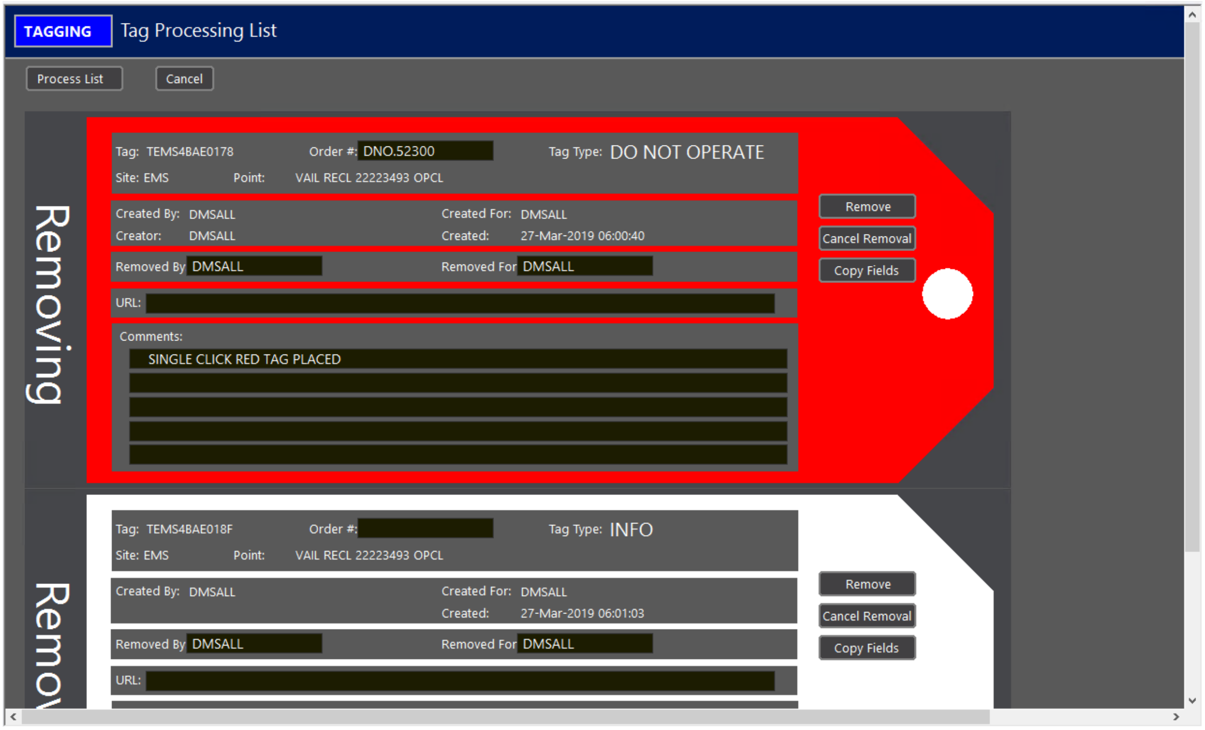
   The Comment field for the created tags shows "Single-Click [color] Tag Placed", where *[color]* is the color of the tag (for example, green).

4.11.3. Removing a Tag

To remove a tag, select the device (not the tag), and then click the Place Tag button on the device-specific toolbar. For SCADA devices, you need to click the Tag button on the SCADA pop-up. This displays the tagging pop-up.

1. Click the View or Remove Tag button.

   The tag processing list is displayed showing all tags placed on the selected point.



2. Identify the required tag on the list.
3. On the tag form, enter the required tag removal information.
4. On the specific tag form, click the Remove button.

   The tag is removed from the device and from the tag processing list.

5. On the tag processing list, click the OK or Cancel button to close the tag removing form.

> **Warning**
>
> Do not click the Process List button as this removes all tags in the list.

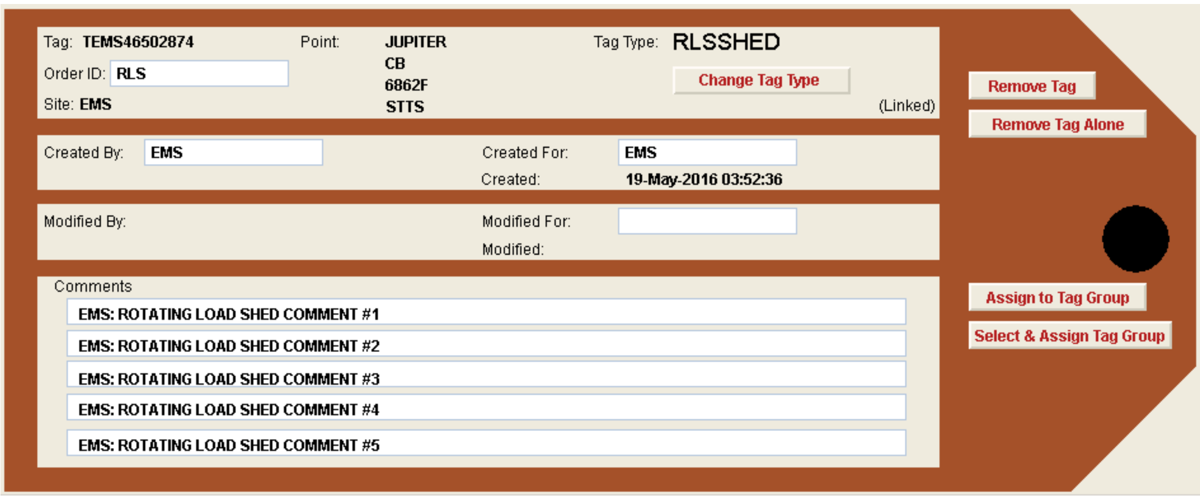
4.11.4. Types of Tags

Each tag type has certain characteristics. At the time a tag is created, it is assigned to a specific tag type. This allows a tag to inherit many of its characteristics from the tag type rather than requiring the operator to define every attribute of each tag.

- **Do Not Operate (Red):** This tag is typically placed on a device involved in a clearance. For example, a tag is placed on a feeder breaker while out on a clearance. When this tag is placed on a switchable device, all control commands, including the operation of manual switches from network displays, are inhibited and a warning message is displayed.
- **Hold Open (Green):** This tag is typically placed when a reclosing relay on a breaker is turned off for line maintenance. For example, a tag is placed on a feeder breaker for a crew issued an RC clearance for a pole replacement and facility transfer job. Only Open/Trip commands are

permitted for devices with Hold Open tags. Close commands are rejected and a warning message is displayed.

- **Info (White):** This tag is for more-permanent notes associated with a specific device. For example, a large industrial customer must be contacted before switching on a particular feeder. This tag imposes no control restrictions on the device. When a device with this tag is selected for operation, a message warning of the tagged condition is displayed.
- **Noshed (Blue):** This tag is available only for SCADA devices. It prevents the device from being operated in load shedding operations.
- **Restrict Control (Yellow):** This tag is available only for SCADA devices. It prevents the Open operations on a selected device, and a warning message is displayed.
- **Sticky (Pink):** This tag is available only for SCADA devices.
- **Hold Close (Light Green):** This tag is available only for SCADA devices.
- **Log Only (Light Orange):** This tag is available only for SCADA devices.
- **Restrict Close (Dark Blue):** This tag is available only for SCADA devices.
- **Restrict M Close (Dark Yellow):** This tag is available only for SCADA devices.
- **Restrict M Open (Yellow):** This tag is available only for SCADA devices.
- **Restrict Open (Cyan):** This tag is available only for SCADA devices.
- **Tag Inhibit (Brown):** This tag is available only for SCADA devices.
- **Tag TestMode (Blue):** This tag is available only for SCADA devices.
- **RLSSHED (Brown):** The Rotating Load Shed tag is only available for SCADA devices. It is automatically placed by the Rotating Load Shed (RLS) application to indicate that a device tripped during a load shed activity.



4.11.5. Managing Clearances

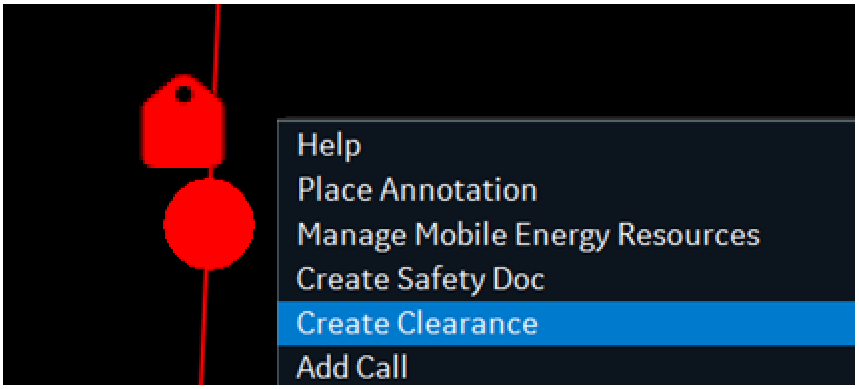
Note

> To use the clearance functionality, you must install the Clearance patch for Scada.

Use a clearance to prevent any operations on a tag (for example, to make sure a tag is not removed until all work on an associated device is completed).

To issue a clearance for a tag:

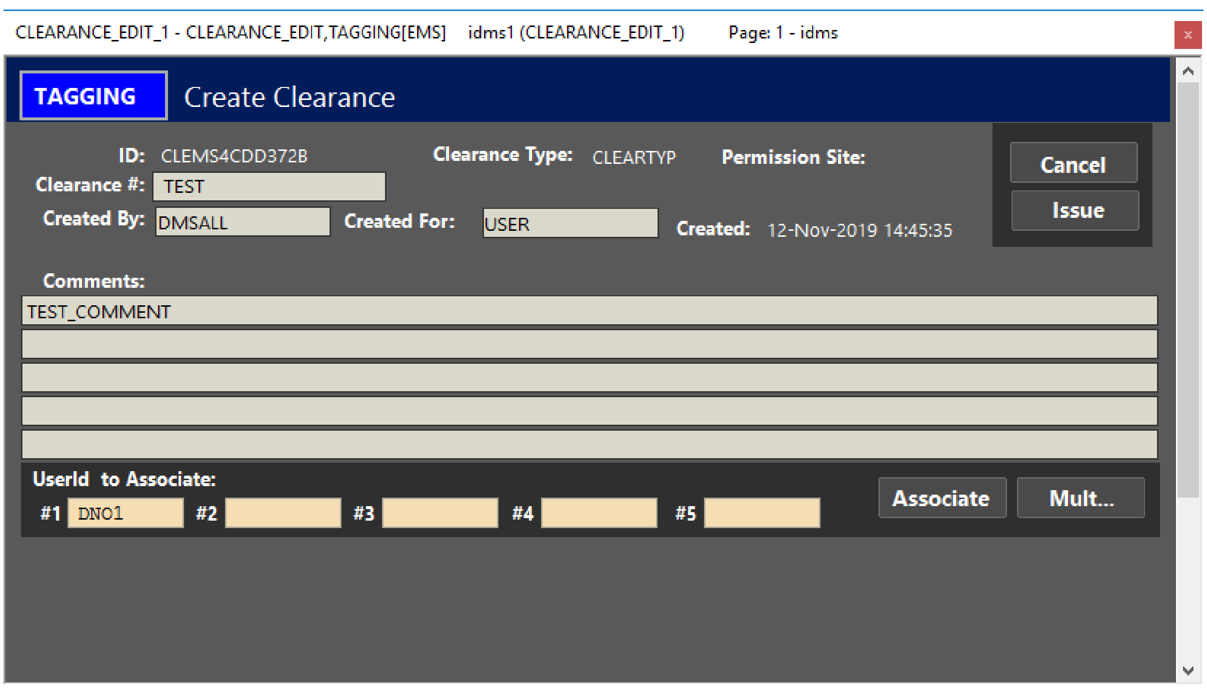
1. From the device toolbar or network view right-click menu, click Create Clearance.

   The Create Clearance display appears.

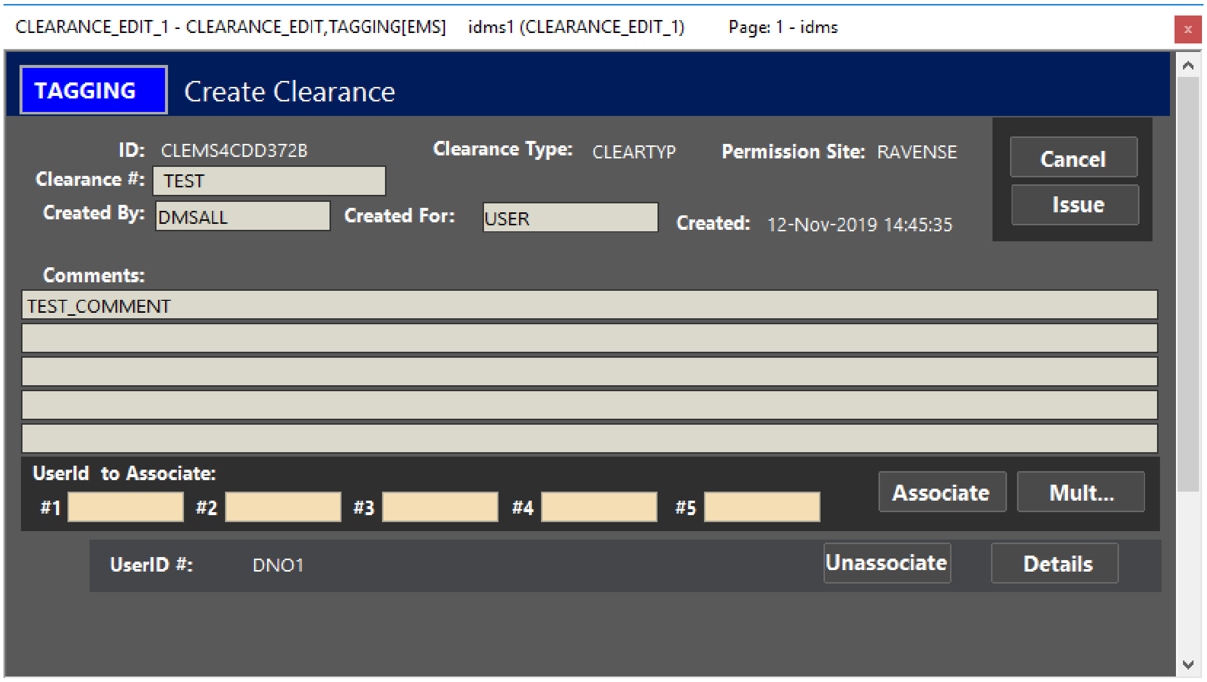
2. Complete the Clearance Number and Created For fields.
3. Enter the tag number in a User ID to Associate field.

   To associate more than five tags with a clearance, click Mult.

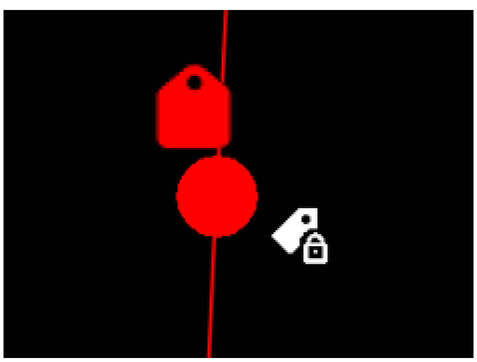
4. Click Associate.

The clearance is associated with the specified tag. The clearance entry appears below the User ID to Associate field.



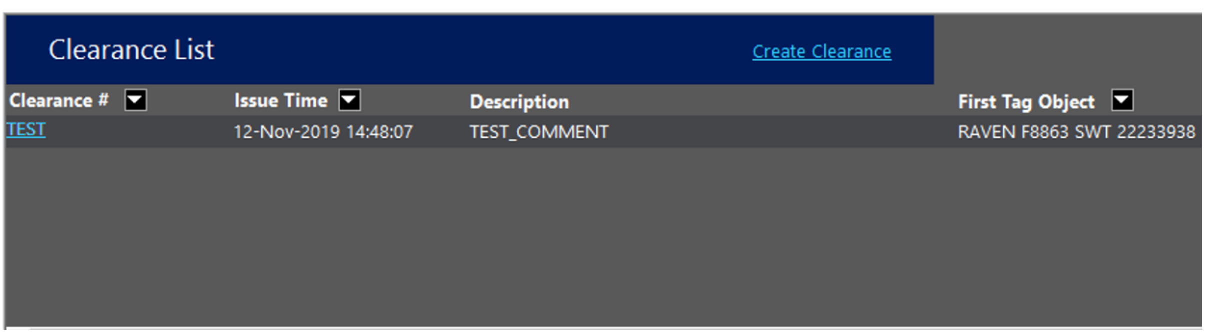
5. Click Issue.

   The clearance is issued. The clearance icon appears near the tag icon.



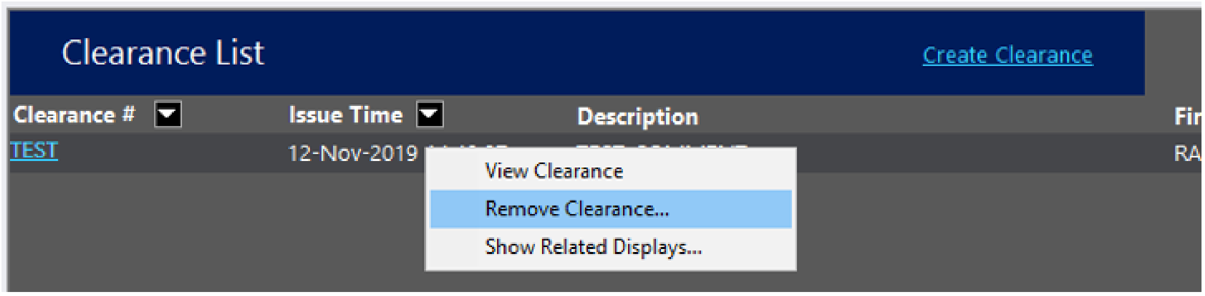
To review the list of clearances:

- Open the Clearance List display (SCADA Applications menu > Tagging > Clearance List).



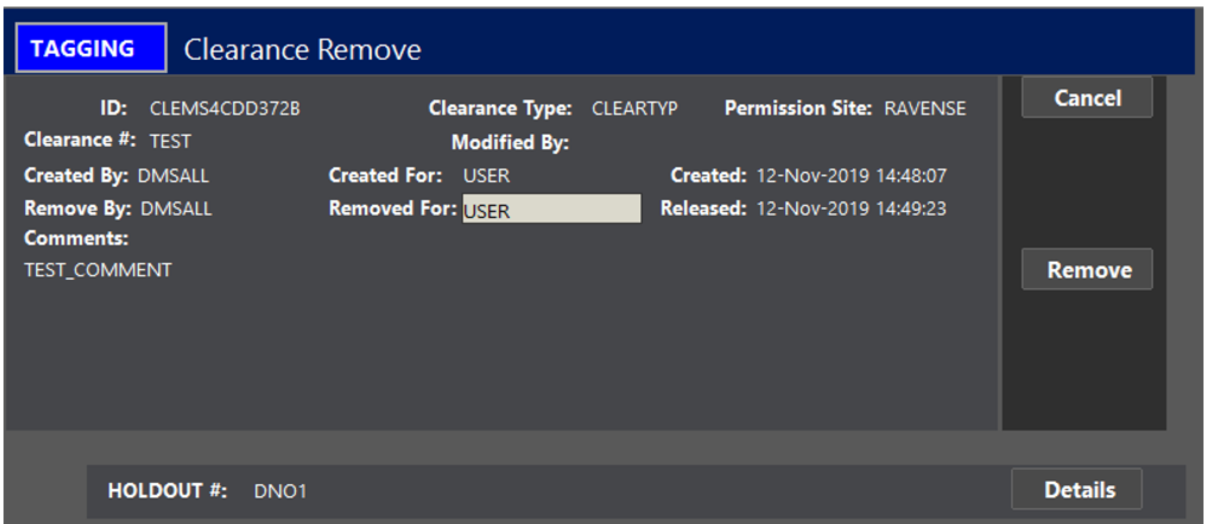
To remove a clearance:

1. Open the Clearance List display.
2. Right-click a clearance entry.
3. Click Remove Clearance.



The Clearance Remove display appears.

4. Enter the Removed For field. (This field must be set with the same value as the Created For field.)



5. Click Remove.

   The clearance is removed.

4.12. Customer Advanced Metering Infrastructure (AMI)

In utilities that have an AMI program, the operator can control customer smart meters remotely.

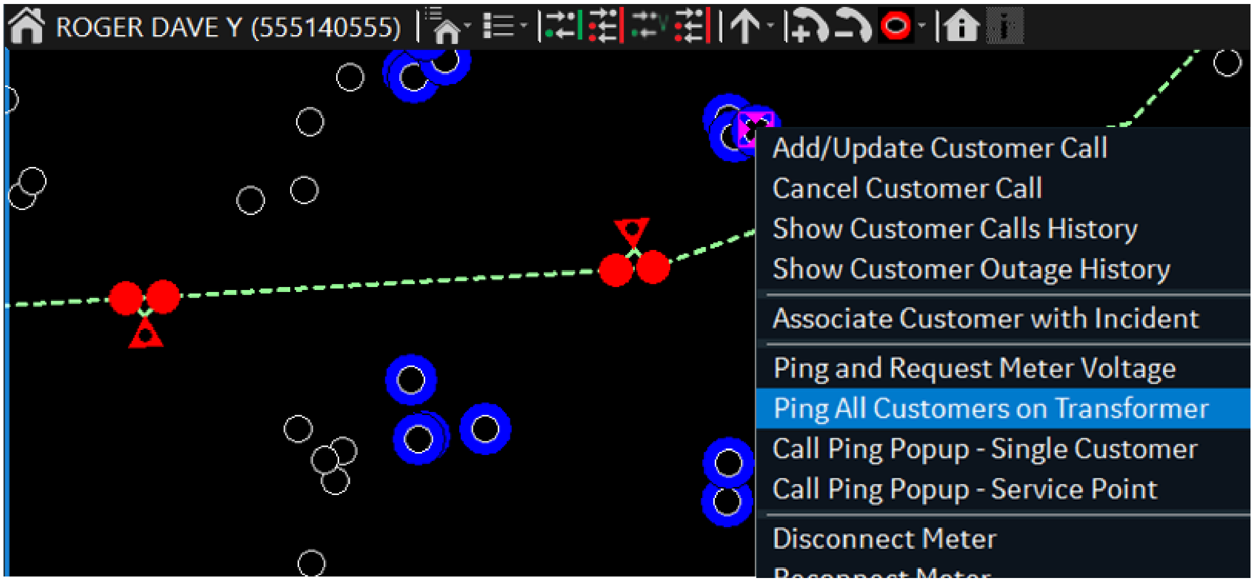


Figure 181. Pinging All Customers Connected to the Transformer

You can perform the following metering operations using the customer toolbar or the context menu:

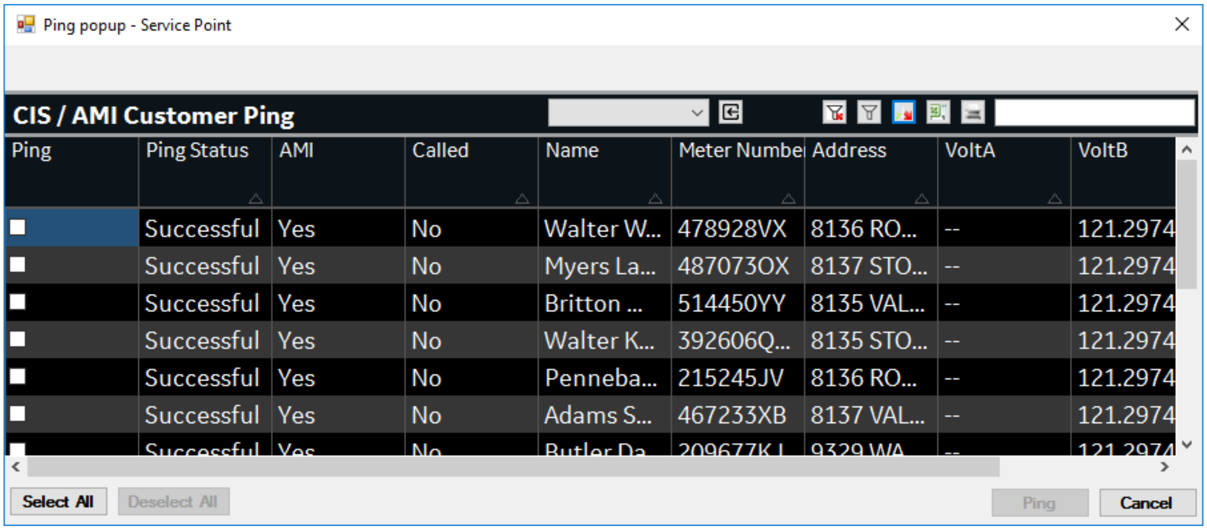
- **Ping and Request Meter Voltage:** Pings the selected customer and requests the meter voltage value.

  On the network view, the ping status is shown as follows:

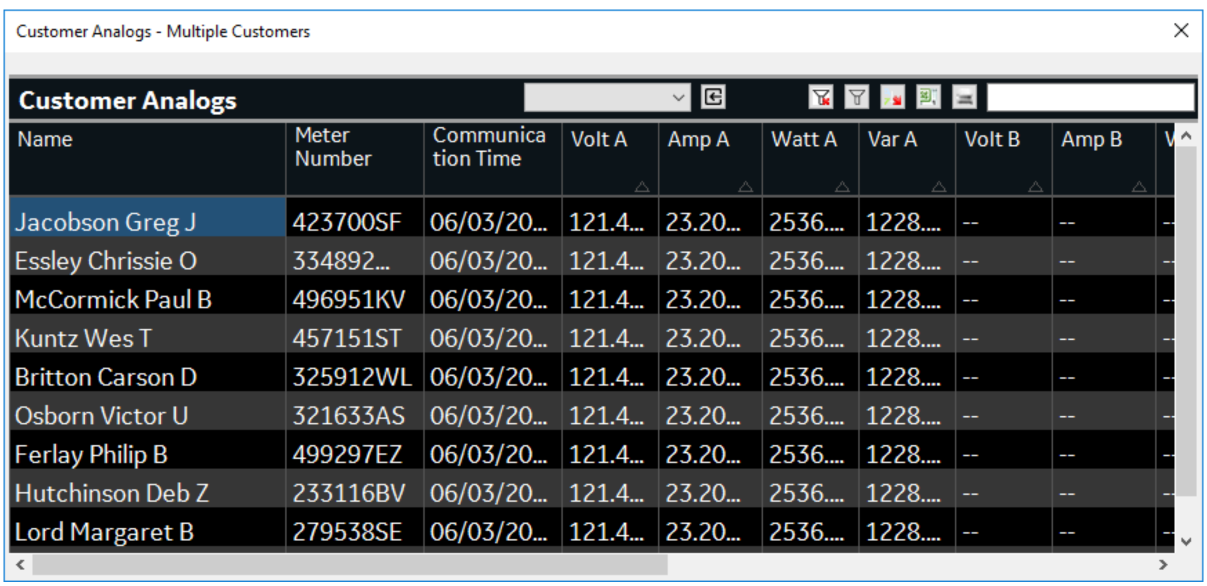
  - A blue circle around the customer symbol indicates that a ping is pending.
  - A green circle indicates that a ping succeeded.
  - A red circle indicates that a ping failed.
  - A gray circle indicates that the ping is undetermined or not tried.
  - A yellow circle indicates a ping error. The error appears if the AMI connection is down, a customer is not equipped with a smart meter, the meter number is invalid, or the meter is temporary out (for example, for servicing).

    The ping color is configurable in the Viewer Configuration Editor.

- **Ping All Customers on Transformer:** Automatically pings all the customers associated with the transformer.
- **Disconnect Meter:** Disconnects the customer's meter.
- **Reconnect Meter:** Reconnects the customer's meter.
- **Call Ping Pop-Up – Single Customer:** Calls up the CIS/AMI Customer Ping display, which shows customer meter data and allows you to ping the selected customer.
- **Call Ping Pop-Up – Service Point:** Calls up the CIS/AMI Customer Ping display, which shows customer meter data and allows you to ping the customers connected to the same transformer.



- **Scan AMI Analogs:** Sends the command for detecting load side voltage, current, and power values for the selected customer.
- **Scan AMI Analogs – Multiple Customers:** Sends the command for detecting load side voltage, current, and power values for the customers connected to the same transformer.
- **Show Analogs – Single Customer:** Calls up the Customer Analogs display, which shows the analog meter data for the selected customer. You must scan AMI analogs before showing them.
- **Show Analogs – Multiple Customers:** Calls up the Customer Analogs display, which shows the analog meter data for the customers connected to the same transformer. You must scan AMI analogs before showing them.



4.13. Display Annotations

Annotations are symbols that can be applied to the network view to indicate the location of a non-electrical item (emergency services, a key customer, so on) or an incident (broken pole), as shown in image below. The placement of these symbols, along with the associated descriptions, provides an effective means of keeping all operators informed of the current situation on the network. Annotations can be placed anywhere on the display.

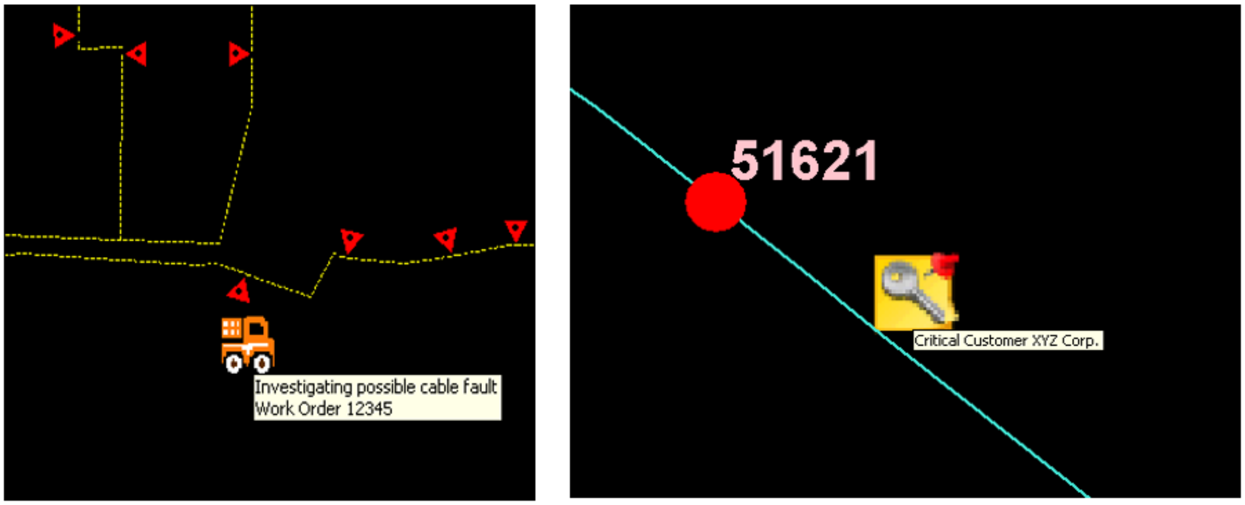


Figure 182. Examples of Annotations

To place an annotation at any point on the network view, position the mouse cursor and do one of the following:

- Press Ctrl+A to display the annotation pop-up dialog box.
- On the right-click menu, select Place Annotation.

  This opens the Annotation Popup window, as shown in image below.

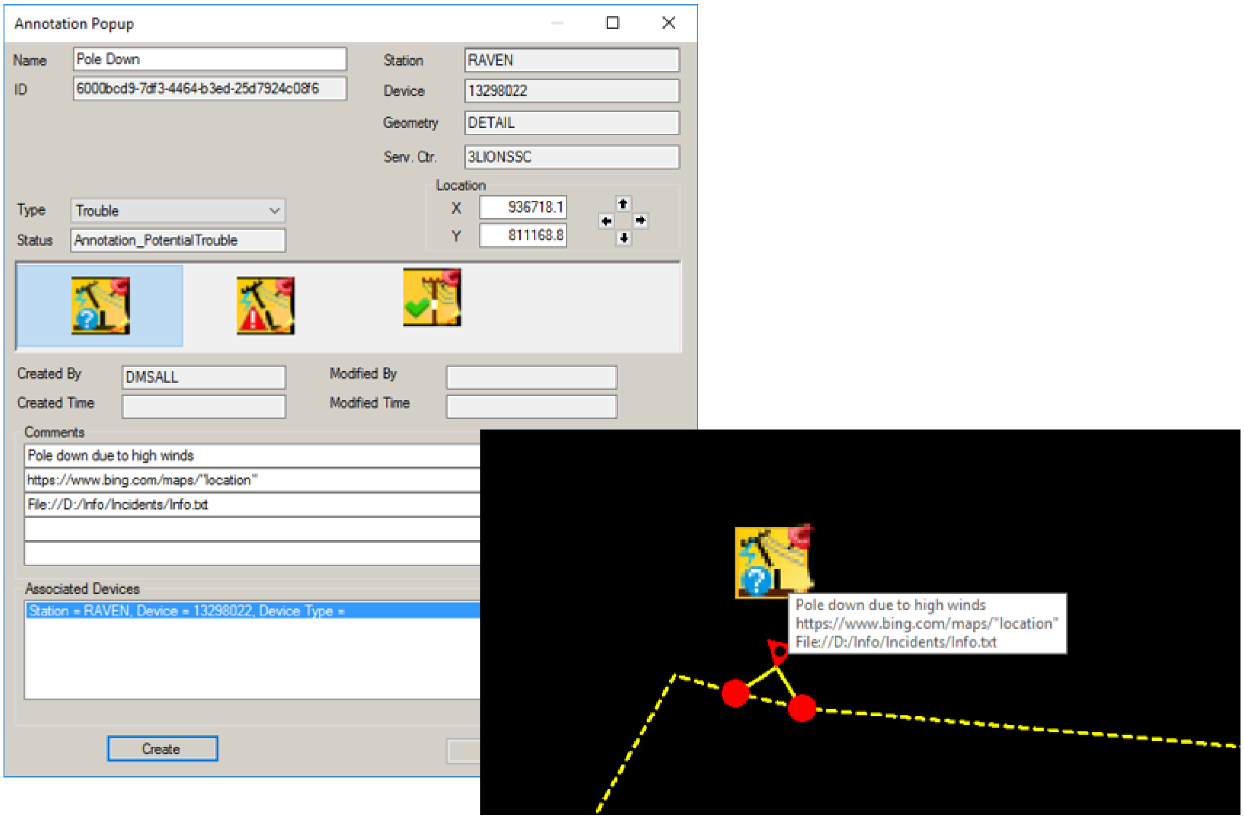


Figure 183. Annotation Pop-up Window and an Example Annotation

Select the annotation type from the drop-down menu, and then select the required annotation from the row of icons displayed. The actual placement location of the annotation icon can be adjusted by shifting the X and Y coordinates. Click any of the four buttons to adjust the coordinate values. Click the Create button to place the annotation on the network view.

Hovering the mouse cursor over the annotation icon displays the comment text.

To modify or remove an existing annotation, select the icon on the network view, and then click the Navigate to Related Display button on the device-specific toolbar. This displays the annotation pop-up dialog box. Change the details as required, or click the Remove button to remove the annotation.

Once an annotation has been placed, it can also be repositioned by using the move mode.

> **Note**
>
> You can navigate to an internet URL or local file from the Annotation pop-up. When a valid URL format is entered in one of the comment fields, the Globe button becomes enabled. Clicking the Globe button opens this URL in a new instance of the default web browser.

4.13.1. Reviewing Display Annotations

A tabular summary of currently applied annotations is available from the RTDMS menu.

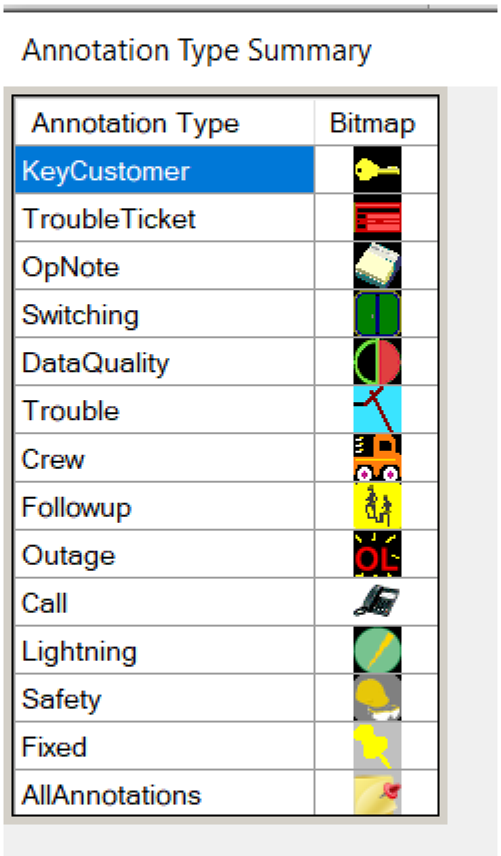


Figure 184. Annotation Type Summary Dialog Box

Click the required annotation type to display a list of annotations. Each annotation is shown with a Locate button to allow navigation back to the geographic network view, centered on the annotation.

4.13.2. Copying and Pasting Annotations

To copy an annotation on the geographic view:

1. Right-click an annotation.
2. Select "Copy Annotation".

   The annotation is copied.

To paste an annotation on the geographic view:

1. Right-click on the geographic view.
2. Select "Paste Annotation".

The annotation is pasted.

> **Note**
>
> You can paste an annotation multiple times.

A new annotation is created based on the original annotation type. The name of the new annotation is the original annotation name suffixed with "-1" (or incremented by 1 if pasting multiple times).

4.14. Moving Objects

This section describes how to move objects on the network view.

> **Note**
>
> Moving a customer or a temporary object on the network view removes all selection marks on devices and incidents.

4.14.1. Moving a Temporary Object

Temporary objects (that is, annotations, tags, jumpers, mobile substations, incidents, crews, damage assessment locations, and unassociated calls) can be repositioned on the screen after they have been placed. This provides a simple means of overcoming the problem of objects obscuring parts of the network or other objects.

- To move a single object, right-click the object and select Move Object.
- To invoke the move mode for all temporary objects, press Ctrl+M.

  The toolbar is replaced by a yellow bar with a Cancel button, and the objects that are available to be moved are highlighted with a large rectangular magenta halo.

To move an object, click it and hold the mouse button while dragging the object to the desired new location. In the move mode, when you release the mouse button, the moved objects are shown in both the new (cyan halo) and original (magenta halo) locations, with a line connecting the locations. The move toolbar is now shown in a salmon/orange color with buttons to Accept or Cancel. Clicking the Accept button updates the display to place the object at the new location and cancel the move mode.

> **Tip**
>
> More than one object can be moved while the move mode is enabled. Click and drag each object, and then click the Accept button.

> **Note**

Even if the incident has been moved manually by the operator, the incident symbol moves to the nearest open device after the partial restoration of customers.

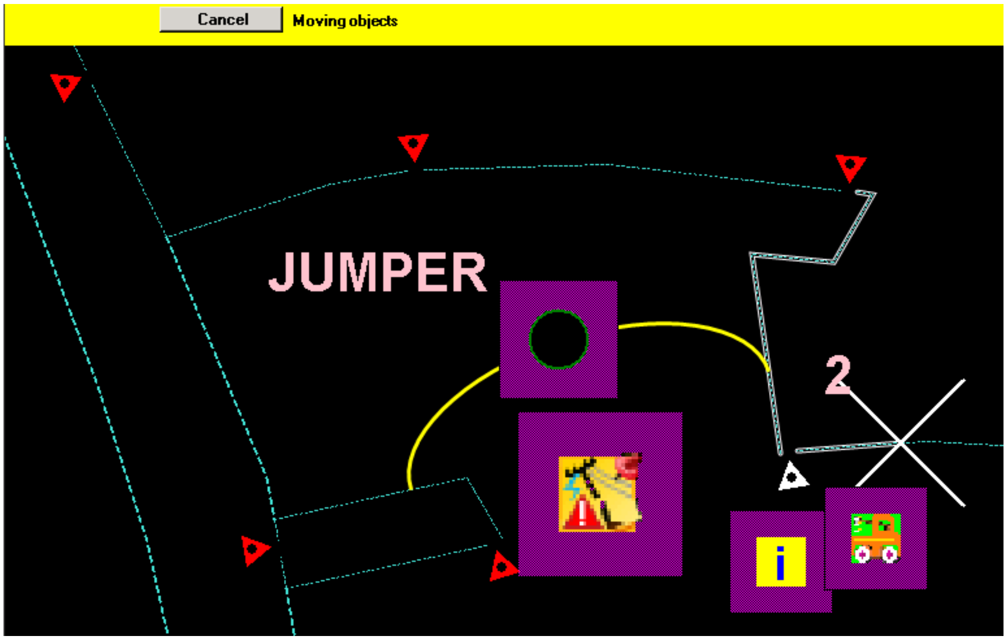


Figure 185. Move Mode Enabled

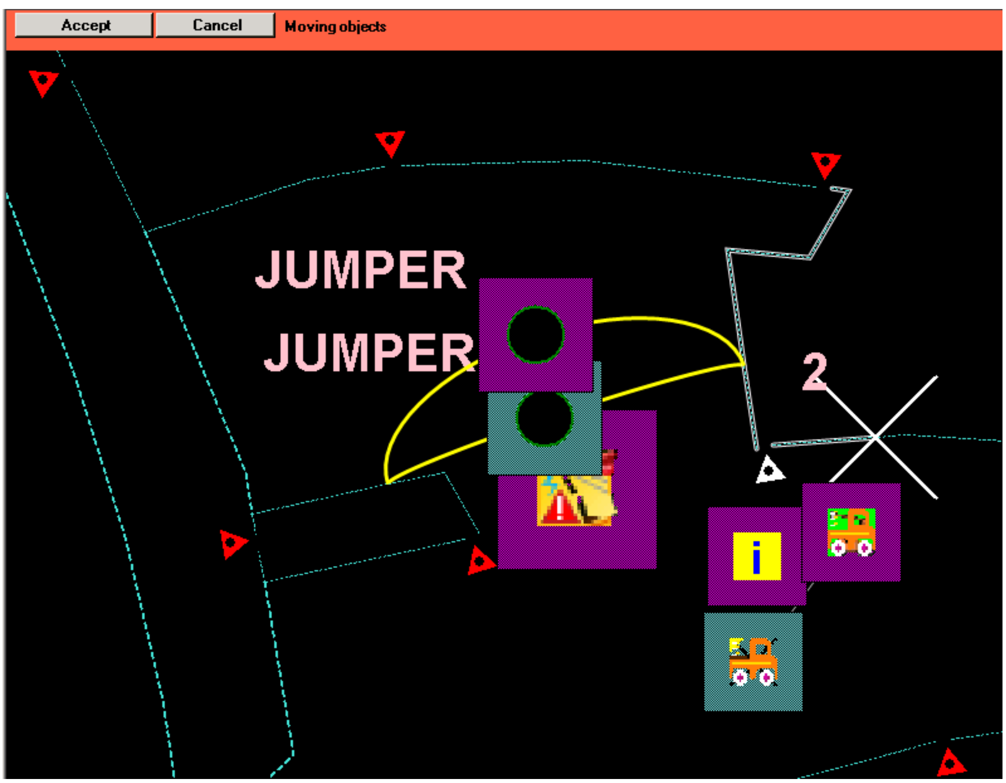


Figure 186. Objects Dragged to New Positions

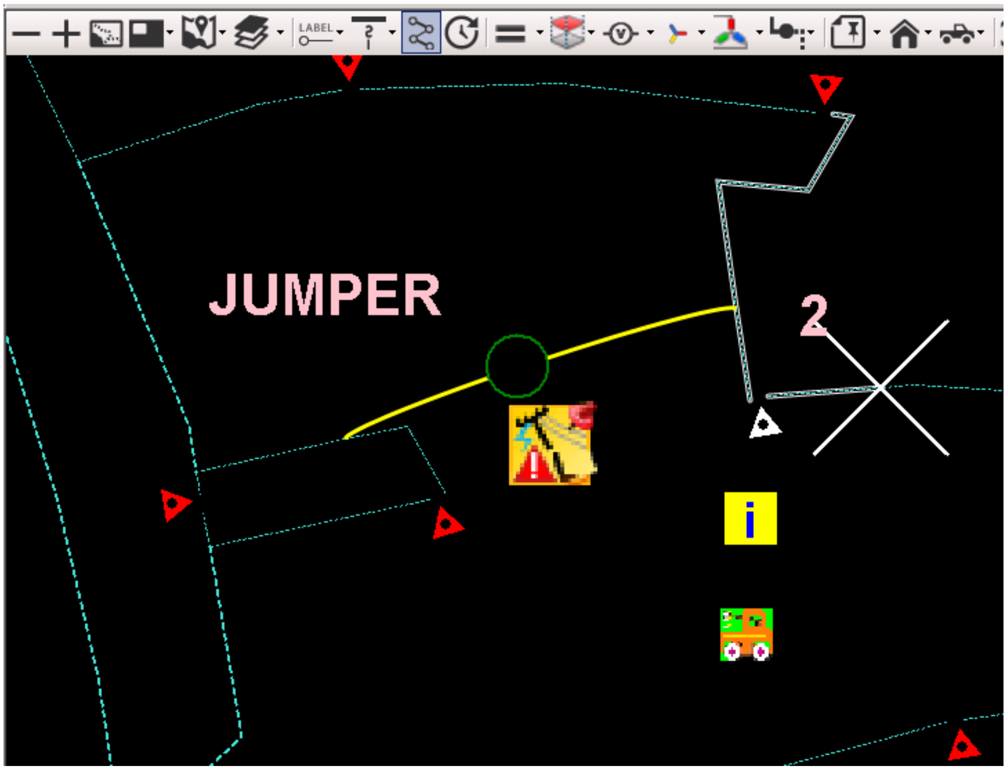


Figure 187. Objects Moved

4.14.2. Moving a Customer

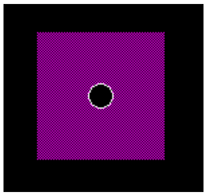
The operator can dynamically change the spatial location of a customer. Moving the customer on the geographic view also updates the X and Y coordinates in Customer Service System (CSS) and Geographic Information System (GIS).

To change the location of a customer:

1. Select a customer on the geographic view.
2. Right-click the customer, and then click Move Customer.

> **Note**
>
> The Move Customer option does not exist in the default configuration. It can be added manually, if needed.

The customer symbol is highlighted with a large rectangular magenta halo, showing that it is ready to be moved.



3. Drag the customer symbol to the new location.

In addition, you can view the list of moved customers and undo moving a customer. These features are available as part of the optional Outage Management functionality. For more information, refer to the *Outage Management System User Guide*.

4.15. Calculating Distance

The network distance function is implemented locally within the ADMS client. It has no effect on the operations model or other clients viewing the network. It enables dispatchers to evaluate distances between network components, such as lines, switches, and sensor locations. This feature can also be used when analyzing fault locations.

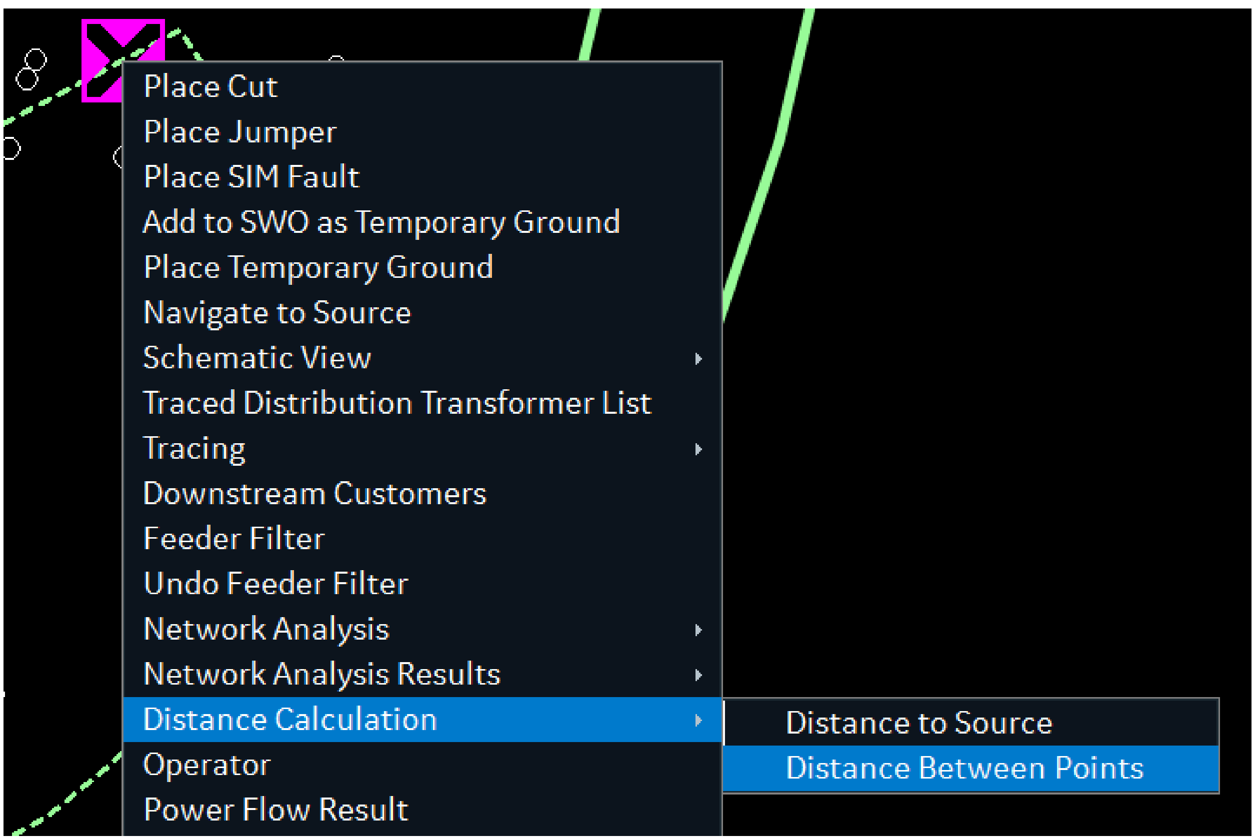


Figure 188. Distance Options

All distance calculations are phase specific because a multi-phase line may branch at some point leading to a different distance for each phase. Only the modeled line lengths are included in the total distance. Permanent jumpers, switches, and other components do not count towards the total length. The network path between selected points is not highlighted.

Distance calculations are available for line, switch, recloser, sectionalizer, sensor, and temporary jumper.

Distance calculation units are feet for station models that support imperial units and meters for station models that support metric units. Miles and kilometers distance units are not supported.

Table 8. Network Distance Type and Calculation

Distance Type	Function
Distance to Source	Calculates the network distance from the selected point to the source. Only modeled line lengths are included. This option is not available for de-energized components.
Distance to Point	Calculates the network distance between two points on the same network. If the selected points are not on the same connected network, no distance is calculated.
Straight Line Distance	Calculates the straight line distance between two selected points. (As the crow flies.)

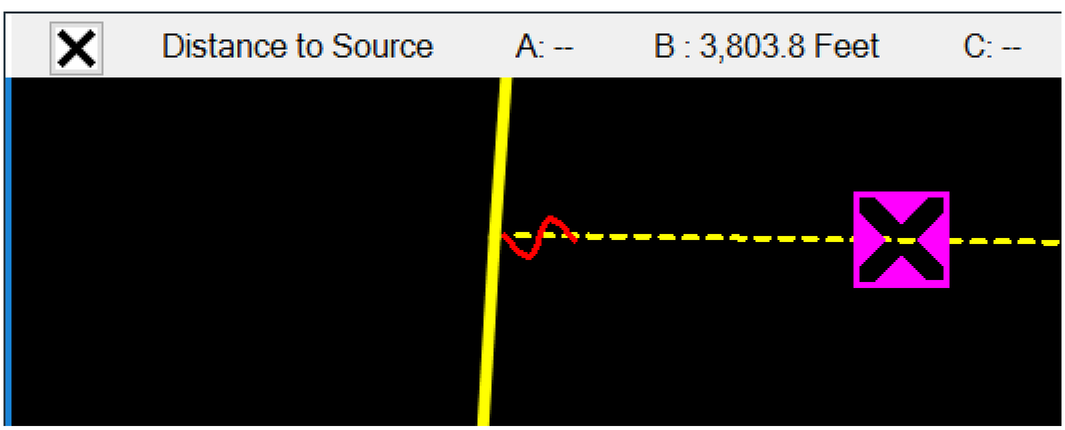
To calculate distance from the selected point to the source:

1. Select a component or a point on a line.

   The point is highlighted with a magenta cursor.

2. Select the Distance to Source option from the component's context menu.

   The Distance toolbar appears showing the distance results per phase.



> **Tip**
>
> Selecting another point on the network updates the values on the toolbar.

3. Click Close to close the distance calculation function.

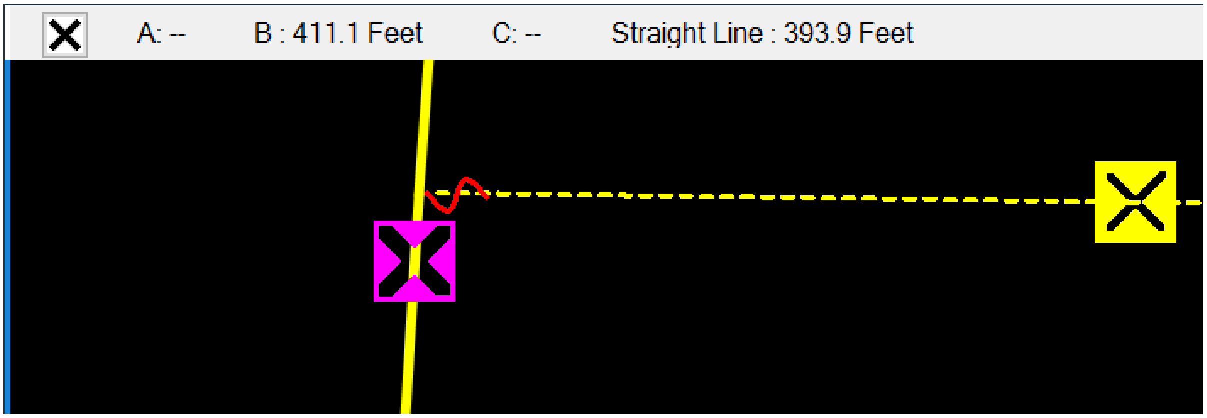
To calculate network distance between points, including the straight-line distance:

1. Select a component or a point on a line.

The point is highlighted with a magenta cursor.

2. Select the Distance Between Points option from the component's context menu.
3. Select the second component or point on a line.

   The Distance toolbar appears showing the network distance between the selected points per phase. In addition, the straight-line distance between the points is shown.



> **Tip**
>
> Selecting another point on the network updates the values on the toolbar.

4. Click Close to close the distance calculation function.

> **Note**
>
> Due to a limitation in determining partial line lengths, the distance calculated between a selected line and an immediately connected (non-line) component is not accurate. To obtain this distance, select both points on the line itself.

4.16. Viewing Static and Dynamic Schedules

The instantaneous real and reactive load values for every load point in a distribution substation are not known, therefore load schedules are used to estimate the load value for different times of the day, different day types, and different seasons. In addition, generation by generators, distributed energy resources, and loads with embedded generation also must be considered.

The DNAF load allocation algorithm adjusts the load values to match the flow measurements. A reasonable initial estimate of the peak or average load and generation values (schedules) can provide a good starting point for the load allocation algorithm and increase the likelihood of load allocation convergence.

The following schedule types are supported for load and generation objects:

- **Static Schedules:** These schedules are specified during modeling. They are not dynamically updated and therefore provide rough load and generation estimates.

- **Dynamic Schedules:** These schedules are periodically loaded by an external system using the DMS Interface server functionality. Dynamic schedules are calculated based on different data, such as solar panel state, weather forecast, and solar irradiance. This helps to obtain an accurate generation forecast for a certain period.

  Using dynamic load profiles, you can calculate power flow for a particular time in the future using the study-mode DNAF functionality.

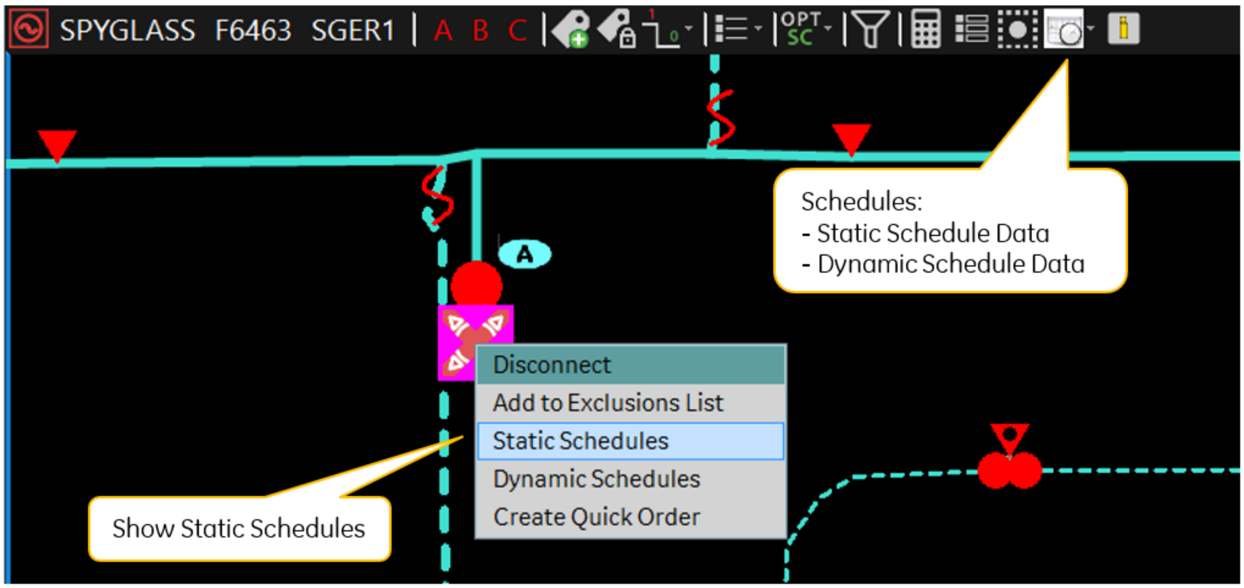
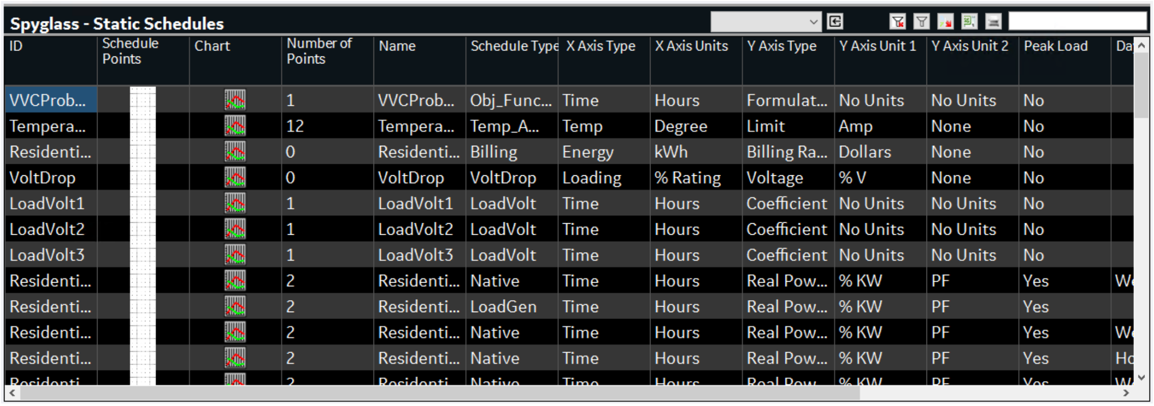


Figure 189. Static and Dynamic Schedule Options

> **Note**
>
> Static and dynamic schedule data is shown only if provided during modeling or loaded via the DMS web service. Otherwise, the displays and charts are blank.

To view schedule data, use the following toolbar and context menu options:

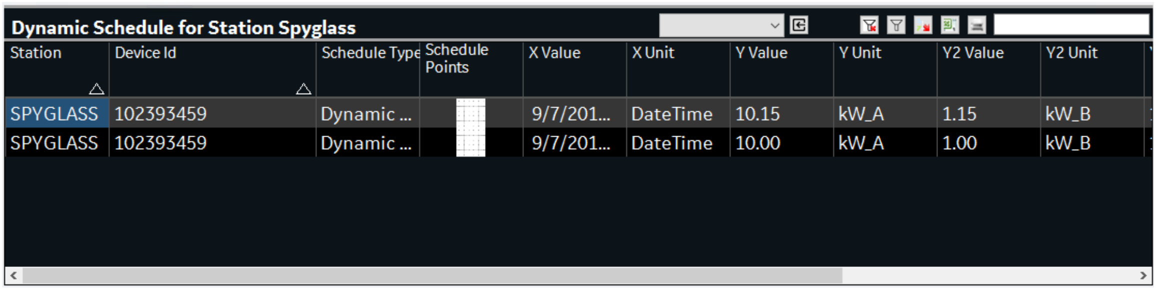
- **Schedules**
  - **Static Schedules:** Shows a tabular display with static schedule data. From the Static Schedules display, you can show static schedule charts.



Spyglass - Static Schedules

ID	Schedule Points	Chart	Number of Points	Name	Schedule Type	X Axis Type	X Axis Units	Y Axis Type	Y Axis Unit 1	Y Axis Unit 2	Peak Load	Da
VVCProb...			1	VVCProb...	Obj_Func...	Time	Hours	Formulat...	No Units	No Units	No	
Tempera...			12	Tempera...	Temp_A...	Temp	Degree	Limit	Amp	None	No	
Residenti...			0	Residenti...	Billing	Energy	kWh	Billing Ra...	Dollars	None	No	
VoltDrop			0	VoltDrop	VoltDrop	Loading	% Rating	Voltage	% V	None	No	
LoadVolt1			1	LoadVolt1	LoadVolt	Time	Hours	Coefficient	No Units	No Units	No	
LoadVolt2			1	LoadVolt2	LoadVolt	Time	Hours	Coefficient	No Units	No Units	No	
LoadVolt3			1	LoadVolt3	LoadVolt	Time	Hours	Coefficient	No Units	No Units	No	
Residenti...			2	Residenti...	Native	Time	Hours	Real Pow...	% KW	PF	Yes	W
Residenti...			2	Residenti...	LoadGen	Time	Hours	Real Pow...	% KW	PF	Yes	
Residenti...			2	Residenti...	Native	Time	Hours	Real Pow...	% KW	PF	Yes	W
Residenti...			2	Residenti...	Native	Time	Hours	Real Pow...	% KW	PF	Yes	H

- ◦ **Dynamic Schedules:** Shows a tabular display with dynamic schedules data.



Dynamic Schedule for Station Spyglass

Station	Device Id	Schedule Type	Schedule Points	X Value	X Unit	Y Value	Y Unit	Y2 Value	Y2 Unit
SPYGLASS	102393459	Dynamic ...		9/7/201...	DateTime	10.15	kW_A	1.15	kW_B
SPYGLASS	102393459	Dynamic ...		9/7/201...	DateTime	10.00	kW_A	1.00	kW_B

- **Show Schedule Chart:** Shows the dynamic schedule chart.

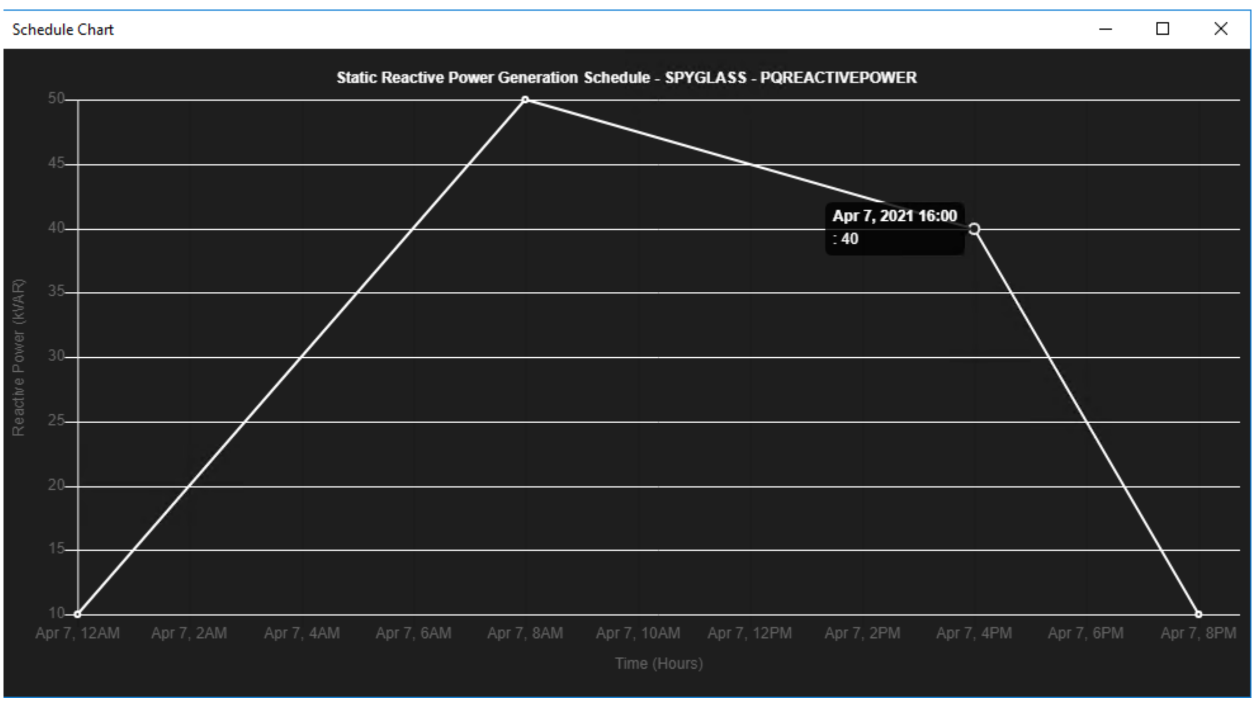


Figure 190. Static Schedule for a Solar Panel

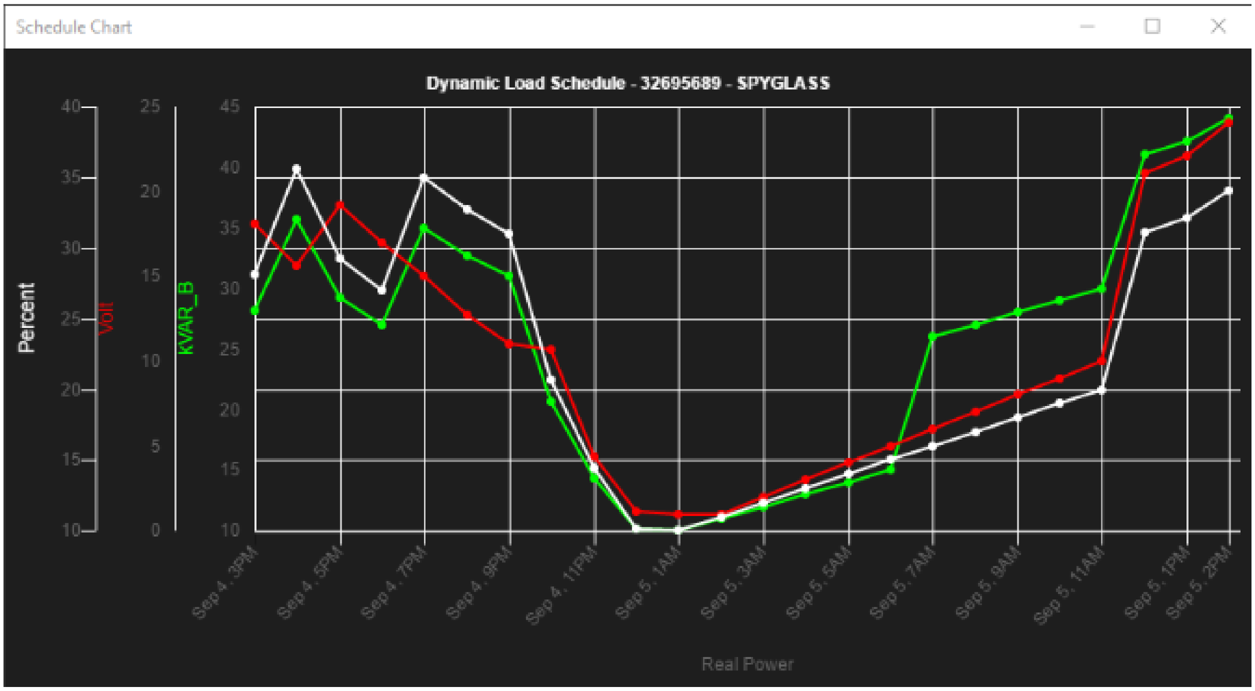


Figure 191. Dynamic Schedule for a Distribution Transformer

5. Working in Study Mode

Study mode allows the network view and network analysis functions to be run on a private instance of the network model on the client workstation.

Study mode allows operators to perform detailed "what if" analysis of proposed network configurations and switching procedures.

Study mode uses the same static models as real-time mode, but with a separate instance of the dynamics. This removes the requirement to separately set up and maintain the study mode environment.

Changes made to the study mode network model are applied only to the user's workstation and are not visible to others. Both real-time and study modes can be run concurrently on the same workstation, allowing an operator to perform a study immediately prior to executing the operation in the real-time environment.

Study mode provides the same toolbars, menus, and displays as real-time operation. Switching can be performed in study mode in the same manner as on the real-time system, and loading conditions can be easily modified.

Typically, study mode network displays have a gray background with pop-ups that have a yellow border, as shown in image below. In addition, study mode grid tabular displays have a yellow background.

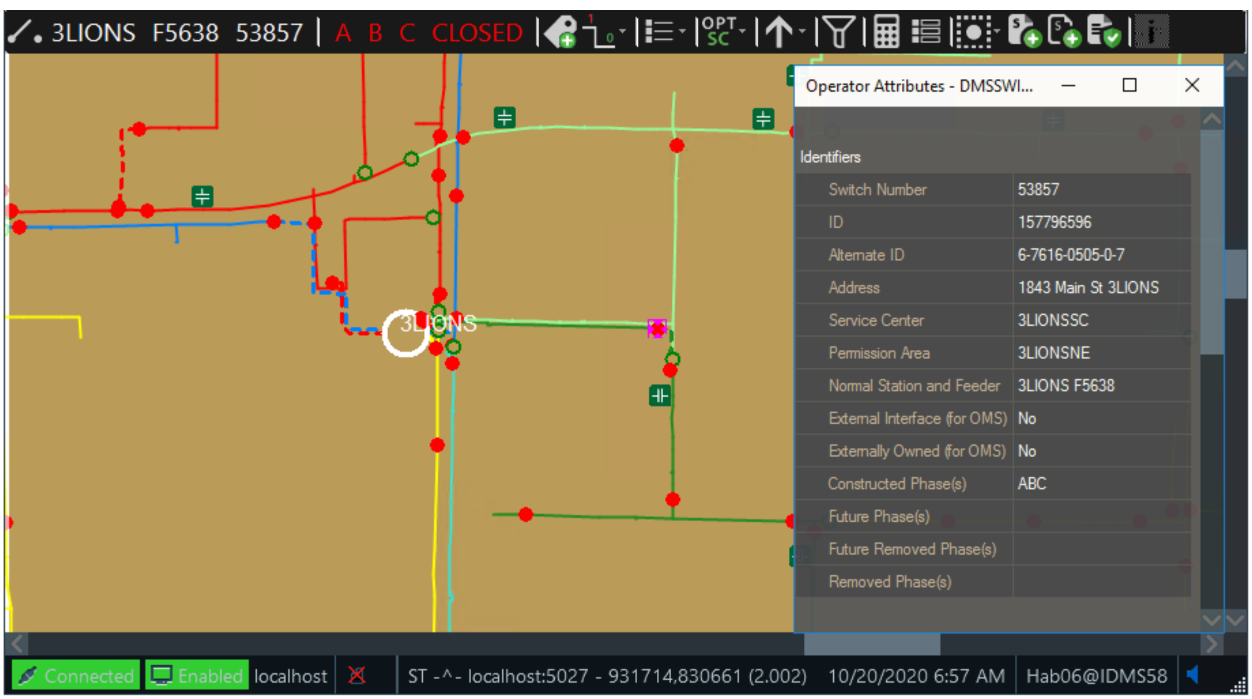


Figure 192. Study Mode Display with Operator Attributes Window

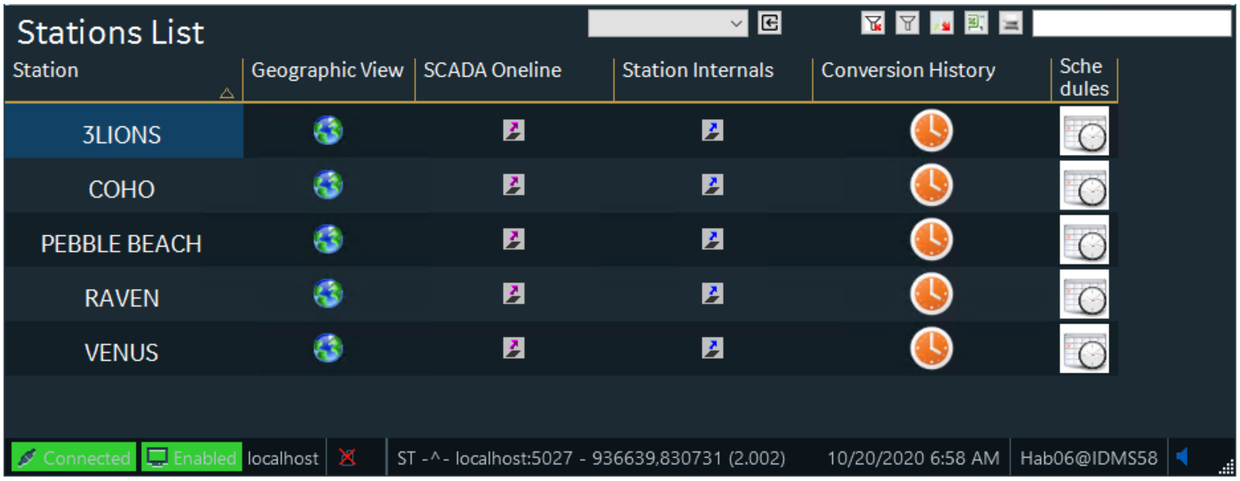


Figure 193. Study Mode Grid Tabular Display

The study environment can be initialized directly from the real-time system, capturing the current status, or it can be initialized to the nominal state or to a previously saved study case.

To use study mode, the local study server must be started, the network model files loaded, and the network dynamics initialized to the required state. These functions are performed using the Study Case Setup dialog.

5.1. Starting the Local Study Server

The local study server is not started by default. It must be started manually on the workstation.

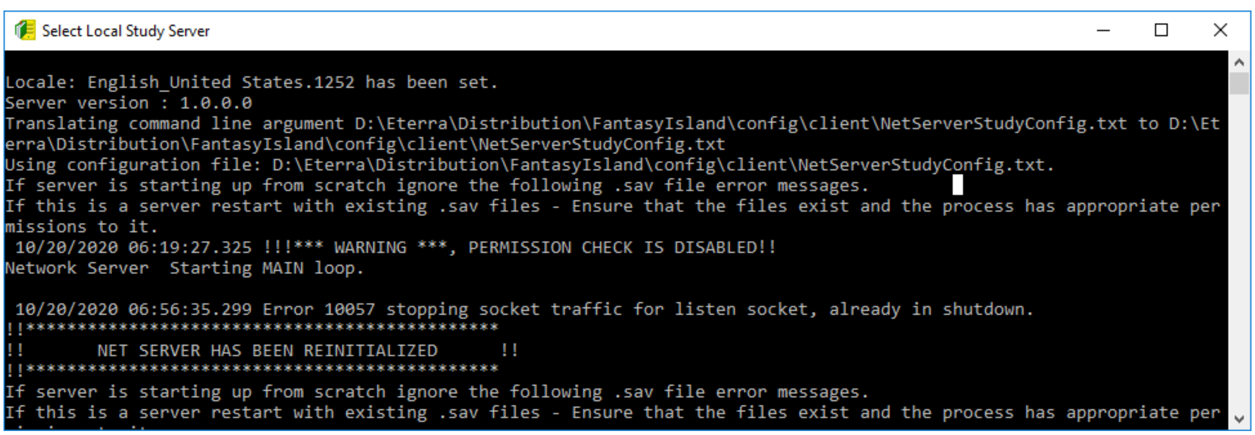


Figure 194. Local Study Server Command Prompt Window

To start the local study server:

- From the Windows Start menu, click Programs > Eterra > Distribution > Client > Local Study Server.

  A command prompt window opens, showing the status messages as the study server starts up. To avoid screen clutter, it is best to minimize this window while running study mode.

To shut down the study server, close the command prompt window.

5.2. Study Case Setup

You can initialize the study mode environment from the nominal or real-time system state, or from a savecase. The Study Case Setup dialog allows you to quickly create the required environment.

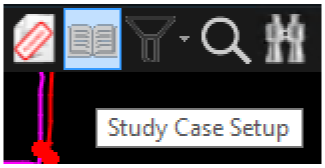


Figure 195. Study Case Setup Icon

The Current Study Case tab of the Study Case Setup dialog indicates the current study case and shows the log of creating the study case.

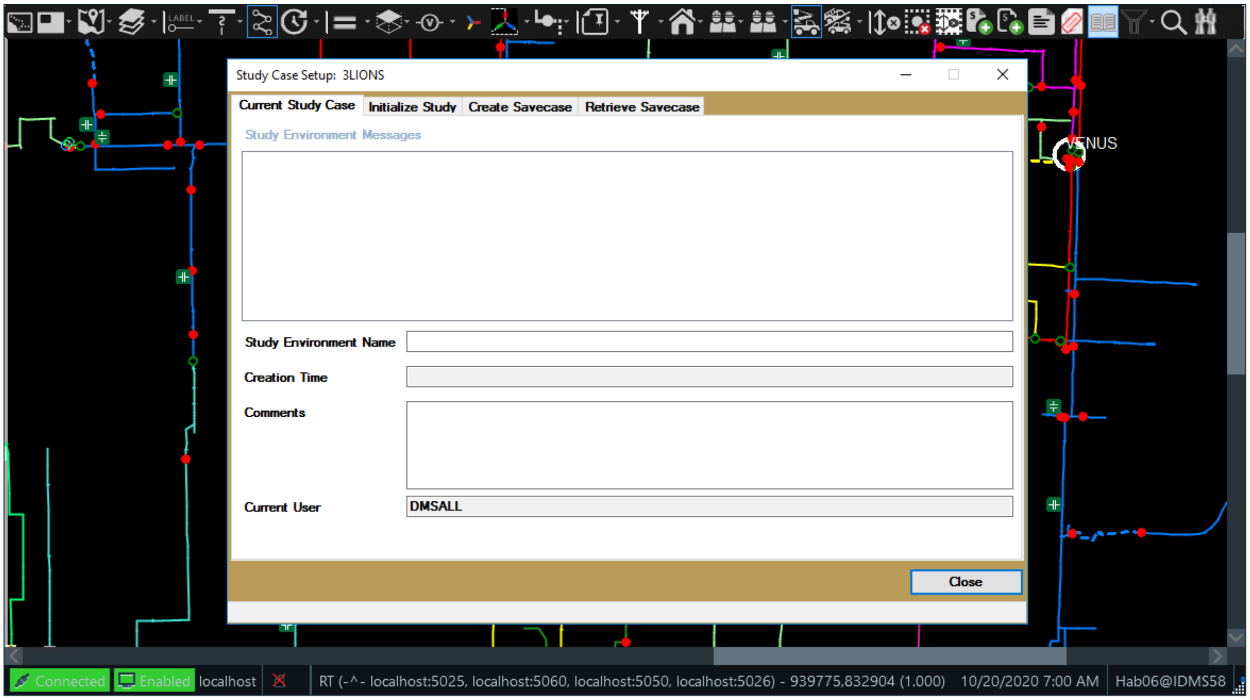


Figure 196. Study Case Setup - Current Study Case Tab

The Initialize Study tab of the Study Case Setup dialog includes options to initialize the study environment from the real-time or nominal system state and to load the selected station and the neighboring stations that make up the partial model view.

To initialize the study case:

1. Open the Initialize Study tab.
2. Select a substation from the Station drop-down list.

3. Optionally, you can:
    - Include the mobile substation usage (mobile dynamics and model files will be copied).
    - Include dynamic schedule data in the study case. The following options are available:
        - **None:** Do not copy any dynamic schedule data.
        - **Hour:** Copy the schedule data for one hour. This data is extended to be used for each hour, day, month, and year.
        - **Day:** Copy the schedule data for one day. This data is extended to be used for each day, month, and year.
        - **Month:** Copy the schedule data for one month. This data is extended to be used for each month and year.
        - **Year:** Copy the schedule data for up to a year. This data is extended to be used for each year.
        - **All:** Copy all relevant schedule data and use it as is.
4. Click one of the following buttons:
    - **Initialize to Nominal:** Initializes the study environment to the nominal state of the system. These functions create a partial model view centered around the selected substation.
    - **Initialize to Real-Time:** Initializes the study environment to the real-time state of the system.

    Both functions create a partial model view centered around the selected substation (the selected substation and the neighboring substations are loaded). When initializing to real-time mode, the partial model view also includes substations that are connected to any neighboring substations via a jumper (open or closed) or a boundary switch (completely or partially closed).
5. After you initialize the study environment, you can click the Load or Unload Adjacent Stations button to select substations that need to be loaded or unloaded. For more details, see Loading or Unloading Adjacent Stations.

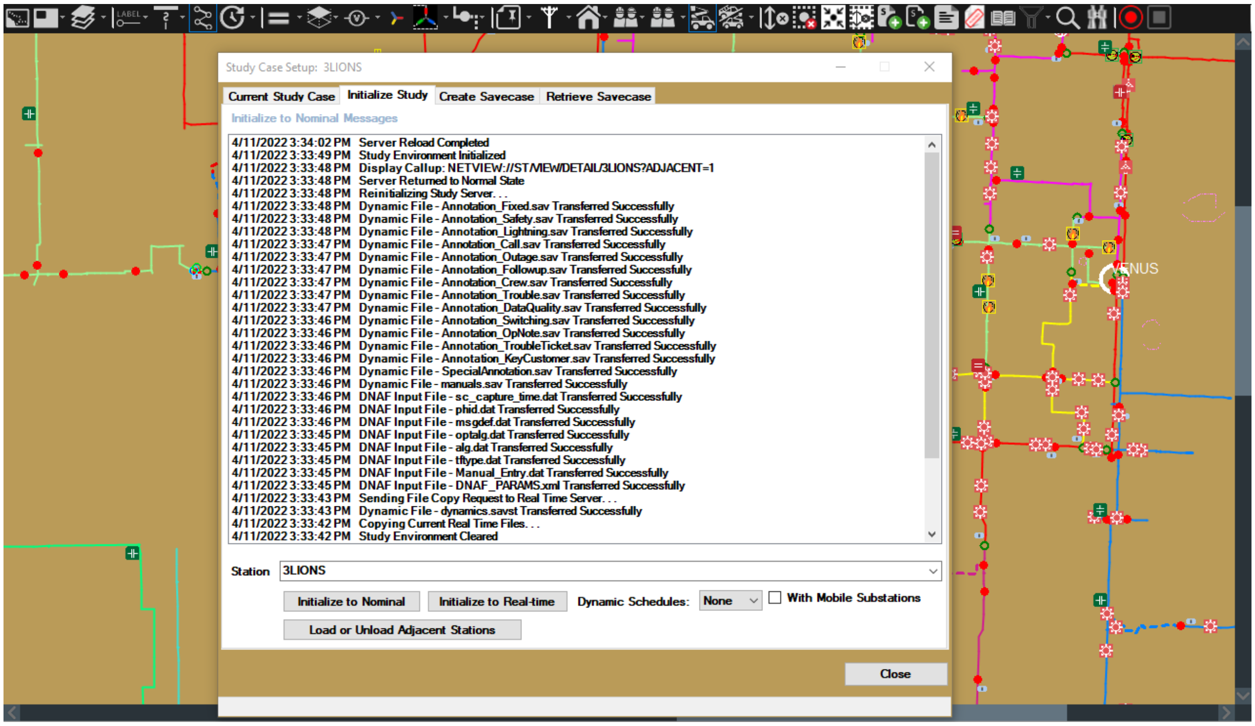


Figure 197. Study Case Setup- Initialize Study Tab

Alternatively, you can initialize the study mode environment using a Netview command:

`NETVIEW://ST/POPUP/SAVECASE?STATION=<Station>&ACTION=<InitializationMode>`

Where:

- STATION: Use this parameter to specify which station to show initially.
- ACTION: Use this parameter to initialize a nominal (INITNOMINAL) or real-time (INITREALTIME) study mode environment.

For example:

`NETVIEW://ST/POPUP/SAVECASE?STATION=3LIONS&ACTION=INITREALTIME`

5.2.1. Loading or Unloading Adjacent Stations

The Load or Unload Adjacent Stations button is enabled on the Initialize Study tab of the Study Case Setup dialog after the study case is initialized to nominal or real-time conditions.

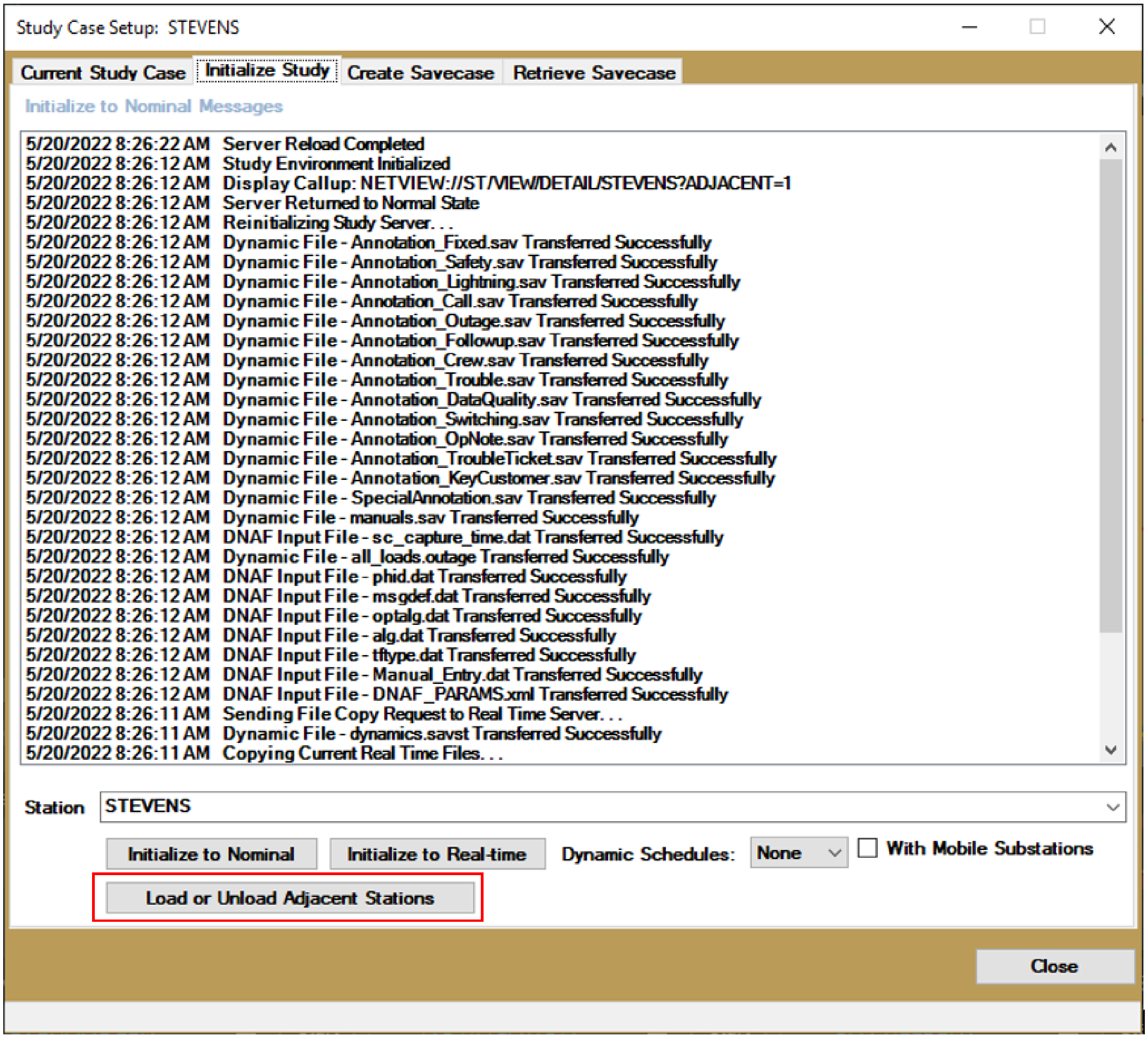


Figure 198. Study Case Setup - The Load or Unload Adjacent Stations Button

Clicking the Load or Unload Adjacent Stations button opens the Study Case Setup - Advanced Initialization popup, where you can create a study case with the required stations. The popup lists stations in a tree view format. The root station is the station that is selected in the Station field. Each level of the tree view lists the stations that are adjacent to the parent station. The tree view includes all stations in the system. The stations in the tree view are not repeated. Initially, stations that were already loaded during the study case initialization are selected. The popup also indicates whether the study case is initialized to nominal or real-time conditions.

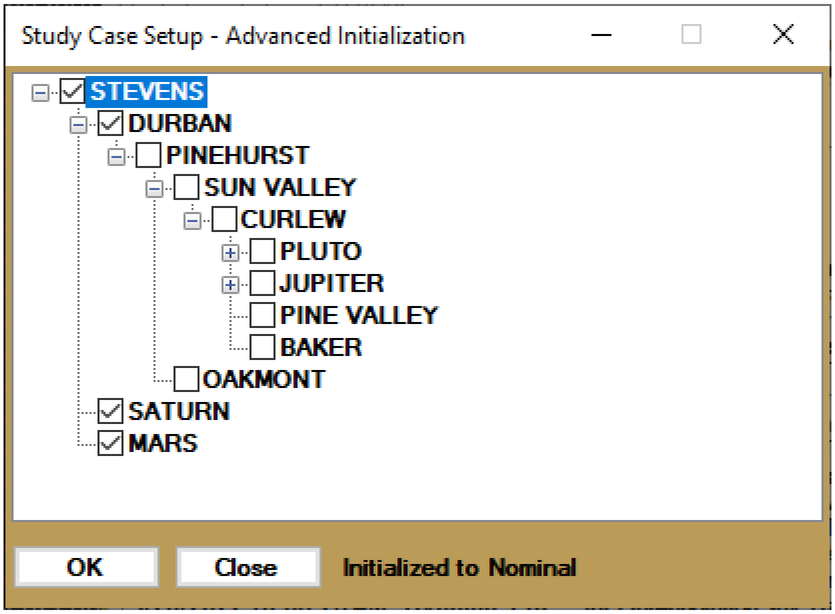
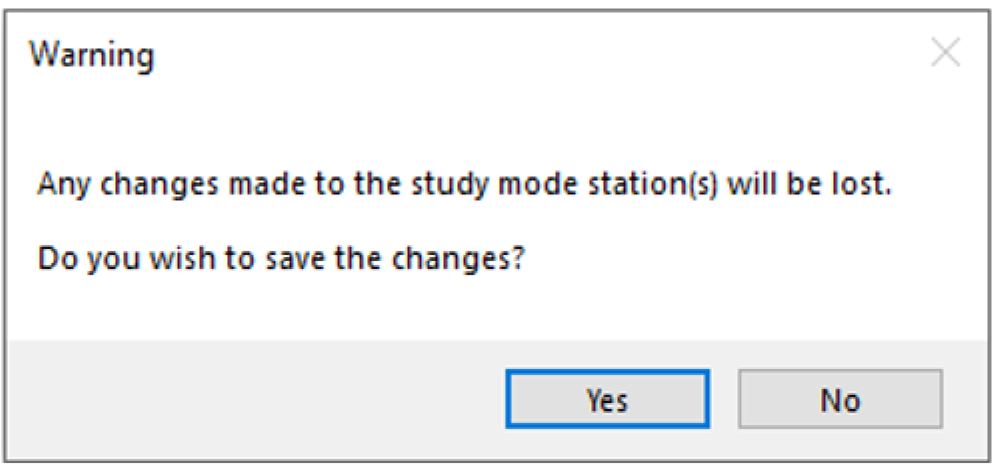


Figure 199. Study Case Setup - Advanced Initialization

To load or unload the required stations:

1. Open the Study Case Setup - Advanced Initialization popup.
2. Check or uncheck the checkboxes next to the required station names.
3. Click OK.



A warning appears asking you whether you want to save changes as a savecase.

- Click Yes to open the Create Savecase tab to create a savecase before re-initializing the study case with the selected stations. For more details, see Creating a Savecase.
- Click No to re-initialize the study case with the selected stations without saving the previous changes.

The study case is reloaded with the selected stations from the tree view, keeping the same initialization option (nominal or real-time).

5.3. Using Study Mode

Once study mode is running, the most important thing to remember is that the study mode model is running exclusively on the current workstation. Changes made to the study network do not affect the real-time system, and they are not visible to any other workstation on the network.

One of the few differences between study mode and real-time is that SCADA switches are operated in the same way as manual distribution switches in study mode.

You can also save a copy of the current study environment as a savecase. All the relevant files to create the network state are saved in an archive, where they can be recalled at a later time. This is done with the savecase creation and savecase retrieval functions on the study case setup form.

5.3.1. Creating a Savecase

Access this functionality from the Create Savecase tab on the Study Case Setup form.

To create a savecase:

1. Open the Create Savecase tab.
2. Complete the Name, Path, and Comment fields.
3. In the Include group, select the data to be saved.

   > **Tip**
   >
   > The Stations field shows the loaded stations.

4. Click Create Savecase.

   This creates an archive that contains all the files necessary to recall the study environment that you created in study mode.

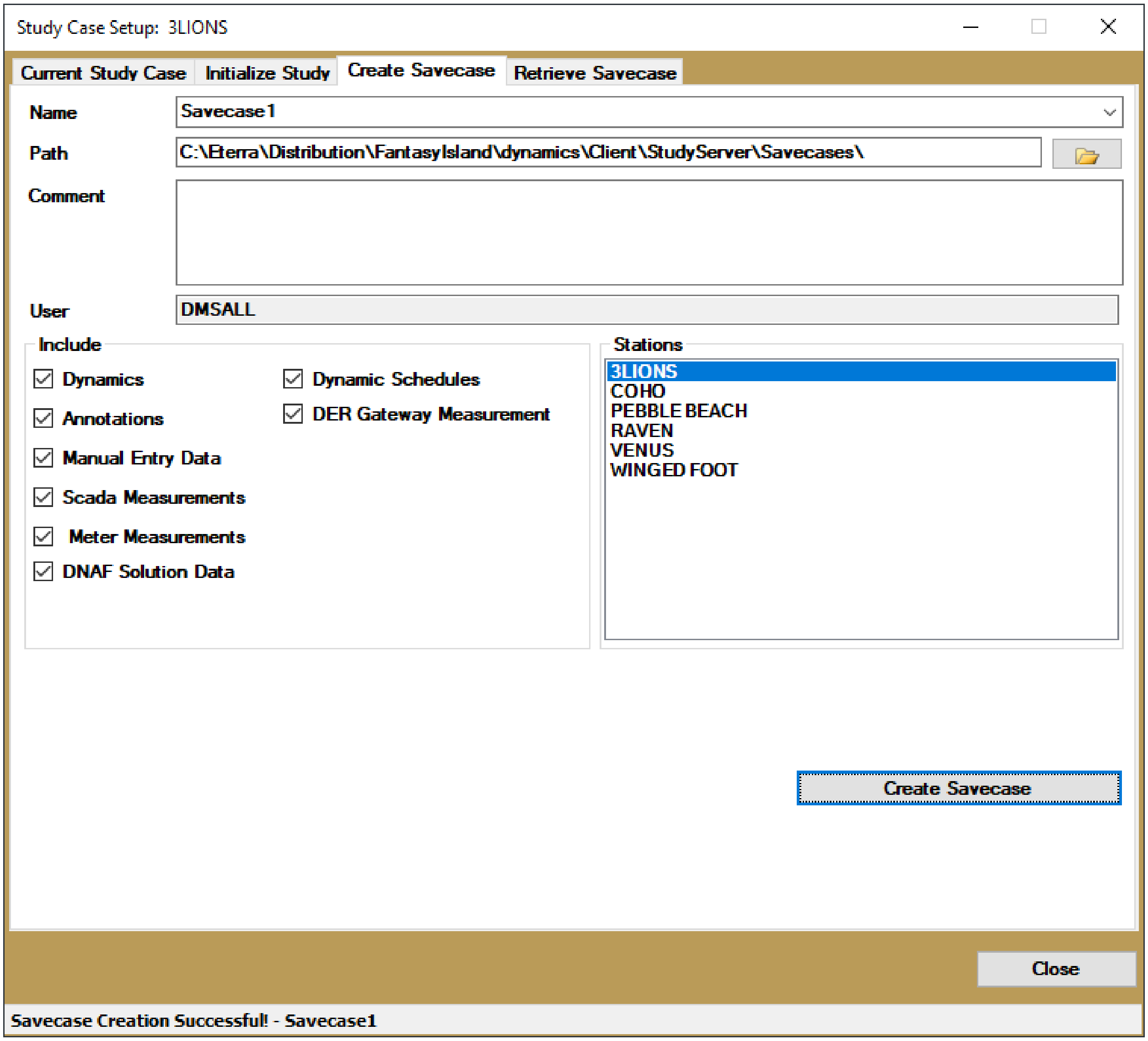


Figure 200. Study Case Setup - Create Savecase Tab

5.3.2. Retrieving a Savecase

Access this functionality from the Retrieve Savecase tab on the Study Case Setup form.

To retrieve a savecase:

1. Open the Retrieve Savecase tab.
2. Select the desired savecase using the Path and Savecase fields.
3. Click Retrieve Savecase.

   The savecase is retrieved.

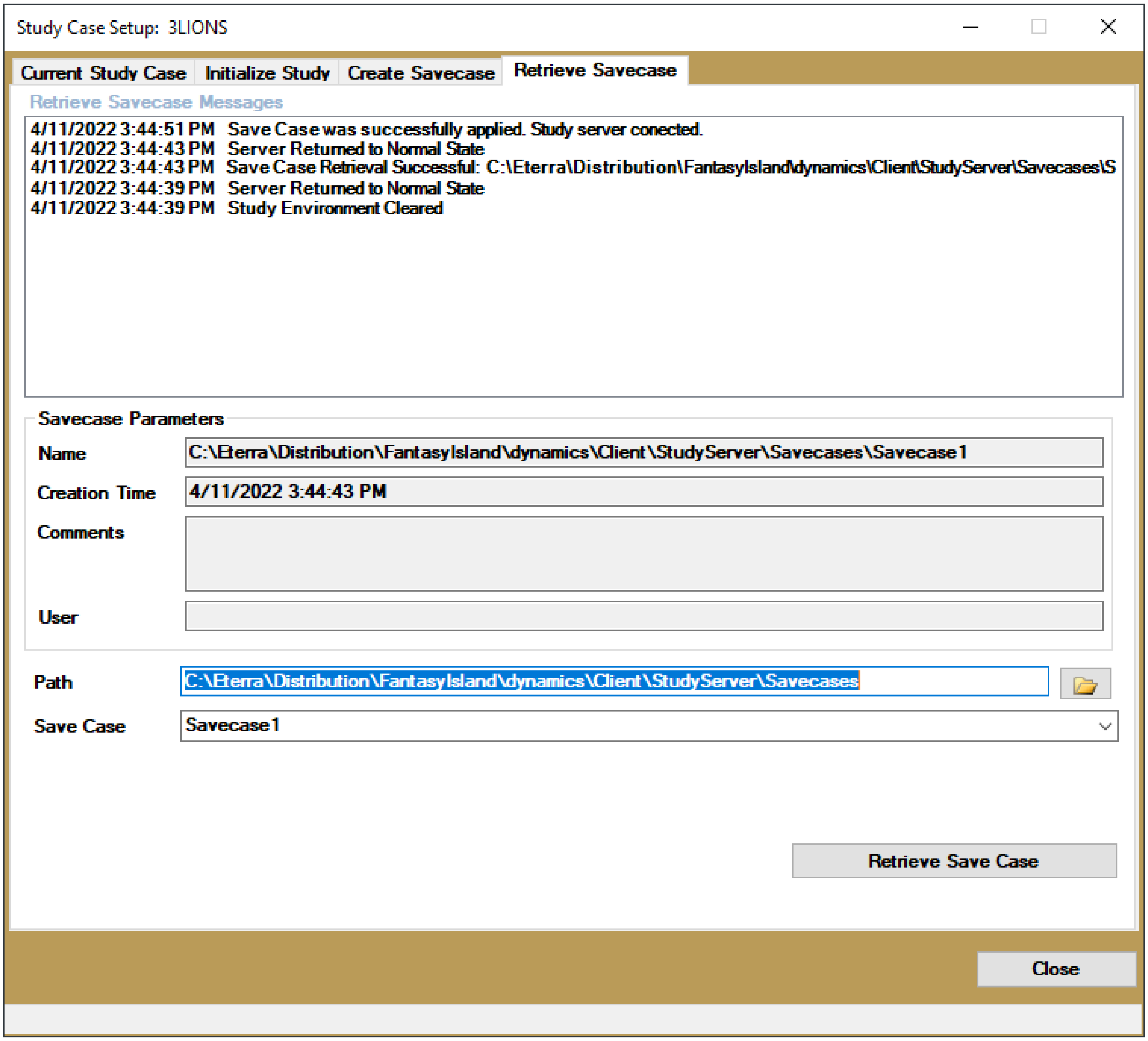


Figure 201. Study Case Setup - Retrieve Savecase Tab

> **Caution**
>
> Retrieving a savecase clears the current environment. Any changes made to the network in study mode are deleted. If you want to save these changes, you must create another savecase.

5.3.3. Running a Power Flow Analysis

Unlike real-time operations, changing the status of a switch or fuse in study mode does not automatically cause the distribution power flow to be run. In all cases, it is necessary to manually run the power flow.

To manually run a power flow, select a line segment or other device that is connected to the station of interest. Click the Run Network Analysis button on the device-specific toolbar to open the Study

DNAF Solution window, and proceed with initiating a power flow analysis.

A more-detailed description of running the power flow and other network analysis functions is given in Network Analysis and Optimization Functions.

6. Network Analysis and Optimization Functions

The Distribution Network Analysis Function (DNAF) provides a range of tools to allow distribution operators and engineers to monitor, control, analyze, optimize, and plan the operation of the distribution network.

The DNAF functions form an integrated package that can be run on either the real-time or the study mode models. The Distribution Network Object Model (DNOM) provides all of the information required by the DNAF functions. The results of the DNAF functions are also saved within DNOM.

The main DNAF functions are listed below. This chapter provides a brief outline of how to use the main functions and how to view the results.

For more information about the use, interpretation, and configuration of the DNAF functions, refer to the following:

- *Distribution Network Analysis Functions (DNAF) User Guide*
- *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst's Guide*
- *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst's Guide*

Table 9. DNAF Functions and Descriptions

Application	Description
Load Model and Estimation	Estimates the loads based on the current time, day type, and season.
Distribution Power Flow (DPF)	Calculates the voltages, currents, phase angles, real, and reactive power for each device in the specified portion of the network.
Limit Monitor (LIM)	Identifies overload and voltage violation conditions.
Bus Load Allocation (BLA)	Provides estimates for the loads (kW and kVAR) at the distribution feeder nodes.
Power Quality Index Evaluation (PQI)	Determines the extent of feeder voltage violations, including voltage imbalance.
Loss Analysis (LA)	Calculates instantaneous technical losses for conductors, transformers, and feeders.
Protection Validation (PRV)	Calculates the maximum and minimum short circuit current for a phase-to-phase or phase-to-ground fault at any selected point in the distribution network. The calculated fault currents are then used to verify that each segment of the feeder is properly protected.
Short Circuit Calculation (SCC)	Calculates single phase-to-ground, double phase-to-ground, phase-to-phase, and three-phase currents for each selected location (power transformer, line, and switch device). Available only in study mode.
Fault Location (FL)	Uses feeder breaker fault amp measurements and faulted phase status (target) measurements to determine potential fault locations along the distribution feeders.
Load and Voltage/VAR Management (LVM)	Recommends settings for capacitors, regulators, and transformers to achieve specified voltage and reactive flow objectives.

Application	Description
Fault Isolation and Service Restoration (FISR)	Generates a switching plan to isolate the nominated faulted section of the network and restore the un-faulted sections of the network that have been de-energized as a result of the fault.
Automated Feeder Reconfiguration (AFR)	Provides switching plans to optimally relieve overloads, voltage violations, and feeder imbalances.
Planned Outage Study (PLOS)	Provides switching plans to prepare for planned outages. This function runs in study mode only.
Fault Isolation and Service Restoration Contingency Analysis (FISRCA)	A study function that simulates faults on each switchable section and determines section contingencies for which not all customers can be restored. This function runs in study mode only.
Surgical Load Shed (SLS)	The load shed functionality is part of a combined DMS and SCADA application. The application can prioritize and shed circuits to obtain a desired target reduction in system load. The actual shedding and rotating of loads is controlled by the Rotating Load Shed (RLS) SCADA application controls. The DMS Surgical Load Shed (SLS) functionality controls configuration to enable or disable circuits for load shed and decide the order in which the circuits are to be shed. The functionality involves such components as throwover schemes, reserved capacity, and DG/DER installations.

6.1. Station Model DNAF Capability

ADMS is capable of simultaneously supporting station models that have DNAF capabilities both disabled and enabled. This is a common occurrence in systems that are still undergoing implementation.

The network models for stations that are considered complete and fully validated have the DNAF capability enabled, whereas other station models that are still under development have their DNAF capability disabled. This is a normal arrangement within ADMS; it does not affect the other functions of the system.

Determining which stations are DNAF-capable and which are not is most easily done by opening the Real-Time DNAF Solution pop-up. Assuming that the system has been running for more than a few minutes and a periodic DPF solution has been run, the stations that are not DNAF-capable are shown as "Unsolved". If an attempt is made to run a DNAF function on these stations, the status shows "Rejected – Station is not DNAF Capable".

Similarly, for devices belonging to a station that is not DNAF-capable, the output pop-up display shows blank DPF results when selected.

If a DNAF function that requires analyzing adjacent stations is run on a DNAF-capable station that is tied to a neighbor that is not DNAF-capable, the request is rejected with a warning message (see image below).

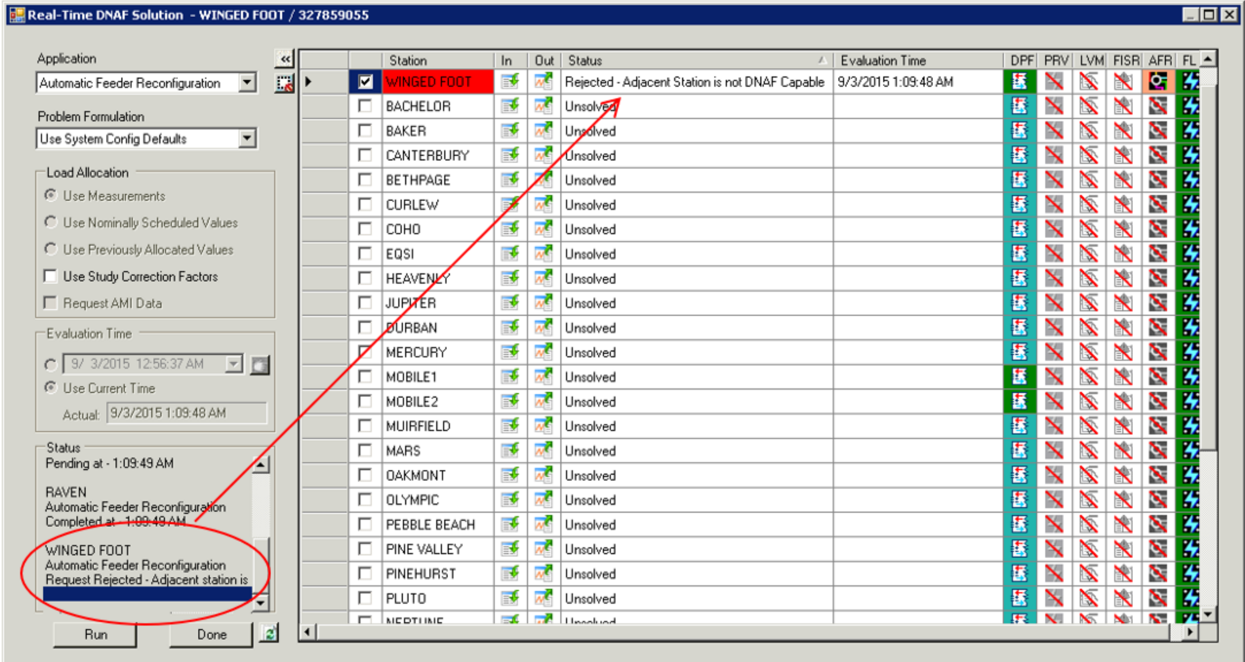


Figure 202. DNAF Warning - Station Is Tied to a Non DNAF Capable Neighbor

As the station models within the system are updated with DNAF-capable versions, the DNAF functions automatically become available.

6.2. Real-Time Power Flow

In real-time mode, many of the network analysis functions, including Distribution Power Flow, execute periodically or based upon user-defined events (for example, topology change). In this way, updated results are always available.

For example, event triggers include the following:

- **Topology Change:** A topology change includes a change in switching device status or the insertion/removal of a temporary modification.

  For a topology event, main and lateral triggers are separated. Substation internal equipment is considered to be part of the circuit main for processing topology event triggers.

- **Sudden Measurement Change:** A feeder head sudden measurement change trigger may also trigger the DPF application. This trigger includes a flow trigger (amps) and a voltage trigger.

  If the measurement change exceeds a user-configurable threshold (in amps for the flow trigger and percent volts for the voltage trigger), and if a specified time interval has elapsed since the last sudden measurement change solution, an application is triggered and solved. The elapsed time interval is used to prevent too many solutions from being triggered if there is an error in the measurement that is causing its value to change dramatically between measurement scans.

  > **Note**

> The SCADA sudden measurement change trigger currently applies only to those voltage and current measurements that are attached to the station internal objects (such as interface, station transformer, breaker, and feeder head). The trigger does not apply to the measurements of any device outside of the station internals (such as reclosers and sensors).

The results of the Distribution Power Flow are most easily viewable on the DPF Results display or on a device output pop-up display (see Attribute and Output Displays).

For more information about the operation of the DPF function, refer to the *Distribution Network Analysis Functions (DNAF) User Guide*.

6.3. Running Network Analysis

The DNAF Solution pop-up display allows you to manually invoke a network application. Depending on the system configuration, different applications can be available in real-time mode or study mode (for example, Planned Outage Study is a study-mode–only application).

The Network Analysis applications can be run from either real-time or study mode. The mode is selected by starting from either a real-time or a study mode display. When in study mode, all solutions must be manually requested by the user.

You can invoke a DNAF application directly from the network view by right-clicking a line and selecting one of the Network Analysis options. The solution status is shown in the DNAF toolbar.

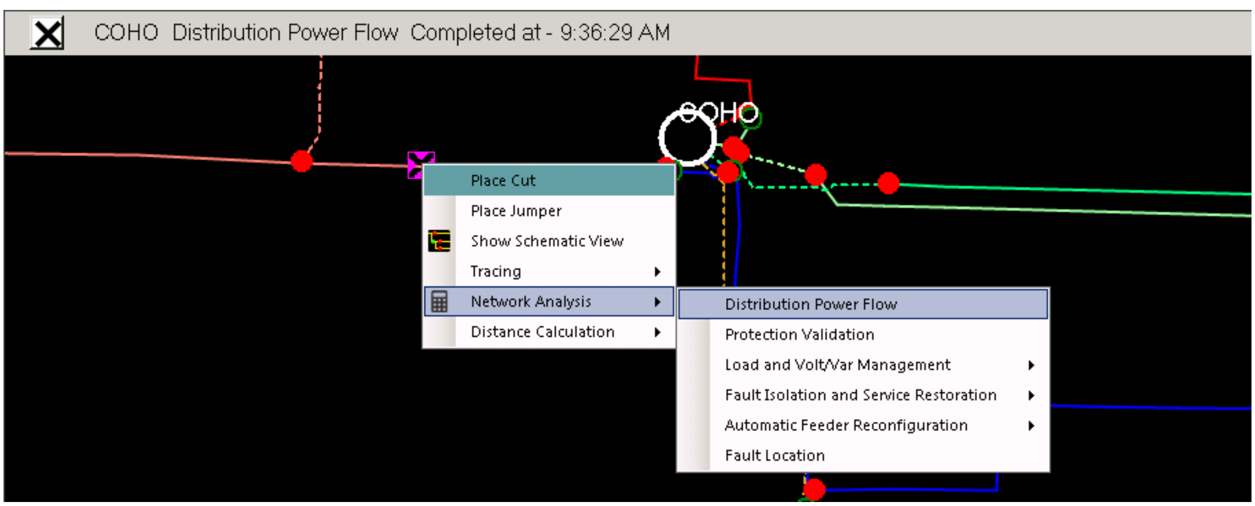


Figure 203. Running DPF from the Network View

To manually invoke a DNAF solution using the DNAF Solution pop-up:

1. On a geographic network diagram, select a substation, device, or line section that belongs to the required station.
2. On the device-specific toolbar, click the Run Network Analysis button.

   This displays the DNAF Solution pop-up, as shown in image below. The selected station is checked in the Station Status list. It is not normally necessary to select any other stations, as the model automatically takes into account the effect of other feeder tie switches and neighboring stations.

3. Click Run.

The solution status is shown in the Status box.

The study mode and real-time DNAF Solution pop-ups are differentiated by the background color of the display. Typically, study mode is yellow and real-time is gray.

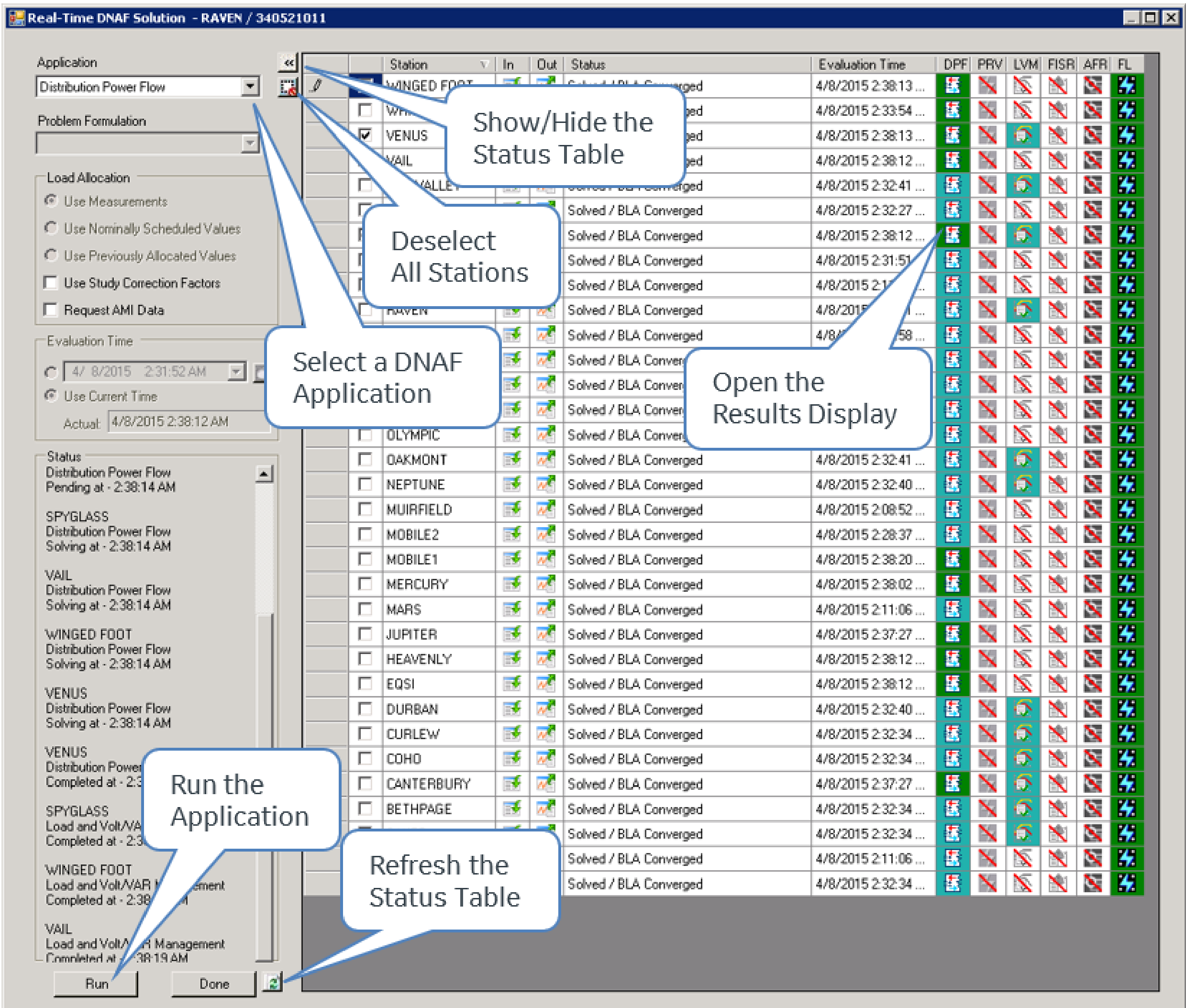


Figure 204. Real-Time DNAF Solution Window

By default, the Distribution Power Flow application is selected. If required, select another DNAF application and problem formulation from the pull-down menus. The status of the Station model is updated, indicating the last time that this station was solved for this application. "Problem Formulation" name is limited to a maximum of 38 characters. If it exceeds this limit, the additional characters will not be displayed.

In real-time mode, the Load Allocation options are set automatically. In study mode, the Load Allocation options can be set as required.

- The Use Measurements option incorporates all valid analog measurements in the load allocation. If no valid measurements are available in study mode, this option is not available.
- If the Use Nominally Scheduled Values option is selected, the load allocation utilizes the nominal customer load schedules for setting the demand values (kW and kVAR).
- If the Use Previously Allocated Values option is selected, the load values that were determined from a previous power flow solution are used. This option is commonly used when performing switching studies (after the loads have been allocated, based on an earlier measurement set using the Use Measurements Load Allocation option).
- If the Use Study Correction Factors check box is selected, the study load correction factors that are calculated and persisted in the real-time power flow are applied during the load allocation. If the Use Previously Allocated Load Values option has been selected, this option does nothing.
- If the Request AMI Data check box is selected, the meter measurement data are used. This option is only available for DPF and LVM applications.

The Evaluation Time option allows setting the time for which the analysis is run. The solution uses this time and data to determine the allocated loads, based on the defined load profiles in the network model. The default setting is Current Time, which means that the solution is based on the conditions that currently exist.

To manually set a date and time, click the upper radio button, and then click the button to the right of the Date/Time field to toggle between the data picker and the up/down control.

To start the analysis, click the Run button; the status of the solution is shown in the Status panel. When the analysis is complete, the Station Status list is updated. The icon for the station and function is updated, to show that a solution is available; to navigate to the appropriate tabular display of the results, click the icon.

To dismiss the DNAF solution pop-up, click the Done button.

6.4. Station Output Summary

The Station Output Summary display provides navigation to DNAF results for a particular station. For example, DPF results for a station can be accessed from this display.

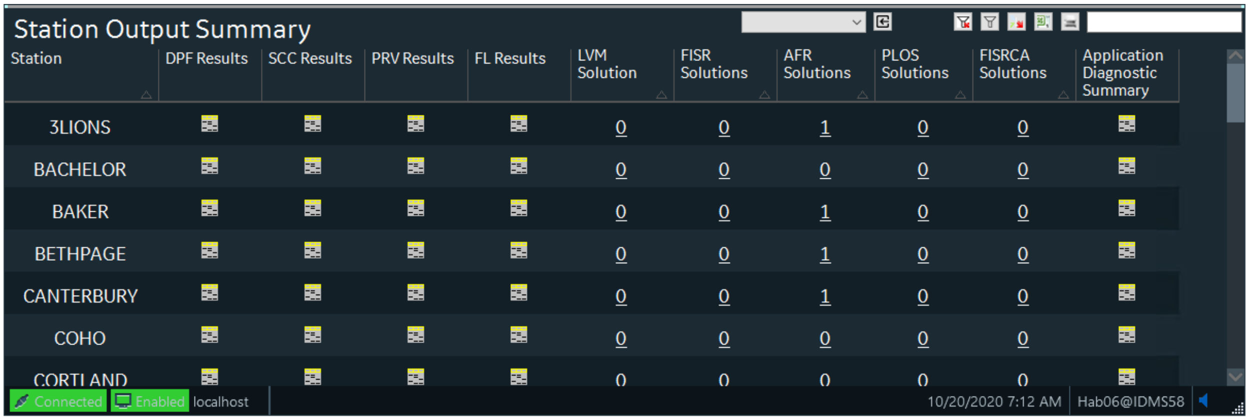


Figure 205. Station Output Summary Display

The following information is shown on the Station Output Summary display:

- **Substation:** The list of stations for which the user currently has operational responsibility. In the case of study mode, it is all the stations currently loaded in the Study server.
- **DPF Results:** Click this icon to navigate to the Distribution Power Flow Summary display for this station (see Distribution Power Flow Summary Display).
- **PRV Results:** Click this icon to navigate to a summary of protection validation results for breakers, reclosers, and fuses depending on the application configuration.
- **FL Results:** Click this icon to navigate to a summary of potential feeder fault locations.
- **LVM Solutions:** This column indicates the number of Load and Volt/Var Management plans that have been generated for each station. A "0" indicates that no plans exist. Click the number to open the LVM Plan Control Moves summary for this station.
- **FISR Solutions:** This column indicates the number of Fault Isolation and Service Restoration plans that have been generated for each station. A "0" indicates that no plans exist. Click the number to open the FISR Results summary for this station.
- **AFR Solutions:** This column indicates the number of Automatic Feeder Reconfiguration plans that have been generated for each station. A "0" indicates that no plans exist. Click the number to open the AFR Results Summary display.
- **Application Diagnostic Summary:** Click this icon to navigate to the DNAF application results summary for this station.

> **Note**
>
> In Study Mode, this display also includes buttons to navigate to SCC, FISRCA, and PLOS results.

6.4.1. Distribution Power Flow Summary Display

The DPF Results display (see image below) provides high-level DPF output for the substation. It can be accessed by clicking the DPF Results icon on the Station Output Summary display, or by selecting it directly from the menu. It contains three tables giving the following information:

- DPF results for the substation with links to detailed displays
- Power flow results for each feeder

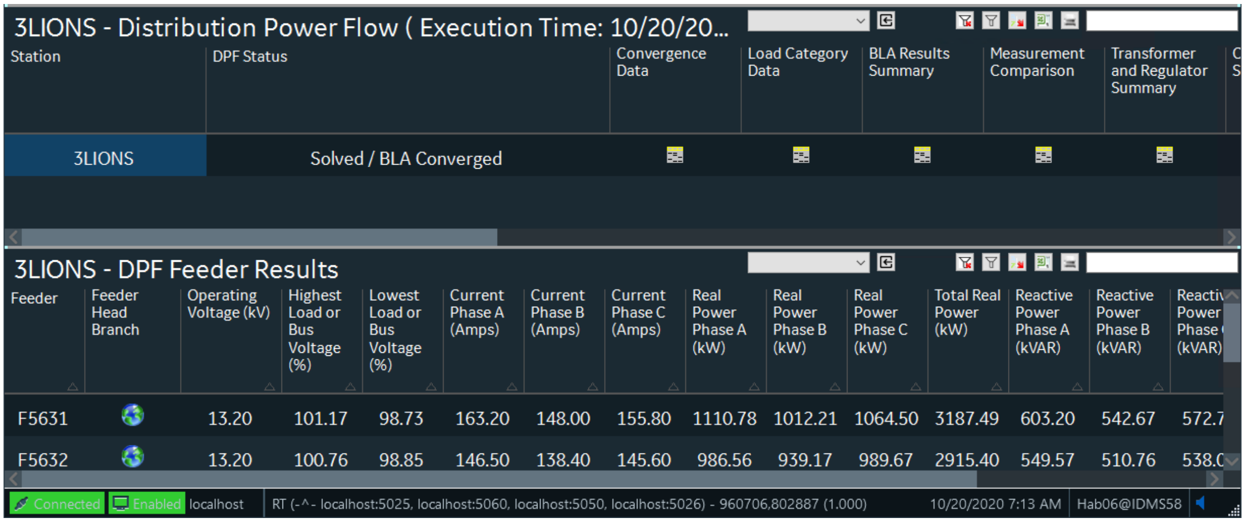


Figure 206. DPF Substation Results Display

For descriptions of all DNAF tabular displays, refer to the *ADMS Display Reference Manual*.

6.4.2. Parameter Entry Displays

The Network Analysis functions allow the user to modify input values used by the Network Analysis solution.

For this purpose, a set of data entry tabular displays is available. These displays are accessed from the Application Input Summary display. The example shown in image below is the Feeder Input display, which is part of the Station Feeder Input display. These displays show the currently entered nominal values and provide for override values to be entered. After an entry has been typed in, select the Enable check box for the override values to be active. To save the newly entered values to the database, press the Enter key.

The Manual Input Parameter Entry display allows an operator to manually set parameters for the whole system, or system sub-elements: service centers, stations, power transformers, and feeders using the tree-like structure.

Device-specific parameter entry displays can also be accessed by selecting a component from the geographic view and clicking the Enter Parameters button on the device-specific toolbar.

For a complete explanation of the use of data input parameters for DNAF solutions, refer to the Manual Inputs section in the *Distribution Network Analysis Functions (DNAF) User Guide*.

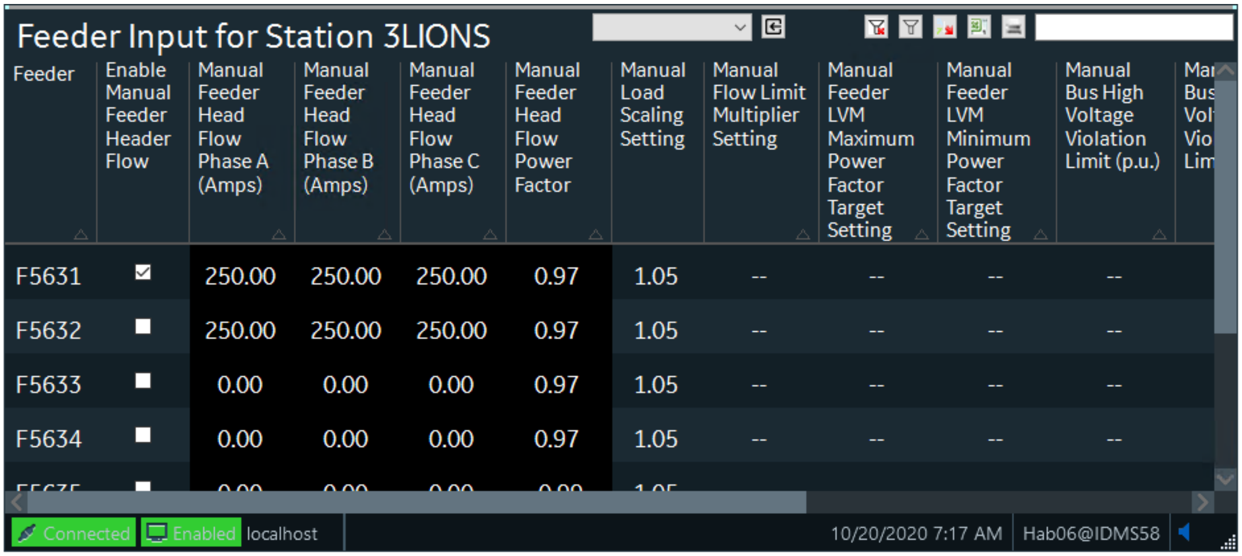


Figure 207. Example of DNAF Data Input Display

6.5. FISR Operation from a Fault Location

This section describes the direct initiation of a Fault Isolation and Service Restoration (FISR) solution and the automatic generation of a switching order in the case of a feeder trip. It applies to real-time operation only.

When a feeder protection trip occurs, the DNAF fault location function is triggered. The fault location function uses the fault current value from the protection relay, if available, and any telemetered fault indicators, if available, to determine the likely points along the feeder where the fault can be located.

After a feeder trip event, selecting the Fault Halo options on the common toolbar shows the fault indicator halos and halos that were generated by the fault location function. Due to the nature of the fault location calculation, a number of zones of the feeder may be shown with halos. The most likely location of the fault is where there is an overlap of the fault location and fault indicator halos.

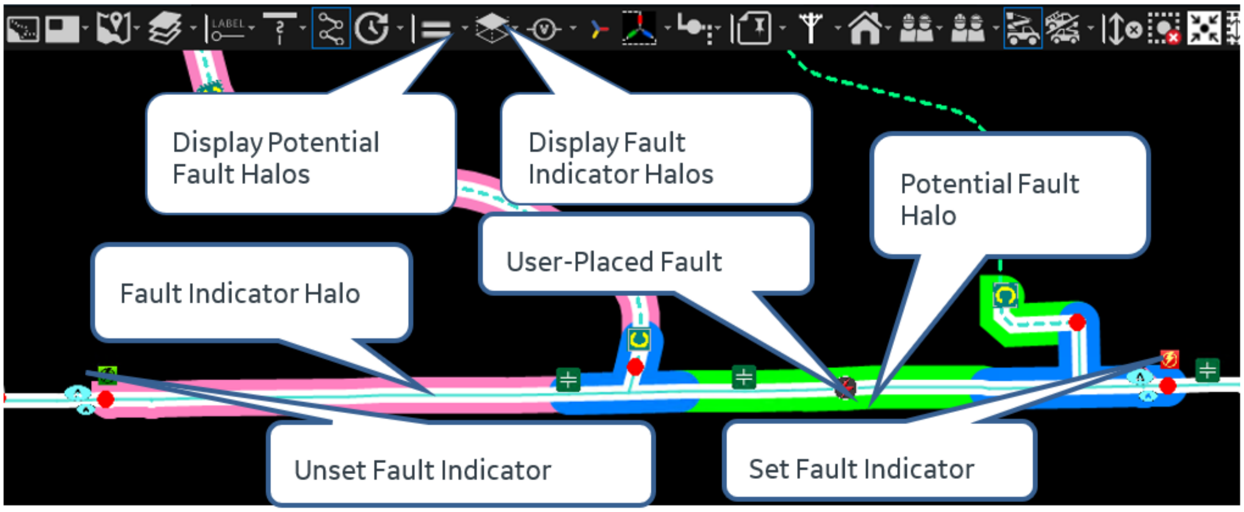
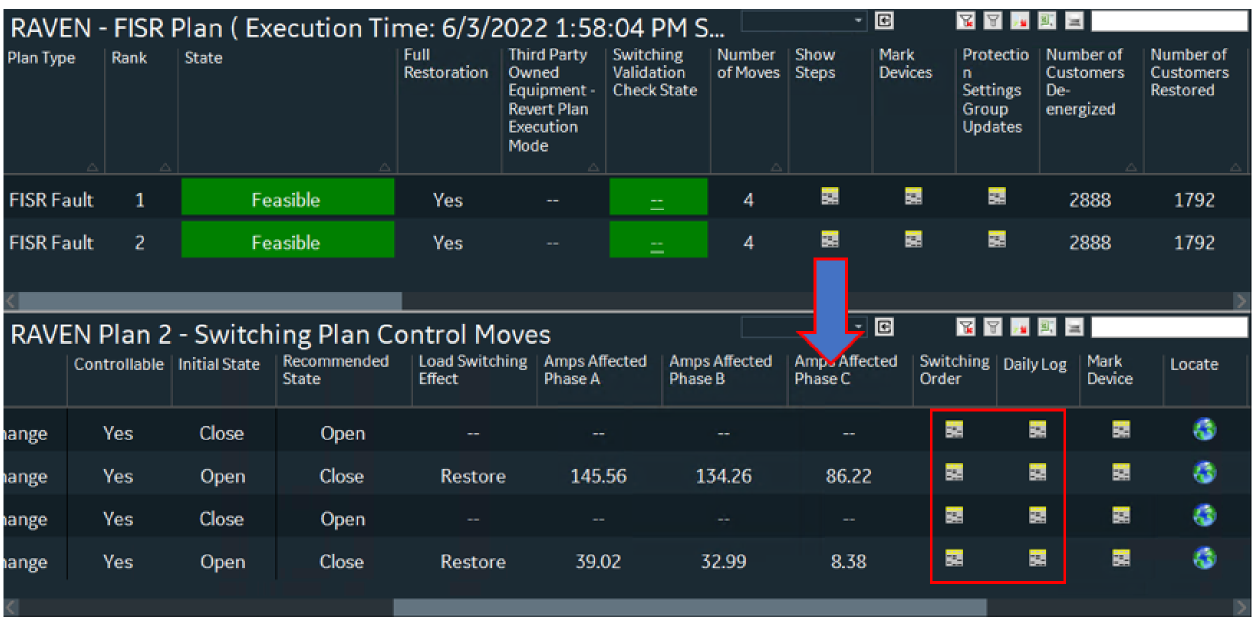


Figure 208. Display of Fault Location Halos and the Likely Position of the Fault

Once the most likely position of the fault has been determined (either from the halos and/or other observations) the position on the line is selected and a fault symbol is placed by the Fault Isolation and Service Restoration application and the application executes. Typically, FISR is configured to search for a SCADA switching-only solution, so that it can be applied immediately without the need for field crews.

Once the FISR solution has been generated (typically 30–40 seconds), select the fault symbol and click Navigate to FISR Plans. One or more plans are normally shown.


RAVEN - FISR Plan (Execution Time: 6/3/2022 1:58:04 PM S...

Plan Type	Rank	State	Full Restoration	Third Party Owned Equipment - Revert Plan Execution Mode	Switching Validation Check State	Number of Moves	Show Steps	Mark Devices	Protection Settings Group Updates	Number of Customers De-energized	Number of Customers Restored
FISR Fault	1	Feasible	Yes	--	--	4				2888	1792
FISR Fault	2	Feasible	Yes	--	--	4				2888	1792

RAVEN Plan 2 - Switching Plan Control Moves

	Controllable	Initial State	Recommended State	Load Switching Effect	Amps Affected Phase A	Amps Affected Phase B	Amps Affected Phase C	Switching Order	Daily Log	Mark Device	Locate
ange	Yes	Close	Open	--	--	--	--				
ange	Yes	Open	Close	Restore	145.56	134.26	86.22				
ange	Yes	Close	Open	--	--	--	--				
ange	Yes	Open	Close	Restore	39.02	32.99	8.38				

Figure 209. FISR Plan Showing the Steps

A summary of the end result of executing each plan is shown in the table, in particular the number of customers not restored and the power demand not restored (unserved kW). The details of each plan can be viewed by clicking on the Show Steps icon.

Once the most appropriate plan has been determined, click the Switch Order or Daily Log icon for any step of the selected plan to automatically generate a formal switching order or add the switching steps to a daily log. You can then execute the switching order or daily log in accordance with standard switching procedures.

6.6. Planned Outage Studies

This section describes how to use Planned Outage Studies (PLOS) to generate switching plans for the isolation of network devices.

The study model is normally initialized to real time to establish an environment that matches the status of the network when the operation takes place. Additional switching can also be manually applied to the network, to reflect other expected conditions.

1. On the Study Mode view, select the device to be taken out of service, and open the DNAF Solution pop-up.

   Note that the ID of the selected device is shown in the title bar of the pop-up.

2. Select the Planned Outage Study application and the required problem formulation.

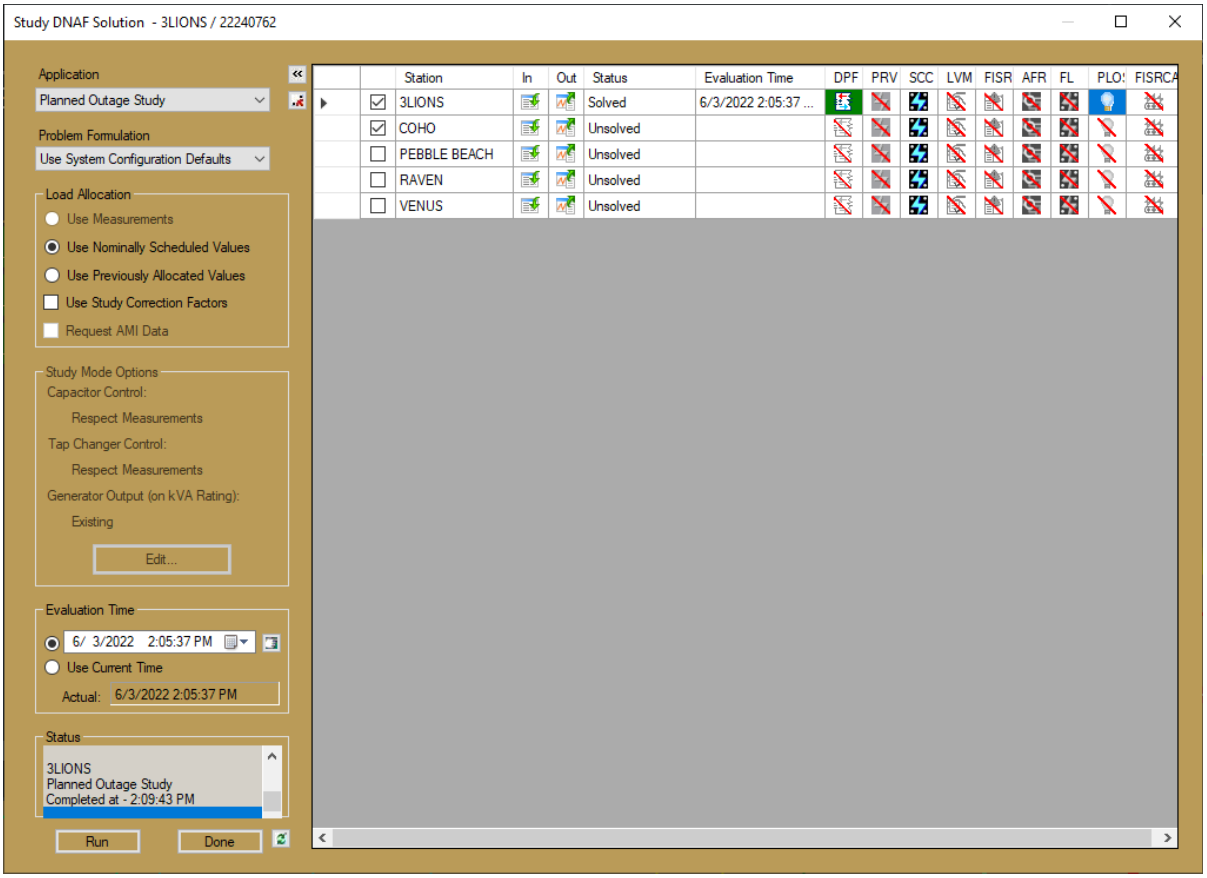
   Typically, the "Minimum Unserved kW (Auto+Manual)" problem formulation is the most appropriate.

3. Select the required Load Allocation options.

   If the study model has been initialized from real time and if the Use Measurements option is available, this uses the valid measurements derived from the real-time environment.

4. Enter the required evaluation time, which is the time when the isolation switching is expected to take place.
5. Click the Run button.

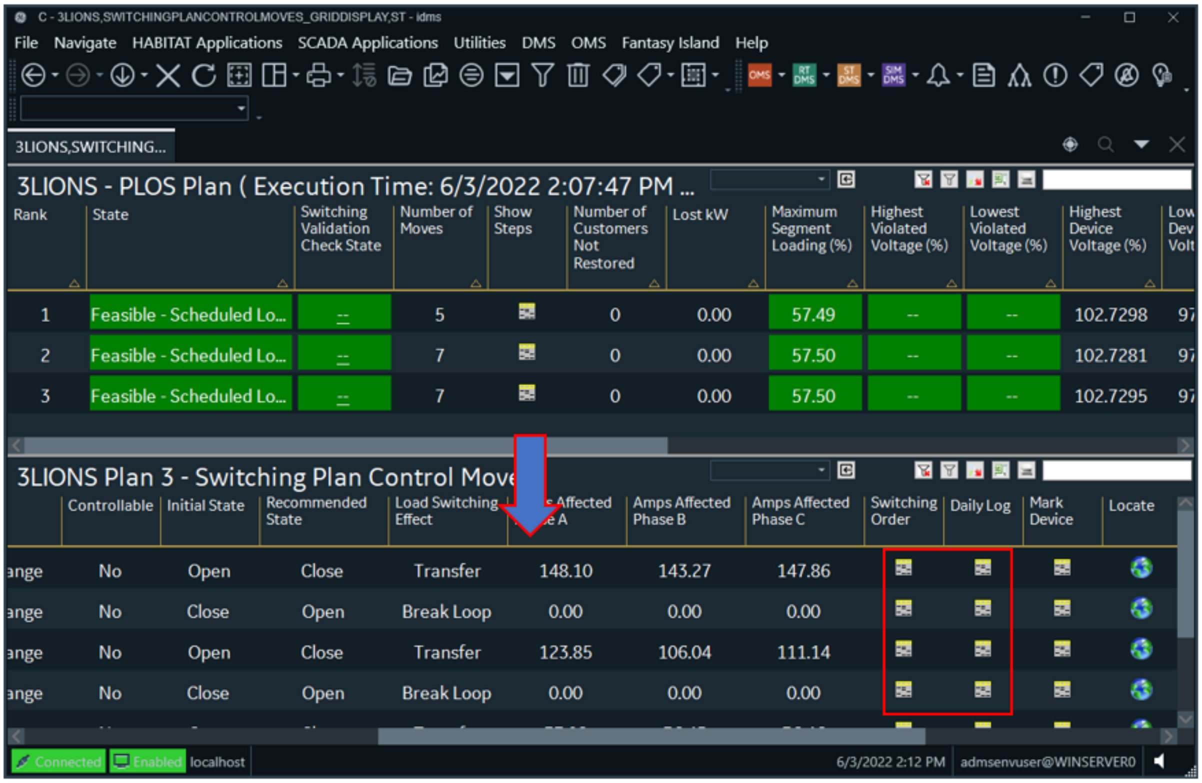
   The status panel shows when the application has completed. On completion, the Station Status summary refreshes and the PLOS icon for the selected station updates, indicating that a solution is available.



6. Click the icon to open the PLOS plan summary.
7. On the DNAF pop-up, click Done to dismiss the pop-up.

   The PLOS Plan display (see the figure below) normally shows a number of plans. Each of these plans is viable, but there are variations in the number of customers and power demand affected due to the different switching.

8. To view the individual switching steps for each plan, click the Show Steps icon.



9. Once the most appropriate plan has been selected, click the Switch Order or Daily Log icon for any step to automatically create a formal switching order or add the switching steps to a daily log. This switching order or daily log can then be executed against the study model to verify that the expected results are achieved.

7. Quick Guide

Table 10. Basic Navigation Functions and Shortcut Keys

Function	Action
Select a Device	Click the left mouse button
Deselect a Device	Click the left mouse button in an empty area
Mark a Device	Click Shift+ left mouse button
Re-center the Display	Click the scroll wheel to center the display on the current cursor position
Zoom In	F4 or Scroll Up
Zoom Out	F5 or Scroll Down
Return to Original Zoom Level	F3
Rubber-Band Zoom	Shift + scroll wheel – click and drag
Pan	Left-click the display background, hold down the mouse button, and drag the cursor in the direction you want to pan. When the mouse button is released, the display is repositioned corresponding with the cursor shift.
Navigational Overview	F6 or Ctrl+N: Displays the navigational overview for the current partial model view. Clicking a point on the navigational overview re-centers the main display on that point.
Screen Annotation	Ctrl+A: Places a screen annotation at the current cursor position.
Find	Ctrl+F or the Find Tool button on the common toolbar
Reload Partial Model View	Ctrl+R or the Load Surrounding Substations (Update Partial Model View) button on the common toolbar
Station Exception Display	Ctrl+S
Station Output Summary	Ctrl+O
Substation List Display	Ctrl+U
Show Geo Schematic View	Ctrl+H
Show Geographic View	Ctrl+G
Create Planned Outage	F7 or the Create Planned Outage button on the device toolbar.
Activate Move Mode	Ctrl+M
Show Picture-in-Picture View	Ctrl+P
Increase/Decrease Label Font Size	Ctrl+Shift+Scroll Up/Down
Show the Switching Order Application	Ctrl+W
Copy the URL of the Current Network View to Clipboard	Click an empty area in a network view display and press Ctrl+C
Copy Grid Tabular Cell Contents to Clipboard	Click a cell or select a cell range in a grid tabular display and press Ctrl+C
Copy Attribute Display Cell Contents to Clipboard	Click a cell in an attribute display and press Ctrl+C
Dynamically Update the Viewer Configuration	Alt+A or F9

Function	Action
Clear Marks	Alt+B or Ctrl+B
Open the Real-Time DNAF Solutions Display for the Closest Object	Alt+F2 or Ctrl+F2
Open the Real-Time DNAF Solutions Display for the Substation	F12
Highlight an incident on the Incident Detail display	Ctrl+click an incident on the network view
Turn on the CNV mode	CTRL+R
Turn off the CNV mode	ALT+CTRL+R

8. Partial Model View - How It Works

To handle potential memory/performance problems, client workstations only load the stations that are directly related to the view that is requested. In the station overview, this subset of stations appears in color.

The image below shows the station overview, showing stations currently loaded in a partial model view.

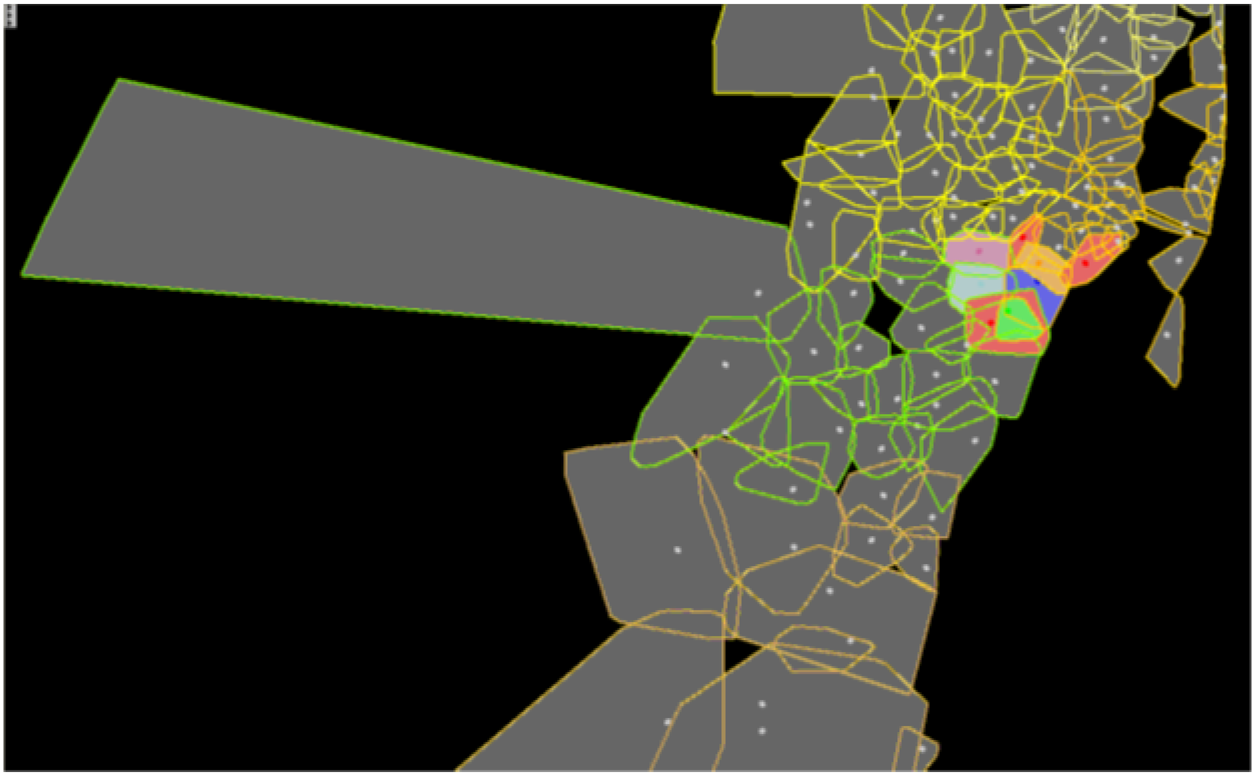


Figure 210. Station Overview

Consider a nominal model with a geographical layout, as shown in image below. Each station is represented by a circle with a letter, and these stations geographically touch each other and also have electrical interconnections.

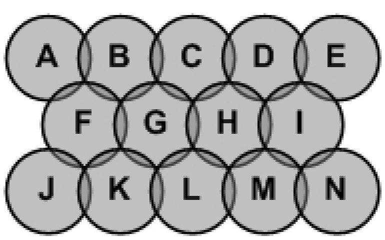


Figure 211. Nominal Model

Assume that there is a navigation overview with the same geographical shape as the nominal model. When the user clicks on this navigation overview (or specifies the display center using any other method), this causes the proper stations to be loaded into memory. The term "proper stations" means all of the stations whose geographical boundaries intersect with the location, plus all of the stations that directly tie electrically with the station(s) from this initial selection.

For example, if the user clicks station G in the overview:

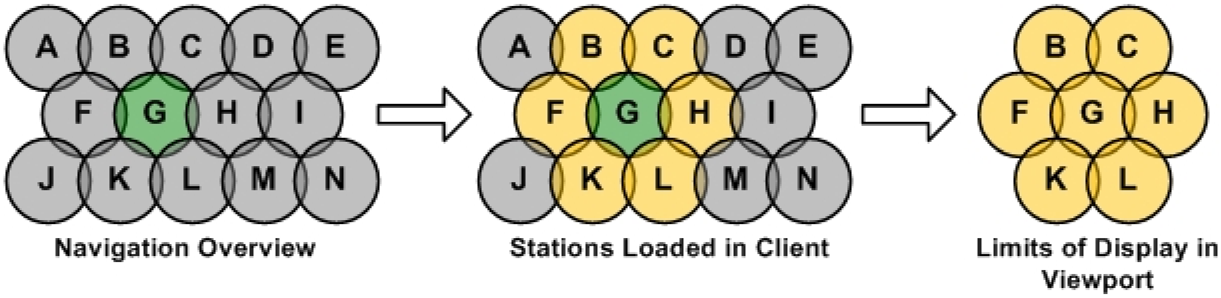


Figure 212. Partial Model View Example (One Level of Adjacency)

The display appears to be centered on the selected point, and the view of the display only includes the highlighted stations. The user can quickly pan around according to the stations that are loaded in this view. So, stations B, C, F, G, H, K, and L are loaded into memory as the first level of adjacency.

Normal panning ceases at the edge of this Partial Model View (PMV). Zooming out also functions as if the stations making up the PMV are all that exist. Thus, no additional stations are loaded even when the zoom level gets to a point where the PMV uses only a fraction of the screen.

To "pan" outside of this partial view, the user has to reload the view by selecting another station on the Station Overview display, or use the re-center function (Load Surrounding Substations (Update Partial Model View) button on the common toolbar).

In addition to the re-center/re-load mechanism, several other navigational means can be used to load additional stations into the partial model view. For example, the find feature can be used to navigate to a particular switch anywhere within the whole model, a particular station can be called up, a specific room can be requested, the user can navigate somewhere from an alarm, so on.